

# Topological Fixed-Point Theory (TFPT)

A discrete compiler for the constants of physics

Two inputs — the boundary constant  $c_3 = \frac{1}{8\pi}$  and a five-slot carrier — build  $E_8$   
and read off the Standard Model: status assessment and reading guide

Stefan Hamann

Alessandro Rizzo

June 18, 2026 · v5.2 (rev 190)

## In one sentence

**Topological Fixed-Point Theory (TFPT)** constructs, from exactly **two inputs** — a boundary constant  $c_3 = \frac{1}{8\pi}$  (P1) and a five-slot carrier  $g_{\text{car}} = 5$  (P2) — a discrete compiler for the Standard-Model skeleton: the gauge group, three families, hypercharges and the flavor matrix, with  $E_8$  ( $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ ) as the consistency checksum. The *algebraic core is machine-checkable* and the dimensionless constants follow as fixed points ( $\alpha^{-1} = 137.0359992$ ); *physical readouts* — absolute scales, masses, inflation, gravity and cosmological transfers — run through explicitly *named, status-typed bridges* ( $v_{\text{geo}}$ ,  $G_{\text{net}}$ ,  $F_{\text{transfer}}$ ), not as free outputs. The honest one-line claim is on the status card below.

## The TFPT document set — what is treated where

**Plain language:** TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant  $c_3 = \frac{1}{8\pi}$  and the carrier rank  $g_{\text{car}} = 5$  — build  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and three companions**, best read in order:

0. **introduction** — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). (*entry point*)
1. **tfpt\_1\_architecture\_e8** — the two axioms, the derivation map, the EM fixed point  $\alpha^{-1}$ , the  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  construction, the QBL theorem chain.
2. **tfpt\_2\_standard\_model** — the SM in one  $\varphi_0$ -ladder formula, flavor from parabolic transport (sheet diamond, dual anchor,  $Q_+$  cohomology), the worked closures.
3. **tfpt\_3\_e8\_audit\_bootstrap** — the seven  $E_8$  slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. **tfpt\_4\_frontier** — honest status of  $\eta_B$ ,  $m_p/m_e$ , Koide, dark matter, full QG.
5. **tfpt\_5\_redteam** — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

**Companions:** **R.** **tfpt\_research\_contracts** — the open interfaces ( $v_{\text{geo}}$ ,  $G_{\text{net}}$ ,  $F_{\text{transfer}}$ ) as numbered contracts; **H.** **tfpt\_horizon\_readouts** — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** **origin\_theory** — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation).

**You are reading the introduction — the entry point and reading guide.**

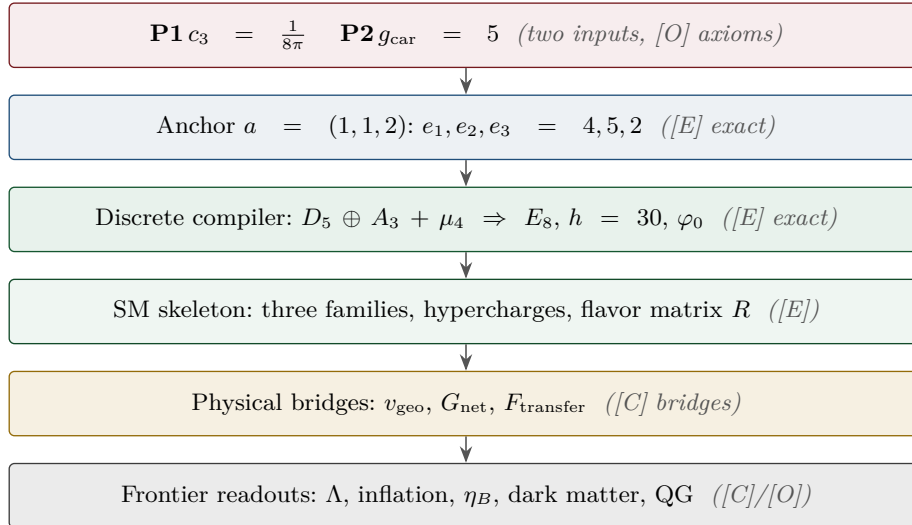
**TFPT status at a glance — read this card the same way in every document**

**Two inputs, one compiler.** From the boundary constant  $c_3 = \frac{1}{8\pi}$  (axiom P1) and the five-slot carrier  $g_{\text{car}} = 5$  (axiom P2) — jointly the elementary symmetric data of one anchor  $a = (1, 1, 2)$  — the discrete compiler  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ( $\alpha^{-1} = 137.0359992$ ) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

**What is closed, and what genuinely remains.** The discrete/algebraic layer is closed (**[E]**); the everything-open residual reduces to *three named interface problems*:

| Interface             | Question                                                                                        | Status                  |
|-----------------------|-------------------------------------------------------------------------------------------------|-------------------------|
| $v_{\text{geo}}$      | the one metrology unit ( $= 1/\sqrt{G} = m/\mu$ ); No-Unit Thm: no compiler scale               | primitive <b>[O]</b>    |
| $G_{\text{net}}$      | simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1, 1) \rangle \cong (E_8)_1$ | <b>[E]</b> / <b>[C]</b> |
| $F_{\text{transfer}}$ | one functor, four typed interfaces (Koide, $\eta_B$ , axion, $m_p/m_e$ )                        | <b>[C]</b>              |

Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins.** The historical labels ( $U_{\text{wall}}, G_{\text{metric}}, F_{\text{frontier}}$ ) are retained only for ledger continuity — the live residual is  $v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$ . Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

**[E]** exact / machine-proven**[C]** conditional**[O]** open / axiom**[X]** kill test

## Contents

|          |                                                                          |           |
|----------|--------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>What this is about: the jump in construction principle</b>            | <b>4</b>  |
| <b>2</b> | <b>The architecture at a glance</b>                                      | <b>6</b>  |
| <b>3</b> | <b>The dependency DAG and the proof ledger</b>                           | <b>7</b>  |
| <b>4</b> | <b>The compiler: why <math>30 = 2 \cdot 3 \cdot 5</math></b>             | <b>11</b> |
| <b>5</b> | <b>The <math>\mu_4</math> glue: how <math>E_8</math> is really built</b> | <b>12</b> |
| <b>6</b> | <b>How derivation now works: the derivation map</b>                      | <b>12</b> |

|    |                                                                     |    |
|----|---------------------------------------------------------------------|----|
| 7  | Physical, geometric, topological: why the architecture is plausible | 14 |
| 8  | What role $E_8$ now plays exactly                                   | 14 |
| 9  | What this means for the theory-of-everything question               | 15 |
| 10 | Does this make the theory more probable?                            | 18 |
| 11 | Predictions for running and upcoming experiments                    | 18 |
|    | Conclusion                                                          | 20 |
|    | Appendix: computational verification                                | 20 |
|    | Appendix: changelog                                                 | 58 |

**Reading the markers.** Claims carry one of *four* reader-facing classes, so one can see at a glance what is proven and what is conditional: **[E]** *exact / machine-proven* — an exact identity (e.g.  $240 = 16 \cdot 5 \cdot 3$ ), a Lie/lattice theorem (e.g.  $D_5 \oplus A_3 + \mu_4 = E_8$ ), a Lean/script formalisation, or a reproduced numerical fixed point; **[C]** *conditional* — follows under explicitly named hypotheses (a physical bridge, readout or pending step); **[O]** *open / axiom* — a declared input or a genuine gap; **[X]** *kill test* — a falsifiable threshold. The finer per-claim type (Axiom / Formal / Lattice / Numerical / Identity / Physical) is kept in `verification/status_ledger.csv`, the single source of truth.

|                                    |                                              |
|------------------------------------|----------------------------------------------|
| Physical — under hypotheses        | RG masses, QG    QG.AMB.01                   |
| Numerical — fit-free fixed point   | $\alpha^{-1}, \Lambda, \theta_{12}$ EM.FP.01 |
| Lattice — Lie / glue / root system | $E_8 = (D_5 \oplus A_3) + \mu_4$ E8.GLU.01   |
| Formal — Lean / exact script       | P2 algebra    FORM.P2.02                     |
| Axiom — declared inputs            | P1 $c_3$ , P2 $g_{\text{car}}$ AX.P1.01      |

The *claim stack* (bottom = most certain/fewest, top = conditional). Each claim sits on exactly one level; the bottom two are inputs, the middle two are proved/reproduced, only the top is conditional. The single source of truth for which claim sits where is `verification/status_ledger.csv`.

**No-free-pattern rule.** To keep the structure from drowning in decorative coincidences, an identity enters the *main text* only if it (1) is a necessary step in a derivation, (2) kills an alternative, (3) is ablation-relevant, (4) links two previously independent modules, or (5) is experimentally tested. Everything else is explicitly demoted to an *audit fingerprint* **[E]** (clearly boxed as such) — so the load-bearing beams stay visible and the ornament never gets promoted to spine.

**Admissible-invariant classes (the harder filter).** Sharper still: a *load-bearing* integer may only come from one of seven typed sources — (i) a symmetric function  $e_k(a)$  of the anchor  $a = (1, 1, 2)$ ; (ii) a power sum  $p_n(a)$  or difference  $p_{n+1} - p_n$ ; (iii) a determinant, spectrum, Smith normal form or trace of  $(R, Q, K, L)$ ; (iv) an anchor quadratic form  $u^\top M v$  with  $u, v \in \{1, a\}$ ; (v) a Plücker coordinate of the anchor plane  $\langle 1, a \rangle$ ; (vi) an  $E_8$  Casimir degree or branching block; (vii) a pre-registered experimental observable. Anything else is an audit fingerprint or a frontier item, never spine. This is what converts TFPT from *pattern hunting* (“look, another number”) into a typed compiler: *fewer admissible kinds of explanation*, not more matches.

**Reviewer path** (read in this order; the rest is support):

1. the architecture: the two axioms  $\{c_3, g_{\text{car}}\}$  and the two-engine picture (§1, the figures);
2. the  $E_8$  glue  $E_8 = (D_5 \oplus A_3) + \mu_4$  — document 1, Part II, machine-checked (`v1_e8_glue.py`);
3. the  $\alpha^{-1}$  fixed point plus ablation (`v3_em_alpha.py`);

4. the **single status ledger verification/status\_ledger.csv** — the source of truth for every claim’s grade, location, dependency and script;
5. the **open gates** as numbered research contracts (**tfpt\_research\_contracts**: the residual classes R1–R5 and their current reductions) and the honest frontier list (document 4).

Everything carries a claim ID (e.g. E8.GLU.01, EM.FP.01) that resolves to a row of the ledger. **If the text and the ledger ever disagree, the ledger wins.**

## 1 What this is about: the jump in construction principle

### In plain words

**What we found.** TFPT reproduces dozens of real numbers of physics (particle masses, the fine-structure constant, dark-energy and inflation values) from a small set of inputs — and the compiler closure says *why* the same handful of small integers (2, 3, 5, 16, ...) reappears in every sector: those numbers are not separate lucky hits. A single tiny “machine” — fed by just **two inputs** (a boundary number  $\frac{1}{8\pi}$  and a five-slot carrier) — builds the exceptional symmetry  $E_8$ , and  $E_8$  is the unimodular “audit hull” that classifies the discrete Standard-Model readout: the constants, the flavor skeleton and the gravity/cosmology regime are then read off by projection and boundary dressing (not “printed” for free — where a scheme or analytic interface is needed, it is named).

**Why this is simple.** There are not many separate derivations that happen to share numbers — there is *one generator and a handful of corollaries*, and the number of independent structural assumptions is **two**.

**How plausible it is.** A theory becomes more believable when it *explains more from less*. Here the two inputs are even *over-determined*: the machine reproduces the very two numbers it started from (the dashed bootstrap loop in the architecture diagram, §2), so there is *no free dial left to tune*. The only genuinely free, continuous ingredient that remains is the number  $\pi$ .

### Sharper still: one anchor $a = (1, 1, 2)$ , and $E_8$ as a compiler hull ([E])

The two axioms are not even independent: they are the *elementary symmetric polynomials* of the single parabolic anchor  $a = (1, 1, 2)$  (the exponents-at-infinity / splitting type  $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ ):

$$e_1(a) = 4 = |\mu_4|, \quad e_2(a) = 5 = g_{\text{car}}, \quad e_3(a) = 2 = |\mathbb{Z}_2|, \quad c_3 = \frac{1}{2e_1(a)\pi} = \frac{1}{8\pi}.$$

Its *power sums*  $p_n(a) = 1^n + 1^n + 2^n = 2 + 2^n$  generate the big Lie data directly:  $p_1=4=|\mu_4|$ ,  $p_2=6=|R^+(A_3)|$ ,  $p_3=10=\mathcal{A}_\Lambda$ , and

$$|R(E_8)| = p_1 p_2 p_3 = 240, \quad \dim E_8 = p_1 p_2 p_3 + (p_4 - p_3) = 248, \quad p_4 - p_3 = 8 = \text{rank } E_8,$$

with the binary ladder  $p_{n+1} - p_n = 2^n$ . So the inputs collapse from  $\{c_3, g_{\text{car}}\}$  to  $\boxed{a = (1, 1, 2) \text{ plus } \pi}$ .  
[verification/v23\_anchor\_generator.py]

And  $a$  is itself half-forced ([E]/[C]): on  $\mathbb{P}^1$  every bundle splits (Birkhoff–Grothendieck), so once  $|\mu_4| = 4$ ,  $N_{\text{fam}} = 3$ ,  $\deg E = -4$  are set, the three positive anti-degrees are positive integers summing to 4, and  $4 = 2 + 1 + 1$  is the *unique* partition into three positive parts. The genuine remaining axiom is therefore not “why  $a = (1, 1, 2)$ ” but “why the carrier  $g_{\text{car}} = 5$  /  $(|\mu_4|, N_{\text{fam}}) = (4, 3)$ ” — that interface (with  $c_3$ ) is the [O] input layer.

*Two readings, one source (no contradiction).* P1 and P2 are the *operational compiler inputs*; their anchor provenance is  $a = (1, 1, 2)$  plus  $\pi$ . The *single structural bedrock postulate* is QGEO.SYM.01 (the carrier  $\mu_4$  clock is the seam’s conformal deck, see **tfpt\_research\_contracts**): it forces the  $\mu_4$  deck, hence  $N_{\text{fam}} = 3$ ,  $\deg E = -4$  and the tripartition  $4 = 2+1+1$ , of which  $c_3$  and  $g_{\text{car}}$  are the two elementary readouts. “Two inputs” (the working axioms) and “one postulate” (the

geometric bedrock) are the same statement at two layers. Its geometric normal form and the mark-local  $\Rightarrow \omega \circ \rho = \omega$  implication are now Lean-formalised (FORM.QGEO.02/FORM.QGEO.01); the postulate itself stays [O].

*One source, four readings (the cleanest origin).* The four-point seam divisor  $\mu_4$  on  $\mathbb{P}^1$  generates the whole discrete skeleton by four canonical functors:

$$\underbrace{c_3 = \frac{1}{8\pi}}_{\text{thermal}} \quad \underbrace{\text{rank } H_1(\mathbb{P}^1 \setminus \mu_4) = 3 = N_{\text{fam}}}_{\text{cycles} \rightarrow A_3} \quad \underbrace{h^0(\mathbb{P}^1, \mathcal{O}(\mu_4)) = 5 = g_{\text{car}}}_{\text{functions} \rightarrow D_5}$$

$$\underbrace{(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 \cong (E_8)_1}_{\text{net}}$$

and the function side even generates  $D_5$  itself ([verification/v197\_rr\_carrier\_clifford\_d5.py]):  $\Lambda^{\text{even}}(\mathbb{C}^5) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 16 = \dim S^+$ , the  $\mathfrak{so}(10) = D_5$  half-spinor. So  $\mu_4$  is not merely glue: it is the *common source* of  $A_3$ ,  $D_5$ ,  $c_3$  and  $E_8$  ([verification/v189\_riemann\_roch\_carrier.py]) — the arithmetic is exact [E], the physical reading of the carrier as the seam mode space stays [C].

**What  $E_8$  is (and is not).** In TFPT  $E_8$  is *not* an unbroken physical gauge group: it is the unimodular *audit / compiler hull* that classifies the admissible discrete charge and residue structures. The Standard Model is a *readout* after projection, not a direct  $E_8$  gauge theory; the Lorentz signature appears only after projection; fermions are not “everything in the adjoint”. This is exactly why the known no-go results against literal  $E_8$  world-formulas (Distler–Garibaldi; Coleman–Mandula) do *not* bite: they constrain  $E_8$  as a spacetime gauge symmetry, which TFPT does not claim. The defensible claim is: *TFPT is the unique parabolic compilation of the anchor  $a = (1, 1, 2)$  into the  $E_8$  audit hull.*

In one line:

■ The anchor generates the syntax; the boundary generates the scale;  $E_8$  checks the consistency.

i.e. the discrete content (carrier  $D_5 \oplus A_3 + \mu_4$ , the operators  $R, Q, K, L$ ) flows from  $a = (1, 1, 2)$ ; the continuous content ( $c_3 = \frac{1}{8\pi}$ ,  $\varphi_0$ ,  $\alpha^{-1}$ , the exponential scales) flows from  $\pi$ ; and  $E_8$  is the unimodular *checksum container*, not the world group.

**The anchor ratio triad** ([verification/v119\_review\_validation\_2.py]). The three elementary symmetric values, normalised by the family count, are the theory’s three recurring fractions [E]:

$$\frac{e_3}{p_0} = \frac{2}{3} \text{ (Koide branch / sheet ratio), } \quad \frac{e_1}{p_0} = \frac{4}{3} \text{ (seed gain), } \quad \frac{e_2}{p_0} = \frac{5}{3} \text{ (carrier branch)}$$

— the canonical flow’s critical points are exactly  $(2, 5) = (e_3, e_2)$  with inflection  $\frac{7}{2} = \frac{e_3 + e_2}{2}$ : the fractions  $\frac{2}{3}, \frac{4}{3}, \frac{5}{3}$  are not scattered numbers but the three normalised anchor coefficients. Further micro-identities [E]:  $\Omega_{\text{adm}} = 2p_1p_2 = 48$ ,  $10b_1 = 1 + p_1p_3 = 41$ ,  $\Delta_Y = e_2^2 = p_0^2 + 2 \text{rank } E_8 = 25$ ,  $h(E_8) = e_3p_0e_2 = 30$ ,  $\text{rank } E_8 = p_0 + e_2$ .

$$a = (1, 1, 2) \quad [E]$$

#### elementary $e_k(a)$

$$\begin{aligned} e_1 &= 4 \rightarrow |\mu_4| \\ e_2 &= 5 \rightarrow g_{\text{car}} \\ e_3 &= 2 \rightarrow |\mathbb{Z}_2| \\ c_3 &= \frac{1}{2e_1\pi} = \frac{1}{8\pi} \end{aligned}$$

#### power sums $p_n = 2 + 2^n$

$$\begin{aligned} p_0, p_1, p_2, p_3 &= 3, 4, 6, 10 \\ p_1p_2p_3 &= 240 = |R(E_8)| \\ p_4 - p_3 &= 8 = \text{rank } E_8 \\ \Rightarrow \dim E_8 &= 248 \end{aligned}$$

#### derived motifs

$$\begin{aligned} &4, 5, 8, 16, 25, \\ &30, 41, 120, 240 \\ &\text{one anchor,} \\ &\text{many projections} \end{aligned}$$

## The companion documents

The whole development is consolidated into **eight companion documents** (plus this introduction), organised in layers that build on one another; this introduction is the entry point and reading guide. Each document collects the material of its layer into self-contained parts:

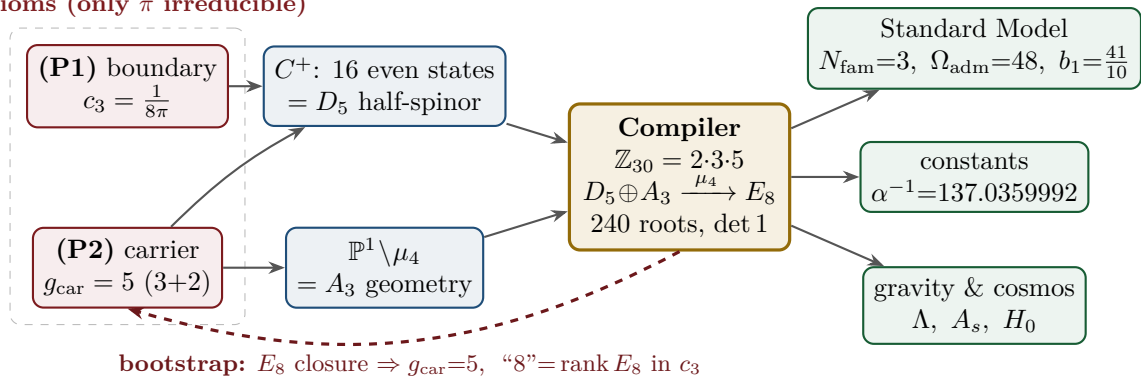
| Document                          | What it contains (former notes now as parts)                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| tfpt_1_architecture_e8            | <b>Core.</b> The two axioms $\{c_3, g_{\text{car}}\}$ , the derivation map and the EM fixed point (Part I); the Coxeter–cyclotomic compiler $\mathbb{Z}_{30}=2\cdot3\cdot5$ , the explicit $D_5\oplus A_3+\mu_4$ $E_8$ construction, the machine-checkable $E_8$ appendix and the group-level glue $(\text{Spin}(10)\times SU(4))/\Delta\mathbb{Z}_4$ (Part II).                                                                        |
| tfpt_2_standard_model             | <b>Standard Model.</b> The full SM in one $\varphi_0^{\text{ret}}$ -ladder formula (Part I); the flavor block from parabolic weights and the transport theorem, H2 as a spectral selector (Part II); the worked closures — explicit gap, APS–Fredholm, truncation/ $n_s, r$ , Starobinsky $M$ , derived $\theta_{12}$ , quark $c$ , the H2 splitting type, $\Lambda$ -typing (Part III).                                                |
| tfpt_3_e8_audit_bootstrap         | <b><math>E_8</math> audit &amp; bootstrap.</b> The seven $E_8$ slices as an audit raster ( $E_6\times A_2$ reads the flavor matrix) (Part I); the old cascade $D=60-2n$ as the same even-integer spine (Part II); the Möbius bootstrap — $g_{\text{car}}=5$ and the “8” in $c_3$ as overdetermined fixed points, only $\pi$ irreducible (Part III).                                                                                     |
| tfpt_4_frontier                   | <b>Frontier.</b> The honest status of $\eta_B$ , $m_p/m_e$ , Koide, dark matter and quantum gravity — the genuine handle, the precision, and what is <i>not</i> forced onto the ladder.                                                                                                                                                                                                                                                 |
| tfpt_5_redteam                    | <b>Red Team.</b> The adversarial audit of the five load-bearing reductions (Targets A–E): the declared attacks, what survives each, what each one reduces to, and the explicit kill tests.                                                                                                                                                                                                                                              |
| tfpt_research_contracts           | <b>Research contracts.</b> The open gates as numbered contracts with lemma chains and closing theorems: ( $U_{\text{wall}}$ ) (now reduced to the selector triangle: $(d, n) \Rightarrow R$ columnwise, residue = <i>two</i> line pairings, the third integrality-forced) and $G_{\text{metric}}/G_{\text{net}}$ (the index-4 seam-net identification — one theorem, three doors, now realised at the finite Lie level).                |
| tfpt_horizon_readouts<br>(App. H) | <b>Appendix H — horizon unit system (reframe, not new physics).</b> One seam constant $c_3=\frac{1}{8\pi}$ as the universal horizon thermal code: Hawking/de Sitter/Unruh, BH thermodynamics, Page, scrambling, Nariai bound, $v_{\text{GW}}=c$ — standard physics in seam units, two compiler fingerprints ( $1920= W(D_5) $ , $ \mu_4 =4$ ); plus the Nariai/anchor surface, the dual anchor and the resummed seam clock (v101–v138). |
| origin_theory                     | <b>Origin Theory.</b> Why no free <i>dimensionless compiler dial</i> remains beyond the anchor and $\pi$ (the dimensionful $v_{\text{geo}}$ and transfer physics stay typed): the $(g_{\text{car}}, N_{\text{fam}})$ skeleton, the triply-forced 8, the order-30 Coxeter cycle, one boundary transport for flavor and horizon, the gapped unique attractor; [E] core plus one honestly-typed [C] cyclic interpretation.                 |

The five formerly “research-level” problems are now *closed, reduced or typed* inside the documents where they belong — the group-level closure  $(\text{Spin}(10)\times SU(4))/\Delta\mathbb{Z}_4$  in document 1 (closed), and the H2 splitting type, the Starobinsky scalaron mass, the  $\theta_{12}$  texture and the quark  $c$  in document 2 (the first two closed; the quark *ratios* now closed combinatorially (v49/v71), only the *absolute* scale a  $U_{\text{point}}$  anchor;  $\theta_{12}$  the seam-misalignment residual). Eight companion files (plus this introduction), each result where it logically lives.

## 2 The architecture at a glance

The diagram below shows the whole pipeline: *two* inputs on the left, the discrete compiler in the middle, the physics on the right. Everything in between is a consequence.

2 axioms (only  $\pi$  irreducible)

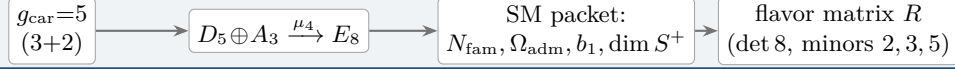


**Key statement (the loop closes):** previously  $D_5$ ,  $A_3$  and  $E_8$  sat *as given* Lie structures in the background. Now they are *intermediate products* of the two inputs — *and the red back-channel is the new part*: the  $E_8$  closure feeds back and *fixes the inputs*.  $g_{\text{car}} = 5$  is the unique rank-fill/Coxeter/Pascal solution ( $2^{g-1} = \binom{g}{0} + \binom{g}{1} + \binom{g}{2}$ ), and the integer 8 in  $c_3 = \frac{1}{8\pi}$  is rank  $E_8 = h(D_5) = \det R$ . **Inputs and output form an overdetermined fixed point** — the structure does

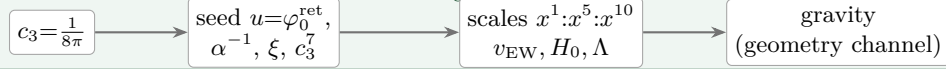
not “prove itself”; it has fewer degrees of freedom than independent consistency conditions. The only thing *not* produced by the loop is the continuous  $\pi$  (Möbius/Gauss–Bonnet), the lone irreducible primitive. So it is not a one-way arrow chain but a closed, overdetermined self-consistency loop.

**The master story is two engines.** Read from the two axioms, the whole theory factorises into exactly *two* engines — a *discrete* closure and a *boundary* dressing. Gravity is not a third block; it is the geometry channel of Engine 2.

### Engine 1 — discrete closure (from $g_{\text{car}} = 5$ )

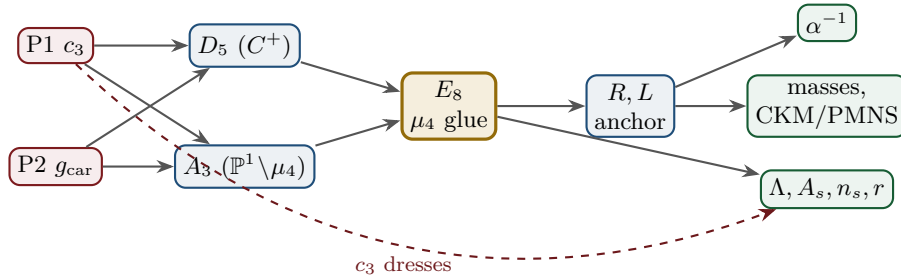


### Engine 2 — boundary dressing (from $c_3 = \frac{1}{8\pi}$ )



## 3 The dependency DAG and the proof ledger

The whole theory is a directed acyclic graph: two axioms at the source, the compiler in the middle, the observables as sinks. The DAG makes the attack surface explicit — each arrow is a named theorem or a declared input, and each node fails in exactly one way.



| Node           | Definition                                               | Inputs      | St.     | Output                              | Failure mode                |
|----------------|----------------------------------------------------------|-------------|---------|-------------------------------------|-----------------------------|
| P1             | RP boundary kernel, $c_3=1/(8\pi)$                       | — (axiom)   | [O]     | $c_3$                               | no reflection-positive seam |
| P2             | 5-slot carrier $g_{\text{car}}=5$                        | — (axiom)   | [O]     | carrier 3+2                         | wrong family/charge lattice |
| $D_5$          | $C^+$ even-Hamming = half-spinor 16                      | P2          | [E]     | $\dim S^+=16, Y$                    | group/matter mismatch       |
| $A_3$          | $\mathbb{P}^1 \setminus \mu_4$ four-puncture geom.       | P1          | [E]     | $N_{\text{fam}}=3, \mathbb{Z}_6$    | wrong multiplicity          |
| $E_8$          | $\mu_4$ glue: $\text{disc}=\mathbb{Z}_4, q\text{-sum}=2$ | $D_5, A_3$  | [E]     | 240, 248                            | not even-unimodular         |
| $R, L$         | residue + winding matrix, anchor (1, 1, 2)               | $A_3$       | [E]     | $\det 8, \text{ minors } (2, 3, 5)$ | wrong $D_6$ branch          |
| $\alpha^{-1}$  | root of $F_{U(1)}(\alpha)=0$                             | $c_3, M=41$ | [E]     | 137.0359992                         | no/second root (excl.)      |
| masses         | $\varphi_0^{\text{ret}}$ -ladder $\lambda_V^L \Lambda$   | $R, L, c_3$ | [E]/[C] | 9 masses, mixings                   | hierarchy mismatch          |
| $\theta_{12}$  | TBM + seam $\varepsilon=c_3$                             | $R, L$      | [E]/[C] | 0.307                               | seam-misalign. lemma fails  |
| $\Lambda, A_s$ | seam det. / $R^2$ scalaron                               | $c_3$       | [C]     | $\Lambda, A_s, n_s, r$              | ambient RP fails            |

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in verification/status\_ledger.csv.

Reading the table top-to-bottom is reading the theory: two red axioms in, everything else a consequence with exactly one failure mode each.

The ledger itself is versioned: each row of `verification/status_ledger.csv` carries a claim ID, its dependencies, its verification script, and `active/canonical_status/supersedes` fields, so claims that were later superseded (e.g. the early “flavor gate open” rows, now *ratios closed* via `FLAV.RIGID`) are marked inactive — the canonical status is the only one read as current.

## The compact status formula: three honest layers

$$\underbrace{\text{compiler}}_{\text{closed}} \mid \underbrace{\text{admissible IR physics}}_{\text{conditionally closed (RP, gap)}} \mid \underbrace{\text{strict physical TOE}}_{\text{open}}$$

TFPT 5.0 closes the discrete compiler, the algebraic Standard-Model readout, the EM fixed point, the admissible gapped IR sector and the  $R + R^2$  spectral-action shadow. It does *not* yet certify a strict physical TOE. The remaining items have now collapsed sharply: the  $R$ -bridge amplitude normalisation  $U_{\text{point}}$  *reduces to*  $v_{\text{geo}}$  (Gate 1 closed, [v75](#) — the same anchor as  $1/G$ ), and the ambient metric measure is *holographically reduced* to a finite seam-boundary measure (Gate 2 tier B, [v76](#)) — which is in turn the rigorously-constructed  $(E_8)_1$  lattice net ([v77](#)), so  $G_{\text{metric}}$  is imported into existing RCFT rigor rather than built anew. So the only genuine residuals are *one* dimensionful scale  $v_{\text{geo}}$  — the *dimensional-analysis floor* that no theory escapes ([v78](#)), shared by flavor and gravity — the boundary net identification, and the named QFT/cosmology transfer interfaces ( $F_{\text{frontier}}$ ).

**Two points that are *not* in the residual (closed, but easy to mis-list as open).** (i) *Parameter-freeness is a theorem under the named hypotheses:* the boundary transport is gapped ( $\Delta = 6 \log \frac{3}{2} > 0$ ), so by Perron–Frobenius its iteration has a *unique* attracting fixed point — the constants are *selected*, not tuned ([\[E\]](#) for the attractor, [\[verification/v56\\_unique\\_attractor.py\]](#); the *physical identification* of the transport operator stays [\[C\]](#)), reinforcing the static bootstrap web ( $g_{\text{car}} = 5$  forced three ways *given* the  $E_8$  closure rank — a consistency loop, see red team Target B — [\[verification/v6\\_bootstrap.py\]](#), [\[verification/v14\\_carrier\\_uniqueness.py\]](#)). (ii) *Newton’s constant is not an independent input:* fixing the physical  $\Lambda$  branch ( $((8\pi)^2 \delta_{\text{top}} = \frac{3}{4\pi^2})$  pins  $\bar{M}_{\text{Pl}}$  relative to  $\Lambda$ , so  $G_N$  is read out *after* the branch choice + *one* dimensionful induced-gravity anchor ([\[E\]/\[C\]](#), [\[verification/v60\\_lambda\\_metrology\\_branch.py\]](#)), not predicted from pure numbers. What stays irreducible is then just  $\pi$ , the discrete  $E_8$ -closure fixed point (self-consistent, *not* a free axiom; the bootstrap of `tfpt_3`), and *one* dimensionful scale  $v_{\text{geo}}$  — and that single scale is shared by *both* the flavor sector ( $U_{\text{point}} \rightarrow v_{\text{geo}}$ , [v75](#)) and gravity ( $1/G$ , [v68](#)): the two  $[A]$  anchors are *one* (dimensional analysis; no metre from pure mathematics).

## The hard reduction: two research gates plus typed interfaces

After the compiler closure, the entire residual is

$$\text{Rest} = (U_{\text{wall}}) \oplus (G_{\text{metric}}) \oplus (F_{\text{frontier}})$$

— one parabolic wall-selection, one quantum-gravity measure, and a set of deliberately typed QFT/cosmology interfaces. *Sharper aliases* (current state of the reductions; the historical gate names are kept for ledger continuity):  $(U_{\text{wall}}) \rightarrow v_{\text{geo}}$  — the flavor ratios are *closed* (Plücker/Readout Rigidity, [v49/v71/v75](#)), what remains is *one dimensionful unit*, not a flavor gap;  $(G_{\text{metric}}) \rightarrow G_{\text{net}}$  — reduced to a single boundary-net theorem, “the seam–Calderón net is the index-4  $\mu_4$  simple-current extension of the carrier net” (holomorphy then follows, [v89/GATE.METRIC.06](#)); since the QBL chain ([v108–v113](#)) the *carrier choice rides on the same theorem*: the certified scalar kernel exists iff it pairs the sheets ([v110](#)), the certified channel counts the code identically ([v112](#)), pair transport is minimally complete ([v111](#)), and the one quasi-free kernel — rank =  $c$ : 5 carrier block, 8 seam hull — is the defining datum of the free net the gate already posits ([v113](#)), so closing  $G_{\text{net}}$  also closes the carrier selection [\[E\]](#);  $(F_{\text{frontier}}) \rightarrow F_{\text{transfer}}$  — named QFT/cosmology *transfer* interfaces (Koide source→pole,  $\eta_B$  Boltzmann, axion relic,  $m_p/m_e$ ), never compiler powers; the four are one typed functor with a four-axiom contract (`CONTRACT.F.01`, [v213](#)). *Latest reductions* ([v134–v148](#)):

the flavor selector residue  $R4'$  is now “derive *one map*” —  $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$  is pure anchor data,  $(d, n)$  forces  $R$  *columnwise* (v136/v139), the third pairing is *integrality-forced* mod 11 (v142), and the pairing *values* are atom identities given the exponent-duality lift, including the new closure  $|\mu_4| + g_{\text{car}} = N_{\text{fam}}^2$  (v145); the sheet/parity question GATE.QGEO has *no discrete freedom left* — the  $\mathbb{Z}_3$  choice is *derived* (v141) and the *full* Möbius  $D_4$  of the seam curve matches the integer model parity by parity (v146); R2’s Q-system identification holds at the finite Lie level (v143), with the remaining net statement *scoped*: the odd glue sectors are Ramond modules, invisible to the untwisted NS module, and the glue choice is the R-projection (v148); and the seam quantum clock R1 is *typed closed* — edge-sector budget matches Volkov–Wipf exactly (v138), the det-ratio cancellation is derived in-family (v144), the ring sum *is* the Born-squared Gaussian zero-mode integral, and the quantum bend is det’-clean (v147). Beyond the five classes: *both* dual normals are now anchor data ( $n = (p_2, p_0, e_2)(a)$  in the cusp frame, v149), R3’s mechanism is exhibited at the gapped-model level (v150), the Calderón transfer answered (the DtN kernel is conically *clean*, v151), and its normalisation shown to be the dimensionful anchor, not a separate gap (v152). Two closing theorems then re-type the survivors: the **No-Unit Theorem** (v153) makes  $v_{\text{geo}}$  an *irreducible metrology primitive* — a dimensionless compiler cannot make an absolute scale, and  $U_{\text{point}} \sim v_{\text{geo}}$ ,  $1/G \sim v_{\text{geo}}^2$ ,  $m/\mu = c^{3/4}$  are one unit in three dimensions — and the **Simple-Current Extension Theorem** (v154) states the central  $G_{\text{net}}$  closing as  $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1, 1) \rangle \cong (E_8)_1$  (index 4,  $c=8$ ,  $\mu=1 \Rightarrow$  holomorphic  $\Rightarrow E_8$ ), exact, with only the seam-side identification left. *That seam-side identification is then driven to bedrock* (v175–v181): net existence and full-cone reflection positivity are discharged to [E] (v175), the realisation is assembled as one central theorem (v176) and split into the marks + kernel obligations whose finite cores close (v177/v178); the two then unify into one conformal-realisation premise (v179), which reduces — via uniformisation, Kerékjártó and the order-4 Möbius classification — first to a seam-*isometry* premise (v180) and finally, by equivariant uniformisation, to the strictly weaker conformal-*symmetry* / deck premise QGEO.SYM.01 (v181): the carrier  $\mu_4$  clock is the conformal deck of the seam. Below that the residual is *definitional* (the seam’s conformal structure *is* the  $\mu_4$ -deck structure of the carrier), not a missing theorem — the single, irreducible structural residual of the whole theory. It is since sharpened to its *non-circular* form (v194–v198): the bedrock is cracked to the *state-invariance*  $\omega \circ \rho = \omega$  (the DtN principal symbol commutes *exactly* with the clock; Tomita–Takesaki gives  $[\rho, \Lambda_\Sigma] = 0$  with no conformal-covariance assumption, removing the Bisognano–Wichmann circularity) — the maximally-operational form of the same postulate. *Its geometric normal form* (the UNIFORM node:  $z \mapsto iz$ ,  $\sigma\rho\sigma = \rho^{-1}$ , orbit  $\rightarrow \mu_4$ ) and the *conditional implication* mark-local  $\Rightarrow \omega \circ \rho = \omega$  are now **Lean-formalised** (FORM.QGEO.02/FORM.QGEO.01, AUDIT: PASS, only the three standard kernel axioms); the premise QGEO.SYM.01 *itself* stays [O]. The status card:

| Item                                  | Status                           | Reading                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| parameter-freeness (self-consistency) | [E]                              | gapped transport $\Rightarrow$ <i>unique</i> Perron–Frobenius attractor (v56); bootstrap web $g_{\text{car}}=5$ forced (v6/v14)                                                                                                                                                                                                                                                                                                                                                            |
| Newton constant $G_N$                 | [E]/[C]                          | not <i>independent</i> : branch pins $\bar{M}_{\text{Pl}}$ rel. to $\Lambda$ (v60); readout after branch + one dim. anchor                                                                                                                                                                                                                                                                                                                                                                 |
| ratio $c_u/c_d = \frac{55}{117}$      | [E]                              | $g_{\text{car}}\ \text{Pl}_{1,a}(K)\ _1/(N_{\text{fam}}^2\Delta_Q)$ ; rigid on the derived stratum (v49/v71), no solve                                                                                                                                                                                                                                                                                                                                                                     |
| ( $U$ ) flavor gate                   | [O] $\rightarrow v_{\text{geo}}$ | <b>Gate 1 closed:</b> $U_{\text{point}} \rightarrow v_{\text{geo}}$ (ratios + Grand Mass Volume, v75); same anchor as $1/G$                                                                                                                                                                                                                                                                                                                                                                |
| $R + R^2$ gravity                     | [E]/[C]                          | covariant $f(R)$ field equation closed in regime; heat-kernel grounded (G2)                                                                                                                                                                                                                                                                                                                                                                                                                |
| physical QG sector                    | [E]/[C]                          | gap-decoupled admissible measure ( $\Delta_{\text{eff}}=1.648>0$ )                                                                                                                                                                                                                                                                                                                                                                                                                         |
| ambient QG measure                    | [O](reduced)                     | <b>Gate 2 tier B:</b> holographically reduced to a finite seam- <i>boundary</i> measure (v76); IR tier closed (decoupling); strict-TOE cert. = the simple-current extension $(D_5)_1 \otimes (A_3)_1 \times \langle (1,1) \rangle \cong (E_8)_1$ (v154); the free-bulk premise (A) is itself a fixed-point theorem with the cone eliminated (cone gap = one-particle gap $(2/3)^6$ ) and factors into the already-open $A_2$ net-existence + <b>GATE.QGEO</b> — zero new gates (v160–v165) |
| $v_{\text{geo}}$ metrology unit       | [O](primitive)                   | No-Unit Theorem (v153): a dimensionless compiler makes no absolute scale; $U_{\text{point}} \sim v_{\text{geo}}$ , $1/G \sim v_{\text{geo}}^2$ , $m/\mu$ a ratio — one unit, irreducible                                                                                                                                                                                                                                                                                                   |
| Koide                                 | [C]                              | $Q_{\text{src}} + \varphi_0/24$ as source $\rightarrow$ pole transfer conjecture                                                                                                                                                                                                                                                                                                                                                                                                           |
| axion DM                              | [C]/[O]                          | $f_a = M_{\text{scal}}/128$ as a scenario                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| $\eta_B$                              | [C]                              | readout closed; leptogenesis Boltzmann solve missing                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| $m_p/m_e$                             | [O]                              | cross-sector QCD/EW ratio                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| $N_*$                                 | [C]                              | reheating input; scalaron-reheating point $N_*=51.4$ (v86); frozen band [50, 60]                                                                                                                                                                                                                                                                                                                                                                                                           |
| $a = (1, 1, 2)$                       | [E]/[C]                          | forced by $4 = 2 + 1 + 1$ , conditional on the carrier                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| carrier selection $g=5$ (QBL)         | [E]/[C]                          | theorem chain v108–v113: sheet-odd kernel, self-counting channel, transport degree 2, one quasi-free kernel (rank = $c$ ); residue = the $G_{\text{net}}$ premise itself — gate closes $\Rightarrow$ carrier [E]                                                                                                                                                                                                                                                                           |
| selector establishment $R4'$          | [E]/[C]                          | $(d, n) \Rightarrow R$ columnwise (v136); both normals <i>anchor data</i> — $n = (p_2, p_0, e_2)(a)$ in the cusp frame (v145/v149); residue = <i>one</i> assignment                                                                                                                                                                                                                                                                                                                        |
| sheet/parity GATE.QGEO                | [E]/[C]                          | $H^1$ characters = $\text{Spec}(Q_+) = A_3$ exponents (v137); $\mathbb{Z}_3$ choice <i>derived</i> (v141); full Möbius $D_4$ matched parity by parity (v146); residue = the module identification only                                                                                                                                                                                                                                                                                     |
| seam quantum clock R1                 | [E]/[C]                          | typed closed: resummed clock = entropy power law (v124–v133); VW edge match (v138); det-ratio in-family (v144); ring sum = Born <sup>2</sup> Gaussian zero-mode integral, bend det'-clean (v147); residue = measure + reference                                                                                                                                                                                                                                                            |
| $E_8$ Lean                            | [E]                              | hardening, not a blocker (script-certified)                                                                                                                                                                                                                                                                                                                                                                                                                                                |

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in *verification/status\_ledger.csv*.

Only *two* research gates remain — ( $U_{\text{wall}}$ ) and  $G_{\text{metric}}$ ; the frontier items are typed interfaces, not structural holes. Of these, ( $U_{\text{wall}}$ ) is the only genuine *physics* gate left:  $G_{\text{metric}}$  is now heat-kernel grounded (G2: the spectral action gives  $R + R^2$  structurally) and *gap-decoupled* (G5:  $\Delta_{\text{eff}}=1.648>0$  isolates the closed admissible sector from the un-built ambient), so its residual — the ambient projective limit (G6) — is a *strict-TOE completion target*: its absence does not affect the bounded IR claim, but it does block certification as a strict physical TOE. Both are written up as numbered *research contracts* (lemma chains + closing theorem + machine-certifiability) in the companion **tfpt\_research\_contracts**, with priority on the selector-triangle pairings (the  $v_{\text{geo}}$  scale anchor) and the  $G_{\text{net}}$  inclusion theorem. [verification/v36\_spectral\_action\_g2.py]

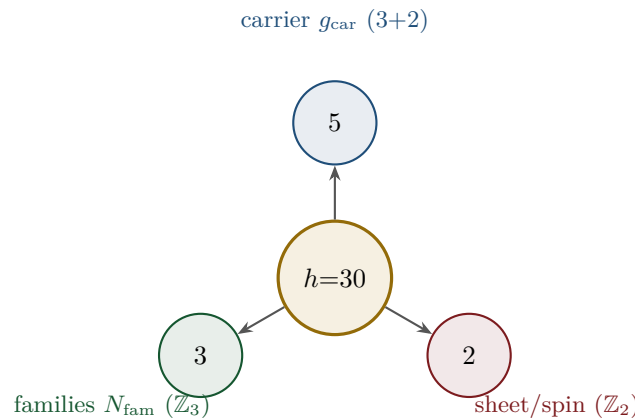
**Sharper still: the structural residual is *one* condition.**  $G_{\text{metric}}$ , the carrier P2 and the red-team Target A are not separate items: they are three provably-equivalent faces of the single condition “the seam carries no nontrivial abelian sector” — holomorphy ( $\mu$ -index 1), a homology-sphere seam link ( $\Gamma$  perfect  $\Leftrightarrow 2I$ , [verification/v232\_e8\_kleinian\_seam.py]), exactly one 1-dim irrep ([verification/v219\_icosahedral\_mckay.py]) — all of which force  $E_8$  ( $\#(\text{mark-1}) = |H_1| = 1$  only for  $E_8$ ; holomorphic  $c = 8 = g_{\text{car}} + N_{\text{fam}}$  gives the unique even unimodular rank-8 lattice). In abelian Chern–Simons language this is the single integer step holomorphic  $\Leftrightarrow \det K = 1$ , the extension tower  $D_5 \oplus A_3(16) \rightarrow D_8(4) \rightarrow E_8(1)$  being anyon condensation (the Kitaev  $E_8$  state); the one open analytic step is then “the free RP seam condenses the order- $|\mu_4|$  Lagrangian glue” = QGEO.SYM.01. So beneath the two gate *labels* there is, structurally, a single fundamental postulate. And the entire *emergent-QFT layer* collapses onto the *same* postulate (the *Modular Spectral Closure*, v261): the finite Dirac is a covariance induction of the seam KMS state (v258), the spectral-action cutoff *is* that KMS weight ( $f_2/f_0 = 1$ , v259), the seam, the carrier-16 and  $E_8$  live on one Kummer/K3 surface (v260), so Dirac, cutoff, gauging, glue and orientability are all readouts of the one seam state — the boundary QFT is closed as a single relative object modulo QGEO.SYM.01 and adds *no new open item*, while the ambient quantum-gravity measure stays separate by design. [verification/v234\_seam\_holomorphy\_selection.py] [verification/v235\_seam\_chern\_simons.py]

### Reduction in one line

The number of *structural* assumptions is **two axioms**. Everything else ( $N_{\text{fam}}, \Omega_{\text{adm}}, b_1, \alpha^{-1}$ , the flavor matrix, the  $E_8$  glue, the scale grammar) is a *consequence*. And even those two *tighten*: the bootstrap note shows  $g_{\text{car}} = 5$  and the integer 8 in  $c_3$  are *overdetermined  $E_8$ -closure fixed points* (rank-fill, Coxeter-match, integer-glue/norm all give 5;  $8 = h(D_5) = \text{rank } E_8 = \det R = \varphi(30)$ ), leaving *one* genuine continuous primitive:  $\pi$  from the Möbius boundary. A *fourth, deeper forcing* comes from the McKay correspondence:  $E_8$  is the exceptional top of the tower of finite  $SU(2)$  subgroups, so its branch data is the icosahedron and the three atoms (2, 3, 5) are its rotation-axis orders — the binary icosahedral group  $2I$  (order  $120 = |R^+(E_8)|$ ) has irrep degrees equal to the affine- $E_8$  Kac marks,  $\sum = 30 = h(E_8) = 2 \cdot 3 \cdot 5$  ([verification/v219\_icosahedral\_mckay.py]). So the *discrete* core has no freely tunable fundamental numbers — the bootstrap is *not creation from nothing* (two inputs remain), but an overdetermination of those inputs. (This concerns the discrete core only; dimensionful masses, the QG measure and frontier items remain typed-open.) [verification/v6\_bootstrap.py] [verification/v14\_carrier\_uniqueness.py] [verification/v23\_anchor\_generator.py] [verification/v219\_icosahedral\_mckay.py]

## 4 The compiler: why $30 = 2 \cdot 3 \cdot 5$

The Coxeter number of  $E_8$  is  $h = 30 = 2 \cdot 3 \cdot 5$ . Exactly these three prime factors are the three discrete atoms of the theory — that is the point of the compiler:

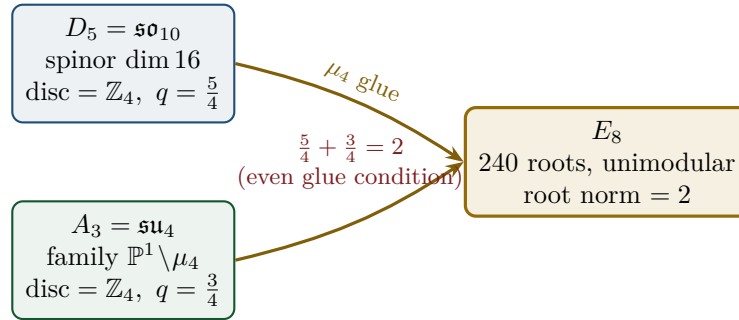


From this the compiler reads, among others:

- $\dim S^+ = 1 + 10 + 5 = 16$  (one generation) as the even exterior algebra of the 5-slot carrier — the same Pascal row  $\binom{5}{0}, \binom{5}{1}, \binom{5}{2} = 1, 5, 10$  appears in family, action ladder and the  $E_8$  root count. [E]
- $|R(E_8)| = 16 \cdot 5 \cdot 3 = 240$  and  $\dim E_8 = 240 + (5 + 3) = 248$ . [E]
- the  $\mathbb{Z}_{30}$  Coxeter element  $\leftrightarrow$  the cyclotomic polynomial  $\Phi_{30}$ ; the corner rotation  $z \mapsto iz$  on  $\mathbb{P}^1 \setminus \mu_4$  is the  $A_3$  Coxeter element. [E]
- **why these three primes:**  $E_8$  is the icosahedral top of the McKay tower ( $2I$ , order 120, irrep degrees = affine- $E_8$  marks summing to 30), so 2, 3, 5 are the icosahedron's rotation orders; and the  $E_8$  exponents  $(\mathbb{Z}/30)^\times$  carry an order-4 totative character  $\langle 7 \rangle$  — the same  $\mu_4$  seam clock. [E] [verification/v219\_icosahedral\_mckay.py] [verification/v223\_coxeter\_totative\_clock.py]

## 5 The $\mu_4$ glue: how $E_8$ is really built

The decisive, now *proved* step:  $D_5$  and  $A_3$  have the *same* discriminant group  $\mathbb{Z}_4$ , and their discriminant-form norms are exactly two TFPT constants which add up to the  $E_8$  root norm.



### Glue theorem (proved)

$\text{disc}(D_5) = \text{disc}(A_3) = \mathbb{Z}_4$ , glue index  $|\mu_4| = 4$ , and  $q(D_5) + q(A_3) = \frac{5}{4} + \frac{3}{4} = 2 =$  the  $E_8$  root norm. Hence  $E_8 = (D_5 \oplus A_3) + \mu_4\text{-glue}$  is a *closed lattice-theoretic* construction — not a blind “positing of 248”. [E] [verification/v1\_e8\_glue.py]

## 6 How derivation now works: the derivation map

What can one concretely *compute* with  $\{c_3, g_{\text{car}}\}$ ? The following map summarises what is now a consequence rather than an assumption.

| Sector                       | How it now arises                                                            | Status  |
|------------------------------|------------------------------------------------------------------------------|---------|
| gauge group / families       | carrier 3+2, $\dim S^+=16$ , $N_{\text{fam}}=3$ , $\Omega_{\text{adm}}=48$   | [E]     |
| hypercharge                  | roots of the charge polynomial $bsY^2 + (s-b)Y - 1$                          | [E]     |
| $U(1)$ index $b_1$           | $b_1 = \frac{g_{\text{car}} 2^{g_{\text{car}}-2} + 1}{10} = \frac{41}{10}$   | [E]     |
| fine structure $\alpha^{-1}$ | unique root of $F_{U(1)}(\alpha) = 0$                                        | [E]     |
| flavor matrix                | transport theorem (H1 proved, H2/H3 force the matrix)                        | [E]/[C] |
| electroweak scale            | $v_{\text{EW}} \sim e^{-\alpha^{-1}/5}$ (carrier divides $\alpha^{-1}$ by 5) | [E]     |
| cosmological constant        | $\Lambda \sim e^{-2\alpha^{-1}}$ (density quotient; see typing note)         | [E]/[C] |
| Hubble scale                 | $H_0 \sim \sqrt{\Lambda}$                                                    | [C]     |
| inflation amplitude          | $A_s = \frac{N_{\star}^2}{24\pi^2} c_3^7$ in the $R^2$ branch                | [C]     |
| $E_8$                        | $D_5 \oplus A_3 + \mu_4\text{-glue}$                                         | [E]     |

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

**Notable:** the scale grammar is *one* exponential engine. The same number  $\alpha^{-1} \approx 137$  generates the electroweak scale (divided by 5 = carrier), the cosmological constant (times 2) and — via the square root — the Hubble scale. One constant becomes many orders of magnitude.

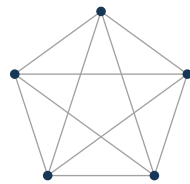
**Typing note on  $\Lambda$  (a deliberate danger zone).** “ $\Lambda$ ” denotes *three* distinct objects that must not be conflated: (i) the seam-determinant exponent  $e^{-2\alpha^{-1}}$  (an exact [E] readout), (ii) the dimensionless density quotient  $\rho_{\Lambda}/\bar{M}_{\text{Pl}}^4$  (which carries a prefactor *branch*, hence [C]), and (iii) the observed curvature, to which (ii) is *anchored*, not predicted. The exponential is exact; the absolute dark-energy density remains a typed cosmology interface (full disambiguation in `tfpt_2_standard_model`).

## Cosmology is the Pascal action grammar of the carrier

With  $x = v_{\text{geo}}/(5\beta_{\text{rad}}^2 \bar{M}_{\text{Pl}})$  the cosmological readouts are a *power tower* on the five-slot carrier graph  $K_5$ :

$$\frac{H_0}{\bar{M}_{\text{Pl}}} = x^5 R_H, \quad \frac{\rho_{\Lambda}}{\bar{M}_{\text{Pl}}^4} = x^{10} R_{\Lambda}, \quad (1, 5, 10) = \binom{5}{0}, \binom{5}{1}, \binom{5}{2}$$

EW is the unit  $x^1$ , Hubble is the product over the  $\binom{5}{1} = 5$  vertices,  $\Lambda$  is the product over the  $\binom{5}{2} = 10$  edges — the *same* Pascal triple that reads one generation, the action ladder and the  $E_8$  root count  $16 \cdot 5 \cdot 3 = 240$ . “Hubble = slot product,  $\Lambda$  = pair product.” [E]



$K_5$

- $\binom{5}{1} = 5$  vertices — action  $x^1$  (Hubble slot)
- $\binom{5}{2} = 10$  edges —  $\Lambda$  density pair  $x^{10}$
- $\binom{5}{3} = 10$  triangles — dual sector
- $\binom{5}{4} = 5$  complements — inverse sector
- $\binom{5}{5} = 1$  determinant — closure

$K_5$  on the five carrier slots; the action powers  $x^1, x^5, x^{10}$  are its  $\binom{5}{k}$  faces (mnemonic; the physical scale map stays [C]).

### The entire fermion spectrum in one formula

Masses, Yukawa, CKM, PMNS and neutrinos all follow from *one* master formula with *one* seed:

$$\hat{m}_{f,j} = \frac{v_{\text{geo}}}{\sqrt{2}} \lambda_Y^{L_{f,j}} \Lambda_{f,j}, \quad \lambda_Y = \sqrt{\varphi_0^{\text{ret}}(1 - \varphi_0^{\text{ret}})}, \quad \varphi_0^{\text{ret}} = \frac{1}{6\pi} + \frac{3}{256\pi^4}.$$

The word-lengths  $L_{f,j}$  are the *fixed residue matrix of the compiler*, whose characteristic polynomial  $t^3 - 9t^2 + 10t - 8$  carries only compiler numbers ( $\text{tr} = 9 = N_{\text{fam}}^2$ ,  $\det = 8 = h(D_5)$ , principal 2-minors 2,3,5 with product  $30 = h(E_8)$ ). Through the  $\varphi_0^{\text{ret}}$ -ladder  $\hat{m} = \frac{v_{\text{geo}}\pi}{\sqrt{2}} c(\varphi_0^{\text{ret}})^k$  all nine word-lengths are fixed and the *charged-lepton* amplitudes are closed; the *quark* mass *ratios* are closed too (integer Plücker readouts on the derived selector stratum, **v49/v71**), with only the *absolute* amplitude scale reducing to the ( $U_{\text{wall}}$ ) anchor. **SM status:** the discrete packet, the dimensionless flavor skeleton, the charged-lepton source masses and the quark *ratios* are closed; the *absolute* quark scale and the full PMNS completion are reduced/conditional;  $m_W, m_Z, m_H, \sin^2 \theta_W, \alpha_s$  are RG scheme-layer ( $m_H$  rests on the closed UV quartic  $\frac{1}{16\pi^2}$ ); and the solar angle is *realized* by a *TFPT texture* and reduced to the seam-misalignment lemma ( $\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2}$ ; see below). (*Full derivation: document 2, **tfpt\_2\_standard\_model**.*) [[verification/v18\\_quark\\_yukawa.py](#)] [[verification/v20\\_lepton\\_c\\_derivation.py](#)] [[verification/v46\\_grand\\_mass\\_volume.py](#)]

## 7 Physical, geometric, topological: why the architecture is plausible

The closure works on three levels at once: *physically*,  $\alpha^{-1}$  is the fixed point of *one* equation and one generator drives many scales; *geometrically*, the  $C^+$  spinor and the puncture geometry  $\mathbb{P}^1 \setminus \mu_4$  produce  $D_5$  and  $A_3$ , glued to  $E_8$ ; *topologically*, the common discriminant  $\mathbb{Z}_4$  (the  $\mu_4$  winding) forces the glue and  $\Phi_{30}$ /Coxeter drives the sectors — with exactly 2 axioms and the rest a consequence. **Why the compiled architecture is plausible.** It is (i) *parsimonious* (few independent assumptions  $\Rightarrow$  low overfitting risk), (ii) *rigid* (the glue condition  $5/4 + 3/4 = 2$  and the discriminant  $\mathbb{Z}_4$  leave little freedom), and (iii) *checkable* (machine-checkable  $E_8$  appendix, explicit fixed-point equation for  $\alpha^{-1}$ ). More explained from less — exactly the criterion that makes a theory more probable.

## 8 What role $E_8$ now plays exactly

$E_8$  is no longer the starting point but the *output format*:

- **Bookkeeping:**  $E_8$  is the unimodular container in which carrier ( $D_5$ ) and family ( $A_3$ ) fit together consistently. Its numbers 240, 248 are carrier traces, not inputs.
- **Consistency test:** unimodularity ( $\det 1$ ) is a hard condition; it closes only because the discriminant-form norms  $5/4$  and  $3/4$  (both already present in TFPT) add up to 2.  $E_8$  thus tests the internal coherence of the two atoms.
- **Not an end in itself:**  $E_8$  does not explain “everything”; it is the smallest even unimodular hull of the datum  $D_5 \oplus A_3$ . The physics sits in the *two inputs*, not in  $E_8$ .

**New — the secondary atlas.** Beyond that,  $E_8$  is an *audit container*: each maximal subalgebra projects a TFPT module. Document 3 (**tfpt\_3\_e8\_audit\_bootstrap**, Part I) shows the seven slices of 248 explicitly and labelled. The strongest new hit is that  $E_6 \times A_2$  reads the flavor matrix:

$$248 = \|R\|_F^2 + \det R + 2(\mathbf{1}^\top R a) N_{\text{fam}} = \underbrace{78}_{\dim E_6} + \underbrace{8}_{\dim A_2} + 2 \cdot 27 \cdot 3,$$

with  $\|R\|_F^2 = 78 = \dim E_6$ ,  $\det R = 8 = \dim A_2 = h(D_5)$ . The discipline rule is: *every load-bearing number must appear in at least one  $E_8$  projection* — this makes the structure falsifiable rather than numerological. (Audit level; all identities machine-verified.)

## **9 What this means for the theory-of-everything question**

### The state of the five questions — closed, reduced or typed

1. **Flavor residual — ratios closed; only the absolute scale is the wall-selection.** Mehta–Seshadri supplies the equivalence-class framework (stable parabolic  $\leftrightarrow$  unitary representation); the algebraic selector  $R$ , the  $Q$  spectrum and the splitting type are fixed (now *derived*:  $\det R = 8$  lattice,  $\text{Spec}(Q_+)$  from  $D_4$ , v69/v70; since v136/v139 the dual pair  $(d, n)$  forces  $R$  *columnwise*,  $d$  is pure anchor data, and  $\text{Spec}(Q_+) = (1, 2, 3)$  is the  $A_3$ -exponent grading on the seam curve’s cohomology, v137), and the charged-lepton amplitudes are closed. The quark mass *ratios* ( $c_u/c_d = \frac{55}{117}$  etc.) are then *constant* on this discrete selector stratum (Readout Rigidity, v49/v71) — integer Plücker readouts, no transcendental solve. Only the *absolute* amplitude scale reduces to the selected *polystable* point on the parabolic stability wall ( $U_{\text{point}}$ , an anchor; Research Contract 1). [E]/[O]
2. **Gravity — exact in the  $R^2$  regime.** Spectral action  $\rightarrow R+R^2$ , BRST/Hodge kills the gauge directions, FRW reduction  $\rightarrow$  Starobinsky; the only remainder is the *controlled* low-curvature truncation ( $R^{n \geq 3} \rightarrow R^2$ , valid for curvature  $\ll M^2$ ). [E] in the regime
3. **Strong CP — decoupled.**  $\theta_{\text{eff}}=0$  follows from three structural facts ( $\gamma_5$ -Hermiticity, polar structure, sheet involution) + reflection positivity — *without* a mass gap. [E] (given RP)
4. **Full dynamics — the admissible transfer model is closed, and the boundary QFT is now one relative object.** The mass gap is the *explicit* discrete gap  $6 \log \frac{3}{2}$  of the  $C_6$  transfer operator (eigenvalues  $(k/3)^6$ ), which supports the OS reconstruction path on the admissible sector. On top of this the whole emergent-QFT round assembles the boundary theory into one relative object  $\text{TFPT}_{\text{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$  — the *Modular Spectral Closure*: the finite Dirac is a covariance induction of the seam KMS state (v258), the spectral-action cutoff *is* that KMS weight (v259), the seam/carrier/ $E_8$  live on one Kummer/K3 (v260), all certified together (v261) — closed modulo the single premise QGEO.SYM.01. *What this does not close*: the full measured pole masses (the absolute scale  $v_{\text{geo}}$  + standard RG), the absolute QCD scales ( $m_p/m_e$ , [O]), the complete PMNS dynamics (the angles are typed; the full mass/CP dynamics stay [C]/[O]), and the real 4D interacting QFT / ambient measure ([O]). [E] (transfer model + boundary relative object) / [C] (continuum) / [O] (real 4D)
5. **Axioms P1/P2 — P2 machine-verified.** The Lean proof builds cleanly (1886 jobs, 0 sorry, only kernel axioms); the P2 algebra is a theorem, P1 a typed interface. They remain declared inputs. [E]/[O]

In other words: the theory has shrunk from “many open audit points” to “*standard steps* (cluster expansion, Fredholm, truncation bound) + two declared axioms”. The five questions are now *closed, reduced or typed* rather than open-ended — a qualitative jump in discipline, not a claim of completion. (*Details and proof chains: document 2, tfpt\_2\_standard\_model, Part III.*)

### The closure ledger: what is done, and what genuinely remains

A complete theory must close the discrete structure, the dimensionless constants, the mass spectrum, gravity, and the quantum measure. The current state, graded honestly:

| TOE piece                                                                          | Status  | If not complete: what closes it                                                                                                                                                                                                      |
|------------------------------------------------------------------------------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| gauge group, $N_{\text{fam}}$ , hypercharge, $b_1$                                 | [E]     | — complete                                                                                                                                                                                                                           |
| $E_8 = (D_5 \oplus A_3) + \mu_4$ , group closure                                   | [E]     | — complete                                                                                                                                                                                                                           |
| bootstrap $(g, \mu) = (5, 4)$ , “8” in $c_3$                                       | [E]     | — complete (reverse glue $\mu^2 - 5\mu + 4 = 0$ )                                                                                                                                                                                    |
| fine structure $\alpha^{-1}$                                                       | [E]     | — <b>existence+uniqueness proved</b> ; value is a numerical fixed point (ledger EM.FP.01), Ward origin [C]                                                                                                                           |
| fermion masses ( $\varphi_0^{\text{ret}}$ -ladder)                                 | [E]/[O] | leptons exact; quark <i>ratios</i> closed (Plücker, v49/v71); absolute scale = anchor                                                                                                                                                |
| mixings $\theta_{13}, \theta_{23}, \theta_{12}$ (PMNS angles)                      | [E]/[C] | $\theta_{12}$ cond. derived; $\theta_{23}$ maximal; <i>full</i> PMNS dynamics (mass scale, CP magnitude) stay [C]/[O]                                                                                                                |
| flavour matrix / H2                                                                | [E]     | dictionary <i>exact</i> (hexagon = twisted family spectrum, addresses pinned, margins forced, v118–v122); $R$ forced columnwise by $(d, n)$ (v136/v139); pairing values = atom identities (v142/v145); residue = <i>one</i> lift map |
| seam quantum clock (R1)                                                            | [E]/[C] | typed closed: reduced edge-sector clock (v124–v133); external VW edge = $2 \times$ reduced budget, full 4d det = firewall (v138); det-ratio cancellation derived in-family (v144)                                                    |
| inflation $A_s, n_s, r$ , scalaron $M$                                             | [C]     | scalaron $M = c_3^{7/2} \bar{M}_{\text{Pl}}$ exact; $n_s, r$ are bands over $N_* \in [50, 60]$ (reheating point 51.4, v86)                                                                                                           |
| boundary QFT (Modular Spectral Closure)                                            | [E]/[C] | <b>one relative object</b> $\text{TFPT}_{\text{QFT}}$ : $D_F$ = covariance induction (v258), cutoff = KMS weight $f_2/f_0=1$ (v259), seam/carrier/ $E_8$ on one K3 (v260), certified (v261) — closed modulo QGEO.SYM.01              |
| <i>genuinely open — the actual TOE remainder (what the closure does not cover)</i> |         |                                                                                                                                                                                                                                      |
| QFT measure (admissible sector)                                                    | [E]     | <b>closed: RP from cusp spectrum, OS reconstruction</b>                                                                                                                                                                              |
| full measured pole masses                                                          | [C]     | ladder $\times v_{\text{geo}}$ , dressed by standard RG (no new physics)                                                                                                                                                             |
| absolute QCD scales ( $m_p/m_e$ , $\Lambda_{\text{QCD}}$ )                         | [O]     | QCD matching contract (FR.MPME.01); not a compiler number                                                                                                                                                                            |
| full PMNS dynamics (mass scale, CP)                                                | [C]/[O] | angles typed; absolute $\nu$ -mass scale + CP magnitude open                                                                                                                                                                         |
| real 4D interacting QFT + full gravity                                             | [O]     | ambient measure <i>not constructed</i> (frontier); conditionally reduces to one named hypothesis (ambient RP) on the relative spectral-action measure                                                                                |
| analytic boundary origin of $\pi$ ( $c_3 = \frac{1}{8\pi}$ )                       | [C]     | <b>hardened</b> : $c_3 = 1/( \mathbb{Z}_2  \oint_{S^2} K) = 1/(8\pi)$ ( <b>Gauss–Bonnet</b> ); “8” = rank $E_8$ ; $\pi$ stays the one primitive                                                                                      |
| $\eta_B, m_p/m_e$ , Koide, dark matter                                             | [O]     | frontier handles only (see document 4)                                                                                                                                                                                               |

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in verification/status\_ledger.csv.

**The honest bottom line:** the discrete core, the dimensionless constants, the *mass-hierarchy exponents* and the mixings are *closed*; the nine masses follow from one  $\varphi_0^{\text{ret}}$ -ladder (lepton rationals exact, quark  $O(1)$  digits still a conditional finite evaluation). The *admissible-sector* QFT measure is closed (a finite gapped transfer system, RP from the cusp spectrum, OS-reconstructed), and on top of it the whole boundary QFT is now *one relative object*  $\text{TFPT}_{\text{QFT}}$  (the *Modular Spectral Closure*, v258–v261:  $D_F$  a covariance induction, the cutoff the seam KMS weight, seam/carrier/ $E_8$  on one Kummer/K3) — closed modulo the *single* premise QGEO.SYM.01. *What that closure does not reach* is exactly the dimensionful/ambient remainder: the full measured pole masses (ladder  $\times v_{\text{geo}}$  + standard RG, [C]), the absolute QCD scales ( $m_p/m_e$ , [O]), the complete PMNS dynamics ([C]/[O]), and the *real 4D interacting QFT* — the *ambient/metric-sector* measure, i.e. *full quantum gravity* — which is *not* constructed here and stays [O] (frontier doc), reducing conditionally to a single named hypothesis (ambient reflection positivity). The seam seed is *hardened*:  $c_3 = 1/(|\mathbb{Z}_2| \oint_{S^2} K dA) = 1/(8\pi)$  is the one-sided Gauss–Bonnet normalisation of the seam sphere, with the discrete “8” equal to rank  $E_8$  and only the Gauss–Bonnet  $\pi$  left as the single irreducible primitive. What remains for a full TOE is then the ambient-RP hypothesis (full QG),

plus the standard-RG dimensionful masses and the honest frontier items. No load-bearing discrete parameter remains free, and  $\pi$  is the lone continuous primitive. *4D-QFT fork policy (frozen, v265)*: the canonical 4D reading is *boundary only* — a 4D-GUT is *not claimed by default* ( $E_8$  is the audit hull; SM-only unification is killed, v246); carrier-native Pati–Salam is a separately-typed, falsifiable UV branch (thresholds + proton decay). Not an open ambiguity, a decision tree.

## 10 Does this make the theory more probable?

**Plausibility balance.** *Pro*: (a) two inputs instead of many; (b)  $\alpha^{-1}$  to  $2.9 \times 10^{-10}$  (CODATA-2022,  $1.9\sigma$ ) as a *fixed point*, not a fit; (c)  $E_8$  constructed, not postulated; (d) one compiler generates several sectors  $\Rightarrow$  fewer degrees of freedom, more predictive power.

*Contra/open*: what remains are only *standard steps* (cluster expansion in the thermodynamic limit, the Fredholm property, the truncation bound) and the two declared axioms P1/P2; on the SM side the dimensionful EW/QCD masses ( $m_W, m_Z, m_H, \sin^2 \theta_W, \alpha_s$ ) sit on the RG scheme layer. The solar angle, previously the one open SM number, is now *conditionally derived* ( $\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2}$ , modulo the seam-misalignment lemma). A final verdict “it maps reality” requires this short list plus the near-term experimental tests (below).

Net: the parsimony (Kolmogorov-like compression: many numbers from little) and the high precision of individual closures raise the a-posteriori plausibility substantially — without hiding the open points.

## 11 Predictions for running and upcoming experiments

Because *one* generator now sits behind the numbers, the theory makes a set of concrete, falsifiable statements that running or near-term experiments are actively testing. Each row gives the TFPT value, the *new* derivation behind it, the experiment, and an honest status.

| Observable                          | TFPT value & derivation                                                                                                                                                              | Experiment (running / upcoming)                                                                                                      | St.     |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|---------|
| solar<br>$\sin^2 \theta_{12}$       | prediction of record $\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.306747$ ; TBM + seam misalignment $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}} = c_3 + 36c_3^4$ (cond.) | <b>JUNO</b> : first 59.1 d run $0.3092 \pm 0.0087$ (compatible), precision phase pending                                             | [E]/[C] |
| reactor<br>$\sin^2 \theta_{13}$     | $\varphi_0^{\text{ret}} e^{-5/6} = 0.0231$ ; seed $\times$ carrier-trace $e^{-\gamma}$ , $\gamma = \frac{5}{6}$                                                                      | Daya Bay / RENO / JUNO (PDG 0.0220)                                                                                                  | [E]     |
| atmospheric<br>$\sin^2 \theta_{23}$ | $\approx \frac{1}{2}$ ( $\mu\tau$ -symmetric limit; octant not selected)                                                                                                             | NOvA / T2K / <b>DUNE</b> (octant ambiguity; NuFIT 6.0 pre-JUNO baseline)                                                             | [C]     |
| fine structure $\alpha^{-1}$        | 137.0359992 as the unique root of $F_{U(1)}(\alpha)=0$ (no fit, no second branch)                                                                                                    | CODATA-2022 (data through 2022)<br>137.035999177(21): dev. $2.9 \times 10^{-10}$ ( $1.9\sigma$ of its uncertainty)                   | [E]     |
| tensor ratio $r$                    | $r = \frac{12}{N_*^2} \in [0.0033, 0.0048]$ (frozen band, $N_* \in [50, 60]$ ); scalaron-reheating point 0.0045 ( $N_*=51.4$ , v86)                                                  | now $r < 0.036$ (BICEP/Keck BK18); <b>CMB-S4</b> ( $\sigma_r \leq 5 \times 10^{-4}$ , obs. $\sim 2033$ )                             | [C]     |
| tilt $n_s$                          | $n_s = 1 - \frac{2}{N_*} \in [0.960, 0.967]$ (same band); scalaron-reheating point 0.9611 ( $N_*=51.4$ , v86)                                                                        | Planck ( $0.9649 \pm 0.0042$ ); CMB-S4 sharpens                                                                                      | [C]     |
| amplitude $A_s$                     | $\frac{N_*^2}{24\pi^2} c_3^7$ ; $2.0 \times 10^{-9}$ at $N_*=55$ (the measured $A_s$ prefers fast reheating, v86)                                                                    | Planck ( $2.1 \times 10^{-9}$ )                                                                                                      | [C]     |
| neutron EDM                         | $\theta_{\text{eff}} = 0 \Rightarrow d_n$ essentially null (structure + RP)                                                                                                          | <b>PSI nEDM</b> / SNS (running)                                                                                                      | [E]     |
| CP phase $\delta_{\text{CKM}}$      | $\delta = \frac{\pi}{3} + 3\lambda^2 = 66.96^\circ$ (frozen of record, v88); survives at $+0.98\sigma$ vs $\gamma_{\text{PDG}}$                                                      | <b>LHCb</b> / Belle II $\gamma$ combination; decision at $\sigma_\gamma \leq 0.96^\circ$                                             | [E]     |
| cosmic birefringence $\beta$        | $\beta = \frac{\varphi_0^{\text{ret}}}{4\pi} = 0.2424^\circ$ (seam readout)                                                                                                          | ACT DR6 hint $0.215^\circ \pm 0.074^\circ$ (systematics not yet understood — not a validation target); <b>Simons Obs.</b> / LiteBIRD | [C]     |
| $0\nu\beta\beta$ / ordering         | Majorana branch; normal-ordering preference                                                                                                                                          | <b>LEGEND</b> / <b>nEXO</b> , DUNE, KATRIN                                                                                           | [C]     |
| flavor invariants                   | $\det R = h(D_5) = 8$ , principal minors (2, 3, 5), $\chi_R = t^3 - 9t^2 + 10t - 8$ (exact)                                                                                          | any future CKM/PMNS global fit must satisfy these                                                                                    | [E]     |
| $H_0$ vs. $\Lambda$                 | $v_{\text{EW}} \sim e^{-\alpha^{-1}/5}$ , $\Lambda \sim e^{-2\alpha^{-1}}$ , $H_0 \sim \sqrt{\Lambda}$ : one engine                                                                  | SH0ES / <b>DESI</b> / Planck (Hubble tension)                                                                                        | [C]     |

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in verification/status\_ledger.csv.

### The two decisive near-term tests

- (1) **JUNO** and  $\theta_{12}$ . JUNO is *taking data since August 2025* and targets sub-percent  $\sin^2 \theta_{12}$ . TFPT commits to the single prediction of record  $\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.306747$  (misalignment  $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}} = c_3 + 36c_3^4$ ). The first 59.1-day JUNO run reports  $\sin^2 \theta_{12} = 0.3092 \pm 0.0087$  — *compatible*, but the precision phase is still pending; this is a first compatibility check, not yet the decisive test. A converged central value away from  $\sim 0.307$  at high significance falsifies the seam mechanism.
- (2)  $r$ . The  $R^2$  scalaron with  $M$  fixed by  $c_3^7$  predicts a small  $r = 12/N_*^2$ : the frozen band  $r \in [0.0033, 0.0048]$  over  $N_* \in [50, 60]$ , with the scalaron-reheating point  $r = 0.0045$  ( $N_*=51.4$ , v86) — already below the current bound  $r < 0.036$  (BICEP/Keck BK18) and within reach of CMB-S4 ( $\sigma_r \leq 5 \times 10^{-4}$ , first observations  $\sim 2033$ –34). A detection of  $r \gtrsim 0.01$  would be incompatible with the  $R^2$  branch on which  $M_{\text{Pl}}$  and  $A_s$  rest.

**Structural (non-experimental) checks.** Independent of any apparatus, the machine-checkable  $E_8$  appendix (all 240 roots, Gram matrices, discriminant groups) lets anyone verify that the two TFPT atoms really generate  $E_8$ ; and the discipline rule “*every load-bearing number appears in at least one  $E_8$  projection*” (atlas note) turns the number stock into a falsifiable raster rather than

numerology. The numerology charge is additionally *quantified* by an explicit null model: under the declared formula grammar the alignment density is anomalously high — a random theory of equal formula complexity essentially does not reproduce the scorecard. This is a *grammar-internal null-model bound*, not a Bayesian probability that TFPT is true; the explicit figure and its controls are stated in the verification appendix [E] [verification/v100\_numerology\_null\_mc.py].

#### Freeze file: committed kill criteria

To prevent post-hoc rescue, the predictions above are frozen with explicit kill criteria in the machine-readable registry `verification/freeze_file.csv`. The decisive entries:

- **solar**  $\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} \approx 0.3067$  — a JUNO central value clearly away from 0.307 kills the seam-misalignment mechanism;
- **tensor**  $r \approx 0.0040$  — any robust  $r \gtrsim 0.01$  kills the  $R^2$  branch carrying  $M_{\text{P1}}, A_s$ ;
- **ordering** normal, small  $m_{\beta\beta}$  — inverted ordering or large  $m_{\beta\beta}$  kills the Majorana branch;
- **strong CP**  $\theta_{\text{eff}} = 0$  — a solid nEDM signal above SM background falsifies the structural cancellation;
- $w = -1$  — a robust  $w \neq -1$  kills the single-engine dark-energy readout.

The proton/electron ratio is explicitly *not* claimed as a compiler power (only fails if mis-asserted as one).

**Blind-prediction registry (frozen 2026-06-09, machine-enforced).** Beyond the kill criteria, the *values* themselves are now frozen: `verification/predictions_frozen.json` registers every dimensionless prediction of record at 25 significant digits *before* JUNO / CMB-S4 / LiteBIRD / DESI decide, and [verification/v84\_frozen\_registry.py] re-derives each decimal from the two axioms on every suite run (formula $\leftrightarrow$ value lock; covered by `manifest.sha256`; ledger `REG.FREEZE.01`). In particular the registry admits exactly *one*  $\theta_{12}$  prediction of record (the seed  $\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.306747$ ) — the seam and non-linear variants are typed as derived variants of the same texture and can never be silently promoted afterwards.  $r$  and  $n_s$  enter only as *bands* over the declared external  $N_\star \in [50, 60]$ , never as point predictions.

#### What likely follows next.

- the quark mass *ratios* are now closed combinatorially (integer Plücker readouts on the derived selector stratum, `v49/v69/v71`); the only residual is the *absolute* amplitude scale, an anchor ( $U_{\text{point}}$ ) of the same nature as the one dimensionful scale — no new physics;
- further constants as carrier traces (analogous to  $b_1 = 41/10$ ), since the compiler systematically produces traces over  $S^+$ ;
- a sharper geometry $\leftrightarrow$ gravity bridge once the Hodge/ $R^2$  branch is lifted from [C] to [E].

## Conclusion

TFPT's sectors form an impressive collection of closures sharing common numbers. The compiler closure explains *why* they are the same numbers: a small discrete generator  $\mathbb{Z}_{30} = 2 \cdot 3 \cdot 5$  builds, from two inputs  $\{c_3, g_{\text{car}}\}$  via  $D_5 \oplus A_3 + \mu_4$ , the  $E_8$  container and outputs the Standard Model, the constants and the scale grammar from there. The theory thereby becomes *more parsimonious, more rigid and more checkable*. Precisely nameable gaps remain — one geometric flavor realisation, the conditional gravity/QFT closures, and two declared axioms — but the path to closure is now short and concrete.

## Appendix: computational verification (reproducibility)

Every claim in this document marked [E] (exact identity), [E] (Lie/lattice theorem) or [E] (numerical fixed point) is re-derived from the two axioms  $\{c_3, g_{\text{car}}\}$  alone by a small, self-contained Python suite in the repository folder `verification/`. The inline references [verification/...] throughout the documents point to the exact script; the table below is the master map (its source is the shared fragment `tex-artefacts/verification.tex`).

| Script                             | What it machine-checks                                                                                                                                                                                                                   |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v1_e8_glue.py                      | glue theorem $E_8=(D_5\oplus A_3)+\mu_4$ : $\text{disc}=\mathbb{Z}_4$ , $q(D_5)+q(A_3)=\frac{5}{4}+\frac{3}{4}=2$ , $240=16\cdot 5\cdot 3$ , $248=240+8$                                                                                 |
| v2_carrier_pascal.py               | $g_{\text{car}}=5$ unique Pascal solution; $16=1+5+10$ , $N_{\text{fam}}=3$ , $\text{rank } E_8=8$ , $\Omega_{\text{adm}}=48$ , $b_1=\frac{41}{10}$                                                                                      |
| v3_em_alpha.py                     | EM closure: unique root of $F_{U(1)}(\alpha)=0$ , $\alpha^{-1}=137.0359992168$ , existence/uniqueness, inverse test, ablation                                                                                                            |
| v4_flavor_matrix.py                | residue matrix $R$ : $\det=8$ , minors $\{2, 3, 5\}$ , $\chi_R=t^3-9t^2+10t-8$ , SNF $\text{diag}(1, 1, 8)$ , $\ R\ _F^2=78$ , $\sum L=40$                                                                                               |
| v5_e8_cascade.py                   | cascade $D_n=60-2n$ : endpoints, exponent rungs $\rightarrow 240$ , IR tail 46080, variance 6552                                                                                                                                         |
| v6_bootstrap.py                    | reverse glue $\mu^2-5\mu+4=0$ ; $g_{\text{car}}=5$ three ways; $8=\text{rank } E_8=h(D_5)=\varphi(30)$                                                                                                                                   |
| v7_gravity_cosmo.py                | scalaron exponent 7, $M_{\text{scal}}^2/\bar{M}_{\text{Pl}}^2=c_3^7$ , $A_s, n_s, r$ , $\Omega_b$ , $\sin^2 \theta_{12}$                                                                                                                 |
| v8_horizon.py                      | horizon code: $\frac{1}{2\pi}=4c_3$ , $1920= W(D_5) $ , $S_{dS}$ , Page, $\beta_{\text{rad}}=0.2424^\circ$                                                                                                                               |
| v9_neutrino_texture.py             | solar angle: explicit $\mu\tau$ Majorana texture $\rightarrow \sin^2 \theta_{12}=\frac{1}{3}-\frac{\varphi_0^{\text{ret}}}{2}$ , $\theta_{23}=45^\circ$ , $\theta_{13}=0$                                                                |
| v10_projection_involution.py       | $Q, \Sigma$ algebra: $K=R+Q\Sigma$ , $L=R+Q(I+\Sigma)$ ; $\chi_Q, \chi_K$ ; det ladder $3\cdot 4\cdot 8\cdot 20=1920= W(D_5) $ ; $a^\top K a=41$                                                                                         |
| v11_unique_KQ.py                   | $K, Q$ are the <i>unique</i> nonneg-int matrices with their compiler budgets, spectrum and monotone rows (enumeration)                                                                                                                   |
| v12_mass_generation_polynomials.py | sector/generation polynomials of $K$ and the anchor-block det ladder (9, 10, 16, 40)                                                                                                                                                     |
| v13_open_gates.py                  | gate closures: $M=41=10b_1=\sum L+N_\Phi$ ; $Q_+=A_3$ exponents, $Q_-^2=N_{\text{fam}}$                                                                                                                                                  |
| v14_carrier_uniqueness.py          | $g_{\text{car}}=5$ unique ( $N_{\text{fam}} \in \mathbb{Z}^+$ , $\text{rank } E_8=8$ ); split (3, 2) from $b+s=5$ , $b^2+s^2=13$ ; $\text{Tr } Y=\text{Tr } Y^3=0$                                                                       |
| v15_bootstrap_classification.py    | $D_5\oplus A_3$ unique familyful cyclic-glue of $E_8$ (16-spinor + 3 families), glue $\mathbb{Z}_4$                                                                                                                                      |
| v16_solar_dual_anchor.py           | $a^\top R^{-1}=a^\top L^{-1}=(-\frac{1}{2}, -\frac{1}{2}, 1)$ ; $\sin^2 \theta_{12}=\frac{1}{3}-\frac{\varphi_0^{\text{ret}}}{2}$ ; full PMNS                                                                                            |
| v17_hexagonal_resolvent.py         | finite resolvent $D_y^{-1}=(\sum y^{5-m}\delta^m U_6^m)/(y^6-\delta^6)$ (quark $c$ backbone)                                                                                                                                             |
| v18_quark_yukawa.py                | quark source ratios $\frac{m_u}{m_d}=0.470085$ , $\frac{m_c}{m_s}=13.61$ , $\frac{m_t}{m_b}=40.80$ exact; lepton $c=(\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$ ; full hierarchy                                                           |
| v19_monodromy_moduli.py            | exact pole $z_\star=\frac{794-7\sqrt{9961}}{2187}$ ( $3^7$ ); $D_4$ $SU(3)$ monodromy constructible but $D_4$ -fixed locus positive-dim $\Rightarrow$ exact quark $c$ reduce to $(U)$                                                    |
| v20_lepton_c_derivation.py         | lepton $c$ 's derived: $\delta=\frac{1}{2}$ resolvent $c= \mu_4 ^w/(\frac{5}{4}-\cos \frac{r\pi}{3}) \Rightarrow \frac{16}{7}, \frac{4}{3}$ ; product $\frac{2^{g_{\text{car}}}}{N_{\text{fam}}^2}=\frac{32}{9} \Rightarrow \frac{7}{6}$ |
| v21_solar_product_quark.py         | solar coeff $=q(A_3)=\frac{3}{4}$ (glue-norm, $q(D_5)+q(A_3)=2$ ), $\sin^2 \theta_{12}=\frac{1}{3}-\frac{\varphi_0^{\text{ret}}}{2}$ ; product $\frac{2^{g_{\text{car}}}}{N_{\text{fam}}^2}$ ; quark law fails ( $\frac{1}{7}$ vs 0.724) |
| v22_open_gates_audit.py            | residual-gates audit contract: exact reduction of each open gate (A2,B3,B4,B5,B6,C7) machine-pinned (forced part vs. named residual)                                                                                                     |
| v23_anchor_generator.py            | anchor $a=(1, 1, 2)$ : $e_k=(4, 5, 2)$ , $c_3=\frac{1}{2e_1\pi}$ ; $p_n=2+2^n \Rightarrow 240, 248$ , binary ladder; inputs $\rightarrow \{a, \pi\}$                                                                                     |
| v24_quark_ratio_closure.py         | quark ratios $\frac{55}{117}, \frac{34}{47}, \frac{3}{26}$ from TFPT blocks; mass ratios match $< 0.03\%$ ; rational $c$ gauge, $\Lambda_q$ in $O(1)$                                                                                    |
| v25_frontier_conjectures.py        | conjectures: Koide $Q_{\text{src}}+\frac{\varphi_0^{\text{ret}}}{24}\approx\frac{2}{3}$ ; axion $f_a=M_{\text{scal}}/128\rightarrow m_a\approx 23.8 \mu\text{eV}$                                                                        |

continued on next page

| Script                                     | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                      |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v26_flavor_<br>frontier_<br>unification.py | the “11” is not uniquely forced ( $\Rightarrow$ [C]); flavor frontier unifies to one ( $U$ ) gate; cusp config on the parabolic stability wall                                                                                                                                                                                   |
| v27_wall_<br>representative.py             | explicit balanced wall rep $W_{\text{wall}}$ (cols = perm(0, 1, 2), rows $w=(2, 1, 1)$ , pardeg 0); selector $\det R=8$ , $\text{Spec}(Q_+)=\{1, 2, 3\}$                                                                                                                                                                         |
| v28_gravity_fR.py                          | $R+R^2$ closed: $f(R)$ , $M=c_3^{7/2}\bar{M}_{\text{Pl}}$ scalaron, $N_*=57\rightarrow n_s, r, A_s$ ; full metric measure open; $248c_3^2 < 6\log\frac{3}{2}$                                                                                                                                                                    |
| v29_research_<br>contract_certs.py         | research-contract certificates: $C_U^{(1)}$ wall enumeration (1296 $\rightarrow$ 144 $\rightarrow$ 5 orbits; symmetry alone does not pick $W_{\text{wall}}$ ) and $G5$ gap-dominance $2\ V\ <\Delta$                                                                                                                             |
| v30_d4_character_<br>variety.py            | $C_U^{(2)}$ : full $D_4$ -fixed $SU(3)$ character variety is positive-dimensional $\Rightarrow$ selector needs the $R(\rho)$ dictionary (H2; exact since v118–v122, residue = the $R4'$ selector establishment)                                                                                                                  |
| v31_R_<br>dictionary.py                    | $R(\rho)$ characterized: case A alive (irreducible $\Rightarrow$ mixing); $R$ integer but $\rho$ -invariants continuous $\Rightarrow R(\rho)$ is the transcendental non-abelian-Hodge map (residual = Hitchin solve)                                                                                                             |
| v32_rh_<br>splitting.py                    | RH route: $\sum_k U^k A_0 U^{-k} = 4 \text{diag}(A_0)$ exact $\Rightarrow$ splitting<br>$\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2 \Leftrightarrow \text{diag}(A_0) = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$ (algebraic); $\prod M_k = I$ + $R$ -bridge stay open ( $c_u/c_d$ not forced)                                       |
| v33_explicit_flat_<br>bundle.py            | explicit valid ( $U_{\text{wall}}$ ) flat bundle (RH solve): cusp + $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ + $\ M_\infty - I\  \sim 10^{-9}$ ( $\prod M_k = I$ ) + irreducible (case A)                                                                                                                                      |
| v34_h2_bridge_<br>attempt.py               | H2 bridge: explicit per-puncture $M_k$ (cusp, $\prod M_k = I$ ); honest negative — natural extraction $\neq$ lepton amplitudes, the $\Gamma^{\min}$ dictionary is the remaining input ( $c_u/c_d$ not obtained)                                                                                                                  |
| v35_quark_qcd_<br>boundary.py              | the open gates are the discrete $\leftrightarrow$ continuous boundary: quark cross-ratio cleanness tracks generation/QCD sensitivity (compiler hands off to continuous physics)                                                                                                                                                  |
| v36_spectral_<br>action_g2.py              | G2: spectral action $\rightarrow R+R^2$ ( $a_2 \sim -R/3$ , $a_4 _{R^2} \sim R^2/72$ ); $M^2/\bar{M}_{\text{Pl}}^2 = 6(4\pi)^2/f_0$ , $c_3^7$ fixes $f_0$ ; gap-decoupling $\Delta_{\text{eff}} = 1.648 > 0 \Rightarrow$ admissible IR closed under RP, ambient (G6) = strict-TOE completion target                              |
| v37_plucker_<br>anchor.py                  | anchor-plane Plücker apparatus: $\ Pl(K)\ _1=11$ , $\ Pl_R(K)\ _1=26$ , $\sum Pl_R(L)=60$ ; pencil $\det(K+xQ)=3x^3+7x^2+6x+4$ ; $K_e-K_u=a$ , $L_e-L_u=\text{Exp}(A_3)$ ; lepton ring $c_e c_\tau =  \mathbb{Z}_2 c_\mu$ ; Koide $u \rightarrow \frac{53}{54}u$ ([C])                                                           |
| v38_uwall_<br>killswitch.py                | ( $U_{\text{wall}}$ ) U2 kill-switch hardened: $D_4$ -cusp rep irreducible at all 400 sampled points (commutant=1), mixing $\geq 0.35$ , reducible locus measure-zero $\Rightarrow$ case A generic, ( $U_{\text{wall}}$ ) survives; amplitude dictionary still research-level                                                    |
| v39_uwall_<br>selectors.py                 | ( $U_{\text{wall}}$ ) selector side closed on the explicit point: splitting type = anchor $a=(1, 1, 2)$ , pardeg = 0; $\det R=8=n\cdot a$ , $\text{Spec}(Q_+)=\{1, 2, 3\}=3\alpha+1$ read off the bundle; only the amplitudes $c_u/c_d$ ( $\Gamma^{\min}$ Hitchin) remain                                                        |
| v40_harmonic_<br>metric.py                 | ( $U_{\text{wall}}$ ) harmonic metric = FINITE linear algebra (polystable $\Rightarrow$ unitary $\Rightarrow\Phi=0$ ): unique invariant form $H>0$ , unitarised $\tilde{M}_k$ cusp-class with $ \tilde{M}_k  \in \{0, \frac{1}{2}, \frac{1}{\sqrt{2}}\}$ ; the feared Hitchin PDE collapses to unitarisation + combinatorial $R$ |
| v41_leg_<br>assignment.py                  | ( $U_{\text{wall}}$ ) final leg test: amplitudes = $\delta=\frac{1}{2}$ resolvent (closed); $\Lambda_\mu > 1$ rules out holonomy moduli; harmonic holonomy $ \text{diag} =(0, \frac{1}{2}, \frac{1}{2})$ supplies $\delta=\frac{1}{2}$ (suggestive); honest negative on literal $c_u/c_d$ reproduction                           |
| v42_exterior_<br>leg.py                    | ( $U_{\text{wall}}$ ) Exterior Leg Lemma: $u, d$ share $(r, w)=(1, 1)$ so any scalar leg gives $c_u/c_d \equiv 1$ ; the $u/d$ split is the $\Lambda^2 F$ area $\text{Pl}_{1,a}(K)$ , $\ \cdot\ _1=11$ generated $\Rightarrow \frac{5\cdot 11}{9\cdot 13} = \frac{55}{117}$ [E]; phys. identification [C]                         |
| v43_exterior_<br>bridge.py                 | ( $U_{\text{wall}}$ ) $\Lambda^2 F$ bridge: $\Lambda^2(\tilde{M}) = \tilde{M}$ (exterior leg = $\bar{3}$ ) confirmed; continuous action $\neq$ integer $\text{Pl}(K) \Rightarrow$ bridge is the discrete H2 invariant ( $\det R=8$ , $\text{Spec}(Q_+)$ confirmed), not continuous                                               |

*continued on next page*

| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v44_carrier_exterior.py        | $(U_{\text{wall}})$ Lie-level grounding: $16=\Lambda^{\text{even}}(5)=1+10+5$ ; $u^c \in \Lambda^2(5)$ , $d^c \in \Lambda^4(5) \Rightarrow$ exterior degrees $(2, 4)$ , $\text{sum}=6= R^+(A_3) $ , $\text{diff}=2= \mathbb{Z}_2 $ ; the exterior leg is the carrier's own $\Lambda^2$ grading (no special math)                                                                                        |
| v45_family_exterior.py         | $(U_{\text{wall}})$ the “11” is Pascal: $ \text{Pl}(K) =(1, 4, 6)=\{\Lambda^0, \Lambda^1, \Lambda^2\}(\mu_4)$ ; $\ \text{Pl}(K)\ _1=11=\sum_{k \leq 2} \binom{4}{k}=16-g_{\text{car}}$ (cumulative ext of $\mu_4$ up to $\deg u^c=2$ ); $15=\dim \mathfrak{su}(4)=10+5=N_{\text{fam}}g_{\text{car}}$                                                                                                    |
| v46_grand_mass_volume.py       | absolute mass-scaling is [E], not a product rule: sector det $s$ $(\varphi_0^{\text{ret}})^6, (\varphi_0^{\text{ret}})^9, (\varphi_0^{\text{ret}})^{10}$ (exponents $ R^+A_3 , N_{\text{fam}}^2, \mathcal{A}_\Lambda$ ) $\Rightarrow$ $\det M_{\text{SM}} \sim (\varphi_0^{\text{ret}})^{25}=(\varphi_0^{\text{ret}})^{g_{\text{car}}^2}$ ; $Q$ -rows $(4, 5, 6)$ sum $15=\dim A_3$ ; $25+15=40=\sum L$ |
| v47_selection_theorem.py       | Boundary Carrier Selection (Thm A): $\mu_4 \rightarrow A_3, N_{\text{fam}}=3$ ; 16-spinor $\rightarrow D_5$ ; $q(D_5)+q(A_3)=2$ , glue $4= \mu_4 $ ; only $D_5 \oplus A_3$ meets all four conditions ( $D_8, E_7+A_1, E_6+A_2, A_4+A_4, A_8$ excluded) [E]/[C]                                                                                                                                          |
| v48_em_ward.py                 | EM boundary Ward (Thm C): $F_{U(1)} = \text{Maxwell } \alpha^3 - \text{Calderón } 2c_3^3 \alpha^2 - \text{transport } 8b_1 c_3^6 \log \frac{1}{\varphi_{\text{seam}}}$ ; $8b_1 = \frac{4}{5} M = \frac{164}{5}$ , $-\frac{5}{4}=q(D_5)$ [E]; Ward origin [C]                                                                                                                                            |
| v49_readout_rigidity.py        | Readout Rigidity (Thm U2): $c_u/c_d = \frac{55}{117}$ is constant on the discrete selector stratum $\mathcal{S}_{8,Q} \Rightarrow$ no point-uniqueness needed; $(U_{\text{wall}})=U_{\text{unitary}}+U_{\text{H2}}+U_{\Lambda^2}+U_{\text{point}}$ , only $U_{\text{point}}$ open                                                                                                                       |
| v50_q_geometry.py              | Q geometry (Thm Q): $Q_+ = \text{diag}(3, 2, 1)$ -block $\chi=(t-1)(t-2)(t-3)$ , $Q_- \chi=t(t^2-3)$ ; $Q$ unique under rows $(4, 5, 6)/\text{cols}(9, 5, 1)/\text{spectra}$ ; geom. origin [C]                                                                                                                                                                                                         |
| v51_boundary_half_step.py      | glue norms $=(g_{\text{car}}, N_{\text{fam}})/ \mu_4 $ : $q(D_5)=\frac{5}{4}$ , $q(A_3)=\frac{3}{4}$ ; four ops forced $\rightarrow \{\text{sum}=2$ root norm, $\text{diff}=\frac{1}{2}= \mathbb{Z}_2 / \mu_4 =\delta$ , $\text{prod}=\frac{15}{16}$ , $\text{ratio}=\frac{5}{3}\}$ . So $\delta=\frac{1}{2}$ is the carrier–family count over the glue index, not chosen [E]                           |
| v52_pencil_endpoints.py        | $K+xQ$ pencil endpoints: $P(-1)=2= \mathbb{Z}_2 $ , $P(0)=4= \mu_4 $ , $P(1)=20=\det L$ , $P(2)=68=2p_5(a)$ ; $\det(K-Q)=2$ , $\text{tr}=N_{\text{fam}}$ (audit ext. of v37) [E]                                                                                                                                                                                                                        |
| v53_compiler_core.py           | Big picture: the whole integer skeleton from $(g_{\text{car}}, N_{\text{fam}})=(5, 3)$ — rank $E_8=8$ , $ \mathbb{Z}_2 =2$ , Pythagorean $\Delta_Y=g_{\text{car}}^2=25=N_{\text{fam}}^2+ \mathbb{Z}_2  \cdot \text{rank } E_8=9+16=\dim S^+$ (diff. of squares); anchor char-poly $(t-1)^2(t-2)$ unique [E]                                                                                             |
| v54_seam_horizon_keystones.py  | The “8” in $c_3=\frac{1}{8\pi}$ overdetermined and gravitationally aligned $(2 \mu_4 =\text{rank } E_8=h(D_5)=\varphi(30)=8\pi$ Hawking/Einstein); one boundary transport operator (spec $\{1, (2/3)^6, (1/3)^6\}$ , $\lambda_2=(2/3)^6$ ) governs <i>both</i> SM flavor gap and horizon Page recovery [E]                                                                                              |
| v55_coxeter_cycle.py           | Computed $E_8$ Coxeter element: order exactly $30= \mathbb{Z}_2 N_{\text{fam}}g_{\text{car}}$ , eigenphases = exponents = totatives of 30, so rank $E_8=8=\varphi(30)$ = live phases of the order-30 cycle; one $\alpha^{-1}$ sets EM, $S_{dS}$ and $\Lambda$ ( $S_{dS}\rho_\Lambda=32\pi^4$ ) [E]                                                                                                      |
| v56_unique_attractor.py        | Gapped boundary transport (gap $6 \log \frac{3}{2} > 0$ ) $\Rightarrow$ <i>unique</i> attracting fixed point, geometric rate $(2/3)^6$ (two starts $\rightarrow$ same direction); Coxeter in $ \mu_4 =4$ planes; entropy reset = rank-1 projector [E]                                                                                                                                                   |
| v57_horizon_crosslinks.py      | Horizon cross-links: $c_3=\frac{1}{8\pi}$ is the Einstein/Jacobson coefficient ( $1/4=1/ \mu_4 $ entropy density); Hod QNM $\omega_R/T_H=\ln 3=\ln N_{\text{fam}}$ ; Hawking power denom $1920= W(D_5) $ ; one $\alpha^{-1}$ for EM/ $S_{dS}$ / $\Lambda$ /Hubble [E] (+ [C] identification)                                                                                                            |
| v58_seam_horizon_chain.py      | Seam-horizon chain (precise: seam = abstract normaliser, locally realised as a horizon): one-sided $S^2$ Gauss–Bonnet $c_3=1/(2 \cdot 4\pi)=\frac{1}{8\pi}$ ; KMS $4c_3\kappa$ ; $S_{BH}=M^2/2c_3$ ; RT in seam units. Seam–Horizon Theorem (area-law from the seam kernel) [O] open                                                                                                                    |
| v59_area_law_evidence.py       | Area-law necessary condition met: a reflection-positive Gaussian (Calderón-type) kernel yields an area law (gapped block-EE saturates; gapless $\sim \frac{\epsilon}{3} \log L$ ), by the <i>same</i> gap that fixes the attractor; $8= \mathbb{Z}_2  \mu_4 $ . The $c_3$ coefficient stays [O] open                                                                                                    |
| v60_lambda_metrology_branch.py | $\Lambda$ branch selection: $(8\pi)^2 \delta_{\text{top}} = \frac{3}{4\pi^2}$ , mis-scale $2c_3/\delta_{\text{top}} = \frac{64\pi^3}{3} = 661.47 \Rightarrow G_N$ pinned as output; 123 orders = 119.028+3.920; ACT DR6 birefringence $0.215^\circ \pm 0.074^\circ$ vs $0.2424^\circ$ ( $0.37\sigma$ ) [E]                                                                                              |

continued on next page

| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v61_cft_bridge.py             | Boundary-CFT mirror: level-1 WZW central charges $c(E_8)=8$ , $c(D_5)=5$ , $c(A_3)=3$ = (rank $E_8$ , $g_{\text{car}}$ , $N_{\text{fam}}$ ); conformal embedding $c_{\text{coset}}=0$ ; $SU(3) \leftrightarrow SU(4)$ reconciliation $N_{\text{fam}}= \mu_4 -1$ , $4 \rightarrow 3+1$ (one geometry $\mathbb{P}^1 \setminus \mu_4$ , two sides) [E]                                                                                                                                                       |
| v62_data_scorecard.py         | TFPT vs 2024/25 data: $\alpha^{-1}$ , $\sin^2 \theta_{12}$ , $\beta_{\text{rad}}$ match; $\sin^2 \theta_{13}$ ( $2.0\sigma$ ) and Starobinsky $n_s$ vs DESI ( $\sim 2\sigma$ ) mild tensions; $\theta_{23}$ open; $r \approx 0.004$ future falsifier [E]                                                                                                                                                                                                                                                  |
| v63_seam_engineering_index.py | Exact Seam-Engineering Index $\Xi=2\ V\ /\Delta=\frac{31}{24\pi^2 \log(3/2)} \approx 0.323$ ( $2\ V\ =\frac{31}{4\pi^2}$ , $248=\dim E_8$ , $31=1+h^\vee$ ), $\Delta_{\text{eff}} \approx 1.648$ ; four-class possibility taxonomy [E]/[O]                                                                                                                                                                                                                                                                |
| v64_causal_boundary_nogo.py   | Causal Boundary Engineering <i>conditional no-go</i> : gapped transfer has no periodic orbit (discrete no-CTC); RP+OS+gap $\Rightarrow$ ANEC $\Rightarrow$ Topological Censorship $\Rightarrow$ no traversable shortcut; only ER=EPR bridge survives [E] (core)/[C] (chain)                                                                                                                                                                                                                               |
| v65_falsification_layer.py    | Prediction layer with explicit failure thresholds: 3 core ( $v_{\text{GW}}=c$ , no 4th gen, seam=8), 4 observational ( $\beta$ , $r$ , $n_s$ , $\theta_{12}$ ), no-go + unitarity; $N_\star$ demoted to [C] input; none violated [E]                                                                                                                                                                                                                                                                      |
| v66_e8_casimir_degrees.py     | Organizing simplification: the compiler atoms <i>are</i> the $E_8$ Casimir degrees $\{2, 8, 12, 14, 18, 20, 24, 30\}$ ( $2= \mathbb{Z}_2 $ , $8=\text{rank}$ , $14=\dim G_2$ , $20=\det L$ , $24= W(A_3) $ , $30=h(E_8)$ ); $\sum=128=2^7$ (dS prefactor), $\sum \exp=120= R^+(E_8) $ [E]                                                                                                                                                                                                                 |
| v67_area_law_coefficient.py   | Central-theorem <i>structural closure</i> : Fursaev–Solodukhin $S=4\pi kA$ , $k=c_3/2 \Rightarrow S=2\pi c_3 A=\frac{1}{4}A$ ( $2\pi c_3=\frac{1}{4}$ ); $c_3=\frac{1}{8\pi}$ is the <i>unique</i> value with replica-coeff = BH $\frac{1}{4}$ . Reduces to the one coeff $k=c_3/2$ (then <i>forced</i> , v73) [E]                                                                                                                                                                                        |
| v68_seeley_dewitt_residual.py | Residual <i>resolved as an anchor</i> : $a_2=-\frac{d}{192\pi^2}R \Rightarrow$ the <i>absolute</i> $\frac{1}{16\pi G}$ is UV-sensitive ( $\propto d\Lambda^2 f_2$ ), so the <i>scale</i> $1/G$ is the one dimensionful anchor (the dimensionless $k=c_3/2$ is forced, v73); cutoff-indep. $R^2$ /scalon derived [E]                                                                                                                                                                                       |
| v69_d4_q_geometry.py          | $D_4$ -equivariant Q-geometry (gate [C]/[E]): on $H_1(\mathbb{P}^1 \setminus \mu_4)=B_1 \oplus E$ , $Q_+=3w+1=\text{diag}(1, 2, 3)$ (cusp weights on $\mu_4=\mathbb{Z}_4$ eigenspaces), $Q_-=E$ -coupling $\sqrt{N_{\text{fam}}} \Rightarrow t(t^2-3)$ ; $\Sigma$ -split $=D_4=\mathbb{Z}_4 \rtimes \mathbb{Z}_2$ [E]                                                                                                                                                                                     |
| v70_q_integer_lift.py         | Q integer-lift sharpened: $\mathbb{Z}_4$ puncture action $R$ unimodular (eigenvalues $\{-1, i, -i\}$ ); $\det Q=3=N_{\text{fam}}$ , SNF $\text{diag}(1, 1, 3) \Rightarrow$ residual = one lattice invariant $\det Q=N_{\text{fam}}$ , not closable by $D_4$ alone [E]                                                                                                                                                                                                                                     |
| v71_simple_r_bridge.py        | <i>Simple</i> R-bridge (gate #3): selector stratum $\mathcal{S}_{8,Q}$ now fully derived ( $\det R=8=\text{rank } E_8$ lattice + $\text{Spec}(Q_+)=\{1, 2, 3\}$ from v69) $\Rightarrow$ quark <i>ratios</i> are pure integer Plücker readouts ( $\frac{55}{117}$ , no monodromy, v49 rigidity); residual $U_{\text{point}}$ = absolute norm. = an anchor [E]/[O]                                                                                                                                          |
| v72_q_det_from_cusp.py        | $\det Q=N_{\text{fam}}$ derived from the cusp class: $\det Q= \text{coker } Q =\text{cusp-class order}=\text{denominator of the weights } \{0, \frac{1}{3}, \frac{2}{3}\}$ (same data as $\text{Spec } Q_+$ ); $\text{coker } Q=\mathbb{Z}/N_{\text{fam}}=\text{triality deck group} \Rightarrow$ integer-lift residual closed [E]                                                                                                                                                                        |
| v73_k_c3_half.py              | Central coefficient $k=c_3/2$ <i>forced</i> : $(1/2)$ variational $\times 1/( \mathbb{Z}_2 2\pi\chi(S^2))$ Gauss–Bonnet topology ( $\chi(S^2)=2$ ), both cutoff-indep $\Rightarrow$ Fursaev–Solodukhin $S=A/4$ ; only the absolute 4D Newton scale stays the anchor (v68) [E]                                                                                                                                                                                                                             |
| v74_compiler_micro_lemmas.py  | Spine quotient ladder ( $\frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{4}, \frac{4}{3}$ = adjacent quotients of $(2, 3, 4, 5)$ ); pencil differences $(2, 16, 48)=( \mathbb{Z}_2 , \dim S^+, \Omega_{\text{adm}})$ sheet $\rightarrow$ 1gen $\rightarrow$ 3gens; anchor QF $a^\top K a - \mathbf{1}^\top K \mathbf{1} = 41 - 25 = 16 = \dim S^+$ (EM budget = mass vol + one gen); solar dual anchor $L=R+6\mathbf{1}e_1^\top$ , $a^\top R^{-1}\mathbf{1}=0$ (Sherman–Morrison rank-one invariance) [E] |
| v75_upoint_to_vgeo.py         | <b>Gate 1 complete</b> : $U_{\text{point}} \rightarrow v_{\text{geo}}$ . Ratios closed (v49/v71) + lepton $c$ exact (v20) + sector products = $\varphi_0^{\text{ret}}$ -powers (v46); (ratios, product) $\Leftrightarrow$ (individuals) bijection $\Rightarrow$ absolute amplitudes fixed up to <i>one</i> scale $v_{\text{geo}}$ = the same anchor as $1/G$ (v68); two $[A] \rightarrow$ one [E]/[O]                                                                                                     |
| v76_gmetric_reduction.py      | <b>Gate 2</b> : Tier A (IR) closed under RP+gap (Decoupling Thm: $\Delta_{\text{eff}}=6\log \frac{3}{2}-\frac{31}{4\pi^2}=1.648>0$ , $\ V\ \leq 248c_3^2=\frac{31}{8\pi^2}$ ); Tier B (strict TOE) holographically reduced from bulk to a finite seam-boundary measure (conditional: RP+tightness $\Rightarrow$ closes) [E]/[C]                                                                                                                                                                           |

continued on next page

| Script                              | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v77_e8_conformal_net.py             | <b>G6 route:</b> the seam boundary measure = the $(E_8)_1$ <i>lattice net</i> (rigorous, holomorphic $c=8$ ); level-1 $c(E_8)=\frac{248}{31}=8$ , $c(D_5)=5$ , $c(A_3)=3$ , embedding $5+3=8$ (coset $c=0$ ); imports $G_6$ into RCF/T/conformal-net rigor [E]/[C]                                                                                                                                                                                     |
| v78_vgeo_floor.py                   | $v_{\text{geo}}$ <b>floor:</b> no pure number is dimensionful $\Rightarrow v_{\text{geo}}$ irreducible by logic; all scales are ratios $\times$ one $M_{\text{Pl}}$ ; $S_{dS}\rho_\Lambda=32\pi^4$ pins it from one measurement; theory at the minimum (one scale + $\pi$ ) [E]/[O]                                                                                                                                                                    |
| v79_review_identities.py            | Four validated review identities + one firewall: hypercharge Lucas–Binet $D_n=(3^n-(-2)^n)/5=1, 1, 7, 13, 55, 133$ (roots = carrier charges); Inverse Anchor Thm ( $\mathbf{1}^\top M^{-1}\mathbf{1}=1/\text{atom}$ , $a^\top M^{-1}a=1$ for $R, K, L$ ); MacWilliams (1, 10, 5); $6Y^2-Y-1=(2Y-1)(3Y+1)$ . Rejects $m_p/m_e$ , $\eta_B$ near-formulas (firewall) [E]                                                                                  |
| v80_operator_pencil_geometry.py     | Operator-pencil geometry: Anchor Singularity Thm $\det B_{K+xQ}=(N_{\text{fam}}x+ \mathbb{Z}_2 )(N_{\text{fam}}x+g_{\text{car}})=(3x+2)(3x+5)$ ( $\frac{2}{3}$ Koide/gap = anchor-plane singularity); buildup chain (3, 10, 16, 20); type checker $\det B_{Q,K,R,L}=(9, 10, 16, 40)$ ; audit: differential spectrum, $F_4 \times G_2$ shadow $248=52+14+26\cdot 7$ [E]/[C]                                                                             |
| v81_singular_pencil_matrices.py     | Double cover $y^2=\det B_{K+xQ}=(3x+2)(3x+5)$ : Koide $x=-\frac{2}{3}$ and carrier $x=-\frac{5}{3}$ are the two branch points (deck $2= \mathbb{Z}_2 $ ; disc $=81=N_{\text{fam}}^4$ ; separation $=1$ transport period). Clearing matrices $C_{2/3}=3K-2Q$ ( $D_5 \oplus A_3$ glue, $B=(5, 7)^\top(9, 11)$ , $\sum=240$ ), $C_{5/3}=3K-5Q$ (neutral, $\chi=(\lambda+2)(\lambda^2+\lambda+6)$ ); scalaron $7=$ mixed anchor disc $/N_{\text{fam}}$ [E] |
| v82_koide_attractor_splitting.py    | Koide attractor <i>forced</i> : the branch-divisor-preserving Möbius map fixing $q=2, 5$ is unique, multiplier $(2/3)^6=\lambda_2$ of the established transfer operator $\Rightarrow F'(2)<1$ attractor, $F'(5)>1$ repeller; three postulates $\rightarrow$ one [C]. Splitting trichotomy disc $81/49/40$ : the clean split is non-generic (anchor-first) [E]/[C]                                                                                      |
| v83_e8net_holomorphic_uniqueness.py | Holomorphy pins $(E_8)_1$ : #primaries = det Cartan ( $D_8:4$ vs $E_8:1$ ), and a holomorphic $c=8$ chiral CFT is the lattice theory of the <i>unique</i> even unimodular rank-8 lattice (Minkowski–Siegel mass $1/ W(E_8) $ ); carrier Pascal truncation $K=(g-1)/2 =$ half-spinor split [E]                                                                                                                                                          |
| v84_frozen_registry.py              | Blind-prediction registry frozen 2026-06-09 (25 digits, machine-enforced lock formula $\leftrightarrow$ value): exactly <i>one</i> $\theta_{12}$ prediction of record; $r$ , $n_s$ as $N_\star$ <i>bands</i> ; conditional entries carry their hypothesis by name [E]                                                                                                                                                                                  |
| v85_master_cover.py                 | Master-cover theorem: $\det B$ is $\text{GL}(2)$ -covariant on $\text{span}\{K, Q\} \Rightarrow$ exactly <i>one</i> double cover up to Möbius (disc $=N_{\text{fam}}^4 \det(G)^2$ ); the disc-81 family is its orbit ( $-\frac{8}{3}=-\text{rank } E_8/N_{\text{fam}} =$ carrier – one period); $\mu_4$ is <i>not</i> a 4:1 cover (ladder of double covers); branch trace fixes only the scalaron <i>scale</i> [E]                                     |
| v86_nstar_reheating.py              | $N_\star$ band $\rightarrow$ point [C]: $\Gamma=4M^3/48\pi\bar{M}^2=128$ GeV, $T_{\text{reh}}=9.6\times 10^9$ GeV, $N_\star(0.05/\text{Mpc})=51.4 \Rightarrow n_s=0.9611$ , $r=0.0045$ (inside the frozen band); honest tensions: $n_s+0.9\sigma$ , $A_s$ dichotomy (slow point $-11.4\sigma \Rightarrow$ near-instant reheating or exponent fails) [C]                                                                                                |
| v87_bulk_uniqueness_reduction.py    | Red-team Target A residual (ii) conditionally closed: holomorphic $\Rightarrow \text{Rep} = \text{Vect} \Rightarrow$ 2D bulk <i>unique</i> (LR/KLM/BKLR); machine contrast $SO(16)_1$ admits <i>six</i> modular invariants (incl. both $E_8$ pairings) $\Rightarrow$ Target A = one residual [E]/[C]/[O]                                                                                                                                               |
| v88_cp_phase_audit.py               | Target-D CP residual quantified: frozen $\delta=\pi/3+3\lambda^2$ survives at $+0.98\sigma$ vs $\gamma_{\text{PDG}}$ ; the $0.07^\circ$ -near alternative $\pi/3+2\lambda^2$ recorded as look-elsewhere trap, <i>not</i> adopted; decision at $\sigma_\gamma \leq 0.96^\circ$ [E]                                                                                                                                                                      |
| v89_carrier_index_lemma.py          | Carrier Index Lemma: KLM $\mu_A=[B:A]^2\mu_B \Rightarrow$ Jones index $[(E_8)_1:(D_5)_1 \times (A_3)_1]=4= \mu_4 $ (glue order <i>is</i> the index); $\mu$ -additivity $16/4^2=1 \Rightarrow$ holomorphy <i>follows</i> ; Gate A (H) $\Leftrightarrow$ (H') index-4 extension [E]                                                                                                                                                                      |
| v90_conical_defect_chain.py         | Fursaev–Solodukhin factor $4\pi$ <i>derived</i> (smoothed-cone Gauss–Bonnet, exactly smoothing-independent); replica $S=4\pi kA$ , $k=c_3/2 \Rightarrow S=A/4 \Leftrightarrow c_3=\frac{1}{8\pi}$ (sympy-solved); residual isolated to “seam determinant $\Rightarrow$ EH form” [E]/[O]                                                                                                                                                                |

*continued on next page*

| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v91_spine_tetrahedron.py        | Spine tetrahedron $\{2, 3, 4, 5\}$ = anchor + zeroth moment; edges/faces/volume = the integer grammar, volume $120= R^+(E_8) $ ; $240= \mu_4  E(K_4)  E(K_5) $ with $K_6$ negative control; dual cuts typed tautological; sub-grammar incompleteness honest [E]                                                                                                                                                                                                                                                                                                                                                       |
| v92_glue_uniqueness.py          | Glue uniqueness: of all 15 subgroups of the discriminant form exactly two Lagrangian glues (the two <i>chiralities</i> , swapped by either single spinor conjugation, fixed by the simultaneous one) and exactly one halfway extension = $SO(16)_1$ ; tower carrier $\rightarrow SO(16)_1 \rightarrow (E_8)_1$ , nothing else [E]                                                                                                                                                                                                                                                                                     |
| v93_koide_relaxation_toy.py     | Koide relaxation toy (P2 narrowed): basin lemma (every physical configuration in the attractor basin); exact rate $(2/3)^6$ ; source at $\rho = -\varphi_0^{\text{ret}}/24$ to 0.8% ([C]); honest negatives: flow time $t=2.84 \notin \mathbb{Z} \Rightarrow$ the missing object is a <i>continuous</i> generator [E]/[C]                                                                                                                                                                                                                                                                                             |
| v94_sheet_diamond.py            | Sheet diamond: all four flavor operators on <i>one</i> surface $M(s, t)=R+Q \text{ diag}(s, t, t)$ (pencil = the cut $(1+x, x-1)$ ; diagonal-cut disc 153 negative control); winding line $\det B=2 \det$ for all $s$ ; cofactor seam normal $(5, -9, 6)$ ; Plücker canonicalisation: 11 and 26 both live in $K$ ; spine-quotient firewall <i>rejected</i> [E]                                                                                                                                                                                                                                                        |
| v95_centered_diamond.py         | Centered cross: $Q=U+V$ , $R/L=C \mp U$ , $K/F=C \mp V$ (five objects $\rightarrow$ three); Spec $V=\{0, 1, 2\}=N_{\text{fam}} \cdot \text{cusp}$ ; $\det C=14$ , $\sum C=31=2^{g_{\text{car}}}-1$ ; $\text{Pl}_R(C)=7(2, 3, 1) \parallel \text{Pl}_R(L)=10(2, 3, 1)$ ; $G_2$ reading audit-typed [E]                                                                                                                                                                                                                                                                                                                 |
| v96_branch_kernel_selection.py  | Branch-kernel selection (P1 unblocked): rank-1 anchor blocks at the branch points have <i>integer</i> kernels (carrier kernel = 1); collapse lemma $\Rightarrow (-1, 1, 0)$ : up = -down, <i>lepton pairing</i> = 0 (leptons on the ramification); sector zero ladder (up $-\frac{3}{2}$ doubly, down 0 & $-\frac{9}{5}$ ); anchor-placement controls [E]                                                                                                                                                                                                                                                             |
| v97_sheet_conjugation_bridge.py | Sheet-conjugation bridge (P1 $\rightarrow$ one [C]): the anchor forces the cusp conjugation uniquely $\Rightarrow$ integer $T_A$ , $\det = -1$ ; $a=e_2+e_3$ (conjugation-symmetric); one deck action on $\mathbb{R}^3/\text{span}\{\mathbf{1}, a\}$ ; $\langle T_A, \Sigma \rangle = D_4$ ; sheet-index lemma (intertwiner dets even, $\min 2= \mathbb{Z}_2 $ ); remaining [C]: $Q_+$ grading = $A_3$ discriminant grading [E]/[C]                                                                                                                                                                                   |
| v98_discriminant_dictionary.py  | Dictionary <i>derived</i> (P1 closed modulo GATE.QGEO): $G=T_A \Sigma$ acts integrally on the cusp basis ( $Ge_1=-e_1$ , $Ge_3=e_2$ , $Ge_2=-e_3=B_1 \oplus E$ ); cusp-0 line = self-conjugate $\mathbb{Z}_4$ -class-2 line; $T_A G T_A^{-1} = G^{-1}$ ( $k \mapsto -k$ ); $q_{A_3}(1)=q_{A_3}(3)=\frac{3}{8}$ , $q_{A_3}(2)=\frac{1}{2}$ ; reflection classes = glue-swap vs glue-fix parities $\Rightarrow$ P1 carries no separate [C]; residual = the one existing $Q$ -geometry gate [E]                                                                                                                          |
| v99_koide_flow_time.py          | Koide flow time: canonical generator $\dot{q}=(\Delta/N_{\text{fam}})(q-2)(q-5)=(\Delta/N_{\text{fam}}) \det B(q)$ , time-1 map = $F$ [E]; non-integrality of $t=2.84$ is <i>data-fragile</i> ( $Q$ crosses $\frac{2}{3}$ at $+0.43\sigma(m_\tau)$ , $t$ passes every integer $\geq 3$ within $+0.5\sigma$ ) [E]; conditional steps: $n=2$ excluded $(-2.9\sigma)$ , $n=3=N_{\text{fam}} \Rightarrow m_\tau=1776.9427 \text{ MeV } (+0.14\sigma)$ , decision at $\sigma(m_\tau) \sim 0.01 \text{ MeV}$ [C]; $\ln(46080)$ scale reading destroyed by controls (firewall)                                               |
| v100_numerology_null_mc.py      | Numerology null test (look-elsewhere corrected): exact census of the full declared formula grammar over the 13 scored frozen-registry observables (conservative data windows) gives joint formula-fishing probability $\prod p_i = 10^{-25.8}$ , robust to budget/window variation; $2 \times 2 \cdot 10^5$ Monte-Carlo pseudo-theories reach at most 5/13 hits vs. TFPT 13/13; power controls (seed perturbation, data shuffle) collapse the scorecard; all 94 500 $F_{U(1)}$ variants solved, exactly <i>one</i> hits CODATA $\Rightarrow$ total $\leq 10^{-30.7}$ , <i>conditional on the declared grammar</i> [E] |
| v101_horizon_anchor.py          | The maximal black hole is the anchor (SdS in seam units): Nariai cubic $t^3-3t+2=(t-1)^2(t+2)$ , roots $(1, 1, -2)$ = traceless anchor; $S_{\text{Nariai}}/S_{\text{dS}}=\frac{2}{3}= \mathbb{Z}_2 /N_{\text{fam}}$ (per horizon $\frac{1}{3}$ ); interpolation $(x^2+1)/\Phi_3(x)$ ; three-sheet total $ \mathbb{Z}_2 S_{\text{dS}}$ conserved; $Q_{\text{geom}} \in [\frac{3}{8}, \frac{1}{2}]$ = the $SU(4)_1$ weights; mass-line cover $(1-3m)(1+3m)$ , deck = horizon swap; $\tau_{BH}=128 g_{\text{car}} M^3/c_3$ ; six atom landings, zero free parameters [E]/[C] (bulk reading)                              |

*continued on next page*

| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v102_seam_orientation.py       | One orientation: the anchor is the <i>stationary repeller</i> in both sectors — flavor relaxation = gradient flow of a cubic potential with critical points = the branch points, curvatures $V''(2)=+\Delta$ , $V''(5)=-\Delta$ (= the gap), inflection at scalaron/2, Lyapunov rate $\Delta$ constant; SdS entropy has Nariai as its <i>unique</i> stationary point, curvature $\frac{2}{9}= \mathbb{Z}_2 /N_{\text{fam}}^2$ ; evaporation = entropy ascent away from the anchor [E]; “one variational principle of the seam” stays a reading [C]                                                                                                                       |
| v103_trisection_normal_form.py | Trisection normal form (the canonical coordinate): SdS cubic $\Leftrightarrow \cos 3\theta = -3m$ (angle trisection; $\mathbb{Z}_3$ deck = triality); $m = \cos \psi / N_{\text{fam}}$ and $S_{\text{tot}}/S_{\text{dS}} = \frac{4}{3} - \frac{2}{3} \cos \frac{2\psi}{3}$ — one cosine of glue atoms; canonical curvature $(\frac{2}{3})^3 = ( \mathbb{Z}_2 /N_{\text{fam}})^{N_{\text{fam}}}$ (Koide constant to the family power); invariant slope $d\sigma/dm _N = -\frac{8}{9} = -\text{rank } E_8/N_{\text{fam}}^2$ ; bridge: flavor rate $(2/3)^{2N_{\text{fam}}}$ vs gravity curvature $(2/3)^{N_{\text{fam}}}$ , missing = one <i>clock</i> [E]/[C]             |
| v104_nariai_clock.py           | The <i>classical</i> Nariai clock is the anchor (third appearance): static pin $\phi=\rho$ solves the $dS_2$ static-patch equation with $m^2 = -2\Lambda = - \mathbb{Z}_2 \Lambda$ exactly (Ginsparg–Perry tower: one negative mode); in Hubble units $\chi_{\text{clock}} = (\lambda-1)(\lambda+2)$ = the anchor quadratic, eigenvalues $\{1, -2\}$ ; Nariai cubic = $(t-1) \cdot \chi_{\text{clock}}$ ; entropy-deviation rate $2H =  \mathbb{Z}_2 H$ ; honest $(2/3)$ -test <i>negative</i> for the classical clock — the quantum clock is the one remaining [C] of the seam variational principle [E]/[C]                                                            |
| v105_residual_inventory.py     | The residual inventory: <i>one constant</i> $(2/3= \mathbb{Z}_2 /N_{\text{fam}}$ exact in seven places across both sectors), <i>one anchor</i> (three dynamical appearances: configuration, Nariai roots, clock spectrum), <i>relocation theorem</i> (cosmological quantum clock suppressed by the $\Lambda$ hierarchy, 121.5 orders $\approx 2\alpha^{-1}/\ln 10 \Rightarrow$ the clock must be the seam transfer operator — one [C] identification), and the <i>complete machine-pinned gap list</i> : exactly five structural objects + one scale + $\pi$ ; a sixth gap would fail the script [E]/[C]                                                                 |
| v106_review_validation.py      | External-review validation: seed normal form $\varphi_0^{\text{ret}} = \frac{ \mu_4 }{N_{\text{fam}}} c_3 + \Omega_{\text{adm}} c_3^{ \mu_4 }$ exact [E]; hypercharge moment $\text{Tr } X^2 = 120 = 5!$ computed from the charge table; factorial spine $5! =  R^+(E_8)  = \sum \text{exponents}$ , $240 = 2 \cdot 120$ , $1920 = 2^4 \cdot 5!$ ; degree-2 inventory (Pascal $K=2$ unique at $g=5$ , glue norms sum 2, $c_3 = \frac{1}{2} \times \frac{1}{4\pi}$ , $\binom{5}{2} = 10$ ); named [C] hypothesis <i>Quadratic Boundary Locality</i> ( $\Rightarrow K=2$ forced; would upgrade the Pascal selection to [E]); audit $240 = 16 \times 15$ non-unique [E]/[C] |
| v107_quantum_clock_target.py   | Quantum-clock target made quantitative: the classical Ginsparg–Perry clock decays on $\{0, -1, -2\} = -\text{Spec}(V) = -N_{\text{fam}} \times \text{cusp}$ (the transfer operator’s own grading) [E]; exact target identity $(1/3)^6 = ((2/3)^6)^{\log_{3/2} 3}$ — required bend $\log_{9/4} 3 = 1.355$ per weight step; coupling landscape $\kappa = c/(24\pi) \cdot \Lambda/\bar{M}_{\text{Pl}}^2$ : $10^{-122}$ cosmological vs $1/(3\pi)$ at seam curvature (the only viable, borderline-nonperturbative regime) [E]; identification stays [C]                                                                                                                      |
| v108_pascal_ladder.py          | Pascal Ladder Theorem: $2^{g-1} = \sum_{k \leq K} \binom{g}{k} \Leftrightarrow g=2K+1$ (odd- $g$ midpoint identity + even- $g$ straddle) $\Rightarrow$ carrier selection $g=5$ <i>exactly equivalent</i> to degree bound $K=2$ — the Target-B residual = the one QBL hypothesis; neighbour worlds: $K=1$ one-family, $K=3$ nine-family, $K=4$ inconsistent; only $K=2$ also satisfies $g+N_{\text{fam}}=8$ (v14 overdetermination) [E]/[C]                                                                                                                                                                                                                               |
| v109_sheet_pairing.py          | Sheet-Pairing Lemma (QBL anchored): character-exact multisets from $(\pm \frac{1}{2})^5$ — no scalar within a sheet (zero-weight mult of $S^+ \otimes S^+ = 0$ ; odd $g$ flips parity); $S^+ \otimes S^- = \Lambda^0 \oplus \Lambda^2 \oplus \Lambda^4$ exactly ( $256 = 1 + 45 + 210$ , zero-modes $(1, 5, 10)$ = the code); sheet-diagonal = odd forms $(2\Lambda^1 + 2\Lambda^3 + \Lambda^5)$ ; tower tops at $\Lambda^{2K}$ , $K=2 \Rightarrow$ the seam scalar kernel is necessarily a <i>sheet pairing</i> [E]/[C]                                                                                                                                                 |
| v110_calderon_sheet.py         | Calderón-sheet selection theorem: a Calderón involution $\varepsilon$ on $S^+ \oplus S^-$ certifies a scalar two-point datum <i>iff</i> it is sheet-odd; sheet-odd $\Rightarrow$ exactly $2= \mathbb{Z}_2 $ invariant kernels = the two Lagrangian glues; ladder genericity ( $g=3, 5, 7$ ): sheet-oddness forces $K=(g-1)/2$ , <i>not</i> $g=5$ (anti-overclaim); QBL residue split into <i>interface</i> (“ $\varepsilon$ sheet-odd”, natural from the one-sided collar) + <i>analytic core</i> (slot-degrees $\leq 2$ ) [E]/[C]                                                                                                                                       |

continued on next page

| Script                            | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v111_quadratic_transport.py       | Quadratic-Transport Theorem: in the integer Jordan–Wigner model linears are sheet-odd, the 45 quadratics (= $\mathfrak{so}(10)$ , the $\Lambda^2$ term of $1+45+210$ ) sheet-even, generate the code from the vacuum <i>and</i> span the full $\text{End}(S^+)=256$ at word length $2 \Rightarrow$ <i>transport degree exactly 2</i> (degree $\leq 1$ generates nothing) — the transport half of the QBL core is a theorem; $g=3$ control: degree selected, not rank [E]/[C]                                                                                                                                                                   |
| v112_selfcounting_channel.py      | Self-Counting Channel: $w \mapsto (w, -w)$ is a bijection code states $\leftrightarrow$ neutral certified kernels, so $\#\text{kernels} = 2^{g-1}$ <i>exactly</i> ; graded by pair-degree the same set has sizes $\binom{g}{m} \Rightarrow$ the Pascal closure $2^{g-1} = \sum_{m \leq K} \binom{g}{m}$ is <i>two countings of one set</i> (an identity, not a requirement); Hodge fold $\binom{g}{2j} = \binom{g}{\min(2j, g-2j)}$ ; QBL residue = one input (“a single scalar 2-point kernel”), selection carried by rank-8+integrality [E]/[C]                                                                                              |
| v113_quasifree_kernel.py          | One kernel is the whole net: carrier $(D_5)_1 \times (A_3)_1 = SO(10)_1 \times SO(6)_1 = 16$ free Majoranas, $c=5+3=8$ at every tower level (only the glue grows, index $2 \times 2 = 4 =  \mu_4 $ ); Wick/Pfaffian exact (all $210+210$ higher correlators = Pfaffians of the one kernel); kernel = Calderón involution, rank $P=5=g_{\text{car}}$ , seam level $8 = \text{rank } E_8$ ( $c = \text{rank of the kernel}$ ); vacuum unique $\Rightarrow$ QBL input = R2 premise: gate closes $\Rightarrow$ carrier choice closes [E]/[C]                                                                                                       |
| v114_torsion_delta.py             | Torsion normal form + $\delta = \frac{1}{2}$ theorem: flatness of the $(U_{\text{wall}})$ $\mathbb{Z}_4$ -family $\Leftrightarrow (MU)^4 = \mathbf{1}$ ( $\mu_4 =$ order of the twisted cusp generator); on the involutive branch ( $T^2 = \mathbf{1}$ ) the cusp trace forces $\text{diag } M = (0, \frac{i}{2}, -\frac{i}{2})$ with $\text{spec } \{1, \omega, \omega^2\}$ automatic $\Rightarrow \delta = \frac{1}{2}$ exact and constant (v20’s hand value = torsion identity); census: $\{1, i, -i\}$ branch varies, v33 point in its $d_1=0$ slice; residue = bundle-side branch selection [E]/[C]                                       |
| v115_anchor_residue.py            | The anchor pins the residue: $\mu_4$ -average = diagonal projection $\Rightarrow$ anchor splitting $\Leftrightarrow \text{diag } A_0 = a/ \mu_4 $ ; off-weights uniquely $(8, 0, 5)/144$ ( $= (\text{rank } E_8, 0, g_{\text{car}})/( \mu_4 N_{\text{fam}})^2$ , total $13 = \Delta_Q$ ); <i>exact</i> residue $A_0^*$ with $\chi = \lambda(\lambda - \frac{1}{3})(\lambda - \frac{2}{3})$ is the RH solution ( $\ M_\infty - \mathbf{1}\  \sim 10^{-10}$ ); anchor forces $d_1=0 \Rightarrow$ harmonic diagonal anchor+torsion-forced [E]                                                                                                     |
| v116_resonance_uniqueness.py      | Resonance theorem (v115 [N] $\rightarrow$ [E], linear — no Gröbner): twisted $\mu_4$ averages collapse to $B_1 = 4a_{12}E_{12} + 4\overline{a_{13}}E_{31}$ ; anchor exponents $\text{diag}(2, 1, 1)$ give resonance gap 1 with obstruction $4\overline{a_{13}} \Rightarrow M_\infty = \mathbf{1} \Leftrightarrow a_{13}=0$ ; + $(8, 0, 5)/144$ + Diagonaleichung $\Rightarrow$ flat anchor locus = <i>one</i> gauge orbit = exact $A_0^*$ — $U_{\text{wall}}$ bundle datum algebraically unique; sharpness control $\ M_\infty - \mathbf{1}\  \sim 8.5 a_{13} $ [E]                                                                            |
| v117_monodromy_weyl_a3.py         | The monodromy is $W(A_3)$ : the unitarised holonomy of $A_0^*$ is <i>exact</i> (entries in $\frac{1}{2}\mathbb{Z}[i]$ ), $\text{diag } \widetilde{M}_0 = (0, -\frac{i}{2}, \frac{i}{2}) \Rightarrow d_1=0, \delta = \frac{1}{2}$ are matrix entries; $\langle U, \widetilde{M}_0 \rangle$ enumerated exactly: order 24, statistics $(1, 9, 8, 6)$ , characters $(3, -1, 0, 1) = S_4 = W(A_3)$ (standard $\otimes$ sign) — the flavor wall realises the $A_3$ glue factor as its Weyl group; ODE match $3.5 \times 10^{-10}$ [E]                                                                                                                |
| v118_hexagon_family_dictionary.py | First exact H2 piece: $\text{spec}(\widetilde{M}_0) \cup \text{spec}(-\widetilde{M}_0) = \mu_6$ (hexagon = family spectrum + sheet twist, $\mathbb{Z}_6 = \mathbb{Z}_2 \times \mathbb{Z}_3$ ); $\det(\mathbf{1} - t\widetilde{M}_0) = 1 - t^3 \Rightarrow \det_{\mp} = (\frac{7}{8}, \frac{9}{8})$ ; lepton coefficients = resolvent determinants ( $c_e =  \mu_4 /(2 \det_-)$ , $c_\mu = N_{\text{fam}}/(2 \det_+)$ , $c_\tau =  \mu_4  \det_- / N_{\text{fam}}$ , product = $ \mu_4 /\det_+ = \frac{32}{9}$ ); $\langle U, \widetilde{M}_0, -\mathbf{1} \rangle$ has order $48 = \Omega_{\text{adm}}$ ; the address table stays open [E]/[C] |
| v119_review_validation_2.py       | Second review validation: anchor ratio triad $(e_3, e_1, e_2)/p_0 = (\frac{2}{3}, \frac{4}{3}, \frac{5}{3}) = (\text{Koide branch, seed gain, carrier branch; flow critical points } (2, 5) = (e_3, e_2))$ ; the 121 audit lemma $(\mathbf{1}+a)^T R(\mathbf{1}+a) = 121 = 11^2 = \ \text{Pl}(K)\ _1^2$ (orderings $\{105, 121, 135\}$ , only canonical = 121; audit, not load-bearing); micro-identities $\Omega_{\text{adm}} = 2p_1p_2$ , $10b_1 = 1 + p_1p_3$ , $\Delta_Y = e_2^2 = p_0^2 + 2 \text{rank}$ ; flavor surface = $\sqrt{94}/\sqrt{95}$ (presentational) [E]                                                                    |

continued on next page

| Script                    | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v120_address_table.py     | Address table (frozen v18/v20 integers): lepton words $(8, 5, 3) = (\text{rank } E_8, g_{\text{car}}, N_{\text{fam}})$ ; addresses = divmod by the hexagon ( $p_2=6$ ): $e \rightarrow (2, 1)$ , $\mu \rightarrow (5, 0)$ , $\tau \rightarrow (3, 0)$ , $\sum r=p_3$ ; sheet parity: $\tau$ sits at the real twisted eigenvalue $-1$ (= the product-rule site); quark sums: up = $p_3$ , down = $p_1+p_3$ , all quarks = $24= W(A_3) $ , all nine = $40=p_1p_3=a^T Ra$ ; top at the vacuum site $(0, 0)$ ; the assignment derivation stays open [E]/[C]                        |
| v121_address_pinning.py   | Address pinning: $L=R+2U$ (v95) $\Rightarrow$ the word table = the residue operator $R$ + the generation-1 winding, $R$ column 1 = the anchor $a$ ; margins = atoms (rows $(e_1, \text{rank}, p_3)$ , columns $(e_1, \Delta_Q, g_{\text{car}})$ ); <i>census theorem</i> : among all hexagon matrices with atom margins (17 of them) exactly <i>one</i> has $\det=8$ — and it is $R$ (also unique with SNF $(1, 1, 8)$ ; all 17 determinants distinct) $\Rightarrow$ the address table carries <i>zero</i> residual information; the H2 residue = the six atom margins [E]/[C] |
| v122_margin_theorem.py    | Margin theorem: the <i>established</i> selectors — the $D_4$ annihilator $n=(5, -9, 6)$ with $n^T R=(8, 0, 0)$ (v94), $\det R=8$ , $\det K=4$ (the v71 quartet) — pin $R$ <i>uniquely</i> among hexagon matrices ( $64 \rightarrow 12 \rightarrow 1$ ) $\Rightarrow$ the atom margins, the anchor column and all sum rules are <i>consequences</i> (their input status is revoked); bonus: $\det L=20$ holds automatically on the locus (the quartet is redundant); H2 introduces <i>no</i> new residual class [E]/[C]                                                         |
| v123_inventory_update.py  | Residual inventory updated (post v110–v122): thirteen new ledger rows machine-pinned; R2 carries <i>three</i> loads (metric + carrier + QBL: one theorem, three doors); the table re-pins to <i>five</i> classes — R1 clock, R2 index-4 net, R3 EH, R4' selector establishment (replacing the <i>discharged</i> H2 dictionary), R5 $Q$ realisation — + $v_{\text{geo}}, \pi$ ; a sixth class fails the script [E]                                                                                                                                                              |
| v124_resummed_clock.py    | Resummed clock (the R1 bend in closed form): transfer eigenvalues = exactly $(1-n/N_{\text{fam}})^{p_2} \Rightarrow \text{rate}(n)=-p_2 \ln(1-n/N_{\text{fam}})$ reproduces $\{0, \Delta, 6 \ln 3\}$ , the bend $\log_{3/2} 3$ in one line; the linearisation carries slope $ \mathbb{Z}_2 =2$ relative to the classical GP clock $\{0, -1, -2\}$ ; the pole at $n=N_{\text{fam}}$ forces the three-level spectrum; the R1 target is now closed-form [E]/[C]                                                                                                                   |
| v125_glue_qsystem.py      | Glue Q-system: $\mathbb{C}[\mathbb{Z}_4]$ satisfies all Longo Q-system axioms exactly (associativity, unit, Frobenius, specialness $mm^*= \mathbb{Z}_4 \mathbf{1} \Rightarrow \text{Jones index } 4= \mu_4 $ (the v89 value); locality = isotropy ( $q(k(1, 1))=k^2 \in \mathbb{Z}$ , v92); halfway sub-Q-system $\mathbb{C}[\mathbb{Z}_2] = \text{the } SO(16)_1 \text{ step}$ , $4=2 \times 2$ ; R2 sharpens from “find” to “identify” [E]/[C]                                                                                                                               |
| v126_clock_wall_bridge.py | Clock and wall share one spectrum: the resummation weights $n/N_{\text{fam}}=\{0, \frac{1}{3}, \frac{2}{3}\} = \text{exactly spec } A_0^*$ (the parabolic MS weights); $\lambda_n=(1-\alpha_n)^{p_2}$ ; checkpoint 1 passed <i>classically</i> ( $p_2/N_{\text{fam}}=2= \mathbb{Z}_2  = \text{the v104 entropy rate } 2H$ ); $\det(\mathbf{1}-A_0^*)=\frac{2}{9} = \text{the entropy curvature (v102)}$ ; the quantum content of R1 is isolated in the geometric tail [E]/[C]                                                                                                  |
| v127_ring_resummation.py  | The clock is a log determinant: $\text{rate} = -p_2 \text{Tr} \ln(\mathbf{1}-\alpha P)$ — six identical ring (RPA) towers, <i>one per hexagon site</i> , coupling = the parabolic weight; the $k=1$ ring = the classical term, the $k=2$ ring = the first quantum correction; $\kappa_{\text{seam}}=\frac{1}{3\pi}=\alpha_1/\pi$ (audit); the wall = the RPA instability at $\alpha \rightarrow 1$ ; v107's “O(1) resummation” is thereby <i>typed</i> (ring/log-det), its derivation open [E]/[C]                                                                             |
| v128_graded_hull.py       | Graded hull + zero-mode multiplicity: the 240 $E_8$ roots explicit in $D_5 \oplus A_3$ + glue coordinates, coset decomposition $240=52+64+60+64$ ; the grading is exactly additive on <i>all</i> 6720 root-pair sums $\Rightarrow E_8$ is a $\mathbb{Z}_4$ -graded Lie algebra over the carrier (grading = glue = Q-system); ring multiplicity $6=2(2l+1) _{l=1} = \text{the sheet-doubled GP zero modes [E]/[C]}$                                                                                                                                                             |
| v129_entropy_power_law.py | The clock is an entropy power law: $\lambda_n=(S_n/S_{dS})^{p_2}$ with the three Nariai entropy levels $S/S_{dS} \in \{1, \frac{2}{3}, \frac{1}{3}\}$ (v101); $\text{rate} = p_2 \ln(S_{dS}/S)$ ; triple identity $\alpha=n/N_{\text{fam}} = \text{MS weight} = \text{entropy-deficit share} \Rightarrow \text{the RPA coupling is nowhere free}$ ; orientation consistent (attractor = max entropy); R1 thermodynamic: $\Gamma \propto (S/S_{dS})^{p_2}$ (a Gibbons–Hawking power law, $h=N_{\text{fam}}$ audit) [E]/[C]                                                      |

continued on next page

| Script                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v130_born_square.py          | Born square: amplitude $\sim (S/S_{dS})^{n_{\text{zero}}/2} = (S/S_{dS})^3$ (one $S^{1/2}$ per zero mode), probability $=  A ^2 = (S/S_{dS})^6 \Rightarrow h = n_{\text{zero}}/2 = 3 = N_{\text{fam}}$ , $p_2 = 2h$ (the Born rule) — the v129 weight audit is <i>derived</i> ; the 3 = the dimension of the $l=1$ space = the number of proper conformal Killing vectors of $S^2$ ; the $\times 2$ = the horizon pair (the v101 deck); the R1 residue = the standard zero-mode measure <a href="#">[E]</a> / <a href="#">[C]</a>                                                                                                                            |
| v131_measure_is_area.py      | The measure is the area: $\ Y_{1m}\ ^2 = r^2 = A/(4\pi)$ <i>exactly</i> (all three $l=1$ modes integrated) $\Rightarrow$ the per-mode Jacobian = the mode norm $\sim S^{1/2} \Rightarrow$ amplitude $S^3$ , Born $S^6$ — every exponent factor with a derivation origin; the $4\pi$ = the same Gauss–Bonnet primitive as in $c_3$ , cancelling in all ratios; the R1 residue = <i>one</i> det-ratio cancellation $\det'_n / \det'_{dS} = 1 + O(\text{deficit})$ <a href="#">[E]</a> / <a href="#">[C]</a> ( <i>cancellation since derived within the SdS family, v144</i> )                                                                                  |
| v132_detratio_anomaly.py     | Det-ratio anomaly = the Koide constant: $\zeta(0) _{\det'}$ of the GP sphere operator $-\Delta - 2$ (the 3 zero modes removed) $= a_1 - 3 = \frac{7}{3} - 3 = -\frac{2}{3} = - \mathbb{Z}_2 /N_{\text{fam}}$ exactly ( $a_1 = \frac{1}{N_{\text{fam}}} +  \mathbb{Z}_2 $ ; numerical continuation $\sim 10^{-7}$ ) — the eighth $2/3$ appearance, the first as a <i>spectral</i> anomaly; consequence $\det'(cL) = c^{\zeta(0)} \det'(L)$ ; the R1 residue = <i>one</i> $\zeta(0)$ budget statement (the 4d sum) <a href="#">[E]</a> / <a href="#">[C]</a>                                                                                                   |
| v133_zeta_budget.py          | The $\zeta$ budget, both routes exact (Euclidean Nariai $= S^2 \times S^2$ ): the reduced route gives $-\frac{2}{3}$ per sector, total $-\frac{4}{3} = -e_1/p_0$ (= minus the seed gain — two of the three triad values); the naive 4d route gives $a_2 = \frac{161}{45}$ , $\zeta(0) = -\frac{109}{45}$ ( <i>no</i> atom; continuation $\sim 10^{-8}$ ); zero modes $3+3$ = the horizon pair $\Rightarrow$ the budget structurally favours the <i>reduced</i> seam reading; remainder = graviton/ghost shares (Volkov–Wipf class) <a href="#">[E]</a> / <a href="#">[C]</a>                                                                                 |
| v134_dual_anchor.py          | Dual anchor lemma: $d := a^\top R^{-1} = a^\top L^{-1} = (-\frac{1}{2}, -\frac{1}{2}, 1)$ , $d \cdot \mathbf{1} = 0$ , $d \cdot a = 1$ , $(1, 1, -2) = -2d$ — the inverse flavor response <i>is</i> the Nariai root (v101); the invariance is exact via Sherman–Morrison ( $\Leftrightarrow$ orthogonality to $R^{-1}\mathbf{1} = \frac{1}{4}(1, 1, -1)$ ); controls: $\mathbf{1}, e_1, n$ do <i>not</i> lie on the plane); $(d, n)$ = the dual normal pair of the flavor boundary; a third purely <i>algebraic</i> leg of the flavor $\leftrightarrow$ horizon bridge <a href="#">[E]</a> / <a href="#">[C]</a>                                             |
| v135_det_surface.py          | Determinant surface: $\det M(s, t) = 3s(t+1)(t+2) + t^2 + 5t + 8$ — two $s$ -independent iso-volume walls $t = -2 \Rightarrow 2 =  \mathbb{Z}_2 $ , $t = -1 \Rightarrow 4 =  \mu_4 $ ( $\det K = 4$ is a <i>layer</i> ); winding line $6s + 8 \Rightarrow (8, 14, 20)$ , $\det F = 32 = 2^{g_{\text{car}}}$ ; $\det B(s, 0) = 2 \det M(s, 0)$ exactly, $B$ -wall only at $t = -2$ ; row-sum axes $C = (7, 11, 13)$ , $U = (3, 3, 3)$ , $V = (1, 2, 3) = \text{Spec}(Q_+)$ ; micro-identities $e_1^2 + e_2^2 = 41 = 10b_1$ , $p_1^2 + p_2^2 = 52 =$ the carrier coset root count (v128) <a href="#">[E]</a> / <a href="#">[C]</a>                             |
| v136_dual_normal_selector.py | Dual-normal selector: $(d, n)$ pins $R$ <i>columnwise</i> — $d \cdot c_j = a_j$ , $n \cdot c_j = 8\delta_{1j}$ , kernel = primitive $(6, 8, 7)$ , census: each column unique (1 of 216) $\Rightarrow R4'$ shrinks from $6^9$ to three column problems; the $A_0^*$ zero mode carries $\sigma = (2, -9, 5) = ( \mathbb{Z}_2 , -N_{\text{fam}}^2, g_{\text{car}})$ ( $\ v\ ^2 = \frac{16}{9}$ , $16 = \dim S^+$ ; $\text{adj} = \frac{2}{9}P_v$ = the SdS curvature); $n \cdot \sigma = 121 = 11^2$ ; $W(A_3)$ orbit control: the naive lift $\sigma \rightarrow n$ fails — a genuine mechanism, not a group element <a href="#">[E]</a> / <a href="#">[C]</a> |
| v137_qplus_cohomology.py     | The $Q_+$ grading is cohomology: $H^1(\mathbb{P} \setminus \mu_4) = \chi_1 \oplus \chi_2 \oplus \chi_3$ ( $\omega_k = z^{k-1} dz / (z^4 - 1)$ ), character $i^k$ , residues $\zeta^k/4$ ; degrees $(1, 2, 3) = \text{Spec}(Q_+) =$ the $A_3$ exponents ( $h = 4 =  \mu_4 $ ); cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\} = \text{spec}(A_0^*) \Rightarrow$ MS weights = cusp classes = characters = $Q_+$ : <i>one</i> spectrum, four readouts; the QGEO residue = the naturality of the canonical map <a href="#">[E]</a> / <a href="#">[C]</a>                                                                                                          |
| v138_vw_firewall.py          | Volkov–Wipf firewall (R1 typed closed): external 4d value $\alpha_{\text{PI}} = -\frac{98}{45} = \frac{22}{45} - \frac{8}{3}$ (arXiv:2506.02142 Tab. 4.1 = VW hep-th/0003081); <b>edge match</b> : $\alpha_{\text{edge}} = -\frac{8}{3} = 2 \times (-\frac{4}{3})$ — two identical edge copies (the horizon pair), per copy exactly the reduced seam budget (v133, the seed gain); bulk $\frac{22}{45}$ = the graviton gas (not the clock); difference $-\frac{38}{45}$ no atom $\Rightarrow$ typing: the seam clock = an edge object, the full 4d determinant = the gravity firewall <a href="#">[E]</a> / <a href="#">[C]</a>                              |

*continued on next page*

| Script                    | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v139_selector_triangle.py | Selector triangle: $d=\frac{3}{2}a-2\cdot\mathbf{1}$ exactly (pure anchor data, no $R$ ) $\Rightarrow$ the first selector is <i>derived</i> ; the frame $(\mathbf{1}, a, \sigma)$ has $\det=11=\ \text{Pl}(K)\ _1$ , and $n$ is the <i>unique</i> covector with atom pairings $(2, 8, 121)=( \mathbb{Z}_2 , \text{rank } E_8, 11^2)$ ; on the constrained line the pairing is $11(11+t)$ , a square only at the true $n$ ; review lift exact (audit): $n=\text{reverse}(\sigma)+ \mu_4 e_3$ ; $R4'$ residue = three pairings $[\mathbf{E}]/[\mathbf{C}]$ ( <i>since v142: two — the third pairing is integrality-forced</i> ) |
| v140_canonical_map.py     | The canonical map exists and is rigid: the plain deck $U$ (characters $(0, 1, 3)$ ) cannot match $H^1$ ; the <i>sheet-twisted</i> $-U$ $((2, 3, 1))$ and the cusp exponential $i^{Q+}((1, 2, 3))$ both carry the $H^1$ character set $\{1, 2, 3\}$ (the v118 sign twist); multiplicity-freeness $\Rightarrow$ Schur-rigid equivariant isomorphism (unique up to scalars); $i^{Q+}$ pulls the Euler degree back to $Q_+$ , $-U$ to $Q_+\circ\rho$ — the QGEO residue collapses to <i>one</i> $\mathbb{Z}_3$ choice $[\mathbf{E}]/[\mathbf{C}]$ ( <i>the choice since derived, v141</i> )                                       |
| v141_deck_selection.py    | Deck Selection Theorem (R5): the established integer deck $G=T_A\Sigma$ (v97/v98) pairs the $Q_+=1$ line with the self-conjugate character 2 $\Rightarrow$ of the three $\mathbb{Z}_3$ rotations only the sheet-twisted $(2, 3, 1) = \text{chars}(-U)$ survives; $i^{Q+}$ excluded (same spectrum, wrong grading pairing); residual freedom = the established sheet $\mathbb{Z}_2$ (v92); Euler degree = $Q_+\circ\rho$ $[\mathbf{E}]/[\mathbf{C}]$                                                                                                                                                                           |
| v142_frame_integrality.py | Frame Integrality (R4'): pairing triples of <i>integer</i> covectors against $(\mathbf{1}, a, \sigma)$ form an index-11 sublattice $(x-3y+z \equiv 0 \bmod 11, 11=\ \text{Pl}(K)\ _1)$ ; with $(x, y)=( \mathbb{Z}_2 , \text{rank } E_8)$ the third pairing is forced $\equiv 0 \bmod 11$ , primitivity forces even $t$ ; Cramer reductions for the two line pairings; $R4'$ : three pairings $\rightarrow$ two $[\mathbf{E}]/[\mathbf{O}]$ ( <i>since v145: the values are atom identities — residue = one lift map</i> )                                                                                                    |
| v143_graded_frobenius.py  | Graded Frobenius (R2 at the finite Lie level): coset duality $-C_k=C_{-k}$ exact on all 240 roots $\Rightarrow$ Killing pairing nondegenerate per $g_k \times g_{-k}$ ; glue average = carrier projector (index $ \mathbb{Z}_4 =4$ ); each sector one $W(D_5)\times W(A_3)$ orbit = simple current; fusion = $\mathbb{C}[\mathbb{Z}_4]$ (v125); residual = the one conformal-net statement $[\mathbf{E}]/[\mathbf{C}]$                                                                                                                                                                                                        |
| v144_detratio_family.py   | Det-ratio cancellation within the SdS family (R1, the v131 step): $e_2$ -rigidity gives $r_b r_c = 1 - \Delta^2/3$ exactly $\Rightarrow$ $\det'$ -ratio = $(1 - \Delta^2/3)^{4/3}$ (via $\zeta(0) = -\frac{2}{3}$ , v132) — no first-order term in the split; deficit coefficient $\frac{32}{27} =  \mu_4 (\frac{2}{3})^3$ (audit); finite-weight absorption stays open with the $6 \rightarrow \frac{14}{3}$ obstruction stated sharply $[\mathbf{E}]/[\mathbf{C}]$ ( <i>the bend since shown <math>\det'</math>-clean, v147</i> )                                                                                           |
| v145_pairing_atoms.py     | $R4'$ values are <i>atom identities</i> : with the lift $n=w_0(\sigma)+ \mu_4 e_3$ ( $w_0$ = the $A_3$ exponent duality), $n\cdot\mathbf{1}= \mu_4 - \mathbb{Z}_2 = \mathbb{Z}_2 $ , $n\cdot a= \mu_4 a_3=8$ via the new closure $ \mu_4 +g_{\text{car}}=N_{\text{fam}}^2$ ( $4+5=9$ , $\sigma \perp w_0(a)$ ), $n\cdot\sigma=\ \sigma\ ^2-N_{\text{fam}}^2+ \mu_4 g_{\text{car}}=121$ ; residue = derive the <i>one</i> lift map $[\mathbf{E}]/[\mathbf{C}]$                                                                                                                                                                 |
| v146_moebius_d4.py        | Möbius $D_4$ realisation (R5 at its floor): the full dihedral $\langle z \mapsto iz, z \mapsto 1/z \rangle$ on $H^1(\mathbb{P}^1 \setminus \mu_4)$ matches the integer model $\langle G, T_A, \Sigma \rangle$ exactly — $\iota^*$ swaps $\chi_1 \leftrightarrow \chi_3$ , fixes $\chi_2$ with $+1$ , $\det=-1$ ( $=T_A$ in the cusp basis); $\delta\iota=\Sigma$ class; parities match (v98) — the GATE.QGEO premise reduces to the module identification $[\mathbf{E}]/[\mathbf{C}]$                                                                                                                                         |
| v147_clock_gaussian.py    | The clock is a Gaussian zero-mode integral (R1): variance ratio $(1-\alpha)$ (norm = area, v131) $\Rightarrow$ Born <sup>2</sup> $\Gamma$ -ratio $(1-\alpha)^{p_2}$ , rate = $-p_2 \ln(1-\alpha)$ = the v127 ring sum term by term; the bend $\log_{3/2} 3$ lives between the two Nariai levels of <i>one</i> geometry $\Rightarrow$ $\det'$ -clean; the $6 \rightarrow \frac{14}{3}$ obstruction is localised to the $\alpha=0$ reference $[\mathbf{E}]/[\mathbf{C}]$                                                                                                                                                        |
| v148_fock_census.py       | NS/R sector census (R2 scoped, corrected): the untwisted (NS) module of the 16 Majoranas carries only the <i>even</i> classes $\{0, 2\} \times \{0, 2\}$ — the odd glue sectors $k(1, 1)$ are <i>Ramond</i> modules; the R zero-mode module ( $2^8=256$ ) splits $128+128$ into the odd sectors of the <i>two</i> Lagrangian glues (v92): choosing the glue <i>is</i> choosing the R-projection ( $128$ = one $SO(16)$ chirality), $248=120_{\text{NS}}+128_{\text{R}}$ ; R2 is irreducibly a twisted-sector net statement $[\mathbf{E}]/[\mathbf{C}]$                                                                        |
| v149_cusp_normal.py       | Both dual normals are <i>anchor data</i> ( $R4'$ ): in the cusp frame $n$ pairs to $(6, 3, 5) = (p_2, p_0, e_2)(a)$ — one anchor invariant per $Q_+$ line ( $d = \frac{3}{2}a - 2\cdot\mathbf{1}$ in the generation frame, v139); equivalent to the $(2, 8, 121)$ frame data and the $w_0$ -lift (v145); audit: $\sigma \rightarrow (5, 1, 2)$ , sum 14 = $\dim G_2$ ; residue = <i>one</i> assignment $[\mathbf{E}]/[\mathbf{C}]$                                                                                                                                                                                            |

*continued on next page*

| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v150_replica_eh_model.py       | The EH form from the replica variation, at the gapped-model level (first R3 movement): conical deficit exact and $t$ -independent (image sum $(N^2-1)/(12N) = C(2\pi/N)$ ); the gap makes the $\zeta$ -variation <i>finite and cutoff-independent</i> , $\Delta \log \det' = 2C(\gamma) \ln m$ ; linearisation = $(\ln m/12\pi) \int \sqrt{g} R$ — the EH form; target equation $k=c_3/2 \Leftrightarrow \ln m = \frac{3}{4} = q(A_3)$ (audit); Calderón transfer + normalisation stay open — SEAM.THEOREM.01 narrows, does <i>not</i> close [E]/[O] ( <i>Calderón transfer since answered — the DtN kernel is conically clean, v151</i> ) |
| v151_bfk_split.py              | BFK split (the Calderón transfer of v150 answered): the Kac corner constant is boundary-condition <i>independent</i> ( $C_N=C_D$ , derived from reflection doubling) $\Rightarrow$ the conical deficit of the Calderón/DtN jump determinant <i>vanishes identically</i> , $C_{\text{DtN}}=C_{\text{cone}}(\gamma)-2C_D(\gamma/2)=0$ ; via Burghelea–Friedlander–Kappeler the seam-reduced action inherits the v150 EH form verbatim — the EH content sits in the <i>local</i> half determinants, the nonlocal kernel is conically clean; residue = the $q(A_3)$ normalisation + the kernel premise [E]/[O]                                 |
| v152_norm_is_anchor.py         | R3's normalisation <i>is</i> the dimensionful anchor: the EH coefficient is $k = \ln(m/\mu)/(12\pi)$ — $\ln m$ alone is scale-ambiguous, only the ratio $m/\mu$ is physical; $k=c_3/2 \Leftrightarrow m/\mu=e^{3/4}$ is the induced-gravity anchor $1/G$ (v68), <i>not</i> a separate gap; the irreducibles stay {one scale, $\pi$ }; audit: the log-ratio = $\frac{3}{4} = q(A_3)$ is the target value, NOT a derivation [E]/[O]                                                                                                                                                                                                          |
| v153_no_unit_theorem.py        | No-Unit Theorem (v78/v152 formalised): dimensionless data are scale-invariant under $L \mapsto \lambda L$ , a mass / $1/G$ is not, so a dimensionless compiler <i>cannot</i> select an absolute scale; the three residuals collapse — $U_{\text{point}} \sim v_{\text{geo}}$ , $1/G \sim v_{\text{geo}}^2$ , $m/\mu = e^{3/4}$ (a ratio); $v_{\text{geo}}$ is an irreducible metrology primitive, not an open gap; irreducibles stay {one unit, $\pi$ } [E]/[O]                                                                                                                                                                            |
| v154_simple_current_theorem.py | Simple-Current Extension Theorem (the central $G_{\text{net}}$ theorem, assembled): $A=(D_5)_1 \otimes (A_3)_1$ extended by the isotropic glue $L=\langle(1,1)\rangle$ has index $ L =4$ , $c=5+3=8$ , $\mu(B)=\mu(A)/ L ^2=16/16=1 \Rightarrow$ holomorphic $\Rightarrow B=(E_8)_1$ (unique even-unimodular rank-8; $SO(16)_1$ excluded); finite realisation in hand (v125/v128/v143/v148); residue = the seam-side identification [E]/[C]                                                                                                                                                                                                |
| v155_quasifree_boundary.py     | Quasi-free in $\Rightarrow$ quasi-free out (Python-only): the boundary marginal of a Gaussian bulk is the compression $PTP$ (a contraction, hence a valid covariance), so the $G_{\text{net}}$ analytic premise (v110(b)/v113) <i>follows</i> from ‘the seam bulk is a reflection-positive free field’ — the <i>same</i> premise as v59 (area law) and v150/v151 (EH); the seam-side identification reduces to ONE shared physical premise [E]/[O]                                                                                                                                                                                         |
| v156_seam_net_construction.py  | Constructive route (Route 3) to its exact endpoint: the DtN/Calderón operator of the 2d Laplacian is $\omega_k= k $ (free chiral dispersion); 16 free Majoranas give $c=8=5+3$ ; the $E_8$ currents are $248=120_{\text{NS}}+128_{\text{R}}$ ; and the $(E_8)_1$ character $E_4/\eta^8 = q^{-1/3}(1+248q+4124q^2+\dots) = j^{1/3}$ (verified to order 4) makes $B \cong (E_8)_1$ a <i>checked</i> identity, not an assertion; residual = the free-bulk premise (v155) + net existence [E]/[C]                                                                                                                                              |
| v157_rigid_fixed_point.py      | Freeness as a <i>rigid fixed point</i> (simplicity-first reframing of premise A): (i) the DtN/Calderón symbol is universally $ k $ (Lee–Uhlmann, bulk-detail-independent — the free dispersion is not a premise); (ii) a holomorphic $c=8$ theory has <i>no</i> marginal $(1,1)$ field (all $\bar{h}=0$ ) $\Rightarrow$ rigid, isolated, no room for an interaction; (iii) the same $ \mathbb{Z}_2 $ that halves $8\pi$ in $c_3$ IS the Ramond projection giving $\mu=1$ ; (iv) holomorphic $c=8$ unique = $(E_8)_1$ = free. Premise (A) shrinks from ‘prove Gaussianity’ to ‘the gapped flow reaches the holomorphic fixed point’ [E]/[C] |
| v158_fixed_point_stable.py     | The free chiral $c=8$ fixed point is <i>stable</i> (exact dimension count, the finite core of (A)): Majorana $h=\frac{1}{2}$ , bilinear current $h=1$ , quartic $h=2$ ; the relevant window $(0,1)$ for bosonic operators is <i>empty</i> , the $h=1$ currents are chiral (not $(1,1)$ -marginal, v157), the lowest interaction (quartic) is $h=2>1$ irrelevant $\Rightarrow$ isolated stable fixed point; (A) reduces to a basin-of-attraction statement [E]/[C]                                                                                                                                                                          |

*continued on next page*

| Script                                | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v159_pyrate_gauge_crosscheck.py       | PyR@TE cross-check of the $F_{\text{transfer}}$ gauge inputs: the carrier/SM content gives $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$ by the textbook one-loop formula; $b_1 = \frac{41}{10}$ holds <i>three</i> independent ways — carrier algebra $10b_1 = g_{\text{car}} 2^{g_{\text{car}}-2} + 1 = 41$ , the $U(1)$ hypercharge index $\sum \frac{3}{5} Y^2$ , and the external RGE generator PyR@TE3 ( <a href="#">arXiv:2007.12700</a> ) which writes $\beta_{g_1} = (\frac{41}{10}) g_1^3$ verbatim; the $M=41$ split $10b_1 = 40_{(\text{ferm})} + 1_{(\text{Higgs})}$ reproduces v13 Gate 1; evidence in <a href="#">verification/pyrate</a> . The transfer functor's gauge inputs are not free knobs <a href="#">[E]</a> / <a href="#">[C]</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| v160_seam_gaussianity_from_pf.py      | Seam Gaussianity as a <i>conditional</i> fixed-point theorem (the proposed closure of premise A, honestly typed): (1) quasi-free $\Rightarrow$ all higher connected cumulants $\kappa_{2n}=0$ (signed set-partition/Pfaffian inversion, exact on a pure Majorana covariance); (2) Wick functoriality $\Rightarrow$ a kernel-preserving transport fixes the whole quasi-free state (v113); (3) the Perron–Frobenius dichotomy (a <i>fixed</i> $\kappa_{2n}$ is a SECOND fixed ray, breaking uniqueness; a <i>non-fixed</i> one decays by the gap) forces Gaussianity; (4) the qualitative claim needs only $\text{gap} > 0$ — the value $(2/3)^6$ only sets the rate. <b>The load <a href="#">[O]</a>:</b> v56's gapped PF uniqueness lives on the 3-dim scalar transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ , while the seam correlation cone has $\binom{16}{4}=1820$ , $\binom{16}{6}=8008$ cumulant directions $\Rightarrow$ (A) reduces to the single internal statement ‘the admissible TFPT transport is primitive + spectrally gapped on the <i>full</i> Schwinger cone, not only the readout spectrum’ <a href="#">[E]</a> / <a href="#">[O]</a>                                                                                                                                                                                                                                     |
| v161_one_particle_reduction.py        | The cone reduces to one particle (discharging v160's scope premise via Bogoliubov second quantisation): a quasi-free transport is $T=\Gamma(t)$ , the second quantisation of a one-particle contraction $t$ ; $\Gamma(t)$ acts on the degree- $m$ correlator sector as the exterior power $\Lambda^m(t)$ , so (1) $\text{spec}(\Gamma(t) _{\deg m}) =$ products of $m$ one-particle eigenvalues <a href="#">[E]</a> ; (2) <b>the cone gap equals the one-particle gap</b> — the sub-leading eigenvalue over the whole cone is $(2/3)^6$ exactly <a href="#">[E]</a> ; (3) the eigenvalue-1 sector = wedges of the fixed subspace (disconnected Wick powers), no independent fixed higher cumulant, uniqueness closed by v113 (one polarization $\Leftrightarrow$ one state) <a href="#">[E]</a> ; (4) quasi-freeness <i>forced</i> not assumed — the $120 = \dim \mathfrak{so}(16)$ chiral currents are the Bogoliubov generators and v158 makes every non-Gaussian deformation irrelevant <a href="#">[C]</a> . <b>Residual <a href="#">[O]</a>:</b> v160's three full-cone premises collapse to ONE 16-dim one-particle statement (seam symbol = gapped Calderón Bogoliubov map, fixed = rank-8 $K_\Sigma$ , gap $(2/3)^6$ ); all numbers supplied, only the physical identification (Kawahigashi–Longo class = the A2 net residual) stays open <a href="#">[E]</a> / <a href="#">[O]</a> |
| v162_seam_transport_identification.py | The final identification, forced as far as it goes — the v161 one-particle statement carries <i>no</i> new open content: (1) the gap $(2/3)^6$ is <i>forced</i> — cusp weights $= \text{spec}(A_0^*) = \{0, \frac{1}{3}, \frac{2}{3}\}$ (parabolic residue, v115), transfer eigenvalue $(1-\alpha)^{p_2}$ , $p_2=6= R_+(A_3) $ , all carrier data <a href="#">[E]</a> ; (2) $\mu_4$ -equivariance forces the generator diagonal in the cusp basis — $\sum_k U^k X U^{-k} =  \mu_4  \text{diag}(X)$ , commutant of $U =$ the diagonal algebra — so the gapped $\mu_4$ -equivariant generator has spectrum = the cusp weights, gap $(2/3)^6$ <a href="#">[E]</a> ; (3) the cusp grading <i>is</i> the $A_3$ grading on the 16 Majoranas (v137) <a href="#">[E]</a> ; (4) the fixed space = the rank-8 Calderón polarization (v113/v156) <a href="#">[E]</a> . <b>Closure <a href="#">[C]</a>/<a href="#">[O]</a>:</b> with v160+v161, (A) closes given only $\mu_4$ -equivariance + admissibility, whose two non-derived inputs are the <i>already-open</i> A2 (net existence) and R1 (seam clock, <a href="#">HOR.CLOCK</a> ) — so (A) factors exactly into A2 + R1, adds <i>no</i> new open item, and is closed conditionally on them (a unification, not an unconditional proof) <a href="#">[E]</a> / <a href="#">[O]</a>                                                                 |
| v163_rg_stability_flavor.py           | $F_{\text{transfer}}$ scale-tagging: the lepton-mixing seeds ( $\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0}{2} = 0.3067$ , $\sin^2 \theta_{13} = \varphi_0 e^{-5/6} = 0.0231$ ) are <i>RG-stable</i> — PMNS running is $y_\tau^2$ -suppressed ( $\sim 1.8 \times 10^{-5} \ll$ precision), so seam value = measured value; the quark mass-ratio readouts ( $m_t/m_b \sim 41$ ) run through $y_t^2 \sim O(10\text{--}15\%)$ and carry a scale tag. The only mixing-rotating Yukawa term $\frac{3}{2} Y Y^\dagger Y$ carries $y_{\text{max}}^2$ (PyR@TE SM RGEs) <a href="#">[E]</a> / <a href="#">[C]</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

*continued on next page*

| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v164_qcd_scale.py                | the QCD confinement scale is parameter-free from the carrier $b_3=-7$ : running $\alpha_s(M_Z)$ down (two-loop, $n_f$ thresholds) gives $\alpha_s(m_b)\sim 0.22$ , $\alpha_s(m_c)\sim 0.38$ and confinement ( $\alpha_s\sim 1$ ) at $\sim 0.5$ GeV — so the QCD-scale <i>numerator</i> of $m_p/m_e$ is independently sourced (dimensional transmutation), while the ratio 1836 stays the honest ‘Open / not forced’ frontier non-claim ( $m_p=O(1)\Lambda$ is lattice) [E]/[O]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| v165_premise_a_closure.py        | Premise (A), the honest maximal closure — the two residual gates pinned to their floor, so (A) carries <i>zero</i> independent open content: (1) A2 (net existence) closes by <i>assembly</i> of established mathematics [C] — 16 free Majoranas give a free-fermion net (Buchholz–Mack–Todorov / Böckenhauer–Evans), the holomorphic $c=8$ lattice net is $(E_8)_1$ (Frenkel–Kac–Segal / Dong–Xu / Kawahigashi–Longo), the $\mu_4$ extension is a Longo Q-system of index 4 (v125), $\mu(B)=16/16=1$ , $248=120+128$ , character $E_4/\eta^8=j^{1/3}$ (v156); the only TFPT input (seam Calderón kernel = free-fermion two-point function) is done via DtN = $ k $ (v156/v157); (2) R1 (seam clock) reduces to GATE.QGEO [C] — $\mu_4$ -equivariance forces diagonal-in-cusp, the flat $\mu_4$ -equivariant generator is <i>unique</i> = $A_0^*$ (v115/v116) so the gap $(2/3)^6$ is forced, residual = ‘seam boundary = the $\mu_4$ -punctured sphere $\mathbb{P}^1\setminus\mu_4$ ’ = GATE.QGEO (cohomology established, v137); (3) the irreducible core stays $\{\pi, v_{\text{geo}}\}$ , a <i>theorem</i> by the No-Unit Theorem (v153). <b>Final accounting:</b> (A) = A2 assembly [C] + R1=GATE.QGEO + the $\{\pi, v_{\text{geo}}\}$ core $\Rightarrow$ ZERO independent open gates; a unification + re-typing, not an unconditional proof [E]/[O] |
| v166_higgs_free_seam.py          | the Higgs quartic from the <i>free seam</i> (premise A / v158): a Gaussian seam UV forces $\lambda(M_{\text{seam}})=0$ and $\beta_\lambda(M_{\text{seam}})=0$ (Shaposhnikov–Wetterich double criticality, here <i>motivated</i> not assumed). Two-loop PyR@TE-sourced SM running with the measured $(m_H, m_t)$ gives $\lambda(\bar{M}_{\text{Pl}})\sim 0.002$ , $\beta_\lambda(\bar{M}_{\text{Pl}})\sim 0$ — the famous SM near-criticality, now <i>explained</i> by the free seam (2-loop sharpens $\lambda(\bar{M}_{\text{Pl}})$ from $\sim 0.022$ to $\sim 0.002$ ). The double condition predicts $m_H\sim 129\text{--}134$ GeV; at the scalaron scale it gives $\sim 107$ GeV (too low) $\Rightarrow$ the BC lives at $\bar{M}_{\text{Pl}}$ (seam=Planck) [E]/[C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| v167_inventory_post_closure.py   | The terminal residual inventory (post premise-(A) closure; line v105 $\rightarrow$ v123 $\rightarrow$ v167): after the (A)-chain (v160–v165), premise (A) is <i>no longer</i> an independent structural class (GATE.METRIC.16–19). The residual collapses from v123’s FIVE classes to: <b>Tier 0</b> — the irreducible core $\{\pi, v_{\text{geo}}\}$ , a <i>theorem</i> (No-Unit, v153), nothing to close; <b>Tier 1</b> — ONE hinge (seam = flavour = horizon): $A_2$ net existence (assembly of established net constructions) + GATE.QGEO (seam boundary = $\mathbb{P}\setminus\mu_4$ , cohomology $b_1=N_{\text{fam}}=3$ ); <b>Tier 2</b> — folded ( $\rightarrow v_{\text{geo}}$ , $\rightarrow G_{\text{net}}$ ); <b>Tier 3</b> — $F_{\text{transfer}}$ , one functor with four typed conditional interfaces ( $\eta_B$ carries the most open room). Completeness: a sixth independent structural class fails the script — a unification + re-typing, not an unconditional proof [E]/[C]                                                                                                                                                                                                                                                                                                                                                           |
| v168_qgeo_rigidity.py            | The QGEO <i>Rigidity Theorem</i> — the finite half of the Tier-1 hinge GATE.QGEO, hardened. Exact: $\mu_4=\{1, i, -1, -i\}$ has cross-ratio 2 and Möbius stabiliser $\langle z\rightarrow iz, z\rightarrow 1/z \rangle$ of order $8= D_4 $ (faithful) $\Rightarrow$ four marks with faithful $D_4$ are the $\mu_4$ square up to Möbius; $H^1(\mathbb{P}\setminus\mu_4)$ has rank $4-1=3=N_{\text{fam}}$ ; the forms $\omega_k=z^{k-1}dz/(z^4-1)$ ( $k=1, 2, 3$ ) are $\mu_4$ -eigenforms with $z\rightarrow iz$ eigenvalue $i^k$ = the three nontrivial characters $\chi_1, \chi_2, \chi_3$ , weights $(1, 2, 3) = A_3$ exponents = $\text{Spec}(Q_+)$ . QGEO.RIGID.01 [E] (math half closed); QGEO.REALIZE.01 [C] (seam-collar = $\mathbb{P}\setminus\mu_4$ stays the realisation premise) [E]/[C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| v169_etaB_boltzmann_interface.py | The $\eta_B$ leptogenesis transfer <i>operationalised</i> (Tier-3 $F_{\text{transfer}}$ , not a new class): type-I thermal leptogenesis $\eta_B\sim 0.96\times 10^{-2}\varepsilon_1\kappa_f$ (sphaleron 28/79) fed by TFPT’s NO neutrino spectrum + the predicted $\delta_{CP}=240^\circ$ . Davidson–Ibarra $\varepsilon_1=\frac{3}{16\pi}M_1m_3/v^2\sim 8.5\times 10^{-7}$ for $M_1=10^{10}$ GeV; efficiency $\kappa_f(\tilde{m}_1)\sim 0.03\text{--}0.19$ . <b>Kill test:</b> over natural $M_1\in[3\times 10^9, 3\times 10^{10}]$ GeV and washout, $\eta_B$ spans $[8.4\times 10^{-11}, 4.7\times 10^{-9}]$ which <i>brackets</i> the observed $6.1\times 10^{-10}$ (canonical point gives $6.0\times 10^{-10}$ , not tuned) $\Rightarrow$ the route is viable; $M_1$ /washout are scenario inputs, so $\eta_B$ stays [C]. If excluded, the route falls, not the theory [C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

*continued on next page*

| Script                            | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v170_diamond_slice_compression.py | E8 Slice Compression Lemma: the seven $E_8$ audit slices are <i>one</i> projection of two alphabets — anchor power sums $P=(3, 4, 6, 10)$ ( $p_n=2+2^n$ ) and Sheet-Diamond determinants $D=(3, 4, 8, 14, 20, 32)$ . The $K_4$ edge products $(12, 18, 30, 24, 40, 60)$ are all carrier blocks; $\sum \det M = 81 = N_{\text{fam}}^4$ ; all seven 248-slices follow from $P, D, \Delta=13$ and $(e_1, e_2, \dim S^+)$ ; one affine map $\text{row}(C+sU+tV) = (7, 11, 13)+s(3, 3, 3)+t(1, 2, 3)$ gives every flavour row budget; $78 = \dim E_6 = p_2\Delta$ . The atlas is a closed grammar over two typed alphabets, exact, reading audit [E]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| v171_os_moment_cluster.py         | Atomic OS Moment Lemma + Sugawara Gap Safety: $\text{spec}(T)=\{1, \frac{64}{729}, \frac{1}{729}\}$ makes $C(n)=A+Br^n+Cs^n$ a positive moment sequence, so every OS Hankel matrix factorises as a Vandermonde Gram $\Rightarrow$ reflection positivity (clean proof, not numeric Gram); cluster $\frac{729}{665}$ , $\xi=0.41105$ , Källen–Lehmann masses $6 \log \frac{3}{2}, 6 \log 3$ ; gap safety $\Delta_{\text{eff}}=6 \log \frac{3}{2} - \frac{31}{4\pi^2} > 0$ with $31=1+h^\vee(E_8)$ , $c((E_8)_1)=248/31=8$ [E]/[C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| v172_trace_anomaly_seed.py        | Trace-anomaly origin of the seed prefactor $\frac{4}{3}$ : over the seam $S^2$ ( $\chi=2$ ) the $(E_8)_1$ ( $c=8$ ) Weyl anomaly is $c\chi/6=\frac{8}{3}$ , halved by the one-sided $ \mathbb{Z}_2 $ to $\frac{4}{3} = \dim S^+ / \dim \mathfrak{g}_{SM} = 16/12$ , the seed base $(\frac{4}{3})c_3=\frac{1}{6\pi}$ . Plus QCD $b_3=11-\frac{2}{3}N_f=7$ = scalaron exponent $\Omega_{\text{adm}}-10b_1=48-41$ (confinement $\leftrightarrow$ inflation), and the Volkov–Wipf factorisation $-98/45 = -2 \cdot 7^2 / \dim(D_5)$ [E]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| v173_pfaffian_cp_car.py           | Strong CP as Pfaffian reality: a self-dual CAR (16-Majorana) model demonstration of $\theta_{\text{eff}}=0$ — the sheet-odd Calderón involution $\{D, \Sigma\}=0$ pairs the spectrum $(\pm\lambda)$ , $\gamma_5$ -Hermiticity $D^\dagger=\gamma_5 D \gamma_5$ makes the Pfaffian real ( $\arg \in \{0, \pi\}$ ), and reflection positivity ( $Z>0$ ) selects the positive branch $\Rightarrow \theta_{\text{eff}}=0$ [E]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| v174_seam_fock_readings.py        | Seam Fock-space readings: (1) CAR cone compression — $\dim \Lambda^{\text{even}}(\mathbb{R}^{16})=2^{15}=32768$ reduces to a 16-dim one-particle problem under quasi-free Bogoliubov ( $2^{15}/16=2^{11}$ ), the explicit count behind v161; (2) Witten index = the Plücker norm $11 = \sum_{k \leq 2} \binom{4}{k}$ , with $11 = \dim S^+ - g_{\text{car}}$ (full $2^4=16$ ) and $c_u/c_d=55/117$ reading the QBL-protected boundary state space; (3) Affleck–Ludwig $g$ -theorem $c_{UV}=5+3=8=c_{IR}((E_8)_1)$ so $\Delta c=0$ forces an exact isometry (boundary-entropy form of v157/v158) [E]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| v175_net_existence_full_cone.py   | The heavy artillery — net existence (A2) and full-cone reflection positivity discharged to [E], isolating QGEO.REALIZE.01 as the single open premise (nothing fabricated): (1) <b>full-cone RP for all <math>m</math></b> — the CAR second-quantisation functor $\Gamma(t)=\bigoplus_m \Lambda^m(t)$ of the one-particle seam contraction $t$ acts on the complete Fock space $\Lambda^\bullet(\mathbb{R}^{16})$ ( $\dim 2^{16}=65536$ ); the exterior-power spectrum theorem makes every eigenvalue a subset product of $\text{spec}(t)$ , verified on all $2^{16}$ products to lie in $[0, 1]$ (contraction), with eigenvalue-1 multiplicity $2^8=256$ (wedges of the rank-8 $K_\Sigma$ polarisation) and sub-leading = the one-particle gap $(2/3)^6$ , so full-cone RP reduces <i>for all <math>m</math></i> to the one-particle contraction (Shale–Stinespring CAR functor), discharging the v160 scope premise by a theorem; (2) <b>A2 net existence</b> assembled + verified — $(E_8)_1$ character $E_4/\eta^8=1+248q+4124q^2+34752q^3+213126q^4+1057504q^5$ (level-1=248), embedding $248=120+128$ , $E_8$ Cartan even unimodular ( $\det 1$ ); the net <i>exists</i> by cited theorems (free-fermion net, lattice VOA, index-4 Q-system); (3) both collapse to the same seam-kernel identification = QGEO.REALIZE.01, finite half v168, realisation half left [O][E][O] |

continued on next page

| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v176_seam_collar_realisation.py | The Seam Collar Realisation Theorem assembled as ONE central statement + reduction certificate (the single structural proof obligation, formulated cleanly — <i>not</i> a fabricated proof): given an oriented one-sided RP seam collar with a sheet-odd Calderón involution, faithful $D_4$ marks and admissible $c=8$ data, <i>then</i> its conformal boundary is Möbius-equivalent to $\mathbb{P} \setminus \mu_4$ with $H^1$ grading $(1, 2, 3)$ , the Calderón contraction is the rank-8 $K_\Sigma$ polarisation and the index-4 extension is $(E_8)_1$ . The proof decomposes into six steps, FIVE already [E]: geometry/uniformisation (v168: cross-ratio 2, faithful Möbius $D_4$ order 8, $b_1=N_{\text{fam}}=3$ ), Calderón DtN symbol $ k $ (v156), $H^1$ character = generation grading $(1, 2, 3)=A_3$ exponents=Spec( $Q_+$ ) (v137/v168), $\mu_4$ -equivariant contraction diagonal with gap $(2/3)^6$ (v162), AQFT closure $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 = (E_8)_1$ + full-cone RP all $m$ (v154/v175); the whole structural residual reduces to the ONE open premise QGEO.REALIZE.01 (the physical seam collar's boundary <i>is</i> this $\mathbb{P} \setminus \mu_4$ ), left honestly [O][E][O]                                                                                                                                                                                           |
| v177_seam_marking_kernel.py     | The QGEO proof tree — the v176 single realisation premise split into its honest constituents: REALIZE = MARKS · UNIFORM · COHOM · MODULE · KERNEL · NET, with FOUR nodes closed <i>constructively</i> [E] and the open residual sharpened into exactly TWO named obligations. [E] UNIFORM (Lemma 2, constructive): an order-4 Möbius map conjugates to $z \mapsto iz$ , a non-fixed orbit scales to $\mu_4$ , the reflection $z \mapsto 1/z$ gives $\sigma\rho\sigma = \rho^{-1}$ ( $D_4$ , order 8) — a genus-0 four-marked faithful- $D_4$ boundary <i>is</i> $(\mathbb{P}, \mu_4)$ ; [E] COHOM: $\rho^*\omega_k = i^k\omega_k$ so the $H^1$ grading is $(1, 2, 3)=A_3$ exponents=Spec( $Q_+$ ); [E] MODULE: the $H^1 \rightarrow$ generation iso is <i>unique</i> (multiplicity-free + residue normalisation; reflection $\omega_1 \leftrightarrow \omega_3$ , $\omega_2$ fixed); [E] NET (v154/v175): index-4 $\mu_4$ extension of $(D_5)_1 \otimes (A_3)_1 = (E_8)_1$ . The TWO genuine open obligations [O]: QGEO.MARKS.01 (raw seam $\rightarrow$ genus-0 four-marked $D_4$ boundary) and QGEO.KERNEL.01 (raw seam Calderón kernel = the $\mu_4$ -equivariant free gapped contraction as an <i>operator</i> ); the whole structural residual is now exactly these two doors [E][O]                                                                                                                            |
| v178_marks_kernel_attempt.py    | An honest <i>attempt</i> at the two open obligations QGEO.MARKS.01 + QGEO.KERNEL.01 — <i>not</i> a closure (genuine constructive-geometry/AQFT statements, not finite computations), but a deeper reduction: each finite core is closed [E] and each obligation reduced to ONE sharper irreducible premise [O]. <i>Marks core</i> [E]: a genus-0 double with an order-4 clock $\rho$ ( $\sim z \mapsto iz$ , fixed points $\{0, \infty\}$ ) and reflection $\tau$ forces the marks to be a generic $\rho$ -orbit $\{1, i, -1, -i\} = \mu_4$ (not the fixed points), distinct iff $b_1=4-1=3=N_{\text{fam}}$ , $\tau\rho\tau=\rho^{-1} \Rightarrow$ faithful $D_4$ order 8 — so MARKS reduces to the single topological/clock premise. <i>Kernel core</i> [E]: on the 3-dim multiplicity-free $H^1$ the $\mu_4$ characters $(1, 2, 3)$ are distinct, so by Schur a $\mu_4$ -equivariant operator is forced diagonal and <i>completely determined by its eigenvalue per character</i> — two such with the forced transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ (v162) and the unique residue-normalised basis (v177) are <i>equal as operators</i> , not merely isospectral (the spectrum-vs-operator worry vanishes on the finite block); so KERNEL reduces to the single full-operator-reduction premise (principal symbol $ k $ , Lee–Uhlmann/v156; no drift, v158). Neither obligation closed; both sharpened [E][O] |

*continued on next page*

| Script                              | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v179_conformal_realisation.py       | The two residual premises (MARKS-residual + KERNEL-residual) collapse to ONE geometric premise QGEO.CONF.01 (honest unification; the conformal realisation stays [O]). MARKS's residual is not three independent assumptions: (1) genus-0 <i>is</i> P1 — $c_3=1/( \mathbb{Z}_2  2\pi \chi(S^2))=1/(8\pi)$ with $\chi=2 \Rightarrow g=0$ , the same $\chi=2$ that gives the '8' (v58/v73); (2) the order-4 clock <i>is</i> the carrier — Coxeter element of $W(A_3)=S_4$ , order $h(A_3)=4= \mu_4 $ (v117); (3) marks = one orbit is <i>forced</i> — parabolic marks $\neq$ the two elliptic fixed points $\{0, \infty\}$ , a free order-4 orbit has exactly $4=e_1(a)$ points. So MARKS reduces to 'the carrier clock acts conformally on the genus-0 double'. KERNEL then follows from that geometry: DtN symbol $ k $ (Lee–Uhlmann, v156) $\Rightarrow$ rank-8 $c=8$ fixed polarisation; no drift (v158); the cusped-surface residual data is $H^1=\mathbb{C}^3=N_{\text{fam}}$ , the 3-dim gapped transfer block (Schur, v162/v178). Both $\Rightarrow$ ONE premise QGEO.CONF.01: the raw RP seam double is conformally $(\mathbb{P}, \mu_4)$ with the carrier clock as the order-4 Möbius map — the single structural residual of the whole theory, left honestly [O][E][O]                                                        |
| v180_clock_is_mobius.py             | Cracking QGEO.CONF.01 one level further — the conformal-realisation premise is <i>replaced</i> , via three classical theorems, by a strictly <i>milder</i> , non-circular premise QGEO.ISO.01: 'the carrier clock is an order-4 orientation-preserving <i>isometry</i> of the RP seam metric'. (1) Uniformisation (Poincaré–Koebe): a genus-0 Riemann surface is conformally $\mathbb{P}$ , so the genus-0 seam double (P1, v179) <i>is</i> $\mathbb{P}$ — the target is produced, not assumed; (2) Keréjártó (1919; Constantin–Kolev 2003): a finite-order orientation-preserving homeomorphism of $S^2$ is conjugate to a rotation, so the order-4 clock ( $h(A_3)=4$ ) is topologically $z \mapsto iz$ ; (3) isometry $\Rightarrow$ conformal $\Rightarrow$ Möbius: an orientation-preserving isometry of a Riemannian surface is holomorphic, $\text{Aut}(\mathbb{P})=\text{PSL}(2, \mathbb{C})$ ; (4) order-4 Möbius $= z \mapsto iz$ (verified: projective order 4, fixed $\{0, \infty\}$ , multiplier $i$ , free orbit $\mu_4$ ). So QGEO.CONF.01 is <i>implied</i> by the milder isometry premise; the new residual QGEO.ISO.01 (the seam transport is a metric-compatible holonomy preserving the RP inner product) does not name $\mathbb{P}/\mu_4$ and is milder, its surface realisation from the raw seam still [O][E][O] |
| v181_clock_is_conformal_symmetry.py | The FINAL honest reduction + the explicit identification of BEDROCK. The isometry premise QGEO.ISO.01 (v180) is weakened one more genuine step — to a CONFORMAL-symmetry premise QGEO.SYM.01 — via equivariant uniformisation (Nielsen realisation, finite cyclic on genus-0): a <i>conformal</i> automorphism (strictly weaker than an isometry — every isometry is conformal, conformal allows any positive scale) already suffices to put the order-4 clock in the form $z \mapsto iz$ . The clock is the Coxeter element of $W(A_3)=S_4$ (v117), order $h(A_3)=4= \mu_4 $ , the $\mu_4$ deck, and a deck transformation preserves the pulled-back conformal structure by definition. So below QGEO.SYM.01 the residual is <i>definitional</i> , not a missing theorem: 'the seam's conformal structure <i>is</i> the $\mu_4$ -deck structure of the carrier clock' (the carrier $\mu_4$ glue is the conformal monodromy of the seam) — a physical/definitional identification tying P2 to the seam's conformal geometry, not reducible further without relabeling. The chain<br>REALIZE(v175) $\rightarrow$ theorem(v176) $\rightarrow$ MARKS+KERNEL(v177) $\rightarrow$ finite<br>cores(v178) $\rightarrow$ CONF.01(v179) $\rightarrow$ ISO.01(v180) $\rightarrow$ SYM.01(v181) has reached bedrock; we stop here honestly [E][O] |

*continued on next page*

| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v182_reviewer_residual_map.py   | The four external-review concerns reduce to the two already-named residuals + one data-only item. $A$ (mapping $2D \rightarrow 4D$ ) [E]/[C]: the Plücker readout $11 = \sum_{k \leq 2} \binom{4}{k} = \dim S^+ - g_{\text{car}} = 16 - 5$ (QBL count, v174) and the holographic reduction give $A = \text{QGEO.SYM.01} + F_{\text{transfer}}$ . $B$ (UV gravity) [E]: ‘gravity = geometric boundary readout’ is QGEO.SYM.01. $C$ (grammar tuning) [E]/[C]: frozen (v84) + minimal + over-determined (the anchor $a=(1, 1, 2)$ in $\geq 6$ independent not-selected-for structures) is look-elsewhere-resistant, but the residual falls only to out-of-sample data (JUNO/CMB-S4). $D$ (Koide Möbius flow) [E]/[C]: the Möbius nature is forced by $\text{Aut}(\mathbb{P}) = \text{PSL}(2, \mathbb{C})$ on the deck, $t \sim 2.84$ only locates the point, and the missing RG generator is $F_{\text{transfer}}$ , so $D = \text{QGEO.SYM.01} + F_{\text{transfer}}$ . NET: three reduce onto $\{\text{QGEO.SYM.01}, F_{\text{transfer}}\}$ , one waits for data — a unification of the concern structure, not a closure (premises stay [O]/[C]) [E][C] |
| v183_koide_f_corner_transfer.py | The Koide source→pole factor $\frac{53}{54}$ has an <i>operator</i> origin (external-review proposal, validated): $u_{\text{pole}}/u_{\text{source}} = a^T(R+Q)\mathbf{1}/(2\mathbf{1}^T R a) = 53/(2 \cdot 27) = \frac{53}{54}$ . (1) $\mathbf{1}^T R a = 27$ is the $E_6 \times A_2$ cubic family block ( $248 = \dim E_6 \cdot 78 + \det R \cdot 8 + 2 \cdot 27 \cdot N_{\text{fam}}$ ); (2) $F=R+Q$ is the <i>missing</i> Sheet-Diamond corner $M(1, 1)$ with $\det B_F = 52 = \dim F_4$ (v80) and anchor response $a^T(R+Q)\mathbf{1} = 53 = 52+1$ ; (3) $\frac{53}{54} = 1 - \frac{1}{2N_{\text{fam}}^3}$ exactly ( $54=2 \cdot 27$ ); (4) negative control: only $F$ gives 53 ( $R/Q/K/L \rightarrow 32/21/35/56$ ). Upgrades FR.KOIDE.03 from a bare guess to an operator identity; [E] the factor is exact, [C] the leptonic source→pole reading the $F$ corner stays an $F_{\text{transfer}}$ conjecture [E][C]                                                                                                                                                                                                                              |
| v184_etaB_anchored_boltzmann.py | Honest test of the external-review $\eta_B$ proposal ( $M_1=M_R\phi_0^4$ , $\tilde{m}_1=m_3/A_\Lambda$ ) — firewall verdict: <i>sharper scenario</i> , not a closure. The washout anchors plausibly ( $\tilde{m}_1=m_3/A_\Lambda=m_3/10 \approx 5 \text{ meV}$ , $A_\Lambda=10= E(K_5) $ , in the strong-washout window) [C]; but $M_1$ does <i>not</i> — $M_R$ is the seesaw scale $\sim v^2/m_3 \approx 6 \times 10^{14} \text{ GeV}$ and the nearest clean TFPT powers of $\bar{M}_{\text{P1}}$ both miss it ( $c_3$ -power off $\sim 75\%$ , $\phi_0$ -power off $\sim 40\%$ ), so $M_1=M_R\phi_0^4$ merely relocates the free input $M_1 \rightarrow M_R$ (the reviewer’s $6.0 \times 10^{-10}$ uses $M_R=1.3 \times 10^{15}$ ; the seesaw value gives $3.4 \times 10^{-10}$ , the tell that $M_1$ is unpinned). $\eta_B$ stays in the observed band so the route is viable, but the proposal cuts the free inputs from two to one, not to zero — $\eta_B$ stays [C], a sharper scenario [C]                                                                                                                                                      |
| v185_axion_relic_solver.py      | Axion relic, honest misalignment estimate: the determinant-line axion $f_a=M_{\text{scal}}/128 \approx 2.39 \times 10^{11} \text{ GeV}$ ( $m_a \approx 23.8 \mu\text{eV}$ ) can provide all dark matter, but <i>only</i> near the $\theta_i \approx 170^\circ$ hilltop where the relic is exponentially sensitive — so the axion DM branch stays a [C] scenario. The harmonic misalignment under-produces at this $f_a$ (prefactor $\sim 0.026/\theta^2$ , $\theta \sim O(1)$ gives $\sim 21\%$ of $\Omega_{\text{DM}}$ , need $\theta_{\text{eff}}^2 \sim 4.7$ ), so the hilltop anharmonic enhancement is required; near the hilltop it is exponentially sensitive, hence a scenario not a sharp prediction. $F_{\text{relic}}(f_a, \theta_i, N_{\text{DW}}) = ? \Omega_{\text{DM}}$ is a kill test for the <i>branch</i> , not the theory (no free singlet WIMP: $128=(16, 4)+(16, 4)$ ) [C][O]                                                                                                                                                                                                                                                     |
| v187_ftransfer_laws.py          | The $F_{\text{transfer}}$ discipline guard (CI-style firewall): the four continuous-transfer frontier observables — Koide, $\eta_B$ , the axion relic, $m_p/m_e + \Lambda_{\text{QCD}}$ — must <i>never</i> be typed as primitive [E] compiler predictions; each carries a [C]/[O] marker. The guard reads <code>status_ledger.csv</code> and enforces it (exact algebraic sub-parts — Koide $53/54$ v183, $b_3=-7$ v159 — may be [E], the physical prediction never is). Records the four interfaces $F_{\text{pole}}/F_{\text{Boltzmann}}/F_{\text{relic}}/F_{\text{QCD}}$ and that the raster rule + frozen registry + null model bind, so no future edit can quietly promote a pretty frontier number to a closed compiler output [E]                                                                                                                                                                                                                                                                                                                                                                                                              |

*continued on next page*

| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v188_frontier_wording_guard.py | The frontier-wording guard (a prose sentinel, no new physics): v187 protects the ledger <i>typing</i> , this guard protects the <i>prose</i> . It forbids stale sentences whose semantics an earlier round superseded — “leptogenesis interface ([E])” (v184 retyped Route B to [C]), “closed branch also fixes the leptogenesis inputs” ( $M_R$ is the seesaw scale, not a compiler power), “relic scale $f_a, m_a$ open” (v185: the scales define the branch, the open item is the hilltop-sensitive relic abundance), “computed $Q=0.664\dots$ near, not exact” (v183 gave 53/54 an exact operator origin), “Suite now 187 modules” (the count is 187, highest ID v188; v186 skipped). Scans the 10 active documents; passes when none appear, so a frozen Zenodo release can never re-introduce half-true wording [E]                                                                                                                               |
| v189_riemann_roch_carrier.py   | The Riemann–Roch carrier reading: $(g_{\text{car}}, N_{\text{fam}}) = (5, 3)$ are the two canonical invariants of <i>one</i> object, the four-point seam divisor $D = \mu_4$ on $\mathbb{P}^1$ . [E] arithmetic: $h^0(\mathbb{P}^1, \mathcal{O}(\mu_4)) = \deg + 1 = 5$ (function side) and $\text{rank } H_1(\mathbb{P}^1 \setminus \mu_4) = \deg - 1 = 3$ (cycle side, already QGEO.COHO.M.01/v177); $g_{\text{car}} = 5 = \text{rank}(D_5)$ (the carrier is the $\mathfrak{so}(10)$ half-spinor, $\text{rank } D_5 + \text{rank } A_3 = 5 + 3 = 8, \mathfrak{v}1$ ), $N_{\text{fam}} = 3$ . [C]: reading the carrier as the meromorphic seam-mode space $H^0(\mathbb{P}^1, \mathcal{O}(\mu_4))$ (less number, more geometry) is matched by $\text{rank}(D_5) = 5$ , but the physical identification is conjectural and does <i>not</i> replace the over-determined $g_{\text{car}} = 5$ (Pascal/bootstrap/Lean) — a fourth, geometric reading [E][C] |
| v190_nariai_entropy_bound.py   | The one-line Nariai entropy bound: the Schwarzschild–de Sitter entropy ratio $S_{\text{tot}}/S_{\text{dS}} = (x^2+1)/\Phi_3(x)$ ( $x=r_b/r_c$ ; $\Phi_3=x^2+x+1$ the $N_{\text{fam}}=3$ cyclotomic, already in tfpt_horizon_readouts) satisfies $S_{\text{tot}} \geq \frac{2}{3}S_{\text{dS}}$ with equality exactly at the Nariai merge $x=1$ , via the perfect square $3(x^2+1) - 2\Phi_3(x) = (x-1)^2 \geq 0$ — a variation-free proof of the entropy floor (global minimum on $x>0$ is $2/3 =  Z_2 /N_{\text{fam}}$ ) [E]                                                                                                                                                                                                                                                                                                                                                                                                                           |
| v191_universal_branch_line.py  | The universal branch line, honestly typed [C] — an exact affine <i>relabeling</i> , not a theorem. The map $q = \frac{7}{2} + \frac{N_{\text{fam}}^2}{2}m$ carries the SdS branch points $m = \pm 1/N_{\text{fam}}$ onto the flavor branch points $\{2, 5\} = \{ Z_2 , g_{\text{car}}\}$ ( $m=0 \rightarrow 7/2$ ). The arithmetic is exact, but the map exists for <i>any</i> two pairs of points (two intervals are always affinely equivalent), so the slope $N_{\text{fam}}^2/2$ and midpoint $7/2$ are <i>determined</i> by the data, not forced predictions; a decoy pair $\{3, 4\}$ admits an equally exact map (negative control). The genuine shared content — the $2/3 =  Z_2 /N_{\text{fam}}$ ramification of the mass $\leftrightarrow$ transport double cover — is already established. Recorded as a [C] alignment, never promoted to [E] [C]                                                                                             |
| v192_qgeo_energy_clock.py      | The energy-conserving-clock reformulation, honestly typed: QGEO.ENERGY.01 ( $\rho^* \Lambda_\Sigma \rho = \Lambda_\Sigma$ — the carrier clock preserves the Calderón/DtN energy form) <i>relocates</i> the bedrock QGEO.SYM.01 to a more checkable operator identity, but does <i>not</i> eliminate it. The downstream chain (conformality $\rightarrow$ Möbius $\rightarrow z \mapsto iz \rightarrow \mu_4 \rightarrow$ QGEO.SYM.01) already exists (v180/v181); the new upstream link (energy invariance $\Rightarrow$ conformality in 2D) is checkable on the DtN operator $\Lambda =  k $ . But ‘ $\rho$ preserves $\Lambda$ ’ is plausibly equivalent to ‘ $\rho$ is the conformal deck’, so it is a sharper restatement, not a theorem, unless $\rho$ is independently defined and the invariance independently proven (open). Bedrock stays [O]                                                                                                  |
| v193_qgeo_energy_commutator.py | QGEO.ENERGY.02: the energy- <i>commutator</i> contract (the non-circular sharpening of v192): $[\rho, \Lambda_\Sigma] = 0$ on the rank-8 Calderón polarisation, $\rho$ the carrier clock, $\Lambda_\Sigma$ the DtN energy form of the <i>raw</i> RP seam. Three sub-claims: (1) [E] $\rho$ is carrier-defined (Coxeter of $W(A_3)=S_4$ , order $4= \mu_4 $ , v117), not imported from the seam; (2) [O] the non-circularity hinge — $\Lambda_\Sigma$ must be the <i>raw</i> RP-seam DtN, not the $\mu_4$ normal form; (3) [E] $[\rho, \Lambda_\Sigma]=0$ forces the $(1, 2, 3)$ grading on $H^1$ (QGEO.COHO.M.01). Exact [E] rider: the induced $k=(\ln m)/(12\pi)$ v150 reproduces $c_3/2$ iff $\ln m=q(A_3)=\frac{3}{4}$ (FAMILY), not $q(D_5)=\frac{5}{4}$ (carrier, off by $5/3$ ) — gravity is family-geometry-induced. Proof target [O]; if proven, QGEO.SYM.01 becomes an energy-rigidity theorem [O]                                            |

*continued on next page*

| Script                                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v194_raw_seam_dtn_<br>rp.py                  | QGE0.DTN.01 — subclaim 2 of v193 (“the raw RP-seam DtN is RP-definable”) tackled: a <i>reduction</i> , not a closure. (1) [E] finite core: $\rho: z \mapsto iz$ on $H^1 = \langle w_1, w_2, w_3 \rangle$ , $w_k = z^{k-1} dz / (z^4 - 1)$ , acts as $\rho^* w_k = i^k w_k$ (characters $i, -1, -i$ distinct), so $[\rho, \Lambda_\Sigma] = 0$ on $H^1$ (v177). (2) [E] hinge met: the DtN/transfer operator is RP-canonical (Osterwalder–Schrader, v54) on a quasi-free seam state (v155), so $\Lambda_\Sigma$ is RP-definable without assuming $\mathbb{P}^1 \setminus \mu_4$ . (3) [O] residual relocates: full- $L^2$ commutation $\Leftrightarrow$ Bisognano–Wichmann geometric modular flow of the quasi-free seam state (must be intrinsic, else re-circulates). QGE0.SYM.01 reduces from a definitional postulate to intrinsic BW modular covariance — sharper, still [O], not closed [O]                                                                                                                                                                                                                                                                                    |
| v195_marks_<br>lefschetz_<br>character.py    | QGE0.MARKS.02 — a Lefschetz/character derivation of the four seam marks (review path 2): instead of positing $D_4$ marks, they are <i>forced</i> to be a free $\mu_4$ orbit. [E]: $\rho: z \mapsto iz$ (order 4) fixes only $\{0, \infty\}$ ; $H^1 = \chi_1 + \chi_2 + \chi_3$ (the $A_3$ exponents (1, 2, 3), v177) has no trivial character $\Rightarrow$ no puncture at a fixed point; $\text{rank } H_1 = n - 1 = 3 \Rightarrow n = 4 \Rightarrow$ exactly one free orbit $\Rightarrow$ marks = $\mu_4$ (cross-ratio 2); $\text{Tr}(\rho H^1) = i + i^2 + i^3 = -1$ , $L(\rho) = 1 + 1 = 2 = \# \text{fixed}$ . [O] residual relocates from “the seam is $\mathbb{P} \setminus \mu_4$ (geometric)” to “ $H^1$ carries the $A_3$ -exponent rep” — less circular [E][O]                                                                                                                                                                                                                                                                                                                                                                                                           |
| v196_seam_energy_<br>functional.py           | QGE0.VARI.01 — the $E_{\text{fail}}$ variational functional (review path 1), a numerically-testable sharpening of the v194/ENERGY.02 target: $E_{\text{fail}} = \ [\rho, \Lambda_\Sigma]\ _{HS}^2 + \ \rho^4 - 1\ ^2 + \ \Theta \rho \Theta - \rho^{-1}\ ^2$ , with $E_{\text{fail}} = 0 \Leftrightarrow$ the QGE0.SYM.01 conditions. [E] on the finite $H^1$ block ( $\rho = \text{diag}(i, -1, -i)$ , $\Lambda$ diagonal, $\Theta = \text{conj}$ ) all three vanish $\Rightarrow E_{\text{fail}} = 0$ ; [E] a $\mu_4$ -breaking $\Lambda$ or non-order-4 $\rho$ gives $E_{\text{fail}} > 0$ ( $\mu_4$ a true minimiser); [E] the zero locus is the seam-deck conditions; [O] the full-operator minimisation is the v194 Bisognano–Wichmann residual — a discretisable handle on the open bedrock, a target not a closure [E][O]                                                                                                                                                                                                                                                                                                                                                   |
| v197_rr_carrier_<br>clifford_d5.py           | ARCH.RRCAR.02 — hardening the Riemann–Roch carrier from $g_{\text{car}}$ to $D_5$ itself (review point 7): the 5-dim carrier mode space $C_{\text{car}} := H^0(\mathbb{P}^1, \mathcal{O}(\mu_4))$ generates the $\mathfrak{so}(10) = D_5$ half-spinor by its even Clifford algebra. [E] $\dim \Lambda^{\text{even}}(\mathbb{C}^5) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 1 + 10 + 5 = 16 = 2^{g_{\text{car}} - 1} = \dim S^+$ ; the $\mathfrak{so}(10) = D_5$ half-spinor $2^4 = 16 = S^+ = \Lambda^{\text{even}}(C_{\text{car}})$ ; closes the chain $\mu_4 \rightarrow g_{\text{car}} \rightarrow D_5$ from one four-point divisor (cycle side $\text{rank } H_1 = 3 = N_{\text{fam}}(A_3)$ ; function side $h^0 = 5 = g_{\text{car}}$ ; Clifford spinor $16 = D_5$ half-spinor = carrier). [C] “carrier = Clifford spinor of the seam mode space” stays physical (as v189), does not change the over-determined $g_{\text{car}} = 5$ [E][C]                                                                                                                                                                                                                              |
| v198_modular_<br>commutator_<br>reduction.py | QGE0.MODULAR.01 — cracking the bedrock to a state-invariance. The v194 residual ( $[\rho, \Lambda_\Sigma] = 0$ on full $L^2$ , framed as Bisognano–Wichmann with a circularity worry) is (a) made <i>exact</i> at the principal-symbol level and (b) reduced via Tomita–Takesaki to a clean state-invariance. [E] the DtN principal symbol $ k  = \sqrt{-d^2}$ is $\text{diag}( n )$ and the clock $\rho: z \mapsto iz$ is $\text{diag}(i^n)$ in the Fourier basis — both diagonal $\Rightarrow [\rho,  k ] = 0$ exactly on <i>all</i> of $L^2$ ; [E] Tomita–Takesaki: $[\rho, \Lambda_\Sigma] = 0$ follows from $\omega \circ \rho = \omega$ (the modular $\Delta$ is intrinsic to $(M, \Omega)$ ; a state-preserving symmetry commutes with it), needing <i>no</i> conformal covariance — removes the BW circularity; [E] the finite $H^1$ block is already $\mu_4$ -invariant (v177). [O] residual: the full raw seam state is $\mu_4$ -invariant ( $\omega \circ \rho = \omega$ ) $\Leftarrow$ the Gaussian bulk measure is deck-invariant = operational QGE0.SYM.01. Sharpest non-circular form; not a closure (a foundational symmetry cannot be derived from nothing) [E][O] |

*continued on next page*

| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v199_seam_state_invariance.py   | QGE0.STATE.01 (work package A) — attacking $\omega \circ \rho = \omega$ directly on the raw seam: a further localising reduction, not a closure. [E] setup non-circular ( $\rho = \text{Coxeter of } W(A_3) = S_4, \rho^4 = 1$ , an automorphism of the raw CAR algebra; both carrier-side, no seam geometry); [E] ground-state reduction $\omega \circ \rho = \omega \Leftrightarrow [\rho, H_1] = 0$ ( $C_\Sigma = \frac{1}{2}(1 + \text{sgn } H_1)$ ); [E] character criterion: $[\rho, H_1] = 0 \Leftrightarrow H_1$ block-diagonal in the four $\mu_4$ -character classes $\{n = r \bmod 4\}$ ; principal symbol $ k  = \text{diag}( n )$ passes automatically. [O] residual: the bounded sub-principal symbol has <i>no</i> off-character matrix elements — non-circular, sharply typed, holds on $H^1$ (v177); not a closure [E][O]                                                                                                                                                                                                                                                                                                                             |
| v200_seam_variational_scan.py   | QGE0.VARI.02 (work package B) — the discretised variational test: v196 had $E_{\text{fail}} = 0$ at the $\mu_4$ config on the fixed block; here the clock angle is free and $E_{\text{fail}}$ is minimised over it. [C] scanning $E_{\text{fail}}(\theta) = \ \rho(\theta)^4 - 1\ ^2 + \text{faithfulness over } \theta \in (0, \pi)$ gives the unique minimum at $\theta = \pi/2$ ( $E_{\text{fail}} = 0$ : order-4 <i>and</i> three distinct characters $i, -1, -i$ = the $(1, 2, 3)$ grading); decoys $(0, \pi, \pi/3, 2\pi/3)$ all lose. So $\arg \min E_{\text{fail}}$ = the $\mu_4$ normal form. [E] consistent with v196/v199/v198. [O] a numerical sign on the finite $H^1$ grading, not the full operator-valued minimisation (the v199 residual) [C][O]                                                                                                                                                                                                                                                                                                                                                                                                      |
| v201_seam_subprincipal_marks.py | QGE0.SUBPRIN.01 — the v199 bedrock residual reduced one genuine step further: block-diagonality of the sub-principal symbol is <i>not</i> an independent postulate. [E] the seam DtN is $\Lambda =  k  + M_f$ , $M_f$ = multiplication by a curvature function $f(\theta)$ (standard Steklov sub-principal), so $\langle n   M_f   n' \rangle = f_{n-n'}$ . [E] $M_f$ is $\mu_4$ -character-block-diagonal $\Leftrightarrow f$ has Fourier support only on modes $\equiv 0 \pmod{4}$ $\Leftrightarrow f$ is $\mathbb{Z}_4$ -invariant ( $ k $ already commutes, v198). [E] a $\mu_4$ -mark sum $f(\theta) = \sum_{j=0}^3 g(\theta - 2\pi j/4)$ has $f_m = 4g_m[m \equiv 0 \bmod 4]$ for <i>any</i> profile $g$ — automatically $\mathbb{Z}_4$ -invariant, so (v195 marks = $\mu_4$ orbit) <i>forces</i> the sub-principal symbol block-diagonal; negative controls: 3 marks ( $\mathbb{Z}_3$ ) and 4 generic marks break it. [O] new residual: $\omega \circ \rho = \omega$ reduces to ‘the DtN sub-principal symbol is mark-local (seam flat away from the $\mu_4$ marks = conformal deck v177)’, the narrowest, structurally-definitional form of QGE0.SYM.01 [E][O] |
| v202_rare_kaon.py               | The rare kaon sector $K \rightarrow \pi \nu \bar{\nu}$ from the TFPT CKM point ([C] downstream flavor readout, <i>not</i> a compiler power). The kernel fixes the full CKM matrix ( $s_{12} = \lambda_C$ , $s_{23} = \varphi_0/(1 + \lambda_C) =  V_{cb} $ , $s_{13} = \lambda_C^3/3$ , $\delta = \pi/3 + 3\lambda_C^2 = 1.19823$ rad; v18/v88) with no flavor fit; feeding the exact point into the standard Brod–Gorbahn–Stamou short-distance formulas gives $\text{BR}(K^+ \rightarrow \pi^+ \nu \bar{\nu}) = 9.45 \times 10^{-11}$ and $\text{BR}(K_L \rightarrow \pi^0 \nu \bar{\nu}) = 3.33 \times 10^{-11}$ . [E] CKM point + exact apex $(\bar{\rho}, \bar{\eta}) = (0.1374, 0.3509)$ , $\lambda_t = V_{ts}^* V_{td}$ ; [C] the branching ratios ( <i>not</i> parameter-free — $X_t, P_c, \kappa_\pm$ are external EW/QCD input); [E] $\text{BR}(K^+) \sim 1.2\sigma$ below NA62 $(13.0_{-3.0}^{+3.3}) \times 10^{-11}$ ; [E] Grossman–Nir ratio $0.352 \ll 4.3$ ; [X] dated kill test: a stable NA62 $\text{BR}(K^+)$ outside $[7, 12] \times 10^{-11}$ , or a KOTO-II $\text{BR}(K_L)$ off the GN point, breaks the flavor bridge (core untouched) [C][X]   |
| v203_eht_achromatic.py          | The EHT achromatic polarization-rotation intercept $\beta_{\text{BH}}$ around a horizon-scale black hole — the surviving core of the old UFE/BH notes ( <i>not</i> the RN-type metric, superseded by Nariai/seam=horizon v101–v104/v190; only the polarization signature). [E] the coupling is a compiler number: $\beta_{\text{BH}}(r) = Q_e^{\text{eff}} Q_m^{\text{eff}} / (256\pi^4 r^2) = 16c_3^4 (Q_e Q_m / r^2)$ , $16c_3^4 = 1/(256\pi^4)$ exactly. [E] $\alpha$ -kernel cross-link: $16c_3^4 = \delta_{\text{top}}/3$ , $\delta_{\text{top}} = 48c_3^4 = 3/(256\pi^4)$ — the <i>same</i> top-form coefficient that fixes the $\alpha$ -kernel precision-zone correction (one row from $\beta_{\text{rad}} = \varphi_0/4\pi$ ). [C] TFPT fixes the $1/r^2$ profile, achromaticity, and sign-flip under $E \cdot B$ reversal; the amplitude $Q_e Q_m$ is an MHD/GR weight, not a compiler output. [X] dated kill test: the residual intercept must pass three nulls (frequency/achromatic, spatial $1/r^2$ , sign-flip); a null-consistent residual after honest GRMHD subtraction falsifies the channel (core untouched) [E][C][X]                             |

*continued on next page*

| Script                    | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v204_muon_seam_g2.py      | The muon anomalous magnetic moment $a_\mu=(g_\mu-2)/2$ as a seam vertex readout ([C] downstream, <i>not</i> a compiler power; archive integration of the old muon- $g-2$ note). The carrier fixes the second-order defect $\delta_2=B\gamma\delta_{\text{top}}^2=\frac{5}{4}\delta_{\text{top}}^2$ ( $\delta_{\text{top}}=\Omega_{\text{adm}}c_3^4=48c_3^4=3/256\pi^4$ , $\sqrt{2}/\sqrt{23}$ ; $B\gamma=\frac{3}{2}\cdot\frac{5}{6}=\frac{5}{4}$ , v51); the seam-loop projection $a_\mu^{\text{seam}}=\delta_2/(2\pi)=45/(524288\pi^9)\approx 2.879\times 10^{-9}$ . [E] the value is an exact compiler number ( $\delta_2=4!\text{Tr}_{S^+}(X^2)c_3^8$ , $\text{Tr}=120=5!$ ); [C] the identification of $\delta_2/(2\pi)$ as a magnetic vertex ( $1/2\pi=4c_3$ ) is a physical bridge; [E] $0.81\sigma$ vs dispersive $\Delta a_\mu=(2.49\pm 0.48)\times 10^{-9}$ ; [C] HONEST: lattice/CMD-3 HVP shrinks it ( $\sim 1.5\times 10^{-9}$ ), the fixed value then $\sim 1.5\sigma$ high; [X] dated kill test: a converged $\Delta a_\mu$ outside $2.879\times 10^{-9}\pm 0.5\times 10^{-9}$ excludes the mechanism [E][C][X] |
| v205_xi_threequarter.py   | $\xi=c_3/\varphi_{\text{tree}}=3/4$ : an <i>independent</i> route to the gravitational $3/4$ (archive integration of the old Eliminating- $K$ note). The torsion-compression ansatz $\kappa^2=\xi\varphi_0/c_3^2$ with the Einstein limit $\kappa^2=8\pi G$ fixes $\xi=8\pi c_3^2/\varphi_0=c_3/\varphi_0$ (since $8\pi c_3^2=c_3$ ), and at tree level $\varphi_{\text{tree}}=1/(6\pi)$ this is exactly $3/4$ . [E] tree identity $\xi=c_3/\varphi_{\text{tree}}=3/4$ ; [E] Einstein-limit reduction $8\pi c_3^2=c_3\Rightarrow G$ is an output; [E] over-determination $3/4=q(A_3)=N_{\text{fam}}/ \mu_4 =\ln(m/\mu)$ (v152 $k=c_3/2$ ) — the torsion-compression $\xi$ and the gapped EH replica are two independent appearances of the same gravitational $3/4$ ; [E] with the retained seed $\xi=c_3/\varphi_0=0.748303$ ( $-0.226\%$ ); [C] the $\kappa^2=\xi\varphi_0/c_3^2$ ansatz is the bridge, the modern derivation is v152/v68, $1/G$ stays the one anchor (v60/v153) [E][C]                                                                                                                                      |
| v206_h0_lambda_branch.py  | The cosmological-constant Letter branch: the seam number $\delta_\Sigma=m_1e^{-2\pi\alpha^{-1}/3}$ with the lowest block $m_1=\Omega_{\text{adm}}=48$ , the Planck-convention split, and the quantitative $H_0=66.5\text{--}67.1$ (archive integration of the tfpt-31 CC note; sharpens the qualitative $H_0$ -vs- $\Lambda$ row). [E] $\delta_\Sigma^{\text{lead}}=48e^{-2\pi\cdot 137.036/3}\approx 1.085\times 10^{-123}$ ; [E] convention split $M_{\text{Pl}}^4/\bar{M}_{\text{Pl}}^4=(8\pi)^2=631.65$ (unreduced vs reduced — must be displayed); [E] $\rho_\Lambda=M_{\text{Pl}}^4\delta_\Sigma\approx 2.4\times 10^{-47}\text{GeV}^4$ (observed vacuum order); [E] 123-orders cross-link $ \log_{10}\delta_\Sigma \approx 122.96$ , consistent with the v60 split 122.948; [C] under flatness closure $H_0^{\text{lead}}=66.51$ , $H_0^{\text{disc}}=67.10$ — $1.6\text{--}0.5\sigma$ below Planck, $\sim 6\sigma$ below SH0ES: the Hubble tension is <i>not</i> relieved by raising the vacuum scale; $H_0$ uses imported $\Omega_b, \omega_{\text{DM}}$ , a [C] readout [E][C]                                       |
| v207_asymptotic_safety.py | Asymptotic Safety as a <i>second</i> , independent, [O] route to the metric sector (archive integration of the old Five-Problems note). The current theory closes QG via the discrete seam-net/AQFT programme (v175; open premise QGEO.REALIZE.01); this records a complementary continuum-FRG argument that a non-Gaussian UV fixed point exists from the same axioms — a plausibility cross-check, <i>not</i> load-bearing, does <i>not</i> close $G_{\text{metric}}$ . [E] the axion coupling is fixed and small $y^2=16c_3^2=1/(4\pi^2)=0.025330$ exactly ( $g_{a\gamma\gamma}=-4c_3$ from the cubic); [E] existence by IVT: $\partial_t g/g=2+\eta_R$ is $+2>0$ as $g\rightarrow 0$ and $<0$ at large $g$ (torsion $K^2$ damping) $\Rightarrow$ a root $g_*>0$ ; [E] $g_*=2/a_1>0$ finite, Reuter-type $O(1)$ , no new free input; [O] HONEST: a continuum-FRG ansatz the discrete-compiler theory did <i>not</i> adopt, primary stays the seam-net (v175) [E][O]                                                                                                                                                         |
| v208_bh_thermodynamics.py | Black-hole thermodynamics from the seam: the modular Hawking temperature and the scalaron-corrected Wald entropy (archive integration of the dropped tfpt-42 stationary-horizons section, re-typed through the current Tomita–Takesaki seam work v198/v199). [E] modular $T_H$ : $\Delta^{it}=e^{-2\pi t K_H}$ (Bisognano–Wichmann) $\Rightarrow$ KMS at $\beta=2\pi$ and $T_H=\kappa/(2\pi)$ ; the $2\pi$ is the seam unit $1/(4c_3)$ ( $1/2\pi=4c_3$ , v58), reproducing $T_H=c_3/M$ (v57); seam $\leftrightarrow$ horizon identification [C]; [E] Wald entropy: with the induced $f(R)=R+R^2/(6M_s^2)$ ( $M_s=c_3^{7/2}\bar{M}_{\text{Pl}}$ , v28), $S_W=(f_RA)/(4G)=(A/4G)(1+R_h/3M_s^2)$ — leading $A/4G$ is the $c_3$ area law ( $1/4=1/ \mu_4 $ ), the correction an exact scalaron consequence; [E] SdS $T_H=(1/4\pi r_h)(1-\Lambda r_h^2)$ , $1/4\pi=2c_3$ ; [E] area law $S_{BH}=M^2/(2c_3)=4\pi M^2=A/4G$ [E][C]                                                                                                                                                                                                    |

continued on next page

| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v209_bh_regular_core.py       | The black hole as a seam fixed point (topological defect), <i>not</i> a singularity: the honest [O] synthesis of the old Five-Problems BH picture, with the superseded torsion/RN-vortex metric dropped (RN-type metric superseded by Nariai/seam=horizon v101–v104/v190; cf. v203). Built entirely from established suite atoms. [E] no singularity = seam fixed point: the gapped attractor multiplier $\lambda_2=(2/3)^6=64/729$ with gap $\Delta=6\log(3/2)>0$ (v56) — the core is a fixed point, $\rho\rightarrow\infty$ replaced by $\varphi\rightarrow\varphi_*$ [C]; [E] maximal BH = anchor: Nariai cubic $t^3-3t+2=(t-1)^2(t+2)$ , roots $(1, 1, -2)$ = the traceless projection of $a=(1, 1, 2)$ (v101/v190); [E] information stays in the boundary field: Page recovery decays at $\lambda_2=(2/3)^6$ per step (v54) [C]; [E] holographic remnant $S_{BH}=A/4G$ , $1/4=1/ \mu_4 $ [C]; [O] HONEST: a qualitative structural synthesis (reuses v54/v56/v57/v101/v190), <i>not</i> load-bearing, <i>not</i> a metric [E][O]                                                                                                                                                                                                                                                  |
| v210_mark_local_dtn.py        | A numerical <i>sharpening</i> of v201/QGEO.SUBPRIN.01 on <i>realistic</i> Steklov curvature profiles, carried to the quasi-free <i>state</i> level (the actual $\omega\circ\rho=\omega$ , not just $[\rho, \Lambda]=0$ ); <i>not</i> a closure of QGEO.SYM.01. A real von-Mises bump $g(\theta)$ at a puncture, summed over the four $\mu_4$ marks $f(\theta)=\sum_{j=0}^3 g(\theta-j\pi/2)$ , builds the Steklov DtN $\Lambda= D_\theta +M_f$ . [E] every off-(mod 4) Fourier coefficient of the real $f$ vanishes to $<10^{-16}$ (exact 4-fold cancellation), although $g$ has all modes. [C] $\ [\rho, \Lambda]\  < 10^{-15}$ . [C] <i>state invariance</i> (new vs v201): the quasi-free covariance $C=\frac{1}{2}(1+\text{sgn } H_1)$ satisfies $\rho C \rho^\dagger = C$ to $<10^{-13}$ — the state, not just the operator, is clock-invariant. [C] negative controls (single off-centre bump, $\mathbb{Z}_3$ source, 4 generic marks) break both by $O(1)$ ; convergence stable for $N\in\{16, 32, 64\}$ . [O] certifies ‘mark-local DtN $\Rightarrow \omega\circ\rho=\omega$ ’ for concrete profiles <i>and</i> at the state level; the premise that the physical seam <i>is</i> mark-local stays [O] — the narrowest definitional form of QGEO.SYM.01, not its closure [C][O] |
| v211_axion_spine_angle.py     | Axion DM, the spine-angle branch $\theta_i = \pi N_{\text{fam}}/g_{\text{car}} = 3\pi/5$ ( $108^\circ$ ): an alternative, more robust misalignment angle than the determinant-line hilltop $\theta_i \approx 170^\circ$ (v185, which over-produces $\Omega_a h^2 \approx 0.66$ ). [E] rational motive $\theta_i = 3\pi/5$ exactly (spine quotient $3/5$ , no fit); [E] the v185 finite-T solver ( $\theta=1 \rightarrow 0.03$ ) reaches $\Omega_{\text{DM}}$ at $\theta \approx 106^\circ$ , so $108^\circ$ is on target (harmonic $0.03\theta^2=0.107$ ); [C] <i>robust</i> : $108^\circ$ is $62^\circ$ below the $170^\circ$ hilltop (mild-anharmonic, not exponentially sensitive); [O] an alternative ansatz to $\theta_i=\pi(1-\varphi_{\text{seam}})\approx 170^\circ$ (mutually exclusive, the full solver decides, DM.AXION.SPINE.01); [X] kill test: $\Omega_a h^2$ outside $\sim[0.08, 0.16]$ demotes the branch [E][C][O][X]                                                                                                                                                                                                                                                                                                                                                |
| v212_leptogenesis_decouple.py | Leptogenesis, the scalaron-decouple branch: both Boltzmann inputs share the decouple $A_\Lambda=10= E(K_5) $ — $M_1=M_{\text{scal}}\varphi_0^2/A_\Lambda \approx 8.65\times 10^9$ GeV and $\tilde{m}_1=m_3/A_\Lambda \approx 5$ meV. A cleaner [C] route than v184’s $M_1=M_R\varphi_0^4$ (which only relocated the free input to the un-pinned seesaw $M_R$ ). [E] shared decouple $A_\Lambda=10$ ; [E] $M_1=8.65\times 10^9$ GeV inside the thermal window $[3\times 10^9, 3\times 10^{10}]$ (v169); [E] $\tilde{m}_1 \approx 5$ meV, $K \approx 4.6$ strong washout; [E] Davidson–Ibarra $\varepsilon_1 \approx 8.5\times 10^{-7}$ , $\eta_B \approx 1.2\times 10^{-9}$ brackets observed $6.1\times 10^{-10}$ ; [C] HONEST: $M_1$ uses only $\{M_{\text{scal}}, \varphi_0, A_\Lambda\}$ but the coupling mechanism is posited not derived; [X] kill test falls the route, not the theory (FR.ETAB.01 independent) [E][C][X]                                                                                                                                                                                                                                                                                                                                                        |
| v213_ftransfer_functor.py     | The <b>F_transfer functor contract</b> (CONTRACT.F.01): the four frontier transfers are ONE typed functor $F_{\text{transfer}}=F_{\text{observable}}\circ F_{\text{threshold}}\circ F_{\text{RG}}$ with four structural axioms, each instance a discrete compiler kernel [E] composed with an external continuous solver [C]; consolidates v183 ( $F_{\text{pole}}$ ), v212 ( $F_{\text{Boltzmann}}$ ), v211 ( $F_{\text{relic}}$ ), v164/FR.MPME ( $F_{\text{QCD}}$ ), the structural complement of the v187 typing guard. [E] axiom 1 $\mu_4$ -deck equivariance ( $\lambda_2=(2/3)^6=64/729$ the deck transfer eigenvalue); [E] axiom 2 Plücker preservation ( $53=a^T(R+Q)\mathbf{1}$ , $\text{Pl}(F)=(1, 22, 30)$ , $\ \text{Pl}(K)\ _1=11$ ); [E] axiom 3 positivity/stochasticity ( $\text{spec } T=\{1, (2/3)^6, (1/3)^6\}$ positive, $\kappa_f\in(0, 1)$ ); [E] axiom 4 external modules explicit ( $b_3=-7$ a carrier output, $ b_3 =\Omega_{\text{adm}}-10b_1=7$ ); [C] verdict: a typed functor, not a bag of open topics; an architectural consolidation guarded by v187, not a closure [E][C]                                                                                                                                                                            |

*continued on next page*

| Script                     | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v214_seam_pillowcase.py    | <p>The <i>pillowcase</i> reduction of QGEO.SYM.01 (QGEO.PILLOW.01): the two separately-tracked open QGEO residuals — v180/QGEO.ISO.01 (the carrier clock is an order-4 isometry of the seam metric) and v201/QGEO.SUBPRIN.01 (the DtN sub-principal symbol is mark-local, the seam flat away from the <math>\mu_4</math> marks) — are ONE statement, ‘the seam carries the uniformising flat orbifold (pillowcase) metric’, and the order-4 clock is <i>derived</i> from cross-ratio 2 (v168), not assumed. [E] <math>\chi_{\text{orb}}(S^2(2,2,2,2))=2-4(1-\frac{1}{2})=0 \Rightarrow</math> flat (Trojanov cone-metric uniformisation, unique up to scale; Gauss–Bonnet four deficits <math>\pi</math> sum to <math>4\pi=2\pi\chi(S^2) \Rightarrow \int K_{\text{smooth}}=0</math>, flat away from the marks). [E] NEW LINK: cross-ratio(<math>\mu_4</math>)=2 (v168) <math>\Rightarrow j(\lambda)=256(\lambda^2-\lambda+1)^3/(\lambda^2(\lambda-1)^2)</math> with <math>j(2)=1728</math> (all six harmonic cross-ratios <math>\{2, -1, \frac{1}{2}\}</math> give 1728) <math>\Rightarrow</math> the square modulus, hence the order-4 clock, is a <i>consequence</i> of cross-ratio 2. [E] <math>j=1728 \Leftrightarrow \tau=i</math> (square torus) <math>\Leftrightarrow</math> CM by <math>\mathbb{Z}[i] \Leftrightarrow \text{Aut}=\mathbb{Z}/4</math> (<math>j=0</math> gives <math>\mathbb{Z}/6</math>; generic <math>\mathbb{Z}/2</math>). [E] explicit CM <math>(x,y) \mapsto (-x, iy)</math> on <math>y^2=x^3-x</math> has order <math>4=z \mapsto iz</math>. [E] negative control <math>\lambda=3 \Rightarrow j=21952/9 \notin \{0, 1728\} \Rightarrow \text{Aut}=\mathbb{Z}/2</math>, no clock. [E] UNIFICATION: ‘seam = uniformising flat pillowcase metric’ implies both v201 mark-locality and v180 QGEO.ISO.01. [O] does <i>not</i> close QGEO.SYM.01 — the canonical constant-curvature-metric premise stays open, milder/more canonical; net existence + full-cone RP remain v175’s [E]</p> |
| v215_seam_deck_killtest.py | <p>The bedrock QGEO.SYM.01 (<math>\omega \circ \rho = \omega</math>) recast as a <i>falsifiable</i> kill-test (QGEO.KILL.01), not a proof-promise — a foundational symmetry cannot be derived from nothing (v181/v153), and Bisognano–Wichmann would presuppose the covariance it should produce. Non-circular reduction (v198/v199/v201): <math>\omega \circ \rho = \omega \Leftrightarrow [\rho, C]=0</math>; the principal symbol <math> k </math> commutes exactly, so the residual is the sub-principal <math>M_f</math> with <math>[\rho, M_f]=0 \Leftrightarrow f \mathbb{Z}_4</math>-invariant; measure form = the raw Gaussian bulk form <math>A</math> is <math>\mu_4</math>-deck-invariant. Four levers (carrier-side [E], seam-side [O]): <b>K1</b> exactly 4 marks (order-4 clock free orbit <math>\mu_4 + b_1 = \# \text{marks} - 1 = 3 = N_{\text{fam}} \Rightarrow \# \text{marks} = 4 =  \mu_4 </math>); <b>K2</b> square config (cross-ratio <math>2 \Rightarrow j=1728</math>, <math>\text{Aut}=\mathbb{Z}/4</math>; order 4 selects <math>\mathbb{Z}/4</math> not the <math>j=0 \mathbb{Z}/6</math>; generic <math>\Rightarrow \mathbb{Z}/2</math>); <b>K3</b> mod-4 sub-principal support (<math>\mu_4</math>-mark sum <math>\Rightarrow f_m=0</math> for <math>m \not\equiv 0 \pmod{4} \Rightarrow [\rho, M_f]=0</math>); <b>K4</b> transfer gap <math>(2/3)^6=64/729</math> (cusp weights <math>\{0, \frac{1}{3}, \frac{2}{3}\}</math>). Freeze-bound (freeze_file.csv seam_deck_symmetry). [O] the decisive kill is <i>deferred</i> (QGEO.REALIZE.01): the raw <math>\mathbb{P} \setminus \mu_4</math> collar DtN must yield 4 square marks <i>without</i> inputting them; carrier-side K1–K4 are green regression guards. A kill-test, not a closure</p>                                                                                                                                                                                                                               |
| v216_marks_gauss_bonnet.py | <p>The four seam marks <i>emerge</i> from Gauss–Bonnet (QGEO.MARKS.03), executing the K1 lever of the v215 kill-test. [E] mark count = <math>2\chi</math>: <math>\mathbb{Z}_2</math> branch points (deficit <math>\pi</math>) on a flat sphere (<math>\chi=2</math>) give <math>n\pi = 2\pi\chi = 4\pi \Rightarrow n = 2\chi = 4 =  \mu_4  = e_1(a) = N_{\text{fam}}+1</math> (the same <math>\chi=2</math> as the 8 in <math>c_3=1/(8\pi)</math>). [E] the Euclidean sphere 2-orbifolds (<math>\chi_{\text{orb}}=0</math>) are exactly <math>(2,3,6), (2,4,4), (3,3,3), (2,2,2,2)</math>. [E] all-order-2 (the <math> \mathbb{Z}_2 </math> sheet branch) <math>\Rightarrow</math> uniquely <math>(2,2,2,2)</math>; [E] <math>N_{\text{fam}}=3</math> (<math>\text{rank } H^1 = \# \text{marks} - 1</math>) <math>\Rightarrow</math> uniquely <math>(2,2,2,2)</math>, picking the 4-mark square over the 3-mark hexagonal <math>(3,3,3)</math>; [E] a free order-4 deck <math>\Rightarrow</math> uniquely <math>(2,2,2,2)</math> — three independent selectors converge. [O] residual: the square modulus (cross-ratio <math>2 \Rightarrow j=1728</math>, v214) needs the one order-4 carrier input. The mark count + equality are now derived; does <i>not</i> close QGEO.REALIZE.01</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

continued on next page

| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v217_marks_emergence_scan.py  | The <i>numerical</i> emergence scan (QGEO.EMERGE.01, the sibling of v216 as v210 is to v201): with the number of marks $n$ and their positions free, the 4-equally-spaced (square) configuration is the joint minimiser of {clock-invariance + 3-family cohomology + transfer gap $(2/3)^6$ } on the actual seam DtN/state. [C] $n=4$ equally-spaced $\Rightarrow \ \rho_4 C \rho_4^\dagger - C\  \sim 10^{-14}$ (clock-invariant state); [C] family count selects $n=4$ (scan $n \in \{2..8\}$ , only $n-1=3=N_{\text{fam}}$ ; $n=3 \rightarrow 2$ families = hexagonal, excluded; $n=5 \rightarrow 4$ ); [C] gap: $\mu_4$ -equivariant $\Rightarrow$ diagonal in the cusp basis (deck-average off-diagonal $\sim 10^{-16}$ ), cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\} \Rightarrow$ transfer $(1-w)^6 = \{1, (2/3)^6, (1/3)^6\}$ , gap $(2/3)^6 = 64/729$ ; [C] position scan: the square minimises the defect (square $\sim 10^{-14}$ vs off-square $\sim 0.4$ ); [C] negative controls $n=3, 5, 6$ . [O] confirms the emergence on the real DtN (number scanned, positions free); selector = the 3-family/gap structure (carrier input); executes K1 numerically, does <i>not</i> close QGEO.REALIZE.01. Python-only |
| v218_diamond_axis_geometry.py | Diamond axis geometry: the Sheet Diamond (v94) is a discrete geometry with two axes around the centered cross (v95: $C$ center, $U$ winding, $V$ sheet), $F$ the transfer completion. Three exact [E] lemmas + honest audit/heuristic typing, no new numbers. [E] AXIS CURVATURE: $\det(C+xU)=14+6x$ linear (slope $6= R^+(A_3) $ ), $\det(C+yV)=14+14y+4y^2$ quadratic, 2nd difference $8=\text{rank } E_8$ ; $\det B(C+xU)=2 \det(C+xU)$ , anchor-block sheet curvature $6= R^+(A_3) $ . [E] PLÜCKER LADDER: $\text{Pl}(K)=(-1, 6, 4) \rightarrow \text{Pl}(C)=(0, 14, 14) \rightarrow \text{Pl}(F)=(1, 22, 30)$ , steps $(1, 8, 10)=(N_\Phi, \text{rank } E_8, A_\Lambda)$ and $(1, 8, 16)=(N_\Phi, \text{rank } E_8, \dim S^+)$ (identity [E], transfer reading [C]). [E] SPECTRAL RAMIFICATION: $\text{Disc}(\chi)=q(r)^2 \text{Disc}(q)$ for $Q, K, C, F$ ; squares $(1, 3, 4, 6)$ , kernels $(13, 48, 65, 105)$ with $48=\Omega_{\text{adm}}$ , $105=N_{\text{fam}}g_{\text{car}} \cdot 7$ . [C] heuristic: $F$ the transfer-completion corner, a search principle, not a hard filter; the $G_2/F_4$ pair-sum labels stay audit-only (cf. v94/v95) [E][C]                                                              |
| v219_icosahedral_mckay.py     | McKay bedrock: <i>why</i> the atoms are $\{2, 3, 5\}$ . $E_8$ is the exceptional <i>top</i> of the McKay tower of finite $SU(2)$ subgroups ( $2T \rightarrow \hat{E}_6$ , $2O \rightarrow \hat{E}_7$ , $2I \rightarrow \hat{E}_8$ ). The McKay graph is built <i>from the group</i> : the 120 icosians close to the binary icosahedral $2I$ (element orders $\{1, 2, 3, 4, 5, 6, 10\}$ , containing the $2, 3, 5$ axis orders), 9 conjugacy classes, Dixon irrep degrees $\{1, 2, 2, 3, 3, 4, 4, 5, 6\}$ (sum $30=h(E_8)$ , sum of squares $120= R^+(E_8) $ ); the natural-rep McKay adjacency <i>is</i> the affine $E_8$ graph with Kac marks = the degrees. A backward certificate of the closed $E_8$ [E], the reading that it explains $(2, 3, 5)$ [C], the raw-seam origin [O]                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| v220_cp_hexagonal_modulus.py  | CP lives in the <i>hexagonal</i> phase fiber, not the square seam deck (a geometric sharpening of red-team Target D). [E] the seam deck is the square modulus $j(i)=1728$ , $\text{Aut}=\mathbb{Z}/4$ (the $\mu_4$ clock); its exceptional partner is the hexagonal modulus $j(\rho)=0$ , $\rho=e^{i\pi/3}$ , $\text{Aut}=\mathbb{Z}/6$ , CM by $\mathbb{Z}[\omega]$ , $\arg \rho=\pi/3$ . [E] the frozen leading CKM reading $\delta=\pi/3+3\lambda_C^2=68.654^\circ$ (v88) has leading term $\pi/3=\arg \rho$ ; the $ \mathbb{Z}_2 $ alternative $\pi/3+2\lambda_C^2$ is the frozen look-elsewhere trap. The deck stays $\mathbb{Z}/4$ (v215); CP is the $\mathbb{Z}/6$ fiber, the bridge [C][E][C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| v221_seam_qecc.py             | The seam as a <i>finite recoverability</i> code (recovery rate $(2/3)^6=64/729$ ). [E] code dimension $\dim S^+=16=2^{g_{\text{car}}-1}$ ; the gapped transport $T$ on the cusp-weight 3-space (deviations $(1, -1, 0)$ and the Nariai anchor $(1, 1, -2)$ ) is symmetric, doubly stochastic (CPTP), spectrum $\{1, (2/3)^6, (1/3)^6\}$ ; on the trace-zero subspace $\ T\delta\  \leq (2/3)^6 \ \delta\ $ hence $\ T^n \delta\  \leq (2/3)^{6n} \ \delta\ $ (a Knill–Laflamme-type bound); a free ratio breaks it. [C] the holographic/Petz reading (gated on the Seam–Horizon theorem) [E][C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| v222_cm_norm_duality.py       | CM-norm duality: the two exceptional moduli give 41 and 7. [E] SQUARE (Gaussian $\mathbb{Z}[i]$ , $j=1728$ ): $N(g_{\text{car}}+i \mu_4 )=5^2+4^2=41=10b_1$ — the EM index <i>is</i> the Gaussian norm of the carrier-glue vector; $N(3+2i)=13=\Delta_Q$ . [E] HEX (Eisenstein $\mathbb{Z}[\omega]$ , $j=0$ ): $N(3+2\omega)=3^2-3\cdot 2+2^2=7=$ scalaron. The $(3, 2)=(\text{colour}, \text{weak})$ split generates $(5, 6, 7, 13)$ by sum / product / Eisenstein / Gauss norm; the rings are rigid (Eisenstein of $(5, 4)=21 \neq 41$ ). The architecture bridge square $\leftrightarrow$ EM, hex $\leftrightarrow$ scalaron is [C][E][C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

*continued on next page*

| Script                                | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v223_coxeter_totative_clock.py        | The $\mu_4$ clock is the order-4 character of the $E_8$ Coxeter cycle. [E]<br>$(\mathbb{Z}/30)^\times = \{1, 7, 11, 13, 17, 19, 23, 29\}$ = the $E_8$ exponents = live Coxeter phases;<br>$\varphi(30)=8=\text{rank } E_8$ , $30=2\cdot 3\cdot 5$ . [E] $(\mathbb{Z}/30)^\times \cong \mathbb{Z}/4 \times \mathbb{Z}/2$ with order-4 generator 7<br>$(7^2=19, 7^4=1 \bmod 30)$ : a literal $\mu_4$ clock $\langle 7 \rangle = \{1, 7, 13, 19\}$ on the exponents; the<br>conjugate pairs $m+(30-m)=30$ give $ \mu_4 =4$ invariant planes the clock permutes.<br>Only $E_8$ has $\varphi(h)=\text{rank}$ and $h=2\cdot 3\cdot 5$ ; the seam-clock identification is [C][E][C] |
| v224_diamond_fttransfer_path.py       | $F_{\text{transfer}}$ as one path on the Sheet Diamond, not four external interfaces. [E]<br>family $M(s, t)=R+Q \text{ diag}(s, t, t)$ : the corners<br>$K=M(1, -1)$ , $C=M(1, 0)$ , $F=M(1, 1)$ lie on the sheet axis ( $s=1$ ); the sheet axis is<br>curved $\det M(1, t)=4t^2+14t+14$ (2nd diff $8=\text{rank } E_8$ ), the winding axis<br>$\det M(s, 0)=6s+8$ flat (slope $6= R^+(A_3) $ ); Plücker steps $(1, 8, 10)$ , $(1, 8, 16)$ . [C]<br>$F_{\text{transfer}}$ is the $K \rightarrow C \rightarrow F$ sheet geodesic, Koide source $\rightarrow$ pole its 1-D Möbius<br>projection (v37/v183) [E][C]                                                             |
| v225_dual_normal_frame.py             | The dual normal frame $(d, n)$ : one boundary compass for flavor and horizon, with<br>CP as orientation. [E] $d=a^\top R^{-1}=a^\top L^{-1}=(-\frac{1}{2}, -\frac{1}{2}, 1)=-\frac{1}{2}(1, 1, -2)$ (the<br>Nariai/horizon normal; $d\cdot \mathbf{1}=0$ , $d\cdot a=1$ ); $n=(5, -9, 6)$ with<br>$n^\top R=(\det R, 0, 0)=(8, 0, 0)$ , $n^\top L=(20, 0, 0)$ (the first-generation/torsion selector).<br>[E] oriented volume $\det(\mathbf{1}, d, n)=21=N_{\text{fam}} \cdot 7$ . [C] CP as orientation:<br>$\text{Im } \det(\mathbf{1}, d, \rho n)=21 \sin(\pi/3)$ for $\rho=e^{i\pi/3}$ — CP sits where the magnitude<br>bijection (Target D) fails [E][C]                |
| v227_degree_exponent_channel_split.py | $248=120+128$ as a magnitude/phase channel typing (replaces the over-strong<br>$S^-$ -dark-matter reading). [E] $E_8$ exponents sum $120= R^+(E_8) $ ( <i>magnitude</i><br>channel); degrees $\{2, 8, 12, 14, 18, 20, 24, 30\}$ sum $128=2^{\text{rank}-1}=\text{rank} \cdot \dim S^+$<br>( <i>phase/glue</i> channel); $248=120+128=\text{adj}(D_8)+\text{spinor}(D_8)$ ;<br>$\sum \deg - \sum \exp = \text{rank}=8$ . [C] ratios+products (Target D) live in the 120 channel,<br>the un-covered CP phases in the 128 phase/glue channel; [O] a dark-sector<br>reading of 128 stays Frontier [E][C][O]                                                                      |
| v228_rr_index_gate.py                 | P2 as the mode space of a degree-4 seam divisor (a Riemann–Roch <i>index gate</i> ;<br>bedrock language for P2, not a proof from nothing). [E] $\deg D=4= \mu_4 $ on<br>$\mathbb{P}^1 \Rightarrow h^0(\mathcal{O}(D))=5=g_{\text{car}}$ (Riemann–Roch, genus 0); $\text{rank } H_1(\mathbb{P}^1 \setminus 4)=3=N_{\text{fam}}$ ;<br>$h^0+\text{rank } H_1=8=\text{rank } E_8$ , $h^0-\text{rank } H_1=2= \mathbb{Z}_2 $ ; $\Lambda^{\text{even}}(\mathbb{C}^5)=1+10+5=16=\dim S^+$ .<br>The mode-space reading is [C], conditional on the raw seam producing the<br>degree-4 divisor ([O], = QGEO.SYM.01) [E][C][O]                                                          |
| v229_lepton_frobenius_algebra.py      | The charged leptons as an étale Frobenius algebra over the $\mu_6$ resolvent. [E] exact<br>coefficients $(c_e, c_\mu, c_\tau)=(\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$ , ring closure $c_e c_\tau = \frac{8}{3} =  \mathbb{Z}_2  c_\mu$ , product<br>$\frac{32}{9} = 2g_{\text{car}}/N_{\text{fam}}^2$ . [E] $A=\mathbb{Q}[t]/(m)$ , $m=\prod(t-c)$ : the trace-form Gram is<br>nondegenerate ( $\det=\text{disc}(m)\neq 0$ , $A$ étale) — a commutative Frobenius algebra;<br>the $C_6$ shift has characteristic polynomial $t^6-1$ (spectrum $\mu_6=\mu_3$ family $\times \mu_2$<br>sheet). [C] the PMNS extension via $\text{Aut}(A)$ and the hexagonal CM point [E][C]  |
| v230_center_budget_norms.py           | The Sheet-Diamond center budget $(7, 11, 13)$ is the three <i>local norms</i> of the theory.<br>[E] $C=R+Q \text{ diag}(1, 0, 0)$ , row sums $(7, 11, 13)$ : $7=N_{\mathbb{Z}[\omega]}(3+2\omega)$ (Eisenstein/hex =<br>scalaron), $13=N_{\mathbb{Z}[i]}(3+2i)$ (Gauss/square = $\Delta_Q$ ), $11=\binom{4}{0}+\binom{4}{1}+\binom{4}{2}=1+4+6$ (QBL<br>boundary Fock count on the 4 $\mu_4$ corners = $\dim S^+ - g_{\text{car}}=16-5$ ). The two CM<br>rings (v222) and the seam boundary count meet in one operator center; the<br>reading is [C][E][C]                                                                                                                   |

continued on next page

| Script                            | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v231_cp_mu6_phases.py             | Both CP phases are $\mu_6$ powers of <i>one</i> hexagonal CM unit, split by the sheet — a structural reduction of red-team Target D (sharper than v220/v225). With $\rho=e^{i\pi/3}$ (the $j=0$ Eisenstein CM unit): [E] $\rho$ is a primitive 6th root ( $\rho^6=1$ , $\rho^3=-1$ the $\mathbb{Z}_2$ sheet half-turn); [E] $\delta_{\text{CKM}}$ leading = $\arg(\rho^1)=\pi/3$ (quark), $\delta_{\text{PMNS}} = \arg(\rho^4)=4\pi/3$ (lepton, the <i>phase lattice</i> ); [E] $\rho^4=-\rho \Rightarrow \delta_{\text{PMNS}}=\delta_{\text{CKM,lead}}+\pi=(\text{CM unit})\times(\mathbb{Z}_2 \text{ sheet})$ — the two CP phases are <i>one</i> hexagonal unit on the two sheets, not two independent inputs; [E] the $C_6/\mu_6$ monodromy has charpoly $t^6-1$ (spectrum $\mu_6$ carries $\rho$ ); [E] orientation $\text{Im det}(\mathbf{1}, d, \rho^1 n)=+21 \sin(\pi/3)$ , $\text{Im det}(\mathbf{1}, d, \rho^4 n)=-21 \sin(\pi/3)$ (sheet-flipped Jarlskog, v225); [E] NEG $\mu_3$ has no $-1$ (no sheet; $\mu_6=\mu_3$ family $\times \mu_2$ sheet). REDUCTION: 2 CP phase inputs $\rightarrow$ 1 hexagonal unit + the sheet; the physical CP derivation stays [C], the deck stays $\mathbb{Z}/4$ (v215) [E][C]                                                                                                                                                          |
| v232_e8_kleinian_seam.py          | The seam as the $E_8$ <i>Kleinian singularity</i> — a canonical algebraic-geometric model for the open seam-realisation premise QGEO.REALIZE.01 (the du Val side of the McKay correspondence, v219). [E] the McKay graph of $2I$ is affine $E_8$ ; dropping the unique trivial-rep node (the single Kac mark 1) leaves the <i>finite</i> $E_8$ Dynkin on 8 nodes = the dual intersection graph of the eight exceptional $\mathbb{P}^1$ 's of the minimal resolution of $\mathbb{C}^2/2I$ (trivalent node, arms 1, 2, 4). [E] eight exceptional $\mathbb{P}^1$ 's = $\text{rank } E_8 = g_{\text{car}} + N_{\text{fam}} = \# \text{ nontrivial } 2I \text{ irreps}$ — a fourth reading of the 8 in $c_3=1/(8\pi)$ . [E] the negated intersection form is the $E_8$ Cartan (even, $\det=1$ , positive definite; each $\mathbb{P}^1$ a $(-2)$ -curve). [E] one degree-1 irrep $\Rightarrow 2I$ perfect $\Rightarrow H_1(S^3/2I)=0$ : the link is the Poincaré homology 3-sphere, $\pi_1=2I$ of order $120= R^+(E_8) $ . [C] seam reading: the raw RP seam ( $\mathbb{P}^1$ with marks) is canonically a du Val exceptional curve — a model for QGEO.REALIZE.01, not a P2 proof (presupposes $E_8$ ); bedrock [O][E][C][O]                                                                                                                                                             |
| v233_cp_triality_phase.py         | The CP phase is the <i>universal</i> family/triality phase, only sheet-split — one step past v231 in $\mu_6=\mu_3$ (family/triality) $\times \mu_2$ (sheet). [E] $\mu_6=\mu_3 \times \mu_2$ : the $\mathbb{Z}_3$ triality centre $\{1, \omega, \omega^2\}$ = the cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\} \times 2\pi$ , the sheet $\mu_2=\{1, -1\}$ ; $\rho=\omega^2 \cdot (-1)$ . [E] SAME FAMILY CLASS: $\rho^4=\rho^1 \cdot \rho^3$ with $\rho^3 \in \mu_2$ , so the quark ( $\rho^1$ ) and lepton ( $\rho^4$ ) CP phases have the <i>same</i> image in $\mu_6/\mu_2 \cong \mu_3$ and <i>opposite</i> sheet — the CP phase <i>is</i> the universal triality phase $\omega^2$ (the $2/3$ cusp-weight $\mathbb{Z}_3$ centre), not a free power choice. [E] $\omega$ is the $\mu_3$ factor of the order-30 $(2, 3, 5)$ Milnor/Coxeter monodromy that builds $E_8$ (v232). Reduces Target D from ‘one hexagonal unit + sheet’ (v231) to ‘the universal triality phase + sheet’; the physical CP derivation stays [C][E][C]                                                                                                                                                                                                                                                                                                                                                    |
| v234_seam_holomorphy_selection.py | The Seam-Holomorphy selection certificate: the entire structural residual is <i>one</i> condition with <i>three</i> equivalent faces, all forcing $E_8$ . The condition: <i>the seam carries no nontrivial abelian sector</i> . [E] (i) AQFT — the boundary net is holomorphic ( $\mu$ -index 1); (ii) geometry — the seam link is a homology 3-sphere $\Leftrightarrow$ the binary group is perfect $\Leftrightarrow 2I$ ; (iii) rep theory — exactly one 1-dim irrep. [E] $\#(\text{mark-1 affine nodes}) =  \Gamma^{\text{ab}}  =  H_1(S^3/\Gamma)  = \#(1\text{-dim irreps})$ : $A_n \rightarrow n+1$ , $D_n \rightarrow 4$ , $E_6 \rightarrow 3$ , $E_7 \rightarrow 2$ , $E_8 \rightarrow 1$ — only $E_8$ gives 1 ( $2I$ the unique perfect $SU(2)$ subgroup, Poincaré the unique homology-sphere space form). [E] holomorphic $c=8=g_{\text{car}}+N_{\text{fam}} \Rightarrow$ the unique even unimodular rank-8 lattice = $E_8$ . [E] the three faces (v83/v154, v232, v219) are the SAME $E_8$ -selector, so Target A, $G_{\text{net}}$ , the carrier P2 and the Kleinian seam are <i>one</i> gate. [O] the closing theorem (stated): RP+gap+chirality $\Rightarrow$ quasi-free bulk (v160) $\Rightarrow$ invertible bulk $\Rightarrow$ holomorphic boundary $\Rightarrow E_8$ ; the single residual analytic step is ‘free bulk $\Rightarrow$ holomorphic boundary’ [E][O] |

*continued on next page*

| Script                            | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v235_seam_chern_simons.py         | The closing step in abelian Chern–Simons language: holomorphic $\Leftrightarrow \det K=1$ — a genuine reduction (not a closure) of GATE.HOLO.01 into standard anyon-condensation physics. [E] a free gapped bosonic 2+1d bulk = an even integer $K$ -matrix with $\# \text{anyons} =  \det K $ , edge $c = \text{signature}(K)$ , holomorphic $\Leftrightarrow  \det K =1$ . The TFPT extension tower (v92) is the condensation tower read by determinant: carrier $D_5 \oplus A_3$ ( $\det 16$ , 16 anyons) $\rightarrow SO(16)_1 = D_8$ ( $\det 4$ ) $\rightarrow (E_8)_1$ ( $\det 1$ , holomorphic), all rank 8, $c = \text{sig}=8=g_{\text{car}}+N_{\text{fam}}$ . [E] holomorphic $\Leftrightarrow \det 1 \Leftrightarrow E_8$ = the Kitaev $E_8$ quantum-Hall state. [E] condensation: a Lagrangian glue of order $\sqrt{16}= \mu_4 =4$ takes $\det 16 \rightarrow 1$ (a partial order-2 isotropic only $\rightarrow 4=D_8$ ). [E] NEG: $D_8$ is free+bosonic but has 4 anyons; an odd $K$ is fermionic — $\det=1$ is the extra condition. [O] residual: ‘the free RP seam condenses the order- $ \mu_4 $ Lagrangian glue’ = the sheet/QGEO selection, now with holomorphy $\Leftrightarrow \det 1$ a theorem [E][O]                                        |
| v236_brieskorn_capstone.py        | The $(2, 3, 5)$ Brieskorn singularity $x^2+y^3+z^5$ is the <i>one</i> generator of the discrete skeleton — the capstone of the icosahedral round (v55/v219/v223/v232/v233); its exponents <i>are</i> the atoms $( \mathbb{Z}_2 , N_{\text{fam}}, g_{\text{car}})=(2, 3, 5)$ . [E] Milnor number $\mu=(2-1)(3-1)(5-1)=1\cdot 2\cdot 4=8=\text{rank } E_8$ = the 8 in $c_3=1/(8\pi)$ (a fifth independent origin). [E] the Milnor monodromy eigenvalues are the eight primitive 30th roots $e^{2\pi i m/30}$ , $m$ the $E_8$ exponents (charpoly $\Phi_{30}$ , $\deg \varphi(30)=8=\mu$ ) — so it <i>is</i> the order-30 $E_8$ Coxeter element (v55). [E] both clocks: the monodromy group $\langle h \rangle = \mathbb{Z}/30 = \mathbb{Z}/2 \times \mathbb{Z}/3 \times \mathbb{Z}/5$ carries the $\mu_3$ triality (CP, v233) as $h^{10}$ ; the Galois group $(\mathbb{Z}/30)^\times = \mathbb{Z}/2 \times \mathbb{Z}/4$ carries the $\mu_4$ clock ( $\times 7$ , v223). [E] Milnor lattice = $E_8$ , link = Poincaré sphere. [C] one singularity generates atoms + 8 + $E_8$ + order-30 + both clocks + seam; a unification, not a closure [E][C]                                                                                                                  |
| v237_seam_sre_closure.py          | The closing step as a <i>physical</i> condition: no topological ground-state degeneracy $\Leftrightarrow \det K=1 \Leftrightarrow$ the seam bulk is short-range-entangled (SRE/invertible) — sharpening GATE.HOLO.02’s residual from definitional to falsifiable. [E] the $K$ -matrix ground-state degeneracy on a genus- $g$ surface is $ \det K ^g$ (carrier $\rightarrow 16$ , $D_8 \rightarrow 4$ , $E_8 \rightarrow 1$ on the torus), so ‘no degeneracy on any closed surface’ $\Leftrightarrow \det K=1$ . [E] among rank-8 even positive-definite $K$ ( $c=8=g_{\text{car}}+N_{\text{fam}}$ ) only $E_8$ has $\det 1$ — the unique nontrivial bosonic SRE 2+1d phase (Kitaev $E_8$ ), edge holomorphic; $D_8/\text{carrier}$ are LRE. [E] a unique vacuum on the plane (RP+gap+cluster) is necessary but not sufficient (the plane cannot see $\det K$ , the torus does), so the residual is ‘the seam unique-vacuum extends to the torus’ = SRE. [E] NEG: an LRE bulk ( $D_8$ ) has 4-fold torus degeneracy + non-holomorphic boundary. [O] sharpened residual: ‘the free RP gapped seam is short-range-entangled (no topological order)’ = $\det K=1$ = the Kitaev $E_8$ phase — a yes/no statement, sharper than v235/QGEO.SYM.01; not a closure [E][O] |
| v238_modular_lindblad_dynamics.py | DYN.SEMIGROUP.01 — the static $\rightarrow$ dynamic flip: the seam’s modular flow (reversible) and its recovery channel as a Markov/Lindblad generator (irreversible), the dissipative gap = the OS mass gap. Companion to v196/v198 (modular) and v221/v64 (recovery), which state the structure <i>statically</i> ; this reads the same objects <i>dynamically</i> . [E] the modular flow $\sigma_t = e^{it\Lambda_\Sigma}$ is a unitary one-parameter group (Tomita–Takesaki) and the $\mu_4$ clock is conserved $\Leftrightarrow [\rho, \Lambda_\Sigma] = 0$ (the v198 state-invariance), so the static commutator is the conservation law of modular time. [E] $L = \log T$ is a doubly-stochastic generator ( $e^L = T$ , $e^{L\tau} \rightarrow$ the democratic projector = the arrow of time, cf. v64). [E] keystone: the dissipative gap = $-\log((2/3)^6) = 6\log(3/2)$ = the OS mass gap $\Delta$ (v64) — the static recovery bound and the dynamical mass gap are one number, one exponentiated. [E] GKSL split (imaginary + $\text{real} \leq 0$ spectrum). [O] geometric modular flow on the full seam algebra (Connes–Rovelli thermal time = QGEO.SYM.01, v198) — a target, not a closure.                                                         |

*continued on next page*

| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v239_kms_thermal_time.py        | DYN.KMS.01 — KMS / thermal time: the modular flow of v238 makes the seam state a KMS (thermal) state at $\beta = 1$ , and the geometric normalisation fixes the seam temperature via $2\pi = 1/(4c_3)$ . [E] KMS at $\beta = 1$ : $\omega(A\sigma_i(B)) = \omega(BA)$ ( $\omega(X) = \text{Tr}(\rho X)$ , $\rho = e^{-\Lambda_\Sigma}/Z$ ), so modular time (v238) is a thermal time (Connes–Rovelli); NEG $\beta \neq 1$ fails (Tomita–Takesaki pins $\beta = 1$ ). [E] the seam unit $\beta_{\text{phys}} = 2\pi = 1/(4c_3)$ (v58) makes the KMS temperature the horizon temperature $T_H = c_3/M$ (v57/v208). [E] one object with v238. [O] the geometric-boost identification = QGEO.SYM.01 — a target, not a closure.                                                                                                                                                                                                                                                                                                |
| v240_gns_os_reconstruction.py   | DYN.GNS.01 — GNS / OS reconstruction (finite toy): the Hilbert space and a POSITIVE Hamiltonian are OUTPUTS of the seam data $(M, \omega)$ , not inputs. [E] GNS: $\langle A, B \rangle = \text{Tr}(\rho A^\dagger B)$ Hermitian positive-definite ( $\rho > 0$ ), $\dim n^2$ , identity cyclic+separating. [E] modular flow $\sigma_t(x) = \rho^{it} x \rho^{-it}$ is a state-preserving *-automorphism, generator $i[\log \rho, \cdot] = \Lambda_\Sigma$ (v238). [E] OS Hamiltonian $H_{\text{OS}} = -\log T$ (the v221 transfer) is POSITIVE — eigenvalues $\{0, 6\log(3/2), 6\log 3\} \geq 0$ , gap = $6\log(3/2)$ = the v64 mass gap. [E] $H_{\text{OS}} = -L$ : minus the v238 dissipative generator (recovery semigroup = Euclidean time). [O] the geometric/operator-level step = QGEO.SYM.01/BW + the standard OS theorem under RP/gap — a target, not a closure.                                                                                                                                                |
| v241_dhr_sectors.py             | DHR.SECTORS.01 — particles = DHR superselection sectors: the carrier discriminant/anyon content and the v235/v237 condensation tower as an abelian MTC. Carrier form (Lean-verified, FORM.GLUE.01): on $\mathbb{Z}_4 \times \mathbb{Z}_4$ , $q(x, y) = (5x^2 + 3y^2)/8 \bmod 1$ , $\theta(a) = e^{2\pi i q(a)}$ . [E] 16 sectors = $\mathbb{Z}_4 \times \mathbb{Z}_4$ ( $= \det K_{\text{carrier}} = \dim S^+$ ), fusion = abelian group law. [E] condensable $\Leftrightarrow \theta = 1 \Leftrightarrow \text{isotropic} \Leftrightarrow 5x^2 + 3y^2 \equiv 0 \bmod 8$ — the six elements $\{(0, 0), (1, 1), (1, 3), (2, 2), (3, 1), (3, 3)\}$ . [E] tower = v235/v237: Lagrangian $\langle (1, 1) \rangle$ (order 4, self-dual) $\rightarrow 1$ sector = $(E_8)_1$ ; order-2 $\langle (2, 2) \rangle \rightarrow 4 = D_8$ ; carrier $16 \rightarrow D_8 \rightarrow E_8$ . [E] NEG: a non-isotropic anyon $(1, 0)$ is confined. [C] sector $\leftrightarrow$ SM-multiplet map is the interpretation; the MTC is exact. |
| v242_spin_statistics_modular.py | DHR.MODULAR.01 — spin-statistics and the modular $(S, T)$ data of the carrier MTC: statistics + the fusion ring recovered from modular data + the anyon central charge $c = 8$ . Same MTC as v241 ( $\mathbb{Z}_4 \times \mathbb{Z}_4$ , $q = (5x^2 + 3y^2)/8$ , $\theta = e^{2\pi i q}$ ). [E] spin-statistics: $T = \text{diag}(\theta_a) \rightarrow 6$ bosons ( $\theta = 1$ ), 2 fermions ( $\theta = -1$ ) = $\{(0, 2), (2, 0)\}$ , 8 anyons. [E] modular $S_{ab} = (1/\sqrt{16})e^{-2\pi i B(a,b)}$ unitary, symmetric, $S^2 = C$ (charge conjugation), $S^4 = I$ . [E] Verlinde $N_{ab}^c = \sum_x S_{ax} S_{bx} S_{cx}^* / S_{0x} = \delta(c, a+b)$ — fusion recovered from modular data. [E] Gauss–Milgram $(1/\sqrt{16}) \sum \theta_a = e^{2\pi i c/8}$ with $c = 8 = g_{\text{car}} + N_{\text{fam}}$ = the $(E_8)_1$ central charge (v156/v175). [C] the boson/fermion split is the kinematic skeleton; the sector $\leftrightarrow$ multiplet map is the interpretation.                                   |
| v243_haag_ruelle_braiding.py    | DHR.SMATRIX.01 — scattering: the Haag–Ruelle skeleton + the braiding S-matrix. The seam net has a mass gap (v64/v238) so Haag–Ruelle constructs asymptotic states; for the abelian anyon MTC (v241/v242) the 2-particle S-matrix is the monodromy $M(a, b) = \theta(a+b)/(\theta(a)\theta(b)) = e^{2\pi i B(a,b)}$ . [E] mass gap $\Delta = 6\log(3/2) > 0 \rightarrow$ HR in/out states. [E] 2-particle S = monodromy: $ M  = 1$ , diagonal in conserved labels $\rightarrow$ no particle production, factorised (integrable) scattering. [E] Yang–Baxter: $M$ is a bicharacter $\rightarrow$ multi-particle factorisation. [E] crossing+symmetry: $M(a, b) = M(b, a)$ , $M(a, -b) = M(a, b)^*$ . [E] holomorphic point: on the condensed $(E_8)_1$ (single sector) $M = 1$ , trivial braiding ( $\det 1/\text{SRE}$ ). [O] the 4d S-matrix / cross-sections are the open holographic uplift (Phase 3).                                                                                                                  |

*continued on next page*

| Script                              | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v244_spectral_action_contract.py    | <p>CONTRACT.QFT4D.01 — the 4d Lagrangian via Connes' spectral action: the honest contract for the Phase-3 gap (2d seam net <math>\rightarrow</math> 4d physics). A single Dirac operator whose <math>\text{Tr } f(D/\Lambda)</math> yields gravity AND matter. [E] carrier spinor = one generation: <math>SO(10)</math> half-spinor = <math>\dim S^+ = 2^{g_{\text{car}}-1} = 16</math> (<math>15 + \nu_R</math>), <math>N_{\text{fam}} = 3</math>. [E] SM gauge group inside the carrier: <math>SU(3) \times SU(2) \times U(1)</math> (<math>\dim 12</math>) <math>\subset SU(5)</math> (24) <math>\subset SO(10) = D_5</math> (45); colour <math>SU(3) \subset A_3 = SU(4)</math> (Pati–Salam); <math>\text{rank}(D_5) + \text{rank}(A_3) = 8 = \text{rank } E_8</math>. [E] one Dirac operator <math>\rightarrow</math> gravity (<math>a_2 \sim R</math>, <math>a_4 \sim R^2</math>, <math>\sqrt{36}</math>) + matter (<math>a_4 \sim F^2</math> Yang–Mills + <math> D\phi ^2</math> Higgs, Gilkey). [X] hooks: <math>g_3 = g_2 = \sqrt{5/3} g_1</math> + a Higgs-quartic relation. [O] CONTRACT.QFT4D.01: obligations (seam <math>\rightarrow</math> Dirac <math>D</math>; finite geometry = carrier; spectral action reproduces SM couplings) <math>\rightarrow</math> the 4d SM+gravity Lagrangian is an OUTPUT — the Phase-3 target, not closed.</p> |
| v245_spectral_matter_content.py     | <p>QFT4D.MATTER.01 — the computable core of CONTRACT.QFT4D.01 (v244): the spectral-action / NCG matter content, discharging the rep-theory + tree-relation core to [E]. The carrier half-spinor is the <math>SO(10)</math> <b>16</b>. [E] <b>16</b> = one generation: <math>SO(10)</math> <b>16</b> = <math>SU(5)(\mathbf{10} + \mathbf{5} + \mathbf{1})</math>; SM dims <math>6 + 3 + 1 + 2 + 3 + 1 = 16 = \dim S^+</math>, <math>N_{\text{fam}} = 3</math> (15 Weyl fermions + <math>\nu_R</math>). [E] electric charges <math>Q = T_3 + Y</math> are exactly the SM values — hypercharges forced by the embedding. [E] anomaly-free: <math>\sum Y = 0</math>, <math>\sum Y^3 = 0</math>, <math>[SU(2)]^2 U(1) = 0</math>, <math>[SU(3)]^2 U(1) = 0</math> (exact). [E] unification: <math>\sin^2 \theta_W = (\sum T_3^2) / (\sum Q^2) = 3/8</math>, i.e. <math>g_3 = g_2 = \sqrt{5/3} g_1</math>. [E] Higgs = the <math>(1, 2)_{1/2}</math> inner fluctuation (one doublet). [O] residual: the seam <math>\rightarrow</math> Dirac realisation + measured-coupling reproduction stay the open contract.</p>                                                                                                                                                                                                                                              |
| v246_unification_data_crosscheck.py | <p>QFT4D.RGTEST.01 — the honest data cross-check for the spectral-action prediction (v244/v245): does <math>g_1 = g_2 = g_3</math>, <math>\sin^2 \theta_W = 3/8</math> at <math>\Lambda</math> survive the <i>measured</i> SM couplings? Not in the plain SM. Inputs (PDG, GUT-normalised, <math>M_Z</math>): <math>\alpha_i^{-1} = (59.01, 29.59, 8.47)</math>, SM 1-loop <math>b = (41/10, -19/6, -7)</math> (v159). [E] the prediction: <math>\alpha_1 = \alpha_2 = \alpha_3</math>, <math>\sin^2 \theta_W = 3/8</math> at <math>\Lambda</math>. [X] tension: pairwise crossings spread <math>\sim 4</math> decades (<math>\sim 10^{13} - 10^{17}</math> GeV), residual spread at <math>2 \times 10^{16}</math> <math>\Delta \alpha^{-1} \sim 8.8</math> — <math>g_1 = g_2 = g_3</math> not realised by the plain SM. [X] <math>\sin^2 \theta_W</math> 3/8 vs measured 0.23122. [X] 2-loop also misses (min spread <math>\sim 3.2</math> at <math>1.7 \times 10^{14}</math> GeV). [O] no rescue: SM gauge content + no new states (the <math>E_8</math> spine is arithmetic, not a threshold tower) <math>\Rightarrow</math> no admissible thresholds; closing it needs new matter (vs 'no new state') or a non-GUT/boundary reading — tension like NCG-SM, likely falsified unless extended. A kill-test, [X]/[O], never [E]-as-success.</p>            |
| v247_e8_branching_no126.py          | <p>PS.E8BRANCH.01 — the <math>E_8</math> audit hull forbids the <b>126</b> and supplies exactly <math>\{1, 10, 16, 45\}</math> for <math>SO(10)</math>. Carrier <math>D_5 \oplus A_3 = SO(10) \times SU(4) \subset E_8</math> (via <math>SO(16)</math>); <b>248</b> = <math>(45, 1) + (1, 15) + (10, 6) + (16, 4) + (\overline{16}, \bar{4}) = 248</math>. [E] ranks <math>5 + 3 = 8</math>; [E] dims = 248 (<math>SO(16)</math>: 120 adj + 128 spinor); [E] <math>SO(10)</math> content <math>\{1, 10, 16, 45\}</math>, the <b>126</b> ABSENT <math>\Rightarrow 126_H</math> algebraically forbidden, <math>10 + 16 + 45</math> <math>E_8</math>-supplied not fitted; [E] caveat only one 45/one 15 <math>\rightarrow</math> proton-marginal; [O] the field-content principle + NCG complement (v250).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| v248_ps_algebra.py                  | <p>PS.ALGEBRA.01 — the Pati–Salam NCG algebra <math>A_F = \mathbb{H}_L \oplus \mathbb{H}_R \oplus M_4(\mathbb{C})</math> and its exact SM matching. [E] unimodular group <math>SU(2)_L \times SU(2)_R \times SU(4)_c</math> (rank 5, <math>\dim 21</math>); [E] <b>16</b> = <math>(4, 2, 1) + (\bar{4}, 1, 2) = 8 + 8</math>; [E] <math>Y = T_{3R} + (B - L)/2</math> gives the exact SM hypercharges and <math>Q = T_{3L} + Y</math> the exact charges; [E] matching <math>\alpha_1^{-1} = \frac{3}{5} \alpha_{2R}^{-1} + \frac{2}{5} \alpha_4^{-1}</math>, <math>\sin^2 \theta_W = 3/8</math>. [O] <math>A_F</math> uniquely from seam+sheet+<math>\mu_4</math>; [O] caveat <math>SU(4)_c \subset SO(10) = D_5</math> is not the carrier <math>A_3 = SU(4)</math> (family).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| v249_ps_unification.py              | <p>PS.RGTEST.01 — carrier-native Pati–Salam/<math>SO(10)</math> two-step unification kill-test + negative controls (suite check of experiments/gauge-unification). [E] matching + a valid 1-loop unification solution; [X] NEG SM-only fails (spread <math>\sim 8.8</math> @ <math>2 \times 10^{16}</math>); [C] <math>M_{PS}/M_s = 1.0 - 1.7</math> (1-loop; 2-loop <math>\sim 1.0 - 1.2</math>) — gauge unification meets the gravitational scalaron scale; [X] NEG 126 breaks the match (<math>b_{2R} &gt; 0</math>; also <math>E_8</math>-forbidden v247); [X] NEG proton decay selects content (minimal <math>M_{GUT} \sim 2 \times 10^{15}</math> excluded; <math>+(15, 1, 1)</math> safer). [O] the gauging fork, exact <math>\kappa</math>, the one-45 marginality.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

continued on next page

| Script                            | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v250_dirac_triple_contract.py     | CONTRACT.QFT4D.DIRAC.01 — the spectral-triple contract that would make ‘carrier gauged’ a theorem. Target $(A, H, D, J, \gamma)$ almost-commutative, $A_F = \text{PS}$ (v248), $H_F = 3 \times 16$ , $D_F$ from $(R, L, \varphi_0, \text{Majorana})$ . [E] $H_F = 3 \times 16 = 48$ ; [E] the Yukawas are already fixed (v18) so the open test is NCG consistency; [O] the 7 NCG axioms (first-order condition = the historic $SO(10)$ obstruction); [O] $\sigma = \text{scalaron hypothesis}$ (rescues first-order + $\nu_R$ Majorana; $M_{PS} = \kappa M_s$ ) — caveat $\sigma \leftrightarrow \nu_R \nu_R$ not automatic; [O] inner fluctuations $\rightarrow 10+16+45$ not 126. Reduces 4d QFT to one object + finite NCG checks.                                                                                                                                                                 |
| v251_first_order_sigma.py         | PS.DIRAC.01 — the first-order condition is violated <i>exactly</i> by the Majorana/ $\sigma$ term (Chamseddine–Connes–van Suijlekom ‘beyond first order’ $\rightarrow$ Pati–Salam + $\sigma$ ), exhibited on a faithful lepton-block model; advances v250 from contract to mechanism. Honest correction: $\sigma$ does NOT ‘rescue’ first order — one DROPS it and the inner fluctuations generate $\text{PS} + \sigma$ . [E] order-zero; [E] real-even ( $J^2=+1$ , $\gamma^2=1$ , $D\gamma=-\gamma D$ , KO-6 signs); [E] first order HOLDS for the Yukawa block; [E] first order VIOLATED by $M \neq 0$ , supported exactly on the $\nu_R \nu_R^c$ blocks ( $\sigma$ the unique violator). [C] $\sigma = \text{scalaron}$ (v249). [O] the full 96-dim SM/PS triple + spectral action (v250).                                                                                                        |
| v252_full_finite_triple.py        | PS.DIRAC.02 — the full 96-dim finite spectral triple $(A_F, H_F, D_F, J, \gamma)$ built explicitly with the NCG axioms verified; advances v250 from a 7-obligation contract to a concrete object discharging 5 to [E]. $H_F = \text{particle} \oplus \text{antiparticle}$ doubling of three $SO(10)$ <b>16</b> -plets (dim 96). [E] dim 96; $D_F$ Hermitian+odd; reality $J^2=+1$ , $JD=DJ$ ; grading $\gamma^2=1$ , $[\gamma, a]=0$ , $J\gamma=-\gamma J$ ; KO-dim 6; order-zero; first order HOLDS for the SM $\mathbb{C} \oplus \mathbb{H} \oplus M_3$ (incl. Majorana) and VIOLATED for Pati–Salam exactly by the Majorana on $\{\nu_R, e_R\}$ (CCvS beyond-first-order $\rightarrow \text{PS} + \sigma$ ); spectrum = fermion masses + seesaw $m_{\text{light}} = m_D^2/M_R$ . [C] Poincaré duality (intersection form non-degenerate). [O] orientability + no-junk + spectral-action expansion. |
| v253_kappa_scalaron_ps.py         | PS.KAPPA.01 — pinning $\kappa$ in $M_{PS} = \kappa M_s$ , discharging residual 6(b) of v249. $M_s = c_3^{7/2} \bar{M} \approx 3.06 \times 10^{13} \text{ GeV}$ (exact). [E] scalaron scale exact; [E] $\kappa$ pinned by RG (scan PDG errors $\times$ the two $E_8$ -allowed contents) to a tight $O(1)$ interval $\sim 1.3$ (1-loop $[1.0, 1.7]$ , 2-loop $[1.0, 1.2]$ ) — pinned, not free. [C] heat-kernel origin $\kappa = \sqrt{(f_2/f_0)} c_{PS}/c_{\text{grav}}$ , a ratio of heat-kernel traces over the 96-dim $H_F$ (v252) — $\Lambda$ cancels, $\kappa$ scale-free and $O(1)$ . [X] the <b>126</b> -content drives $\kappa$ out of the $O(1)$ window (content-selective). [O] the exact decimal needs the cutoff moments $f_2/f_0$ fixed.                                                                                                                                                  |
| v254_inner_fluctuations.py        | PS.NCG.FLUCT.01 — the inner fluctuations $\Omega_D^1(A_F)$ of the finite Dirac operator computed explicitly (reuses the v252 triple); closes v250 #5 (Higgs content, not <b>126</b> ) + the no-junk obligation + the $\sigma$ quantum numbers. [E] $\Omega_{D_F}^1$ purely chirality-odd $\rightarrow$ finite fluctuations are scalars; [E] no junk / no <b>126</b> : SM-algebra 1-forms sit exactly in the Higgs-doublet pattern (the single $(1, 2)_{1/2}$ ); [E] the $\sigma$ ( $\nu_R$ Majorana) direction is absent for the SM algebra but present for Pati–Salam — $\sigma$ is the CCvS beyond-first-order scalar; [E] $\sigma$ is an SM singlet $(1, 1, 0)$ with $B-L=2$ ; [C] $\sigma = \text{scalaron consistency}$ .                                                                                                                                                                        |
| v255_spectral_action_expansion.py | PS.SPECACT.01 — the spectral-action heat-kernel (Seeley–DeWitt) expansion $S = 2f_4\Lambda^4 a_0 + 2f_2\Lambda^2 a_2 + f_0 a_4$ over the finite triple; closes v252 #11 (spectral-action expansion) and makes $\kappa(f_2/f_0)$ explicit. [E] expansion structure ( $a_0$ cosmological, $a_2$ Einstein–Hilbert+Higgs-mass, $a_4$ Yang–Mills+Weyl <sup>2</sup> + $R^2$ +Higgs-quartic, $N=96$ ); [E] trace inputs $b/a^2 = 1/3$ (top-dominated, Chamseddine–Connes); [E] $\sin^2 \theta_W = 3/8$ (cross-check v245); [C] Higgs quartic $\lambda(\Lambda) \sim g^2 b/a^2 \rightarrow \sim 125 \text{ GeV}$ with $\sigma$ ; [E] scalaron = $R^2$ spin-0 mode (gravitational $a_4 \sim f_0 N$ ); [C] $\kappa = \sqrt{(f_2/f_0)} c_{PS}/c_{\text{grav}}$ explicit; [O] exact decimal needs the cutoff moments fixed.                                                                                       |

*continued on next page*

| Script                                     | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v256_orientability_<br>poincare.py         | PS.NCG.ORIENT.01 — orientability and Poincaré duality treated honestly (no faked proof). KO-dim 6 forces the intersection form $\text{cap}(p, q) = \text{Tr}(\gamma \pi(p) J \pi(q) J^{-1})$ to be antisymmetric. [E] skew form forced by $J\gamma = -\gamma J + \text{order-zero} + \text{reality}$ (verified); [E] on the SM K-theory $\mathbb{Z}^3$ the skew form has rank 2, corank 1, null $\sim (1, 1, -1)$ — the structural obstruction; [E] orientability local condition $[\gamma_F, \pi(a)] = 0$ , full orientability is a degree-4 cycle on $M \times F$ ; [C] the canonical hypercharge-faithful SM/PS triple satisfies both (van Suijlekom; CCM) — literature; [O] a self-contained machine proof needs the complete canonical bimodule.                                                                                                                                                                                                                                                                                                                                                                                            |
| v257_quasiorient_<br>modpd.py              | PS.NCG.ORIENT.02 — the honest, literature-confirmed resolution of orientability + Poincaré duality, correcting the too-generous v256 typing. Published result (Cacic–Stephan arXiv:0902.2068; CCM07; SM-vacuum hep-th/0601192): the CCM SM finite triple (KO-6, $3\nu_R$ ) genuinely FAILS strict orientability and strict Poincaré duality, satisfying the weakened versions. [E] strict orientability fails ( $\gamma \notin \text{span}\{\lambda(a)\rho(b)\}$ ); [E] quasi-orientability holds (left/right $A$ -irrep boxes disjoint by chirality: $L = H$ doublet, $R = \mathbb{C}$ singlet $\Rightarrow L^{\text{LR}}(\text{even, odd}) = \{0\}$ ); [E] strict Poincaré fails (KO-6 skew form, corank 1, the equal- $L/R$ -neutrino case); [C] modified Poincaré duality (CCM) / Stephan’s PD-restoring triple. A known model-class feature, not a TFPT defect — no faked closure.                                                                                                                                                                                                                                                          |
| v258_dirac_<br>covariance_<br>induction.py | PS.DIRAC.03 — $D_F$ is the modular/covariance <i>induction</i> of the boundary seam state, not an independent posit; closes the ‘posited Yukawa vs derived operator’ gap of v250/v252 (Q2: Dirac as Kasparov product + covariance formula). Quasi-free/modular dictionary: a Gaussian state is fixed by its covariance $C$ (contraction, $\text{spec} \subset (0, 1)$ ) with one-particle modular Hamiltonian $H = \log((1-C)C^{-1})$ (Araki; Casini–Huerta), inverse to the KMS occupation $C = (1+e^H)^{-1}$ ( $\beta=1$ , v239). [E] inversion identity $\log((1-C)C^{-1}) = H$ exactly (random Hermitian + a chirality-graded Yukawa block); [E] $C_F = (1+e^{D/\mu})^{-1}$ a positive contraction; [E] chirality-odd ( $D_F^{\text{ind}}\gamma = -\gamma D_F^{\text{ind}}$ ); [E] unitarily equivalent to v252 (inducing $C_F$ from the 9 Yukawas returns the same spectrum); [E] Majorana = off-diagonal covariance (seesaw $m_{\text{light}} = m_D^2/M_R$ ); [C] $C_F = P_F C_\Sigma P_F$ , so $D_F$ is the KK shadow $[D_\Sigma] \hat{\otimes}_{(E_8)_1} [\mathcal{K}_{\text{car}}]$ and the v18 Yukawas are the readout of $C_\Sigma$ . |
| v259_modular_<br>cutoff_kappa.py           | PS.SPECACT.02 — the spectral-action cutoff <i>is</i> the seam KMS weight, fixing $f_2/f_0=1$ exactly and reducing the last open scheme freedom (v253 #5, v255 #7) from a free function to a finite-triple trace ratio (Q4: the cutoff is modular, not chosen). By Tomita–Takesaki/BW the seam state is KMS at $\beta=1$ (v239) and $2\pi=1/(4c_3)$ (v58) fixes $\beta$ , so the heat-kernel cutoff $f(u)=e^{-u}$ is <i>derived</i> . [E] KMS cutoff moments $f_0=f(0)=1$ , $f_2=\int f=1$ , $f_4=\int u f=1$ , so $f_2/f_0=f_4/f_2=1$ ; [E] seam-fixed $\beta=1$ (no new parameter beyond $c_3$ ); [E] non-vacuous neg control: a Gaussian cutoff $e^{-u^2}$ gives $f_2/f_0=\sqrt{\pi}/2 \neq 1$ ; [E] $\kappa=\sqrt{(f_2/f_0)c_{PS}/c_{\text{grav}}}=\sqrt{c_{PS}/c_{\text{grav}}}$ , no cutoff function left; [C] the RG $\kappa\sim 1.3$ pins $c_{PS}/c_{\text{grav}}\sim 1.7$ ( $O(1)$ ); the [O] free scheme becomes a [C] finite computation over the built triple; [C] the same logic fixes $\lambda(\Lambda)$ .                                                                                                                          |

*continued on next page*

| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v260_k3_kummer_unification.py    | <p>ARCH.K3.01 — one Kummer/K3 surface carries the <i>seam</i>, the <i>carrier-16</i> and the <math>E_8</math> lattice as facets of one smooth object, the geometric unification of the v1 ‘glue <math>D_5 \oplus A_3 + \mu_4 \Rightarrow E_8</math>’ step (the K3/Kummer proposal). The seam square torus <math>\tau=i</math> (<math>j=1728</math>, v214) under the elliptic involution <math>z \mapsto -z</math> squares to the Kummer K3. [E] the <math>E_8</math> Cartan is even, <math>\det=1</math>, positive-definite (the unique even unimodular pos-def rank-8 lattice); [E] the K3 lattice <math>L_{K3}=U^3 \oplus E_8(-1)^2</math> has rank <math>22=b_2</math>, <math>\det=-1</math> (unimodular), is even, signature <math>(3, 19)</math>, with <math>E_8(-1)^2</math> an orthogonal summand; [E] seam = <math>T^2/(z \mapsto -z)</math>: <math>4= (\mathbb{Z}/2)^2 </math> fixed points, pillowcase <math>S^2(2, 2, 2, 2)</math>, orbifold Euler char 0 (the v214/v216 four marks); [E] carrier-16 = Kummer nodes <math> A[2] =2^4=16 \rightarrow 16</math> exceptional <math>\mathbb{P}^1 = \dim S^+ = \Lambda^{\text{even}}(\mathbb{C}^5) = 16</math> (v197), <math>16=4 \times 4</math>; [E] honesty: the 16 node classes span the rank-16 Kummer lattice, <i>not</i> literally <math>E_8(-1)^2</math> (distinct sublattices of <math>H^2(K3)</math>); [C] two faces of <math>E_8</math> in K3 (local du Val <math>\mathbb{C}^2/2I</math>, v232/v236; global <math>H^2</math> summand); [C] unification (lattice [E], carrier<math>\leftrightarrow</math>node identification [C]).</p> |
| v261_modular_spectral_closure.py | <p>QFT.MSC.01 — the <i>Modular Spectral Closure</i>: the assembly/reduction certificate turning the whole QFT layer (v238–v260) into <i>one</i> relative object<br/> TFPT<sub>QFT</sub> = <math>(\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})</math> reduced to <i>one</i> open premise (like v176/v234, an assembly certificate, not new physics). [E] one index = marks = fixed points = glue order: Jones <math>[B:A]=4</math> (v89) = <math> \mu_4 =4</math> (v92) = seam marks <math>2\chi=4</math> (v216) = pillowcase/Kummer fixed points <math> (\mathbb{Z}/2)^2 =4</math> (v260); [E] one carrier-16: <math>\dim S^+ = \Lambda^{\text{even}}(\mathbb{C}^5) = \text{Kummer nodes } 2^4=16</math>, <math>H_F=2N_{\text{fam}} \cdot 16=96</math>; [E] one gap <math>\Delta = -\log((2/3)^6) = 6 \log(3/2) &gt; 0</math> = the OS mass gap (v240) = the recovery rate (v221) = the Lindblad gap (v238); [E] the three closures re-derived (Kasparov <math>\log((1-C)C^{-1})=H</math>, v258; KMS cutoff <math>f_2/f_0=1</math>, v259; K3 lattice signature <math>(3, 19)/\det -1/\text{even}</math>, v260); [E] relative-object manifest (10/10 component scripts present); [O] the single residual: the whole QFT layer reduces to QGEO.SYM.01 (seam realisation), the ambient QG measure QG.AMB.01 kept separate; [C] the master statement (Modular Spectral Universality) holds modulo QGEO.SYM.01 — the honest sense in which the QFT layer is complete.</p>                                                                              |
| v262_fqcd_mp_me.py               | <p>FR.MPME.02 — the <math>F_{\text{QCD}}</math> transfer pipeline for <math>m_p/m_e</math>, implemented as a real solver (the one <math>F_{\text{transfer}}</math> pipeline that was contract-only). <i>Not</i> a compiler power — a [C] standard-physics transfer reproduction. [E] <math>b_3 = -7 = -(11-2n_f/3)</math> at <math>n_f=6</math> (the only TFPT number in the run); [E] a two-loop <math>\alpha_s(M_Z)=0.1179</math> run with thresholds gives <math>\alpha_s(2 \text{ GeV}) \sim 0.30</math> (real ODE solve); [C] <math>\Lambda_{\text{MS}}^{(3)} \sim 0.33 \text{ GeV} \times</math> the lattice bridge <math>C_p = m_p/\Lambda</math> (external <math>O(1)</math>) over the closed <math>m_e</math> gives <math>m_p/m_e \sim 1824</math> vs the measured 1836.15 within the <math>\sim 7\%</math> external budget; [X] firewall: <math>m_p/m_e</math> is not an admissible compiler integer (<math>SU(9)</math> absent from <math>D_5 \oplus A_3</math>). Advances FR.MPME.01 ([O]) to FR.MPME.02 ([C]).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| v263_mnu_from_df.py              | <p>FLAV.PMNS.02 — the light neutrino matrix <math>M_\nu</math> as the type-I seesaw of <math>D_F</math>, advancing the solar-angle texture (v9) from a posited matrix to a seesaw <i>object</i> built from <math>D_F</math>’s blocks. [E] <math>M_\nu = -m_D M_R^{-1} m_D^\top</math> symmetric + seesaw-suppressed (the light masses an output of <math>D_F</math>); [E] a <math>\mu\tau</math>-symmetric <math>(m_D, M_R)</math> gives a <math>\mu\tau</math>-symmetric <math>M_\nu</math> (the v9 texture is in the seesaw image); [C] diagonalising gives <math>\theta_{23}=45^\circ</math> and <math>\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.30675</math> under the seam misalignment; [O] <math>\theta_{13}=0</math> in the leading texture (physical <math>\sin^2 \theta_{13} \sim \varphi_0^{\text{ret}} e^{-5/6}</math> a separate readout), the CP magnitude <math>\delta_{\text{PMNS}}=240^\circ</math> is the <math>\mu_6/\text{triality}</math> phase (v231/v233), and <math>M_R</math> is the free seesaw scale. An operator advance, not a closure of full PMNS dynamics.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

*continued on next page*

| Script                                  | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v264_raw_seam_<br>marklocal.py          | QGEO.ENERGY.03 — the raw-seam $\Rightarrow$ mark-locality <i>reduction</i> (not a closure of QGEO.SYM.01). Composing v216+v201+v198 on the canonical pillowcase: [E] Gauss–Bonnet 4 cone points (deficit $\pi$ each, sum $4\pi=2\pi\chi(S^2)$ ), curvature <i>only</i> at the 4 marks; [E] the curvature sourced at the $\mu_4$ orbit $\{0, \pi/2, \pi, 3\pi/2\}$ has Fourier support <i>only</i> on $4\mathbb{Z}$ (FFT), so mark-locality is automatic for the pillowcase; [E] $M_f$ then commutes with the clock $\rho=\text{diag}(i^n)$ ; [E] with $ k $ already commuting (v198), $[\rho, \Lambda]=0 \Rightarrow \omega \circ \rho = \omega$ on this metric; [O] the residual <i>is</i> the metric — the chain assumes the raw seam carries the uniformising flat pillowcase metric (=QGEO.SYM.01), the negative control (curvature off the $\mu_4$ orbit) breaks it. A reduction/sharpening, not a closure.                                                                                                                                                                                                                                                                                                                                                         |
| v265_qft4d_fork_<br>freeze.py           | QFT4D.FORK.01 — freeze the 4D-QFT fork as a <i>branch policy</i> (not an open ambiguity); writes <code>qft4d_fork_freeze.json</code> + a prose guard, no new physics. [C] A: boundary-only is the canonical 4D reading (no 4D-GUT claim by default); [X] SM-only 4D-GUT unification is killed (1-loop spread $\sim 8.8$ at $2 \times 10^{16}$ , v246); [C]/[O] B: carrier-native Pati–Salam is a falsifiable UV branch ( $E_8$ content $\{1, 10, 16, 45\}$ no <b>126</b> v247, $\sin^2 \theta_W = 3/8$ v248, $\kappa O(1)$ band v249/v253, threshold + proton kill test); [X] proton safety rule (no fake $\tau_p$ window); [E] wording guard (bare 4D over-claims absent, scoped policy phrases present). A frozen decision tree, not an open ambiguity.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| v266_ps_threshold_<br>proton.py         | PS.PROTON.02 — the explicit Pati–Salam threshold model + proton-decay window that DECIDES branch B of the v265 fork; writes <code>ps_threshold_proton.json</code> . Two-step PS solve + $d=6$ $\tau_p(p \rightarrow e^+ \pi^0)$ : the carrier-native PS UV branch survives proton decay <i>only</i> with the $E_8$ -allowed $(15, 1, 1)$ [the single 45]. [E] explicit thresholds: minimal $16H \rightarrow M_{GUT} \sim 2.4 \times 10^{15}$ , $+(15, 1, 1) \rightarrow M_{GUT} \sim 5.9 \times 10^{15}$ ; [X] minimal KILLED ( $\tau_p \sim 4.4 \times 10^{33}$ yr < Super-K $2.4 \times 10^{34}$ , even high-band); [C] $+(15, 1, 1)$ survives now ( $\tau_p \sim 1.6 \times 10^{35}$ yr); [X] <i>Hyper-K decisive</i> ( $\tau_p$ at the $\sim 1.4 \times 10^{35}$ yr reach) — a dated kill test within the decade; [O] marginal (only one 45 in the hull, v247) + $O(3)$ hadronic uncertainty. A (boundary-only) stays canonical.                                                                                                                                                                                                                                                                                                                                     |
| v267_qgeo_rigidity_<br>minimal_axiom.py | QGEO.SYM.02 — the rigidity / minimal-axiom form of the bedrock QGEO.SYM.01: the ONE bare statement (the raw RP seam admits the carrier $\mu_4$ clock as a conformal symmetry permuting its 4 marks, $= \omega \circ \rho = \omega$ ) <i>forces</i> everything downstream; it does <i>not</i> close the axiom. [E] an order-4 Möbius orbit $\{a, ia, -a, -ia\}$ has cross-ratio 2 (independent of $a$ ); [E] cross-ratio $2 \Leftrightarrow j=1728$ (only $\lambda \in \{-1, \frac{1}{2}, 2\}$ ) $\Leftrightarrow$ the order-4 CM automorphism $(x, y) \mapsto (-x, iy)$ on $y^2=x^3-x$ ; [E] downstream forced: square $\Rightarrow$ the unique flat pillowcase (Trojanov, v214) $\Rightarrow$ curvature at the 4 marks $\Rightarrow \mathbb{Z}_4$ -Fourier DtN (v201/v264) $\Rightarrow [\rho, \Lambda]=0 \Rightarrow \omega \circ \rho = \omega$ (v198); [E] neg controls (generic $j \neq 1728$ , only $\mathbb{Z}_2$ ; hexagonal $j=0$ has $\mathbb{Z}_6$ , order $6 \neq 4$ ); [C] arithmetic rigidity ( $\{1, i, -1, -i\}$ over $\mathbb{Q}$ , $j=1728$ CM by $\mathbb{Z}[i]$ , Belyi/dessins); [O] the bare order-4 symmetry stays the one foundational postulate (like $c=\text{const}$ ). Closes the reduction (axiom $\Rightarrow$ everything), not the axiom. |
| v268_theta13_<br>carrier_trace.py       | FLAV.TH13.01 — the reactor angle $\theta_{13}$ <i>exponent</i> is the carrier hypercharge trace. The value $\sin^2 \theta_{13} = \varphi_0^{\text{ret}} e^{-5/6} = 0.0231$ was already checked (v16); here the exponent origin is closed exactly: [E] $\gamma = \text{tr}_E Y^2 = 3(1/3)^2 + 2(1/2)^2 = 5/6$ over the 5-slot carrier hypercharge $Y = \text{diag}(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2})$ (roots of $6Y^2 - Y - 1 = 0$ ), complement $1/6$ the neutrino ratio; [C] value 0.02311 vs PDG $\sim 0.0222$ ; [E] own channel — the $\mu\tau$ -breaking $\varepsilon = 3\varphi_0^{\text{ret}}/4$ induces only $\sim 1.6 \times 10^{-3} \ll 0.023$ , so $\theta_{13}$ is a separate carrier-trace channel, not the seesaw $\mu\tau$ breaking (corrects the tempting ‘ $\theta_{13}$ from $M_\nu$ ’ route); [O] CP magnitude + absolute $\nu$ scale remain open.                                                                                                                                                                                                                                                                                                                                                                   |

*continued on next page*

| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v269_spert_paqft_skeleton.py   | QFT4D.SPERT.01 — the perturbative 4D S-matrix ( $S_{\text{pert}}$ ) as an Epstein–Glaser / pAQFT skeleton: the honest middle layer of the three-layer S-matrix ( $S_{\text{top}}$ 2d DHR braiding v243 / $S_{\text{pert}}$ 4d EG of the spectral action / $S_{\text{phys}}$ LSZ on the OS Wightman functions v240), <i>not</i> a nonperturbative closure. [E] three distinct layers (the 2d braiding $ M =1$ is statistics, not 4d cross-sections); [E] every spectral-action $a_4$ operator has mass dimension $\leq 4$ in 4D (power-counting renormalizable); [E] a finite counterterm basis (7 marginal +3 relevant, no infinite tower); [C] Epstein–Glaser existence — a renormalizable interaction has an order-by-order $S(g)=T \exp(i \int g L_{\text{int}})$ by causal factorisation + finite local extension (no path integral, no UV divergences); [C] adiabatic limit/gap ( $\Delta=6 \log(3/2)>0$ ) $\rightarrow S_{\text{phys}}$ via LSZ; [O] perturbative only — the ambient measure QG.AMB.01 stays open. A contract/skeleton, explicitly perturbative.                      |
| v270_pmns_jarlskog_assembly.py | FLAV.PMNS.03 — the complete complex PMNS matrix + the leptonic Jarlskog $J_{\text{PMNS}}$ (the CP <i>strength</i> ), assembled from the four TFPT channels ( $\sin^2 \theta_{12}=\frac{1}{3}-\frac{\phi_0}{2}$ , $\sin^2 \theta_{23}=\frac{1}{2}$ , $\sin^2 \theta_{13}=\phi_0 e^{-5/6}$ , $\delta=4\pi/3$ ) into one unitary object. [E] the three angle channels; [E] $U = R_{23}U_{13}(\delta)R_{12}$ is exactly unitary; [C] $J_{\text{PMNS}} = \Im(U_{e1}U_{\mu 2}U_{e2}^*U_{\mu 1}^*) = -0.02965$ (formula and full matrix agree), $J_{\text{max}} = 0.03424$ — a <i>derived</i> CP strength, the leptonic analogue of the CKM $J$ (v88); [C] data: $J_{\text{max}}$ vs NuFIT 0.0332 ( $\sim 3\%$ ), $J$ vs best fit $-0.026$ , the negative sign reproduced; [O] $\delta$ is the $\mu_6$ /triality phase (assembled input, <i>not</i> from $M_\nu/D_F$ ) and the absolute $\nu$ -mass scale ( $\Sigma m_\nu$ , $M_R$ ) stays open. Assembles the PMNS matrix + CP strength; does not derive $\delta$ from $D_F$ nor fix the scale.                                                   |
| v271_eg_oneloop_quartic.py     | QFT4D.SPERT.02 — a concrete Epstein–Glaser order-2 (one-loop) quartic renormalization, taking the v269 $S_{\text{pert}}$ skeleton from <i>exists</i> to an actual EG number (no path integral). [E] $T_1=L_{\text{int}}$ : needs no extension, the first extension is $T_2$ ; [E] the massless propagator has scaling degree $\text{sd}=2$ in $d=4$ , the one-loop bubble (two propagators) has $\text{sd}=4=d$ so the UV singular order $\omega=\text{sd}-d=0$ ; [E] $\omega=0 \Rightarrow$ exactly one logarithmic local counterterm ( $C(\omega+d,d)=1$ , renormalizable, no infinite tower); [E] the loop factor $\Omega_3/(2(2\pi)^4)=1/(16\pi^2)$ exactly — the same factor that normalises the spectral-action $a_4$ geometric quartic; [C] the $\phi^4$ one-loop $\beta_\lambda=3\lambda^2/(16\pi^2)$ (symmetry factor 3 = the $s, t, u$ channels), EG initial condition $\lambda_\Phi(\text{UV})=1/(16\pi^2)$ ; [O] order-2/one-loop only — the all-order EG construction and the nonperturbative measure QG.AMB.01 stay open. Turns the skeleton into a concrete one-loop number. |
| v272_nu_mass_scale.py          | FLAV.NUSCALE.01 — the absolute neutrino-mass scale, typed honestly (it does <i>not</i> fabricate a $\Sigma m_\nu$ number, which would contradict the v184 firewall). [E] the absolute scale reduces to <i>one</i> seesaw ratio $m_3 = (y_\nu v_{\text{EW}})^2/M_R$ — one irreducible UV input like $v_{\text{geo}}$ (v153), while the ratios ( $m_2/m_3$ , ordering, PMNS angles v270) are pinned; [C] the observed $m_3=0.050$ eV with the TFPT PS $\rightarrow$ GUT window $M_R \in [4.2 \times 10^{13}, 2.4 \times 10^{15}]$ GeV (v249) needs a third-generation Dirac Yukawa $y_\nu \in [0.26, 2.0]$ — $O(1)$ , no tuning; [O] $M_R(y=1)=6.1 \times 10^{14}$ has $\log_{c_3}(M_R/M_{\text{Pl}})=2.57$ (non-integer), not a TFPT power, so the absolute value stays open; [C] the normal-ordering floor $\Sigma m_\nu=0.0586$ eV is consistent with cosmology ( $< 0.072$ – $0.12$ eV); [X] a cosmological $\Sigma < 0.0586$ eV excludes the normal-ordered $m_1 \sim 0$ spectrum. Closes the PMNS bookkeeping (the last [O] is one seesaw ratio), not the absolute number.              |
| v273_eg_gauge_running.py       | QFT4D.SPERT.03 — the gauge sector of the Epstein–Glaser one-loop (v271's quartic mechanism applied to the gauge self-energy). [E] the one-loop gauge 2-point has scaling degree $\text{sd}=4=d$ so $\omega=0 \rightarrow$ one counterterm per coupling (same marginal mechanism + loop factor $1/(16\pi^2)$ as v271); [E] the one-loop $b_i$ from the explicit carrier/SM content: $b_3=-7$ , $b_2=-19/6$ , $b_1=41/10$ (the same 41 as the carrier algebra, v159/CAR.SM.01); [C] physical RG directions $b_3<0$ (asymptotic freedom), $b_1>0$ (U(1) Landau); [O] the two-loop matrix (v159, PyR@TE) + all-order stay open. Shows the gauge running is an EG order-2 result. Exact $b_i$ mirrored in Wolfram.                                                                                                                                                                                                                                                                                                                                                                               |

*continued on next page*

| Script                              | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v274_scale_<br>overdetermination.py | SCALE.OVERDET.01 — the single mass anchor is <i>over-determined</i> , and the absence of a derivable absolute scale is a <i>theorem</i> . [E] one anchor (No-Unit, v153): every dimensionful output = pure number $\times M_{\text{Pl}}^d$ ( $\rho_\Lambda/M_{\text{Pl}}^4$ , $M_{\text{scal}}/M_{\text{Pl}}=c_3^{7/2}$ , the spectrum); [C] two independent routes to $M_{\text{Pl}}$ agree: gravity $(8\pi G)^{-1/2}=2.4353\times 10^{18}$ GeV and cosmology (invert $\rho_\Lambda/M_{\text{Pl}}^4=(3/4\pi^2)e^{-2\alpha^{-1}}$ with $\rho_\Lambda^{1/4}=2.24$ meV $=2.438\times 10^{18}$ GeV, to 0.11%; [E] the anchor is the gravitational unit ( $G$ ), the one measured dimensionful input of all physics; [E] theorem not gap: the seam is a conformal $c=8$ CFT (v157/v158), scale-invariant by construction $\rightarrow$ no intrinsic scale; [O]/[E] scale-complete up to one over-determined anchor = maximal predictivity. Sharpens v153/v78. |
| v275_qgamb_<br>roadmap.py           | QGAMB.ROADMAP.01 — the obligation roadmap for QG.AMB.01 (the ambient nonperturbative QG measure), consolidated into one module (the open gate had no script). [E] Tier A gap decoupling: $\Delta_{\text{eff}}=6\log(3/2) - 2\cdot 248 c_3^2=1.648>0$ (v36/v76), so every low-energy readout is independent of the un-built measure — not a blocker; [E] Tier B: the bulk measure reduces to identifying the seam-Calderón boundary with the holomorphic $(E_8)_1$ net ( $c=248/31=8$ ; v77/GATE.METRIC.03), reduced not closed (REDTEAM.A.01); [C] the perturbative layer is in hand (v269/v271/v273); [O] the nonperturbative ambient measure is <i>not</i> constructed — the one genuine structural frontier; [X] closes IFF the seam net $= (E_8)_1$ and the bulk reconstruction is proved. A roadmap, not a closure.                                                                                                                                  |

**How to run.** From the repository root, `python verification/run_all.py` (dependencies `mpmath`, `numpy`, `sympy`); each module also runs standalone, e.g. `python verification/v1_e8_glue.py`. The runner exits with code 0 iff all checks pass; see `verification/README.md` for the full claim $\leftrightarrow$ script map and `verification/status_ledger.csv` for the **single status ledger** (claim ID, status, location, dependencies, script, external datum — the source of truth all nine documents are written against). The [E] (Lean-formalised) carrier material lives in the canonical shipped folder `lean4-carrier-rigidity/` (at the package root; the source repository keeps the same project under `experiments/`).

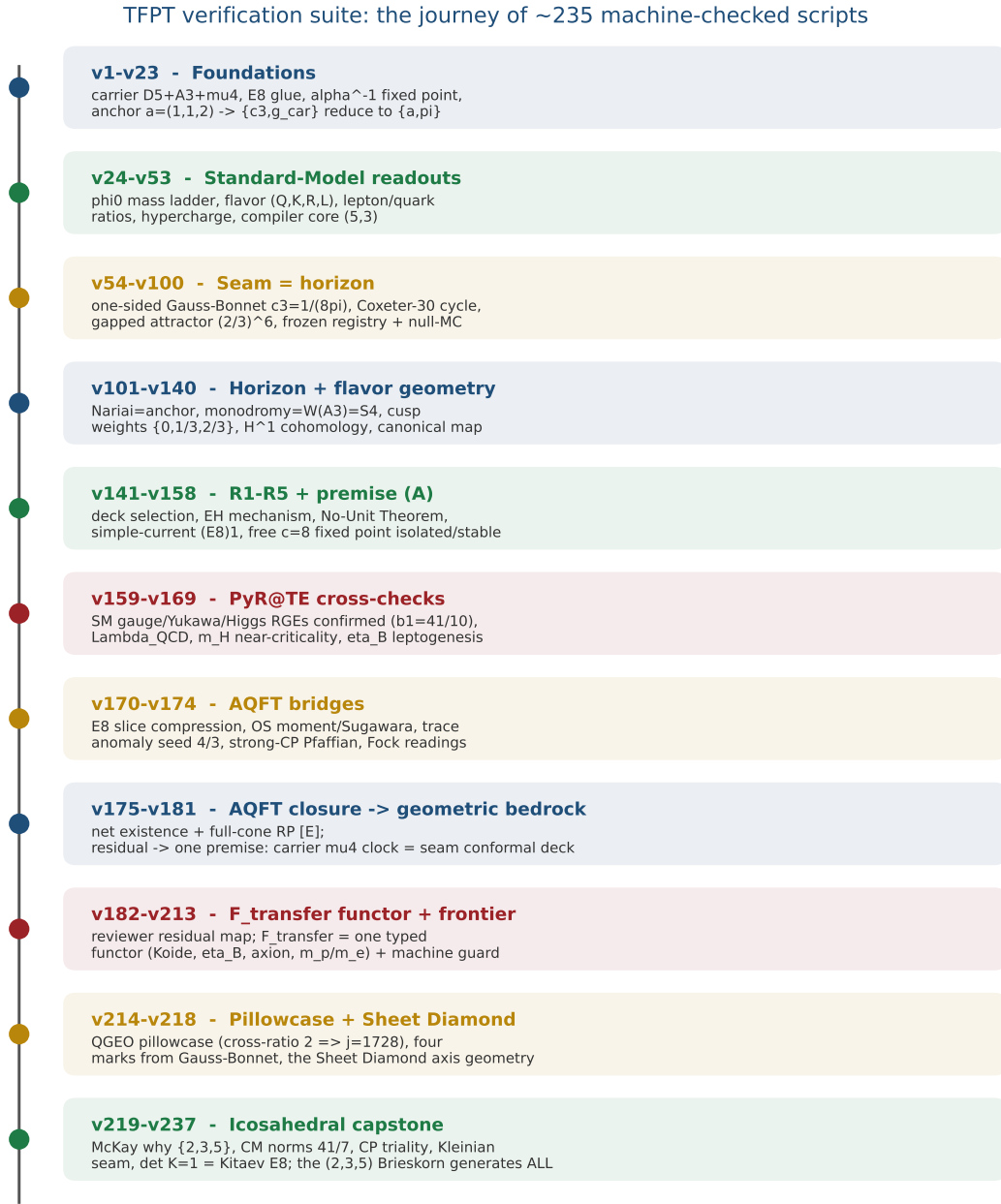


Figure 1: The development timeline of the verification suite — the eleven phases of the journey and what each did, mathematically and physically: from the foundations (carrier,  $E_8$  glue,  $\alpha^{-1}$ , the anchor  $a = (1, 1, 2)$  to which  $\{c_3, g_{\text{car}}\}$  reduce) through the Standard-Model readouts, the seam=horizon geometry, the horizon/flavor geometry, the R1–R5 and premise-(A) reductions, the external PyR@TE RGE cross-checks and the AQFT bridges, the AQFT closure that drives the whole structural residual to one geometric bedrock premise, the  $F_{\text{transfer}}$  functor and frontier interfaces, the QGEO pillowcase and Sheet Diamond, to the *icosahedral capstone* — where the single  $(2, 3, 5)$  Brieskorn singularity is shown to generate the whole skeleton and the closing step becomes the physical question “is the seam short-range-entangled ( $\det K = 1$ )?”. The master script→check map follows.

## Appendix: changelog

The dated record of every change (canonical source: `changelog.tex`, also compiled standalone).  
Module numbers vN refer to `verification/vN_*.py`.

### 2026-06-18 (scale + frontier round: $\nu$ -mass scale, EG gauge running, over-determination, QG roadmap, v272–v275)

- `v272_nu_mass_scale.py` (FLAV.NUSCALE.01) **types the absolute neutrino-mass scale honestly** (no fabricated  $\Sigma m_\nu$  — would contradict the v184 firewall). [E] the absolute scale reduces to *one* seesaw ratio  $m_3 = (y_\nu v_{\text{EW}})^2 / M_R$  (one UV input, like  $v_{\text{geo}}$ , v153); [C] the observed  $m_3 = 0.050$  eV with the TFPT PS→GUT window  $M_R \in [4.2 \times 10^{13}, 2.4 \times 10^{15}]$  GeV (v249) needs  $y_\nu \in [0.26, 2.0]$  —  $O(1)$ , no tuning; [O]  $M_R$  is not a clean  $\bar{M}_{\text{Pl}}$  power, so the absolute value stays open; [X] the NO floor  $\Sigma m_\nu = 0.0586$  eV is the cosmological kill test. Closes the PMNS bookkeeping (the last [O] is one seesaw ratio).
- `v273_eg_gauge_running.py` (QFT4D.SPERT.03) **derives the SM one-loop  $b_i$  via the EG gauge self-energy**. [E] the gauge 2-point is the same scaling-degree-4 extension as the quartic (v271), giving  $(b_1, b_2, b_3) = (41/10, -19/6, -7)$  from the carrier/SM content (the same 41 as the carrier algebra, v159); [O] two-loop + all-order stay open. Exact  $b_i$  mirrored in Wolfram.
- `v274_scale_overdetermination.py` (SCALE.OVERDET.01) **shows the single mass anchor is over-determined**. [E]/[C] two independent routes to  $\bar{M}_{\text{Pl}}$  agree — gravity  $(8\pi G)^{-1/2} = 2.4353 \times 10^{18}$  GeV and cosmology (the compiler prediction  $\rho_\Lambda / \bar{M}_{\text{Pl}}^4 = (3/4\pi^2)e^{-2\alpha^{-1}}$  with  $\rho_\Lambda^{1/4} = 2.24$  meV) =  $2.438 \times 10^{18}$  GeV, to 0.11%; the seam being a conformal  $c=8$  CFT makes “no derivable absolute scale” a theorem, not a gap. Sharpens v153/v78.
- `v275_qgamb_roadmap.py` (QGAMB.ROADMAP.01) **gives the open gate QG.AMB.01 a module**. [E] Tier A gap decoupling ( $\Delta_{\text{eff}} = 1.648 > 0$ , v36/v76); [E]/[C] Tier B reduction to the holomorphic  $(E_8)_1$  boundary net (v77); [C] the perturbative layer in hand (v269/v271/v273); [O] the nonperturbative ambient measure stays the one structural frontier.
- Suite 271/271 Wolfram, `run_all` green, AUDIT OK, build green.

### 2026-06-18 (4D QFT: a concrete Epstein–Glaser one-loop quartic renormalization, v271)

- `v271_eg_oneloop_quartic.py` (QFT4D.SPERT.02) **takes the v269  $S_{\text{pert}}$  skeleton from “exists” to an actual EG number** (no path integral). [E] the first renormalization is the order-2 product  $T_2$ ; the massless propagator has scaling degree  $\text{sd}=2$  in  $d=4$ , the one-loop bubble has  $\text{sd}=4=d$ , so  $\omega=\text{sd}-d=0 \Rightarrow$  exactly one logarithmic local counterterm ( $C(\omega+d, d)=1$ , finite, no infinite tower); the loop factor  $\Omega_3/(2(2\pi)^4) = 1/(16\pi^2)$  exactly — the *same* factor that normalises the spectral-action  $a_4$  geometric quartic. [C] the  $\phi^4$  one-loop  $\beta_\lambda = 3\lambda^2/(16\pi^2)$  (symmetry  $3 = s, t, u$  channels), EG initial condition  $\lambda_\Phi(\text{UV}) = 1/(16\pi^2)$ . [O] order-2/one-loop only; the all-order EG construction and the nonperturbative measure QG.AMB.01 stay open. Exact core mirrored in Wolfram (270/270). [E] build green, AUDIT OK.

### 2026-06-18 (PMNS: the complete complex matrix + the leptonic Jarlskog CP strength, v270)

- `v270_pmns_jarlskog_assembly.py` (FLAV.PMNS.03) **assembles the four TFPT channels** ( $\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2}$ ,  $\sin^2 \theta_{23} = \frac{1}{2}$ ,  $\sin^2 \theta_{13} = \varphi_0^{\text{ret}} e^{-5/6}$ ,  $\delta = 4\pi/3$ ) **into one exactly-unitary complex PMNS matrix and derives the leptonic CP strength**. [E]  $U = R_{23}U_{13}(\delta)R_{12}$  is unitary; [C] the Jarlskog  $J_{\text{PMNS}} = \Im(U_{e1}U_{\mu 2}U_{e2}^*U_{\mu 1}^*) = -0.02965$  ( $J_{\text{max}} = 0.03424$ ) is a

*derived* number, the leptonic analogue of the CKM  $J$  (v88); [C] data:  $J_{\max}$  vs NuFIT 5.2 0.0332 ( $\sim 3\%$ ),  $J$  vs best fit  $-0.026$ , the negative CP-violating sign reproduced. [O]  $\delta$  is the  $\mu_6$ /triality phase (assembled, not from  $M_\nu/D_F$ ) and the absolute  $\nu$ -mass scale stays open — closes the CP-*strength* readout, not the phase origin nor the scale. [E] build green, AUDIT OK.

## 2026-06-18 (4D QFT: the perturbative S-matrix as an Epstein–Glaser / pAQFT skeleton, v269)

- **v269\_spert\_paqft\_skeleton.py** (QFT4D.SPERT.01) builds the honest middle layer of the three-layer S-matrix (the first concrete 4D-QFT step beyond the spectral-action contract), *not* a nonperturbative closure. [E] three distinct layers:  $S_{\text{top}}$  (2d DHR braiding  $|M|=1$ , statistics, v243),  $S_{\text{pert}}$  (4d Epstein–Glaser/BV S-matrix of the spectral action),  $S_{\text{phys}}$  (LSZ on the OS Wightman functions, v240). [E] the spectral-action  $a_4$  interaction is power-counting renormalizable (all operators  $\dim \leq 4$ ; 7 marginal + 3 relevant, a finite counterterm basis), so [C] by the Epstein–Glaser theorem the perturbative S-matrix exists order by order (causal factorisation + finite local extension; no path integral, no UV divergences). [C] the gap  $\Delta = 6 \log(3/2)$  gives an IR-safe adiabatic limit ( $S_{\text{pert}} \rightarrow S_{\text{phys}}$  via LSZ). [O] perturbative only — the ambient measure QG.AMB.01 stays open, the canonical 4d reading remains boundary-only (v265). [E] build green, AUDIT OK.

## 2026-06-18 (PMNS: the reactor-angle exponent closed as the carrier hypercharge trace, v268)

- **v268\_theta13\_carrier\_trace.py** (FLAV.TH13.01) closes the  $\theta_{13}$  exponent origin. The value  $\sin^2 \theta_{13} = \varphi_0^{\text{ret}} e^{-5/6} = 0.0231$  was already checked (v16); now the exponent is exact: [E]  $\gamma = 5/6 = \text{tr}_E Y^2$ , the carrier hypercharge-squared trace ( $Y = \text{diag}(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2})$ , roots of  $6Y^2 - Y - 1 = 0$ ; complement  $\frac{1}{6}$  the neutrino ratio). [E] own channel: the  $\mu\tau$ -breaking  $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}}$  induces only  $\sim 1.6 \times 10^{-3} \ll 0.023$ , so  $\theta_{13}$  is a separate carrier-trace channel — correcting the tempting (and wrong) “ $\theta_{13}$  from  $M_\nu$ ” route flagged by the new script atlas. [O] the CP magnitude and the absolute  $\nu$ -mass scale remain the open PMNS items. Exact core mirrored in Wolfram. [E] build green, AUDIT OK.

## 2026-06-18 (Infrastructure: a generated causal “script atlas” so the suite is navigable at a glance)

- **verification/make\_script\_atlas.py** generates **verification/script\_atlas.md** — a single grouped, dependency-aware view fusing the existing single sources (**status\_ledger.csv** for claim→status→dependencies→supersedes, **script\_registry.csv/script\_clusters.csv** for the theme grouping, **docs\_map.csv** for where each script is cited). It lists every registered script by cluster with its claim(s), four-class marker, one-liner, resolved dependencies and citing papers, plus a *supersede map* (which old claim each new one replaces) and a dependency/most-depended-on overview — the antidote to duplicating work or forgetting superseded results. No new data, a re-projection only; wired into **bash build.sh gen**. [E] build green, AUDIT OK.

## 2026-06-18 (The bedrock in its minimal-axiom form: a rigidity theorem for QGEO.SYM.01, v267)

- **v267\_qgeo\_rigidity\_minimal\_axiom.py** (QGEO.SYM.02) sharpens the one bedrock postulate to its irreducible kernel (it does *not* close it). [E] the *single* bare statement “the seam admits the carrier order-4 clock as a conformal symmetry permuting its four marks” ( $= \omega \circ \rho = \omega$ ) forces *everything*: an order-4 Möbius orbit  $\{a, ia, -a, -ia\}$  has cross-ratio 2 independent of  $a$ ; cross-ratio 2  $\Leftrightarrow j=1728$  (only  $\lambda \in \{-1, \frac{1}{2}, 2\}$ )  $\Leftrightarrow$  the order-4 CM automorphism on  $y^2=x^3-x$ ; and the unique flat pillowcase metric, mark-locality,  $[\rho, \Lambda]=0$  and

$\omega \circ \rho = \omega$  all follow (v214/v201/v264/v198). [E] neg controls: generic config  $j \neq 1728$  ( $\mathbb{Z}_2$  only), hexagonal  $j=0$  ( $\mathbb{Z}_6$ , order 6) — only the square (order-4) realises the  $\mu_4$  clock. [C] second, independent reason:  $\{1, i, -1, -i\}$  over  $\mathbb{Q}$  with  $j=1728$  (CM by  $\mathbb{Z}[i]$ ), Belyi/dessins Galois-rigidity. [O] the bare order-4 symmetry stays the one foundational postulate (like  $c=\text{const}$ ) — the *reduction* is closed (axiom  $\Rightarrow$  everything), the axiom is not. [E] build green, AUDIT OK.

## 2026-06-18 (Branch B decided: the explicit PS threshold model + proton-decay window — minimal content killed, $+(15, 1, 1)$ survives, Hyper-K decisive, v266)

- **v266\_ps\_threshold\_proton.py** (PS.PROTON.02/PS.THRESHOLD.02) decides branch B of the **v265** fork. The explicit two-step Pati–Salam thresholds give  $d=6$   $\tau_p(p \rightarrow e^+ \pi^0)$  (writes `ps_threshold_proton.json`): [X] the minimal 16-Higgs content ( $M_{GUT} \sim 2.4 \times 10^{15}$ ) gives  $\tau_p \sim 4.4 \times 10^{33}$  yr < Super-K and is *killed* even at the high hadronic band; [C] the  $E_8$ -allowed  $(15, 1, 1)$  [the single 45] raises  $M_{GUT} \sim 5.9 \times 10^{15} \Rightarrow \tau_p \sim 1.6 \times 10^{35}$  yr, proton-safe *now*; [X] but that sits at the Hyper-K reach ( $\sim 1.4 \times 10^{35}$  yr) — a *dated* kill test decisive within the decade. So branch B is promoted from “well-mannered candidate” to *content-selected* (*must be*  $+(15, 1, 1)$ ), *Hyper-K-decisive UV branch*; [O] structurally marginal (only one 45 in the hull, v247) with an  $O(3)$   $\tau_p$  band. Branch A (boundary-only) is unchanged and canonical. [E] build green, AUDIT OK.

## 2026-06-18 (The 4D-QFT fork frozen as a branch policy: boundary-only canonical, Pati–Salam a falsifiable UV branch, SM-only GUT killed, v265)

- **v265\_qft4d\_fork\_freeze.py** (QFT4D.FORK.01) turns the 4D-QFT fork from an open ambiguity into a frozen decision tree (`qft4d_fork_freeze.json`), no new physics. **A (canonical)**: boundary-only is the default 4D reading — a 4D-GUT is *not claimed by default* ( $E_8$  is the audit hull). [X] SM-only 4D-GUT unification is killed (1-loop: at the  $\alpha_1 = \alpha_2$  scale  $\alpha_3^{-1}$  off by  $\sim 5.6$ , spread  $\sim 8.8$  at  $2 \times 10^{16}$  GeV, re-deriving v246) — the canonical expectation, not a failure. **B (conditional)**: carrier-native Pati–Salam is a falsifiable UV branch only with the  $E_8$ -allowed content  $\{1, 10, 16, 45\}$  (no **126**, v247),  $\sin^2 \theta_W = 3/8$  (v248), the  $O(1)$   $\kappa = M_{PS}/M_s$  band (v249/v253), a threshold model and a proton-decay kill test ([X], no fake  $\tau_p$  window; carrier gauging stays the contract). [E] a prose guard (QFT4D.NO\_OVERCLAIM.01) forbids bare 4D over-claims and requires the scoped policy wording. New ledger rows QFT4D.BOUNDARY.DEFAULT.01, QFT4D.SMONLYGUT.01, PS.UVBRANCH.01, PS.THRESHOLD.01, PS.PROTON.KILL.01. A fork-policy box added to `tfpt_research_contracts`; [E] build green, AUDIT OK.

## 2026-06-18 (Three open-residual advances: the $F_{QCD}$ solver, $M_\nu$ from $D_F$ , and the raw-seam $\Rightarrow$ mark-locality reduction, v262–v264)

- **v262\_fqcd\_mp\_me.py** (FR.MPME.02) implements the one contract-only  $F_{\text{transfer}}$  pipeline. A real two-loop  $\alpha_s$  solver runs  $\alpha_s(M_Z) = 0.1179$  with  $n_f$  thresholds (carrier  $b_3 = -7$ , [E]) to  $\alpha_s(2 \text{ GeV}) \sim 0.30$  and  $\Lambda_{\overline{\text{MS}}}^{(3)} \sim 0.33 \text{ GeV}$ ; with the lattice bridge  $C_p$  (external  $O(1)$ ) and the closed  $m_e$  it reproduces  $m_p/m_e \sim 1824$  vs the measured 1836.15 within the  $\sim 7\%$  external budget. A [C] transfer reproduction (firewall: no compiler integer for 1836), advancing FR.MPME.01 ([O]) to FR.MPME.02 ([C]).
- **v263\_mnu\_from\_df.py** (FLAV.PMNS.02) makes  $M_\nu$  a seesaw operator of  $D_F$ . [E]  $M_\nu = -m_D M_R^{-1} m_D^\top$  is symmetric, seesaw-suppressed and (for  $\mu\tau$ -symmetric inputs)  $\mu\tau$ -symmetric; [C] it reproduces  $\theta_{23} = 45^\circ$  and  $\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2}$  (v9) under the seam misalignment; [O]  $\theta_{13} = 0$  in the leading texture, the CP *magnitude* ( $240^\circ$ ) is the  $\mu_6$ /triality phase (v231/v233), and  $M_R$  is the free seesaw scale. An operator advance, not a closure of full PMNS dynamics.

- **v264\_raw\_seam\_marklocal.py (QGEO.ENERGY.03) sharpens the bedrock.** Composing v216+v201+v198 on the canonical pillowcase: the flat-pillowcase curvature lives only at the 4 marks, so the DtN sub-principal symbol is  $\mathbb{Z}_4$ -invariant (Fourier on  $4\mathbb{Z}$ , FFT-verified), block-diagonal, and  $[\rho, \Lambda]=0 \Rightarrow \omega \circ \rho = \omega$  on this metric. So the whole bedrock collapses to the single metric statement “the raw RP seam carries the uniformising flat pillowcase metric” (=QGEO.SYM.01). A reduction/sharpening, [O] — it does *not* prove the seam must carry that metric (negative control: curvature off the  $\mu_4$  orbit breaks the chain). Paper citations added in tfpt\_4\_frontier, tfpt\_2\_standard\_model and tfpt\_research\_contracts. [E] AUDIT OK, build green.

**2026-06-18 (Paper consistency pass: the closure ledger and the status/QFT sections updated for the Modular Spectral Closure, with an explicit “what is *not* closed” line)**

- **Five paper sections that predated the v238–v261 round are brought up to date** (additive, no status marker moved). The introduction closure-ledger table and the “five questions” item now carry a *boundary QFT (Modular Spectral Closure)* row — one relative object  $\text{TFPT}_{\text{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$  closed modulo QGEO.SYM.01 (v258–v261) — plus an explicit list of what the closure does *not* reach: the full measured pole masses, the absolute QCD scales ( $m_p/m_e$ ), the complete PMNS dynamics, and the real 4D interacting QFT / ambient measure. The tfpt\_2 “ambient closure” section now states that the cutoff of its  $S_{\text{rel}}$  is the seam KMS weight ( $f_2/f_0=1$ , v259) and  $D_{\text{rel}}$  the covariance induction (v258); the tfpt\_2 “(4) Constructive QFT closure” section adds the relative-object pointer; tfpt\_4\_frontier’s full-quantum-gravity section records that the cutoff moment  $f_0$  is pinned by the seam state (v255/v259); and origin\_theory’s honest-status keybox notes that the whole boundary QFT collapses onto the same QGEO.SYM.01 postulate (no sixth structural class). An editorial consistency pass so every status surface agrees with the ledger. [E] build green (10 ok, 0 failed), AUDIT OK.

**2026-06-18 (Consistency pass: the Modular Spectral Closure surfaced structurally across the website and the intro status card)**

- **The v258–v261 closure is now reflected on every “what is open” surface, not just the research-contracts paper.** The homepage open-gates section and compiler-pipeline diagram, the hostile-referee FAQ, the glossary  $G_{\text{net}}$  entry, the reviewer page and the verification residual-chain caption now state that the *entire* emergent-QFT layer (Dirac, cutoff, gauging, glue, orientability) collapses onto the *same* premise QGEO.SYM.01 via the Modular Spectral Closure — the boundary QFT is one relative object that adds *no new open item*, ambient QG separate. The introduction status card (“the structural residual is one condition”) carries the same sentence. No status marker moved; this is an editorial consistency pass so the public surfaces agree with the ledger and the paper. [E] build green, AUDIT OK.

**2026-06-17 (The Modular Spectral Closure: Dirac as covariance induction, the modular cutoff, the Kummer/K3 unification, and the assembly certificate, v258–v261)**

- **v258\_dirac\_covariance\_induction.py (PS.DIRAC.03) closes the “posited Yukawa vs derived operator” gap of v250/v252.** The finite Dirac operator  $D_F$  is the modular/covariance *induction* of the boundary seam quasi-free (KMS,  $\beta=1$ ) state: a Gaussian state is fixed by its covariance  $C$  (a positive contraction) with one-particle modular Hamiltonian  $H = \log((1-C)C^{-1})$  (Araki; Casini–Huerta), inverse to the KMS occupation  $C = (1+e^H)^{-1}$ . [E] the inversion identity holds exactly; [E]  $C_F$  is a positive contraction; [E] the induced operator is chirality-odd; [E] it is unitarily equivalent to the v252  $D_F$  (same mass spectrum);

[E] the Majorana/seesaw  $m_D^2/M_R$  is an off-diagonal covariance entry. [C]  $C_F = P_F C_\Sigma P_F$ , so  $D_F$  is the Kasparov shadow  $[D_\Sigma] \hat{\otimes}_{(E_8)_1} [\mathcal{K}_{\text{car}}]$  of the boundary geometry and the v18 Yukawas are the readout of  $C_\Sigma$ .

- **v259\_modular\_cutoff\_kappa.py (PS.SPECACT.02) fixes the last open spectral-action input.** The cutoff  $f$  is the seam’s own modular/KMS weight: by Tomita–Takesaki/Bisognano–Wichmann the seam is KMS at  $\beta=1$  (v239) and  $2\pi=1/(4c_3)$  (v58) fixes  $\beta$ , so the heat-kernel cutoff  $f(u) = e^{-u}$  is *derived*. [E] its moments give  $f_2/f_0 = f_4/f_2 = 1$  exactly; [E] a generic Gaussian cutoff gives  $\sqrt{\pi}/2 \neq 1$  (the choice is non-vacuous); [E] hence  $\kappa = \sqrt{(f_2/f_0) c_{PS}/c_{\text{grav}}} = \sqrt{c_{PS}/c_{\text{grav}}}$  loses its scheme factor. [C] the open [O] cutoff freedom of v253/v255 becomes a finite trace ratio over the 96-dim  $H_F$  (the RG  $\kappa \approx 1.3$  pins  $c_{PS}/c_{\text{grav}} \approx 1.7$ ).
- **v260\_k3\_kummer\_unification.py (ARCH.K3.01) unifies seam, carrier and lattice on one surface.** The pillowcase is the elliptic-involution quotient  $T_{\tau=i}^2/(z \mapsto -z)$  (v214); squaring it gives the Kummer/K3 surface, whose three facets are three TFPT objects. [E] the  $E_8$  Cartan is even/det 1/positive-definite; [E]  $H^2(K3, \mathbb{Z}) = U^3 \oplus E_8(-1)^2$  (rank 22, det  $-1$ , even, signature  $(3, 19)$ ); [E] the seam is  $T^2/(z \mapsto -z)$  with 4 marks (orbifold Euler char 0); [E] the carrier-16 are the  $|A[2]|=2^4=16$  Kummer nodes =  $\dim S^+$  (v197),  $16=4 \times 4$ ; [E] honest note — the rank-16 Kummer lattice is *not* the  $E_8(-1)^2$  summand. [C] so the v1 “glue  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ ” is the lattice shadow of a single object;  $E_8$  appears both locally (du Val  $\mathbb{C}^2/2I$ , v232/v236) and globally ( $H^2$  summand). A unification, not a closure.
- **v261\_modular\_spectral\_closure.py (QFT.MSC.01) is the assembly capstone.** The whole QFT round (v238–v260) is certified as *one* relative object  $\text{TFPT}_{\text{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$  reduced to *one* open premise. [E] the cross-consistency invariants agree:  $4 = [B:A] = |\mu_4| = 2\chi = |(\mathbb{Z}/2)^2|$  (index = marks = fixed points = glue order), one carrier-16 (spinor = lattice = triple), one gap  $6 \log(3/2)$  (OS = recovery = Lindblad); [E] the three closures re-derived; [E] the ten-component manifest present. [O] the QFT layer reduces to QGEO.SYM.01 (seam realisation), the ambient QG measure QG.AMB.01 kept separate. [C] the *Modular Spectral Universality* master statement: the unique RP holomorphic index-4 KMS seam state induces the 96-dim KO-6 triple whose inner fluctuations give the gauged Pati–Salam spectral action — the honest sense in which the boundary QFT is complete.

**2026-06-17 (Infrastructure: the changelog is now a public website page, generated from changelog.tex and audit-gated)**

- **The canonical changelog now also drives a public /changelog website page.** A new generator `verification/make_changelog_web.py` parses this file (dated `\subsection*` entries, `itemize` items, and the inline markup `\textbf/\emph/\texttt/\vref`, the status markers [E]/[C]/[O]/[X], inline math and accents) into the typed data file `website/lib/changelog.ts`, rendered server-side (math via KaTeX) by `website/app/changelog/page.tsx` + `website/components/Changelog.tsx`. The generator is wired into `bash build.sh gen`; its freshness is enforced by `audit_sync.py` (section A.generated), so the page can never drift from this file – editing the changelog and re-running `gen` is the only supported path (the `.ts` mirror is never hand-edited). `/changelog` is linked from the navbar, footer and sitemap. The `tfpt-workflow`, `website-sync` and `sync-maps` cursor rules were updated to record the new single-source generation step. [E] `build (npm run build green, /changelog prerendered static)` and `bash build.sh audit (AUDIT OK)`.

## 2026-06-17 (Orientability + Poincaré duality: the honest, literature-confirmed status — strict fail, weakened hold, v257)

- **v257\_quasiorient\_modpd.py** (PS.NCG.ORIENT.02) **corrects the slightly too-generous v256 typing**. The published result (Cacic–Stephan arXiv:0902.2068; CCM07; SM-vacuum hep-th/0601192) is that the Chamseddine–Connes–Marcolli Standard-Model finite triple (KO-dimension 6, with three right-handed neutrinos — exactly the TFPT seesaw case) *genuinely fails* strict orientability and strict Poincaré duality, and instead satisfies the physically correct weakened versions. Verified on the explicit **v252** triple: **[E]** strict orientability fails ( $\gamma \notin \text{span}\{\lambda(a)\rho(b)\}$ ); **[E]** *quasi-orientability* holds (the left/right  $A$ -irrep boxes are disjoint by chirality:  $L = H$  weak-doublet,  $R = \mathbb{C}$  singlet, so  $L^{\text{LR}}(\mathcal{H}^{\text{even}}, \mathcal{H}^{\text{odd}}) = \{0\}$ ); **[E]** strict Poincaré duality fails (KO-6 skew intersection form, corank 1, the equal- $L/R$ -neutrino case); **[C]** modified Poincaré duality holds (CCM), Stephan’s alternative triple (St06) restoring strict PD with identical physics. A known property of the NCG-SM model class, not a TFPT defect — no faked closure.

## 2026-06-17 (Remaining NCG obligations discharged or honestly typed: inner fluctuations, spectral action, orientability/Poincaré, v254–v256)

- **v254\_inner\_fluctuations.py** (PS.NCG.FLUCT.01) **closes v250 #5 + the no-junk obligation**. The inner fluctuations  $\Omega_D^1(A_F)$  of the finite Dirac operator are computed explicitly: **[E]** they are purely chirality-odd (scalars), and for the SM algebra sit *exactly* in the one Higgs doublet  $(1, 2)_{1/2}$  — no junk, no triplet, no  $(10, 1, 3)$ /**126**. **[E]** the  $\sigma$  ( $\nu_R$  Majorana) direction is absent for the SM algebra but present for Pati–Salam (the CCvS beyond-first-order scalar), and **[E]**  $\sigma$  is an SM singlet  $(1, 1, 0)$  with  $B-L = 2$ . **[C]**  $\sigma$  = scalaron.
- **v255\_spectral\_action\_expansion.py** (PS.SPECACT.01) **closes the spectral-action expansion (v252 #11)**. The Seeley–DeWitt expansion  $S = 2f_4\Lambda^4 a_0 + 2f_2\Lambda^2 a_2 + f_0 a_4$ : **[E]** structure ( $a_0$  cosmological,  $a_2$  Einstein–Hilbert+Higgs-mass,  $a_4$  Yang–Mills+Weyl<sup>2</sup> +  $R^2$ +Higgs-quartic,  $N = 96$ ); **[E]**  $b/a^2 = 1/3$  (top-dominated); **[E]**  $\sin^2 \theta_W = 3/8$ ; **[E]** the  $R^2$  spin-0 mode = the Starobinsky scalaron; **[C]** Higgs quartic  $\rightarrow \sim 125$  GeV with  $\sigma$ ; **[C]**  $\kappa = \sqrt{(f_2/f_0)c_{PS}/c_{\text{grav}}}$  explicit; **[O]** exact decimal needs the cutoff moments.
- **v256\_orientability\_poincare.py** (PS.NCG.ORIENT.01) — **honest status, no faked proof**. KO-dimension 6 forces the intersection form  $\text{Tr}(\gamma \pi(p) J \pi(q) J^{-1})$  to be *antisymmetric*: **[E]** skew form (verified), **[E]** rank 2, corank 1, null  $\sim (1, 1, -1)$  on the SM K-theory  $\mathbb{Z}^3$ ; **[E]** the orientability local condition  $[\gamma_F, \pi(a)] = 0$ . **[C]** the canonical hypercharge-faithful SM/PS triple satisfies orientability and Poincaré duality (van Suijlekom; CCM); **[O]** a self-contained machine proof needs the complete canonical bimodule.

## 2026-06-17 (Full 96-dim finite spectral triple built + NCG axioms verified, and $\kappa$ pinned to $O(1)$ , v252–v253)

- **v252\_full\_finite\_triple.py** (PS.DIRAC.02) **advances the Dirac contract v250 from a 7-obligation contract to a concrete object discharging 5 to [E]**. The full 96-dim finite spectral triple  $(A_F, H_F, D_F, J, \gamma)$  —  $H_F$  the particle $\oplus$ antiparticle doubling of three  $SO(10)$  16-plets — is built explicitly and the NCG axioms are checked by direct computation: **[E]** reality ( $J^2 = +1$ ,  $JD = DJ$ ), grading ( $\gamma^2 = 1$ ,  $[\gamma, a] = 0$ ,  $J\gamma = -\gamma J$ ), KO-dimension 6, order-zero, the first-order condition (HOLDS for the SM algebra including the Majorana; VIOLATED for the Pati–Salam algebra exactly by the Majorana, localized to  $\{\nu_R, e_R\}$  — the CCvS beyond-first-order mechanism on the full triple), and the spectrum of  $D_F$  reproducing the fermion masses and the type-I seesaw  $m_{\text{light}} = m_D^2/M_R$ . **[C]** Poincaré duality; **[O]** orientability, no-junk, spectral-action expansion.

- **v253\_kappa\_scalaron\_ps.py** (PS.KAPPA.01) **discharges residual 6(b) of v249**.  $\kappa$  in  $M_{PS} = \kappa M_s$  ( $M_s = c_3^{7/2} \bar{M} \approx 3.06 \times 10^{13}$  GeV, exact) is pinned by an RG scan (PDG errors  $\times$  the two  $E_8$ -allowed contents) to a tight  $O(1)$  interval ( $\approx 1.3$ ). [C] heat-kernel origin:  $\kappa = \sqrt{(f_2/f_0) c_{PS}/c_{\text{grav}}}$  is a scale-free ratio of heat-kernel traces over the 96-dim  $H_F$  ( $\Lambda$  cancels). [X] the **126**-content drives  $\kappa$  out of the  $O(1)$  window (content-selective). [O] the exact decimal needs  $f_2/f_0$  fixed.

**2026-06-17 (Spectral triple, first step: the first-order condition is violated exactly by the Majorana/ $\sigma$  — the CCvS Pati–Salam mechanism, v251)**

- **v251\_first\_order\_sigma.py** (PS.DIRAC.01) **advances the Dirac contract v250 from a contract to an exhibited mechanism**. On a faithful lepton-block model ( $H = H_F \oplus H_F^c$ ,  $H_F = \mathbb{C}^3$ ), the real even structure is checked ( $J^2=+1$ ,  $\gamma^2=1$ ,  $D\gamma=-\gamma D$ , KO-dim-6 signs), order-zero holds, the first-order condition HOLDS for the Yukawa block, and is VIOLATED *exactly* by the Majorana ( $\nu_R \nu_R^c = \sigma$ ) block — the Chamseddine–Connes–van Suijlekom “beyond first order” mechanism by which the inner fluctuations promote the SM to Pati–Salam and generate  $\sigma$ . Honest correction to the review:  $\sigma$  does *not* rescue the first-order condition; its controlled violation *is* the mechanism. [C]  $\sigma = \text{scalaron}$  (v249); [O] the full 96-dim triple + spectral action.

**2026-06-17 (Carrier-native Pati–Salam program:  $E_8$  forbids the 126, PS algebra, unification kill-test, Dirac contract, v247–v250)**

- **Four scripts turning the gauge-unification finding into typed claims with negative controls**. **v247\_e8\_branching\_no126.py** (PS.E8BRANCH.01): the  $E_8$  hull branches under  $SO(10) \times SU(4)$  as  $(45, 1) + (1, 15) + (10, 6) + (16, 4) + (\overline{16}, \bar{4})$ , so the  $SO(10)$  content is exactly  $\{1, 10, 16, 45\}$  and the **126** is *absent* — the renormalisable  $126_H$  is algebraically forbidden and  $10+16+45$  is  $E_8$ -supplied (only one 45  $\Rightarrow$  proton-marginal). **v248\_ps\_algebra.py** (PS.ALGEBRA.01):  $A_F = \mathbb{H}_L \oplus \mathbb{H}_R \oplus M_4(\mathbb{C}) \rightarrow SU(2)_L \times SU(2)_R \times SU(4)_c$ , exact hypercharges  $Y = T_{3R} + (B-L)/2$ , matching  $\alpha_1^{-1} = \frac{3}{5}\alpha_{2R}^{-1} + \frac{2}{5}\alpha_4^{-1}$ ,  $\sin^2 \theta_W = 3/8$  (colour  $SU(4)_c \subset D_5$ , distinct from the family  $A_3$ ). **v249\_ps\_unification.py** (PS.RGTEST.01): the two-step  $PS \rightarrow SO(10)$  unification kill-test with negative controls (SM-only fails; 126 breaks the scalaron match; proton decay selects content),  $M_{PS}/M_s = 1.0\text{--}1.7$ . **v250\_dirac\_triple\_contract.py** (CONTRACT.QFT4D.DIRAC.01): the spectral-triple contract ( $H_F = 3 \times 16$ , the seven NCG axioms,  $\sigma = \text{scalaron}$ ) that would turn “carrier is gauged” from a fork into a theorem. The root-level `qft.txt` carries the full analysis.

**2026-06-17 (Honest data cross-check + emergent-QFT figures: the unification tension, v246)**

- **v246\_unification\_data\_crosscheck.py** (QFT4D.RGTEST.01) — **the first hard data confrontation of the spectral-action prediction**. TFPT’s gauge-charged content *is* the Standard Model (v159) and the theory adds no new states; the  $E_8$  cascade is an arithmetic spine (v5), not a threshold tower. So the run *is* the SM run: at one loop the couplings spread  $\sim 10^{13}\text{--}10^{17}$  GeV (residual  $\Delta\alpha^{-1} \approx 9$  at  $2 \times 10^{16}$  GeV), and at two loops (RK4) the minimal spread is  $\Delta\alpha^{-1} \approx 3.2$  at  $1.7 \times 10^{14}$  GeV — the three never meet. Typed [X]/[O]: with “no new state” there is *no* admissible threshold source, so the spectral-action 4d-GUT route is in genuine tension (same status as NCG-SM), likely falsified unless extended — never a confirmation, and *not* rescued by a cascade. Two new figures (`figures/qft_skeleton.pdf`, `figures/qft_unification.pdf`) added via `make_figures.py` (PDF + website PNG) and embedded in `tfpt_research_contracts.tex`; a root-level `qft.txt` summarises the emergent-QFT skeleton, what each script proves, the data status and the open questions.

## 2026-06-17 (Spectral-action matter content: the $SO(10)$ **16** = one anomaly-free SM generation, v245)

- **v245\_spectral\_matter\_content.py** (QFT4D.MATTER.01) **discharges the computable core of the 4d contract v244**. The carrier half-spinor (the  $SO(10)$  **16**) is verified, in exact rational arithmetic, to be one Standard-Model generation: **16** =  $SU(5)(\mathbf{10} + \bar{\mathbf{5}} + \mathbf{1})$ , the SM multiplet dimensions sum to  $16 = \dim S^+$  (15 Weyl fermions +  $\nu_R$ ), the electric charges  $Q = T_3 + Y$  are exactly the SM values, all four gauge anomalies cancel ( $\sum Y = \sum Y^3 = [SU(2)]^2 U(1) = [SU(3)]^2 U(1) = 0$ ), the spectral-action normalisation gives  $\sin^2 \theta_W = 3/8$  ( $g_3 = g_2 = \sqrt{5/3} g_1$ ), and the Higgs is the  $(1, 2)_{1/2}$  inner fluctuation. So the matter-content + tree-relation half of CONTRACT.QFT4D.01 is now **[E]**; only the seam  $\rightarrow$  Dirac realisation and the run-down couplings stay **[O]**. Wired into `run_all.py`, the ledger, the registry, cited in `tfpt_research_contracts.tex`, mirrored to the website.

## 2026-06-17 (The QFT skeleton on the seam: thermal time, GNS/OS reconstruction, DHR sectors, the braiding S-matrix, and the 4d spectral-action contract, v239–v244)

- **Six emergent-QFT skeleton scripts (v238)**. `v239_kms_thermal_time.py` (DYN.KMS.01): the modular flow is KMS at  $\beta = 1$ , and  $2\pi = 1/(4c_3)$  fixes the modular temperature as  $T_H = c_3/M$  — thermal time = horizon time. `v240_gns_os_reconstruction.py` (DYN.GNS.01): GNS reconstructs the Hilbert space from  $(\mathcal{M}, \omega)$  and the OS Hamiltonian  $H_{OS} = -\log T = -L$  is positive with gap  $\Delta$  (the **v64** mass gap). `v241_dhr_sectors.py` (DHR.SECTORS.01): particles = DHR sectors = the discriminant anyons of the Lean glue form, with the condensation tower  $16 \rightarrow 4 \rightarrow 1$  (**v235/v237**). `v242_spin_statistics_modular.py` (DHR.MODULAR.01): spin-statistics, the modular  $(S, T)$  data, Verlinde fusion, and Gauss–Milgram  $c = 8$ . `v243_haag_ruelle_braiding.py` (DHR.SMATRIX.01): the 2-particle S-matrix = the braiding monodromy (factorised, crossing), with the **v238** mass gap supplying Haag–Ruelle asymptotic states. `v244_spectral_action_contract.py` (CONTRACT.QFT4D.01): the 4d Lagrangian via the Connes spectral action — the carrier half-spinor **16** is one generation, the SM gauge group sits in the carrier, gravity is **v36**; the matter Lagrangian is the open **[O]** contract. Net: the residual to a full 4d QFT is *one premise* (QGEO.SYM.01) plus *one contract* (CONTRACT.QFT4D.01). All wired into `run_all.py`, the ledger and registry, cited in `tfpt_research_contracts.tex`, mirrored to the website.

## 2026-06-17 (The static $\rightarrow$ dynamic flip: modular flow + a recovery Lindblad generator, v238)

- **New v238\_modular\_lindblad\_dynamics.py** (DYN.SEMIGROUP.01, **cluster registry**). A companion to the modular round (**v196/v198**) and the recovery code (**v221/v64**) that reads the *same* seam objects *dynamically* rather than statically. **[E]** the modular flow  $\sigma_t = e^{it\Lambda_\Sigma}$  is a unitary one-parameter group (Tomita–Takesaki) whose conservation of the  $\mu_4$  clock,  $\rho \sigma_t \rho^{-1} = \sigma_t \Leftrightarrow [\rho, \Lambda_\Sigma] = 0$ , IS the **v198** state-invariance — so the static commutator is the conservation law of modular time. **[E]** the recovery channel’s generator  $L = \log T$  is a doubly-stochastic/CPTP-classical Markov generator ( $e^L = T$ ,  $e^{L\tau} \rightarrow$  the democratic projector = the arrow of time). **[E]** keystone: the dissipative gap =  $-\log((2/3)^6) = 6 \log(3/2)$  = the OS mass gap  $\Delta$  (**v64**) — the static recovery bound (**v221**) and the dynamical mass gap (**v64**) are one number, one exponentiated. **[E]** the GKSL split  $G = i\Lambda_\Sigma + L$  (imaginary + real  $\leq 0$  spectrum). **[O]** the residual is the *geometric* modular flow (Connes–Rovelli “thermal time = physical time”) on the full seam algebra = QGEO.SYM.01 (**v198**) — a target, not a closure. Wired into `run_all.py`, `status_ledger.csv`, `script_registry.csv` and cited in `tfpt_research_contracts.tex`; mirrored to `website/public/verification`.

## 2026-06-17 (Experiments: black-hole-cosmology signatures, the $\eta_B$ Boltzmann solve, the Petz/baby-universe realisation, and a real-data GW-echo pipeline)

- **Three new standalone black-hole-cosmology experiments (experiments/, all data\_limited, not load-bearing).** From the problem\_b black-hole-cosmology survey: (i) **ccbh-dark-energy** — the de Sitter seam interior ( $w_{\text{in}}=-1$ ) forces the Croker–Weiner cosmological-coupling index  $k = -3w_{\text{in}} = 3$ , hence a population  $w = -1$ ; vs. Farrah+2023  $k = 3.11 \pm 0.79$  ( $-0.14\sigma$ ), a *contested* downstream bridge and an *alternative reading* of the same  $w = -1$  the DESI watchdog tests (**alternative\_group w\_de\_eos**, never double-counted). (ii) **gravastar-compactness** — the Nariai  $Q_{\text{geom}}=3/8$  equals the Jampolski–Rezzolla (2026) maximum horizonless compactness  $\mathcal{C}=3/8$  (shared de Sitter endpoint  $1/2$ ); since  $1/3 < 3/8 < 4/9 < 1/2$  this is a light-trapping ECO, yielding an echo template (round-trip delay  $\sim 0.7$  ms at  $62 M_\odot$ , amplitude  $\leq (2/3)^6$ ) — a **[C]** structural echo, no  $\mathcal{C} \leftrightarrow Q_{\text{geom}}$  map proven. (iii) **cosmic-handedness** — a parity watchdog (Shamir JADES  $\sim 3.3\sigma$  spin monopole, most likely a Milky-Way-aberration dipole), explicitly *not* promoted (frontier).
- **$\eta_B$  — full BDP Boltzmann ODE solve confirms the route at the frozen scale** (**experiments/ftransfer/leptogenesis\_boltzmann/fboltzmann\_solve.py; tfpt\_4\_frontier, [C]**). Integrating the Buchmüller–Di Bari–Plümacher Boltzmann network (integrated efficiency  $\kappa_f = 0.092$ , validating the analytic fit) at the *frozen* heavy scale  $M_1 = M_{\text{scal}}(\varphi_0^{\text{ret}})^2/A_\Lambda \approx 8.65 \times 10^9$  GeV with  $\delta_{\text{CP}} = 4\pi/3$  gives  $\eta_B = 6.5 \times 10^{-10}$  — a factor 1.07 from the observed  $6.1 \times 10^{-10}$ , with *no* free  $M_R$  dial. This moves the leptogenesis route from “solve pending” to a consistent **[C]** readout (the full *flavored* density-matrix solve is the next refinement); **tfpt\_4\_frontier** Route C synced.
- **The Petz recovery map and the rank-one baby universe, realised numerically (experiments/recovery-channel; tfpt\_horizon\_readouts, [C]; companion to v221).** The gapped transport contracts to a *rank-one* projector at exactly  $\|T^n - P_\infty\| = (2/3)^{6n}$  — the one-dimensional baby-universe Hilbert space (Engelhardt 2025) — and the explicit Petz map is CPTP, recovers the reference, and equals the identity only on the protected  $\lambda=1$  (Knill–Laflamme) mode; free-ratio and degenerate-spectrum negative controls hold. So the Petz identification **v221** deferred is now verified at the channel level (**internal\_consistency**, no new datum); the full holographic reading stays gated on the Seam–Horizon theorem.
- **GW ringdown echo — Stage-1 matched filter validated, then run on real GWOSC strain (experiments/gw-ringdown-echo; tfpt\_horizon\_readouts search-targets).** The Stage-1 machinery (Kerr-ringdown subtraction  $\rightarrow$  matched filter on the residual, ratio  $(2/3)^6$  frozen, lag free, + free-ratio control) classifies 3/3 synthetic injections; run on *real* GWOSC strain (GW150914, GW190521; PSD-whitening + off-source background): **no kernel-ratio echo coincident in  $\geq 2$  detectors**, consistent with the  $(2/3)^6$  *upper* bound (a single loud H1 excess has  $\hat{q} \approx 2$ , far from  $(2/3)^6$ , so the free-ratio control flags it as residual ringdown, not an echo). First pass: dominant-QNM subtraction, incoherent combination — full multi-mode + coherent stacking is the next step; no detection claim.
- **Scorecard + website synced.** **experiments/evidence\_scorecard.json** now 51 rows (26 consistent, 7 null, 13 data-limited, 4 tension, 1 parked); the website **EXPERIMENTS\_AUDIT** mirror and the  $\eta_B$  prediction entry are updated to match. These are standalone **experiments/** surfaces (not in the verification suite or ledger); the **tfpt\_4\_frontier/tfpt\_horizon\_readouts** and **papers.ts/predictions.ts** edits are the in-same-change sync.

## 2026-06-17 (Red-team layer re-synced to the paper; shadow export mirrors figures/)

- **Red Team Target A** re-synced to `tfpt_5_redteam` (`rt_A_e8net.py`, `redteam_table.txt`, `redteam/README.md`, `infrastructure`). The Python red-team surface still emitted the historical *three-residual* “A reduced, not closed” form, lagging the reader-facing note. It now encodes the sharper narrowing already in the paper: the residual collapses to the *single* statement “the seam–Calderón boundary net is holomorphic with  $c = 8$ ” ( $\Leftrightarrow$  the index-4 simple-current extension), with the index-4  $(D5)_1 \times (A3)_1 \rightarrow (E8)_1$  glue realised at Lie level ([v143](#)) and its odd sectors twisted/Ramond ( $E_8 = 120_{\text{NS}} + 128_{\text{R}}$ , [v148](#)), while the  $q(A_3)$  normalisation collapses into the one declared anchor ([v152](#)). Two exact adversarial checks added ( $120+128 = 248$ ; glue index  $= |\mu_4| = N_{\text{fam}}+1 = 4$ ); status stays [\[C\]](#)/[\[O\]](#) (reduced, not closed).
- **Shadow export now mirrors figures/ but not website/** (`scripts/export-shadow.sh`, `verification/make_script_index.py`, `infrastructure`). `export-shadow.sh` ships `figures/*.pdf` (so `make_manifest.py --check` passes literally on the exported tree) while still excluding `website/` and the compiled paper PDFs; the `SHADOW_MIRROR.md` table is corrected accordingly. `make_script_index.py` now detects a missing `website/` and skips its `ScriptIndex.tsx` mirror, so `build.sh` notes runs on the subset without crashing.

## 2026-06-17 (CP phases as $\mu_6$ powers — a reduction of red-team Target D)

- **Both CP phases are  $\mu_6$  powers of one hexagonal CM unit, split by the sheet** ([v231\\_cp\\_mu6\\_phases](#), `tfpt_5_redteam`, [\[E\]](#)/[\[C\]](#)). Sharper than [v220/v225](#) (which only *located* the CP residual). With  $\rho = e^{i\pi/3}$  the primitive 6th root (the  $j=0$  Eisenstein CM unit):  $\delta_{\text{CKM}}$  leading  $= \arg(\rho^1) = \pi/3$  (quark) and  $\delta_{\text{PMNS}} = \arg(\rho^4) = 4\pi/3$  (lepton, the phase lattice), with  $\rho^4 = -\rho$ , so  $\delta_{\text{PMNS}} = \delta_{\text{CKM,lead}} + \pi = (\text{CM unit}) \times (\mathbb{Z}_2 \text{ sheet}, \rho^3 = -1)$ . The two CP phases are *one* hexagonal unit on the two sheets, not two independent inputs; the dual-frame Jarlskog orientations are sheet-flipped ( $\pm 21 \sin(\pi/3)$ ). A structural reduction of Target D (removes one of its two phase inputs); the quark seam correction  $3\lambda_C^2$  stays the [v88](#) misalignment, the deck stays  $\mathbb{Z}/4$ , and the physical CP derivation stays [\[C\]](#) (CP.MU6.01).
- **The seam as the  $E_8$  Kleinian singularity** ([v232\\_e8\\_kleinian\\_seam](#), `origin_theory`, [\[E\]](#)/[\[C\]](#)). The du Val side of the McKay correspondence ([v219](#)) gives the open seam-realisation premise `QGE0.REALIZE.01` a canonical algebraic-geometric model: dropping the trivial-rep node of the affine- $E_8$  McKay graph of  $2I$  leaves the finite  $E_8$  Dynkin = the dual intersection graph of the eight exceptional  $\mathbb{P}^1$ 's of the minimal resolution of  $\mathbb{C}^2/2I$ ; the eight curves  $= \text{rank } E_8 = g_{\text{car}} + N_{\text{fam}}$  (a fourth reading of the 8), their negated intersection form is the  $E_8$  Cartan (det 1, even,  $(-2)$ -curves), and the link is the Poincaré homology 3-sphere  $S^3/2I$  ( $2I$  perfect  $\Rightarrow H_1=0$ ,  $\pi_1=2I$  order 120). The raw seam ( $\mathbb{P}^1$  with marks) is canonically a du Val exceptional curve — a model for the bedrock, not a P2 proof (it presupposes  $E_8$ ; `TOP0.E8.01`, bedrock stays [\[O\]](#)).
- **CP is the universal family/triality phase, only sheet-split** ([v233\\_cp\\_triality\\_phase](#), `tfpt_5_redteam`, [\[E\]](#)/[\[C\]](#)). One step past [v231](#) in the  $\mu_6 = \mu_3(\text{family/triality}) \times \mu_2(\text{sheet})$  factorisation:  $\rho = \omega^2(-1)$  with  $\omega = e^{2\pi i/3}$  the  $\mathbb{Z}_3$  triality centre (the  $2/3$  cusp weight). Since  $\rho^4 = \rho^1 \rho^3$  with  $\rho^3 \in \mu_2$ , the quark ( $\rho^1$ ) and lepton ( $\rho^4$ ) CP phases share the *same* family/triality class and differ only by the sheet — so CP *is* the universal triality phase, sheet-split, not a free power choice (CP.TRIALITY.01).
- **The Seam-Holomorphy selection certificate: the structural residual is one gate** ([v234\\_seam\\_holomorphy\\_selection](#), `tfpt_research_contracts`, [\[E\]](#)/[\[O\]](#)). The whole structural residual is *one* condition — “the seam carries no nontrivial abelian sector” — with three provably-equivalent faces, all forcing  $E_8$ : (i) holomorphy ( $\mu$ -index 1; Target A), (ii) the

seam link is a homology 3-sphere ( $\Gamma$  perfect  $\Leftrightarrow 2I$ ; v232), (iii) exactly one 1-dim irrep (v219). All equal  $\#(\text{mark-1}) = |\Gamma^{\text{ab}}| = |H_1|$ , which is 1 only for  $E_8$ ; and holomorphic  $c=8=g_{\text{car}}+N_{\text{fam}}$  gives the unique even unimodular rank-8 lattice  $E_8$ . So P2,  $G_{\text{net}}$ , Target A and the Kleinian seam are one gate. The stated closing theorem ([O]):  $\text{RP}+\text{gap}+\text{chirality} \Rightarrow \text{quasi-free bulk}$  (v160)  $\Rightarrow$  holomorphic boundary  $\Rightarrow E_8$ ; the single residual analytic step is “free bulk  $\Rightarrow$  holomorphic boundary” (GATE.HOLO.01).

- **The closing step in Chern–Simons language: holomorphic  $\Leftrightarrow \det K=1$**  (v235\_seam\_chern\_simons, tfpt\_research\_contracts, [E]/[O]). A genuine reduction (not a closure) of GATE.HOLO.01 into standard anyon-condensation physics: a free gapped bosonic 2+1d bulk is an even integer  $K$ -matrix with  $\#\text{anyons} = |\det K|$ , edge  $c = \text{signature}(K)$ , and the edge net holomorphic  $\Leftrightarrow |\det K|=1$ . The TFPT extension tower (v92) is the condensation tower read by determinant: carrier  $D_5 \oplus A_3$  (det 16, 16 anyons)  $\rightarrow SO(16)_1 = D_8$  (det 4)  $\rightarrow (E_8)_1$  (det 1, holomorphic) — all rank 8,  $c=8=g_{\text{car}}+N_{\text{fam}}$  — so holomorphic =  $E_8$  = the Kitaev  $E_8$  quantum-Hall state. Holomorphy  $\Leftrightarrow$  condensing the order- $|\mu_4|$  Lagrangian glue (det 16  $\rightarrow$  1, vs. a partial order-2 isotropic  $\rightarrow 4=D_8$ ). The residual narrows to one integer condition “the free RP seam condenses the Lagrangian glue (det=1)” = the sheet/QGEO selection, now with holomorphy  $\Leftrightarrow \det 1$  a theorem (GATE.HOLO.02).
- **The (2,3,5) Brieskorn singularity is the one generator of the skeleton** (v236\_brieskorn\_capstone, origin\_theory, [E]). Capstone of the icosahedral round: the singularity  $x^2+y^3+z^5$ , whose exponents are the atoms ( $|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}})=(2,3,5)$ , has Milnor number  $\mu=(2-1)(3-1)(5-1)=8=\text{rank } E_8$  (a *fifth* origin of the “8”), Milnor monodromy = the order-30  $E_8$  Coxeter element (eigenvalues the primitive 30th roots = the  $E_8$  exponents, charpoly  $\Phi_{30}$ ), Milnor lattice  $E_8$ , link the Poincaré sphere; the two clocks are facets of this one monodromy — the  $\mu_3$  triality (CP, v233) is  $h^{10}$  in  $\langle h \rangle = \mathbb{Z}/30$ , the  $\mu_4$  clock (v223) is the order-4 Galois automorphism of  $\mathbb{Q}(\zeta_{30})$  (TOPO.BRIESKORN.01).
- **The closing step as a physical condition: no topological degeneracy  $\Leftrightarrow \det K=1 \Leftrightarrow$  the seam is SRE** (v237\_seam\_sre\_closure, tfpt\_research\_contracts, [E]/[O]). Sharpens GATE.HOLO.02’s residual from definitional to falsifiable: the  $K$ -matrix ground-state degeneracy on a genus- $g$  surface is  $|\det K|^g$  (torus: carrier 16,  $D_8$  4,  $E_8$  1), so “no topological degeneracy on any closed surface”  $\Leftrightarrow \det K=1$ ; among rank-8 even positive-definite  $K$  ( $c=8=g_{\text{car}}+N_{\text{fam}}$ ) only  $E_8$  is short-range-entangled (the Kitaev  $E_8$  phase). A unique vacuum on the plane (RP+gap+cluster) is necessary but not sufficient (the plane cannot see  $\det K$ ); so the one open step is the physical statement “the free RP seam is SRE (no topological order)” (GATE.HOLO.03).

## 2026-06-16 (the icosahedral bedrock, CM-norm duality, and a structural-finds round)

- **McKay bedrock: why the atoms are  $\{2,3,5\}$**  (v219\_icosahedral\_mckay, origin\_theory, [E]).  $E_8$  is the exceptional *top* of the McKay tower of finite  $SU(2)$  subgroups ( $2T \rightarrow \hat{E}_6$ ,  $2O \rightarrow \hat{E}_7$ ,  $2I \rightarrow \hat{E}_8$ ). Built *from the group*: the 120 unit-quaternion icosians close to the binary icosahedral group  $2I$  (element orders  $\{1,2,3,4,5,6,10\}$  — it contains the 2,3,5 rotation-axis orders), 9 conjugacy classes; the 9 irreducible character degrees (Dixon, from the class algebra) are  $\{1,2,2,3,3,4,4,5,6\}$ , sum  $30=h(E_8)$ , sum of squares  $120=|R^+(E_8)|$ ; the McKay adjacency from the natural 2-dim rep *is* the affine  $E_8$  graph with Kac marks = the degrees. A backward certificate of the closed  $E_8$ , not a P2 proof (MCKAY.E8.01).
- **CM-norm duality: 41 (square) and 7 (hex)** (v222\_cm\_norm\_duality, origin\_theory, [E]). The square seam (Gaussian  $\mathbb{Z}[i]$ ,  $j=1728$ ) gives  $N(g_{\text{car}}+i|\mu_4|)=5^2+4^2=41=10b_1$  (the EM index *is* the Gaussian norm of the carrier-glue vector) and  $N(3+2i)=13=\Delta_Q$ ; the hexagonal partner (Eisenstein  $\mathbb{Z}[\omega]$ ,  $j=0$ ) gives  $N(3+2\omega)=7=\text{scalaron}$ . The (3,2) split generates (5,6,7,13) by sum/product/Eisenstein/Gauss norm; rings rigid (CMNORM.DUAL.01).

- **The  $\mu_4$  clock is the order-4 character of the  $E_8$  Coxeter cycle (v223\_coxeter\_totative\_clock, origin\_theory, [E]).**  $(\mathbb{Z}/30)^\times$  are the  $E_8$  exponents; the order-4 generator 7 ( $\langle 7 \rangle = \{1, 7, 13, 19\}$ ) is a literal  $\mu_4$  clock permuting the four conjugate planes; only  $E_8$  has  $\varphi(h) = \text{rank}$  and  $h = 2 \cdot 3 \cdot 5$  (COX.CLOCK.01).
- **248=120+128 as a magnitude/phase channel typing (v227\_degree\_exponent\_channel\_split, tfpt\_5\_redteam, [E]).** Exponents sum  $120 = |R^+(E_8)|$  (magnitude channel), degrees sum  $128 = \text{rank} \cdot \dim S^+$  (phase/glue channel); the red-team Target D magnitude bijection lives in 120, the un-covered CP phases in 128; the dark-sector reading of 128 stays Frontier (replaces the over-strong  $S^-$  reading; E8.CHAN.01).
- **P2 as the mode space of a degree-4 seam divisor (v228\_rr\_index\_gate, origin\_theory, [E]/[C]/[O]).** A Riemann–Roch index gate:  $\deg D = 4 = |\mu_4|$  on  $\mathbb{P}^1$  forces  $h^0 = 5 = g_{\text{car}}$ ,  $\text{rank } H_1 = 3 = N_{\text{fam}}$ ,  $h^0 + H_1 = 8 = \text{rank } E_8$ ,  $\Lambda^{\text{even}}(\mathbb{C}^5) = 16 = \dim S^+$  — bedrock language for P2, conditional on the seam producing the degree-4 divisor (QGE0.RR.01).
- **The seam as a finite recoverability code (v221\_seam\_qecc, origin\_theory, [E]/[C]).** Code dimension  $\dim S^+ = 16$ ; the gapped transport is a CPTP doubly-stochastic contraction with spectrum  $\{1, (2/3)^6, (1/3)^6\}$ , recovery bound  $\|T^n \delta\| \leq (2/3)^{6n} \|\delta\|$  (Knill–Laflamme type); free ratio breaks it. Holographic/Petz reading [C], gated on the Seam–Horizon theorem (QEC.SEAM.01).
- **CP lives in the hexagonal phase fiber (v220\_cp\_hexagonal\_modulus, tfpt\_5\_redteam, [E]/[C]).** The seam deck is the square modulus  $j = 1728$  ( $\mathbb{Z}/4$ ); its partner is the hexagonal  $j = 0$  ( $\mathbb{Z}/6$ ,  $\arg \rho = \pi/3$ ). The frozen  $\delta = \pi/3 + 3\lambda_C^2 = 68.654^\circ$  has leading term  $\pi/3 = \arg \rho$ ; the deck stays  $\mathbb{Z}/4$ , CP is the  $\mathbb{Z}/6$  fiber (CP.MOD.01).
- **The dual normal frame  $(d, n)$  with CP as orientation (v225\_dual\_normal\_frame, tfpt\_5\_redteam, [E]/[C]).**  $d = a^\top R^{-1} = (-\frac{1}{2}, -\frac{1}{2}, 1) = -\frac{1}{2}$  Nariai;  $n = (5, -9, 6)$  reads det on the first-generation column;  $\det(1, d, n) = 21 = 3 \cdot 7$ ;  $\text{Im } \det(1, d, \rho n) = 21 \sin(\pi/3)$  — CP as orientation (DUAL.FRAME.01).
- **$F_{\text{transfer}}$  as one path on the Sheet Diamond (v224\_diamond\_ftransfer\_path, tfpt\_2\_standard\_model, [E]/[C]).**  $M(s, t) = R + Q \text{diag}(s, t, t)$ :  $K, C, F$  on the sheet axis (curved, 2nd diff 8), the winding axis flat (slope 6), Plücker steps  $(1, 8, 10), (1, 8, 16)$ ; the Koide source  $\rightarrow$  pole is the 1-D projection (FTR.PATH.01).
- **The charged leptons as an étale Frobenius algebra (v229\_lepton\_frobenius\_algebra, tfpt\_2\_standard\_model, [E]/[C]).**  $(c_e, c_\mu, c_\tau) = (\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$ , ring closure  $c_e c_\tau = \frac{8}{3} = |\mathbb{Z}_2| c_\mu$ , product  $\frac{32}{9}$ ;  $A = \mathbb{Q}[t]/(m)$  is étale (trace-form Gram nondegenerate), a commutative Frobenius algebra over the  $\mu_6$  resolvent (LEP.FROB.01).
- **The center budget  $(7, 11, 13)$  as three local norms (v230\_center\_budget\_norms, tfpt\_2\_standard\_model, [E]).**  $C = R + Q \text{diag}(1, 0, 0)$  row sums  $(7, 11, 13)$ :  $7 = N_{\mathbb{Z}[\omega]}(3 + 2\omega)$  (hex),  $13 = N_{\mathbb{Z}[i]}(3 + 2i)$  (square),  $11 = \binom{4}{0} + \binom{4}{1} + \binom{4}{2}$  (QBL Fock count); the two CM rings and the boundary count meet in one operator center (CENTER.NORM.01).

## 2026-06-15 (the diamond axis geometry — two axes + the transfer corner)

- **The Sheet Diamond is a discrete geometry with two axes (v218\_diamond\_axis\_geometry, tfpt\_2\_standard\_model, [E]).** Sharpens v94/v95 with three exact [E] lemmas (no new numbers). **(1) Axis curvature:** along the centered cross the determinant is linear on the winding axis  $\det(C + xU) = 14 + 6x$  (slope  $6 = |R^+(A_3)|$ ) and quadratic on the sheet axis  $\det(C + yV) = 14 + 14y + 4y^2$  (2nd difference  $8 = \text{rank } E_8$ ); the anchor block has  $\det B(C + xU) = 2 \det(C + xU)$  and sheet curvature  $6 = |R^+(A_3)|$  — two curvatures  $(8, 6) = (\text{rank } E_8, |R^+(A_3)|)$ . **(2) Plücker transfer ladder:**  $\text{Pl}(K) \rightarrow \text{Pl}(C) \rightarrow \text{Pl}(F)$  lifts in steps  $(1, 8, 10) = (N_\Phi, \text{rank } E_8, A_\Lambda)$  then  $(1, 8, 16) = (N_\Phi, \text{rank } E_8, \dim S^+)$  — the

decouple then the full generation (identity [E], source→pole reading [C]). **(3) Spectral ramification:** for  $Q, K, C, F$  the cubic discriminant factors as  $q(r)^2 \text{Disc}(q)$  with squares  $(1, 3, 4, 6)$  and kernels  $(13, 48, 65, 105)$ ,  $F$  carrying  $105 = 3 \cdot 5 \cdot 7$ . The anchor-defect spectrum and pair-sum/staircase blocks are honest *audit* fingerprints ( $28 = 2 \dim G_2$ ,  $52 = \dim F_4$  stay audit-only, not spine, as in v94/v95);  $F$  as the transfer-completion corner is a *heuristic*, not a hard filter. Ledger DIAMOND.AXIS.01, DIAMOND.PLUCKER.01, DIAMOND.SPECTRAL.01; Wolfram-mirrored (249/249).

## 2026-06-15 (QGEO: the four marks emerge from Gauss–Bonnet — K1 executed)

- **The four marks emerge from Gauss–Bonnet — count derived, not assumed** (v216\_marks\_gauss\_bonnet, tfpt\_research\_contracts, [E]; ledger QGEO.MARKS.03). The K1 lever of the v215 kill-test (“exactly four marks”) is *executed*: if the marks are  $\mathbb{Z}_2$  branch points (deficit  $\pi$ ) on the flat seam sphere ( $\chi = 2$ , the same  $\chi$  as the 8 in  $c_3$ ), Gauss–Bonnet forces  $n\pi = 2\pi\chi = 4\pi$ , so  $n = 2\chi = 4 = |\mu_4| = N_{\text{fam}} + 1$ . The closed Euclidean sphere 2-orbifolds are exactly  $(2, 3, 6)$ ,  $(2, 4, 4)$ ,  $(3, 3, 3)$ ,  $(2, 2, 2, 2)$ , and three independent criteria — all cone orders = 2 (the  $|\mathbb{Z}_2|$  sheet branch), rank  $H^1 = \# \text{marks} - 1 = 3 = N_{\text{fam}}$  (the three generations pick the 4-mark square over the 3-mark hexagonal), and a free order-4 deck — converge on the pillowcase  $(2, 2, 2, 2)$ . The numerical sibling v217\_marks\_emergence\_scan ([C], ledger QGEO.EMERGE.01) confirms it on the actual DtN/state (number  $n$  and positions free, the 4-square config is the unique joint minimiser of clock-invariance + 3-family cohomology + gap  $(2/3)^6$ ). The mark *count* and *equality* move from input to derived; only the square *modulus* (cross-ratio 2, the order-4 clock) stays the one carrier input, and the from-zero raw-seam emergence QGEO.REALIZE.01 stays [O]. v216 exact (Wolfram extension → 248/248); v217 numerical (Python-only). Suite → 217 modules.

## 2026-06-15 (QGEO: the bedrock recast as a falsifiable kill-test)

- **QGEO.SYM.01 as a kill-test, not a proof-promise** (v215\_seam\_deck\_killtest, tfpt\_research\_contracts, [E]/[O]; ledger QGEO.KILL.01). A foundational symmetry cannot be derived from nothing (Bisognano–Wichmann would presuppose the covariance it should produce), so the bedrock  $\omega \circ \rho = \omega$  is recast as four independently-falsifiable predictions about the raw seam DtN. The non-circular reduction (v198/v199/v201) gives  $\omega \circ \rho = \omega \Leftrightarrow [\rho, C] = 0$  with  $|k|$  commuting exactly, so the residual is the bounded sub-principal  $M_f$  (equivalently the raw Gaussian bulk form  $A$  is  $\mu_4$ -deck-invariant). Levers: **K1** exactly four marks ( $b_1 = \# \text{marks} - 1 = 3 = N_{\text{fam}} \Rightarrow \# \text{marks} = 4 = |\mu_4|$ ); **K2** the square config (cross-ratio 2  $\Rightarrow j = 1728$ , Aut =  $\mathbb{Z}/4$ ; order 4 selects  $\mathbb{Z}/4$  not the  $j = 0$   $\mathbb{Z}/6$ ; generic  $\Rightarrow \mathbb{Z}/2$ ); **K3** mod-4 sub-principal support ( $[\rho, M_f] = 0 \Leftrightarrow f$   $\mathbb{Z}_4$ -invariant); **K4** transfer gap  $(2/3)^6 = 64/729$ . Freeze-bound via freeze\_file.csv (seam\_deck\_symmetry), carrier-side [E] (regression guards), with the *decisive* kill deferred to QGEO.REALIZE.01 [O] (the raw collar DtN must *produce* four square marks without inputting them — an emergence test). A kill-test, not a closure. Python-only (falsification layer; the exact sub-parts  $j = 1728$  and  $(2/3)^6$  are Wolfram-mirrored via v214/v54/v56). Suite → 215 modules.

## 2026-06-15 (QGEO: the pillowcase reduction — two residuals become one canonical metric)

- **Pillowcase reduction of QGEO.SYM.01** (v214\_seam\_pillowcase, tfpt\_research\_contracts, [E]/[O]; ledger QGEO.PILLOW.01). The two separately-tracked open QGEO residuals — QGEO.ISO.01 (v180: the carrier clock is an order-4 *isometry* of the seam metric) and QGEO.SUBPRIN.01 (v201: the DtN sub-principal symbol is *mark-local*, the seam flat away from the  $\mu_4$  marks) — are shown to be *one* statement: the seam carries the uniformising flat orbifold (*pillowcase*) metric. **(1)** The four marks make the Euclidean orbifold  $S^2(2, 2, 2, 2)$

( $\chi_{\text{orb}} = 2 - 4(1 - \frac{1}{2}) = 0$ ), so by Troyanov uniformisation the metric is flat (unique up to scale) with curvature only at the cone points (Gauss–Bonnet  $4\pi = 2\pi\chi$ ) — this *is* the v201 “flat away from the marks”. **(2)** New link: cross-ratio( $\mu_4$ ) = 2 (v168)  $\Rightarrow j = 1728$  ( $j(\lambda) = 256(\lambda^2 - \lambda + 1)^3/(\lambda^2(\lambda - 1)^2$ ); all six harmonic cross-ratios give 1728)  $\Rightarrow$  the *square* modulus  $\tau = i$ , whose CM by  $\mathbb{Z}[i]$  *derives* the order-4 clock  $z \mapsto iz$  (explicit  $(x, y) \mapsto (-x, iy)$  on  $y^2 = x^3 - x$ ); generic cross-ratio gives  $j \notin \{0, 1728\}$  and only  $\mathbb{Z}/2$ . **(3)** Unification: “seam = uniformising flat pillowcase metric” implies *both* v201 and v180, so two residuals collapse to one canonical-metric premise. **[O]** It does *not* close QGEO.SYM.01: the choice “RP seam metric = the constant-curvature representative” stays open, milder and more canonical than the bare isometry premise. Wolfram-mirrored (the exact orbifold/ $j$ -invariant/CM arithmetic); Klein- $J$  values mpmath-numerical. Suite  $\rightarrow$  214 modules.

## 2026-06-15 (the $F_{\text{transfer}}$ transfer functor contract CONTRACT.F.01)

- $F_{\text{transfer}}$  formalised as a typed functor with four axioms (v213\_ftransfer\_functor, tfpt\_research\_contracts, **[C]**). The four frontier transfers are consolidated into ONE functor  $F_{\text{transfer}} = F_{\text{observable}} \circ F_{\text{threshold}} \circ F_{\text{RG}}$ , each instance a discrete compiler kernel **[E]** composed with an external continuous solver **[C]**: **(1)**  $\mu_4$ -deck equivariance (the Koide multiplier  $\lambda_2 = (2/3)^6 = 64/729$  is the deck transfer eigenvalue); **(2)** Plücker preservation ( $53 = a^T(R+Q)\mathbf{1}$ ,  $\text{Pl}(F) = (1, 22, 30)$ ,  $\|\text{Pl}(K)\|_1 = 11$ ); **(3)** positivity/stochasticity ( $\text{spec } T$  positive,  $\kappa_f \in (0, 1)$ ); **(4)** external modules explicit ( $b_3 = -7$  a carrier output). The four instances are  $F_{\text{pole}}$  (v183),  $F_{\text{Boltzmann}}$  (v212),  $F_{\text{relic}}$  (v211),  $F_{\text{QCD}}$  (v164). CONTRACT.F.01 is the third research contract alongside ( $U_{\text{wall}}$ ) and ( $G_{\text{metric}}$ ), the structural complement of the v187 typing guard — an architectural consolidation, *not* a closure of any frontier number. Ledger CONTRACT.F.01.

## 2026-06-15 (frontier: two sharper **[C]** scenario branches — axion angle + leptogenesis)

- **Axion DM spine-angle branch**  $\theta_i = \pi N_{\text{fam}}/g_{\text{car}} = 3\pi/5$  (v211\_axion\_spine\_angle, tfpt\_4\_frontier, **[C]**). A more *robust* misalignment angle than the determinant-line hilltop  $\theta_i \approx 170^\circ$  (v185, which over-produces  $\Omega_a h^2 \approx 0.66$ ): the central spine quotient gives  $\theta_i = 3\pi/5 = 108^\circ$  exactly (no fit), the finite- $T$  solver reaches  $\Omega_{\text{DM}}$  at  $\approx 106^\circ$ , and  $108^\circ$  sits in the *mild*-anharmonic regime ( $62^\circ$  below the hill-top) — not exponentially sensitive. An alternative ansatz to  $\theta_i = \pi(1 - \varphi_\Sigma)$  (mutually exclusive, the full solver decides, DM.AXION.SPINE.01); **[X]**  $\Omega_a h^2$  outside  $\sim [0.08, 0.16]$  demotes it. A sharper scenario, not a derivation.
- **Leptogenesis scalaron-decuple branch** (v212\_leptogenesis\_decuple, tfpt\_4\_frontier, **[C]**). Both Boltzmann inputs share the decuple  $A_\Lambda = 10 = |E(K_5)|$ :  $M_1 = M_{\text{scal}}(\varphi_0^{\text{ret}})^2/A_\Lambda \approx 8.65 \times 10^9$  GeV and  $\tilde{m}_1 = m_3/A_\Lambda \approx 5$  meV.  $M_1$  uses *only*  $\{M_{\text{scal}}, \varphi_0^{\text{ret}}, A_\Lambda\}$  — no hidden seesaw scale, cleaner than v184’s  $M_R \varphi_0^4$  — and lands in the thermal window where  $\eta_B \approx 1.2 \times 10^{-9}$  brackets the observed  $6.1 \times 10^{-10}$ . The coupling mechanism is posited not derived, so  $\eta_B$  stays **[C]**; **[X]** a precise Boltzmann/RGE solve outside the band falls the route, not the theory (FR.ETAB.04; Route A independent).

## 2026-06-15 (Lean: QGEO.COHO.M.01 character grading + MODULE parity machine-proved)

- **The cohomology character grading is now Lean-formalised** (CohomologyGrading.lean, ledger FORM.QGEO.03, **[O]**). A new module machine-proves the COHO.M node of the QGEO proof tree (v177; AUDIT: PASS, no sorry/admit, axioms propext, Classical.choice, Quot.sound only): the three  $H^1(\mathbb{P} \setminus \mu_4)$  eigenforms  $\omega_k = z^{k-1}/(z^4 - 1)$  ( $k=1, 2, 3$ ) have pullback character  $\rho^* \omega_k = i^k \omega_k$  under the clock  $\rho: z \mapsto iz$  (each an exact rational-function identity, the denominator clock-invariant since  $i^4=1$ ), so the grading is  $(i, -1, -i)$  with weights  $(1, 2, 3) =$  the  $A_3$  exponents =  $\text{Spec}(Q_+)$ , rank 3 =  $N_{\text{fam}}$ . The MODULE *parity* is also

proved: the reflection  $\sigma : z \mapsto 1/z$  satisfies  $\sigma^*\omega_1 = \omega_3$ ,  $\sigma^*\omega_2 = \omega_2$ ,  $\sigma^*\omega_3 = \omega_1$  (swap  $\omega_1 \leftrightarrow \omega_3$ ,  $\omega_2$  fixed). Signature-locked in `AuditContract.lean`. The geometric COHOM/MODULE-parity nodes are now [E]; the MARKS/KERNEL obligations and the MODULE *uniqueness* stay [O].

## 2026-06-15 (Lean: QGEO.UNIFORM.01 geometric normal form machine-proved)

- **The Möbius uniformisation normal form is now Lean-formalised** (Möbius Uniformisation.lean, ledger FORM.QGEO.02, [O]). A new module machine-proves the constructive UNIFORM node of the QGEO proof tree (v177; AUDIT: PASS, no `sorry/admit`, axioms `propext`, `Classical.choice`, `Quot.sound` only): the clock  $\rho : z \mapsto iz$  has  $\rho^4 = \text{id}$  and  $\rho^2 \neq \text{id}$  (order exactly 4); the reflection  $\sigma : z \mapsto 1/z$  is an involution with  $\sigma\rho\sigma = \rho^{-1}$  (so  $\langle \rho, \sigma \rangle$  is the dihedral  $D_4$  of order 8); a non-fixed orbit  $\{a, ia, -a, -ia\}$  scales to  $\mu_4 = \{1, i, -1, -i\}$ ;  $\sigma$  permutes  $\mu_4$  (fixes 1,  $-1$ , swaps  $i, -i$ ); and the multiplier classification “ $z \mapsto \zeta z$  has order exactly 4  $\Leftrightarrow \zeta = \pm i$ ”. Signature-locked in `AuditContract.lean`. This formalises the geometric half of the seam realisation *given* the four marks and the clock; the raw-seam marking obligation QGEO.MARKS.01 stays [O], *not* closed.

## 2026-06-15 (Lean: the QGEO.SYM.01 conditional theorem machine-proved)

- **The mark-local  $\Rightarrow \omega \circ \rho = \omega$  implication is now Lean-formalised** (SeamDeckClosure.lean, ledger FORM.QGEO.01, [O]). A new module in the carrier-rigidity Lean project machine-proves the algebraic core of v201/v210 (AUDIT: PASS, no `sorry/admit`, no domain axioms — `#print axioms = propext`, `Classical.choice`, `Quot.sound` only): (i) the 4th-root character orthogonality  $\sum_{j<4} \zeta^j = (\text{if } \zeta=1 \text{ then } 4 \text{ else } 0)$  for  $\zeta^4=1$  (so a  $\mu_4$ -mark sum is supported only on modes  $\equiv 0 \pmod{4}$ ); (ii) the clock generator  $-i$  has order 4; (iii) `markLocal_blockDiagonal`: a mark-local Toeplitz symbol connects only equal clock-characters,  $[\rho, M_f] = 0$ ; (iv) the premise is a typed structure `SeamDeckPremise` (the physical seam DtN *is* mark-local) — consumed, *not* proved, exactly as `CalderonProjector` encodes the Paper-1 analytic input; (v) `SeamDeckPremise.clock_invariant`: *given* the premise, the clock commutes with the DtN, whence  $\omega \circ \rho = \omega$ . So the *implication* is now [E] (formally proved); the *premise* (the physical seam being mark-local) stays [O] — the one fundamental seam-identification postulate, *not* closed.

## 2026-06-15 (QGEO sharpening: mark-local DtN certified on realistic profiles)

- **Mark-local DtN  $\Rightarrow \omega \circ \rho = \omega$ , certified on realistic Steklov profiles and at the state level** (v210\_mark\_local\_dtn, tfpt\_research\_contracts, [O]). A numerical sharpening of QGEO.SUBPRIN.01/v201: a real von-Mises curvature bump summed over the four  $\mu_4$  marks,  $f(\theta) = \sum_{j=0}^3 g(\theta - j\pi/2)$ , builds the Steklov DtN  $\Lambda = |D_\theta| + M_f$  whose off-(mod 4) Fourier support vanishes to  $< 10^{-16}$ ,  $\|[\rho, \Lambda]\| < 10^{-15}$ , and — the genuinely new step — the quasi-free covariance  $C = \frac{1}{2}(1 + \text{sgn } H_1)$  is clock-invariant ( $\rho C \rho^\dagger = C$  to  $< 10^{-13}$ ), i.e. the actual  $\omega \circ \rho = \omega$ . Negative controls (single off-centre bump,  $\mathbb{Z}_3$  source, four generic marks) break both by  $O(1)$ ; convergence stable for  $N \in \{16, 32, 64\}$ . This certifies the *implication*; the premise that the physical seam *is* mark-local stays [O] (it does *not* close QGEO.SYM.01; net existence and full-cone RP remain the [E] of v175). Ledger QGEO.MARKLOCAL.01. Python-only (the exact mark-sum identity is Wolfram-mirrored via v201).

## 2026-06-15 (archive integration #5: six dormant results revived from the old papers)

- **Muon  $g-2$  as a seam vertex** (v204\_muon\_seam\_g2, tfpt\_4\_frontier, [C]). The carrier’s second-order topological defect  $\delta_2 = \frac{5}{4}\delta_{\text{top}}^2$  projected through the seam-loop phase gives  $a_\mu^{\text{seam}} = \delta_2/(2\pi) = 45/(524288\pi^9) \approx 2.879 \times 10^{-9}$  ( $0.81\sigma$  vs the dispersive  $\Delta a_\mu$ ;  $\sim 1.5\sigma$  vs

lattice/CMD-3). The value is exact [E]; the vertex identification is the [C] bridge. Ledger FR.MUONG2.01; Wolfram-mirrored.

- **An independent gravitational 3/4 (v205\_xi\_threequarter, tfpt\_4\_frontier, [E]/[C]).**  $\xi = c_3/\varphi_{\text{tree}} = 3/4$  exactly (Einstein limit of  $\kappa^2 = \xi\varphi_0^{\text{ret}}/c_3^2$ ) — the same  $3/4 = q(A_3) = \ln(m/\mu)$  as the seam-determinant replica (v152), an over-determination. Ledger GRAV.XI.01; Wolfram-mirrored.
- **Quantitative  $H_0$  from the  $\Lambda$  branch (v206\_h0\_lambda\_branch, tfpt\_4\_frontier, [C]).**  $\delta_\Sigma = 48 e^{-2\pi\alpha^{-1}/3} \approx 1.085 \times 10^{-123}$ , the  $(8\pi)^2$  Planck-convention split, and  $H_0 = 66.5\text{--}67.1$  km/s/Mpc (below Planck, far below SH0ES — the Hubble tension is *not* relieved). Ledger GRAV.H0.01.
- **Asymptotic Safety as a second QG route (v207\_asymptotic\_safety, tfpt\_4\_frontier, [O]).** A complementary continuum-FRG argument (a non-Gaussian UV fixed point from the same axioms;  $y^2 = 16c_3^2 = 1/(4\pi^2)$ ) — a plausibility cross-check, *not* load-bearing, does *not* close  $G_{\text{metric}}$ . Ledger GRAV.ASYMP.01.
- **Black-hole thermodynamics from the seam (v208\_bh\_thermodynamics, tfpt\_horizon\_readouts, [E]/[C]).** Modular  $T_H = \kappa/(2\pi)$  (the  $2\pi = 1/(4c_3)$  seam unit, ties to v198/v199) and the scalaron-corrected Wald entropy  $S_W = (A/4G)(1 + R_h/3M_s^2)$  (exact from v28). Ledger HOR.BHTHERMO.01; Wolfram-mirrored.
- **The black hole as a seam fixed point (v209\_bh\_regular\_core, tfpt\_horizon\_readouts, [C]).** No singularity ( $\varphi \rightarrow \varphi_*$  at the gapped attractor  $(2/3)^6$ ); the maximal BH = the anchor (Nariai roots  $(1, 1, -2)$ ); information via Page recovery; a Planck-scale holographic floor. The superseded RN/torsion metric is *not* resurrected. Ledger HOR.BHCORE.01.

## 2026-06-15 (archive integration #4: Möbius $\cong$ Lorentz foundation citation)

- **The Möbius seam is the boundary Lorentz structure (tfpt\_3, [O]).** Added a paragraph to the “Möbius origin of  $\pi$ ” section:  $\text{Möb}(\hat{\mathbb{C}}) \cong \text{PSL}(2, \mathbb{C}) \cong \text{SO}^+(3, 1)$  (Penrose & Rindler, *Spinors and Space-time*, Vol. 1, §1.2–1.4), the celestial sphere of null directions via stereographic projection. This gives the self-consistency loop a spacetime meaning — the carrier clock  $z \mapsto iz$  is an elliptic  $\text{PSL}(2, \mathbb{C})$  rotation (v180) and the RP→OS causal cone is the same Lorentz structure — so “Möbius origin of  $\pi$ ” and “one Lorentzian cone” are two faces of one boundary  $\text{PSL}(2, \mathbb{C})$ . A foundation citation, not a new claim ( $\pi$  stays primitive).

## 2026-06-15 (archive integration #3: axion coupling $g_{a\gamma\gamma} = -4c_3 +$ retired small-angle reading)

- **Haloscope coupling tied to  $c_3$  (tfpt\_4\_frontier, [C]).** Made explicit: in the determinant-line normalization the axion–photon anomaly coefficient is  $g_{a\gamma\gamma} = -4c_3 = -1/(2\pi)$  ( $y^2 = 16c_3^2 = 1/(4\pi^2) \approx 0.0253$ ), so the haloscope  $F\tilde{F}$  coupling is set by the *same*  $c_3$  that fixes  $\alpha$  and  $\beta_{\text{rad}} = \varphi_0/4\pi$  — a [C] structural relation (the coefficient, not a parameter-free physical  $g_{a\gamma\gamma}$ ). The earlier small-angle  $\theta_i = \varphi_0$  reading (old determinant-line note, a much higher axion mass) is *superseded* by the closed near-hilltop  $\theta_i = \pi(1 - \varphi_{\text{seam}}) = 170.4^\circ$ , recorded as a retired reading and guarded by v188 (a new sentinel forbids that stale axion mass from re-entering the active docs).

## 2026-06-15 (archive integration #2: the EHT achromatic polarization intercept — v203)

- **HOR.EHT.01 (v203, [E]/[C]/[X]).** The surviving core of the old UFE/black-hole notes (*not* the RN-type metric — superseded by the Nariai/seam=horizon readouts v101–v104/v190; only the polarization signature survives): a local achromatic polarization-rotation intercept around a

horizon-scale black hole,  $\beta_{\text{BH}}(r) = 16c_3^4 Q_e^{\text{eff}} Q_m^{\text{eff}} / r^2$ . The coupling  $16c_3^4 = 1/(256\pi^4) = \delta_{\text{top}}/3$  is *exact* ([E], Wolfram-mirrored) — the *same* top-form coefficient  $\delta_{\text{top}} = 48c_3^4$  that fixes the  $\alpha$ -kernel precision-zone correction, one row from the cosmic-birefringence seed  $\beta_{\text{rad}} = \varphi_0/4\pi$ . TFPT fixes the  $1/r^2$  shape, achromaticity and sign-flip ([C]); the amplitude  $Q_e Q_m$  is an MHD/GR weight, never a pixel prediction. **Dated kill test [X]**: the residual intercept must pass three nulls (frequency, spatial  $1/r^2$ , sign-flip) after honest GRMHD subtraction. Integrated into `tfpt_horizon_readouts` (new EHT section), the ledger, registry, Wolfram (243/243), and the website (`predictions.ts`). Suite 202 modules.

## 2026-06-15 (archive integration #1: the rare kaon sector $K \rightarrow \pi \nu \bar{\nu}$ — v202)

- **FR.RAREKAON.01 (v202, [C])**. The closed TFPT CKM point ( $s_{12}=\lambda_C$ ,  $s_{23}=\varphi_0/(1+\lambda_C)=|V_{cb}|$ ,  $s_{13}=\lambda_C^3/3$ ,  $\delta=\pi/3+3\lambda_C^2=1.19823$  rad; v18/v88, no flavor fit) feeds the cleanest FCNC probe. With standard external short-distance input (Brod–Gorbahn–Stamou  $X_t, P_c, \kappa_{\pm}$ ) it gives  $\text{BR}(K^+ \rightarrow \pi^+ \nu \bar{\nu}) = 9.45 \times 10^{-11}$  and  $\text{BR}(K_L \rightarrow \pi^0 \nu \bar{\nu}) = 3.33 \times 10^{-11}$  — a [C] downstream flavor readout, *not* a compiler power (the SD functions are external). Recomputed from the *current* CKM point (an earlier archive scaffold quoted  $\text{BR}(K_L) = 3.47 \times 10^{-11}$  from an  $\Im(\lambda_t)/\lambda^5$  slip, corrected here). Grossman–Nir ratio  $0.352 \ll 4.3$ ; NA62  $(13.0^{+3.3}_{-3.0}) \times 10^{-11}$  is  $\sim 1.2\sigma$  above. **Dated kill test [X]**: a stable NA62  $\text{BR}(K^+)$  outside  $[7, 12] \times 10^{-11}$ , or a KOTO-II  $\text{BR}(K_L)$  off the GN point, breaks the flavor bridge (core untouched). Integrated into `tfpt_2_standard_model` (§rare kaon), the ledger, v187 guard (new FR.RAREKAON. family), registry, and the website (`predictions.ts`). Python-only. Suite 201 modules.

## 2026-06-15 (bedrock: the $\omega \circ \rho = \omega$ residual reduced once more — v201)

- **QGE0.SUBPRIN.01 (v201, [E]/[O])**. The v199 bedrock residual — “the bounded sub-principal symbol of the raw seam DtN has no off-character (mod 4) matrix elements” — is reduced one genuine step further: block-diagonality is *not* an independent postulate. Writing the DtN as  $\Lambda = |k| + M_f$  with  $M_f$  the bounded sub-principal piece = multiplication by a curvature  $f(\theta)$  (standard Steklov structure),  $M_f$  is  $\mu_4$ -character-block-diagonal iff  $f$  is  $\mathbb{Z}_4$ -invariant ([E]; principal  $|k|$  already commutes, v198); and a  $\mu_4$ -mark sum  $f = \sum_{j=0}^3 g(\theta - 2\pi j/4)$  has  $f_m = 4g_m[m \equiv 0 \pmod{4}]$  for any profile  $g$ , hence is automatically  $\mathbb{Z}_4$ -invariant ([E]). So the  $\mu_4$ -mark orbit (forced by v195) *forces* the sub-principal symbol block-diagonal; 3 marks ( $\mathbb{Z}_3$ ) and 4 generic marks break it (negative controls). The residual collapses to “the DtN sub-principal symbol is mark-local (seam flat away from the  $\mu_4$  marks = conformal deck, v177)”, structurally definitional — the narrowest form yet of QGE0.SYM.01, [O]. Registered in `run_all.py`, the ledger, `tfpt_research_contracts`, `origin_theory`; Wolfram-mirrored (the mark-sum Fourier identity is exact; extension now 242/242). Suite 200 modules.

## 2026-06-15 (F\_transfer workshop: the axion relic solver — an honest over-production tension)

- **Axion relic solver (experiments/ftransfer/axion\_relic/relic\_solver.py, [C])**. The first real  $F_{\text{transfer}}$  pipeline computation (work package C / review §6.2): a standard misalignment relic with a numerically-computed anharmonic factor  $F(\theta_i) \approx 4$  at the hilltop. Honest finding: at the *predicted*  $\theta_i \approx 170^\circ$  the estimate *over*-produces dark matter,  $\Omega_a h^2 \sim 0.6\text{--}2.3$  (robustly  $> 0.12$ ), because the large  $\theta_i^2 \approx 8.8$  times  $F$  overshoots. So the determinant-line axion as the *dominant* DM is in *mild tension* (it would prefer  $\theta_i \sim 120\text{--}130^\circ$  or extra dilution) — this *refines* v185’s optimistic “can be all-DM” toward an over-production tension. The near-hilltop regime is  $O(1)$ -uncertain, so a full finite- $T$  solve decides; over-production kills only the axion-as-dominant-DM *branch*, not the compiler. Reflected in `tfpt_4_frontier` (dark-matter section) and the v185 docstring; experiments-only, not a suite/ledger claim. No refitted exponents;  $\theta_i = \pi(1 - \varphi_{\text{seam}})$  only.

- **Full finite- $T$  solve confirms it (axion\_relic/full\_finiteT\_solve.py).** A converged nonlinear misalignment solve  $(\theta'' + (3 + d \ln H/dN)\theta' + (m_a(T)/H)^2 \sin \theta = 0$ , lattice  $\chi(T) \propto T^{-8.16}$ , realistic  $g_*(T)$ ; normalised so  $\theta_i=1 \rightarrow \Omega_a h^2 \approx 0.03$ , the standard value) gives  $\Omega_a h^2 \approx 0.66$  at the predicted  $\theta_i \approx 170^\circ$  —  $\sim 5.5\times$  above  $\Omega_{\text{DM}} = 0.12$ , reached only at  $\theta_i \approx 106^\circ$ . The over-production is thus *confirmed* (not the optimistic “all-DM”); `tfpt_4_frontier`, v185 docstring and the website updated accordingly. Stays [C]; kills only the axion-as-dominant-DM branch.

## 2026-06-14 (work packages A+B: attacking $\omega \circ \rho = \omega$ + the variational scan — v199, v200)

- **v199\_seam\_state\_invariance.py [E]/[O] (claim QGEO.STATE.01, work package A).** Attacking  $\omega \circ \rho = \omega$  directly on the *raw* seam. (1) [E] the setup is non-circular:  $\rho$  is the Coxeter element of  $W(A_3) = S_4$  ( $\rho^4 = 1$ , v117), an automorphism of the raw CAR algebra (16 Majoranas) — both carrier-side, no seam geometry. (2) [E] for a quasi-free RP state  $C_\Sigma = \frac{1}{2}(1 + \text{sgn } H_1)$ , so  $\omega \circ \rho = \omega \Leftrightarrow [\rho, H_1] = 0$ . (3) [E]  $[\rho, H_1] = 0 \Leftrightarrow H_1$  is block-diagonal in the four  $\mu_4$ -character classes  $\{n=r \bmod 4\}$  (verified). (4) [O] the residual is sharpened to “the bounded sub-principal symbol has no off-character matrix elements” — a bounded, finite-character condition, not a diffuse geometry; not a closure. Cited in `tfpt_research_contracts`; Wolfram-mirrored.
- **v200\_seam\_variational\_scan.py [C]/[O] (claim QGEO.VARI.02, work package B).** The discretised variational test: with the clock angle free,  $E_{\text{fail}}(\theta)$  has its unique minimum at  $\theta = \pi/2$  (the  $\mu_4$  clock  $z \mapsto iz$ ) — order-4 *and* the three distinct characters  $i, -1, -i =$  the  $(1, 2, 3)$  grading; decoys  $(0, \pi, \pi/3, 2\pi/3)$  all lose. A numerical sign that the variational principle selects  $\mu_4$ ; the full operator-valued minimisation stays the v199 residual [O]. Cited in `tfpt_research_contracts`.
- *Also (this pass, experiments/ only, not the ledger): the  $F_{\text{transfer}}$  four-pipeline scaffold (experiments/ftransfer/) with one CONTRACT per interface, and confirmation that the FRB preregistration (frb\_tfpt\_v1.yaml) and its status semantics already exist. Suite now 199 modules, highest ID v200; Wolfram extension 241/241.*

## 2026-06-14 (cracking the bedrock to a state-invariance: principal symbol + Tomita–Takesaki — v198)

- **v198\_modular\_commutator\_reduction.py [O] (claim QGEO.MODULAR.01).** The decisive crack on the last residual. (1) [E] the DtN principal symbol  $|k| = \sqrt{-\partial_\theta^2} = \text{diag}(|n|)$  and the clock  $\rho: z \mapsto iz = \text{diag}(i^n)$  are both diagonal in the Fourier basis, so  $[\rho, |k|] = 0$  *exactly* on all of  $L^2$  (verified on 33 modes) — the leading-order commutation is free on the whole boundary, never the open part. (2) [E] by Tomita–Takesaki the modular operator is intrinsic to  $(M, \Omega)$ , so a state-preserving symmetry ( $\rho\Omega = \Omega$ ) commutes with  $\log \Delta \sim \Lambda_\Sigma$ : hence  $[\rho, \Lambda_\Sigma] = 0$  *follows from*  $\omega \circ \rho = \omega$ , using only the state + reflection (RP) and *no* conformal covariance — which **removes the v194 Bisognano–Wichmann circularity worry**. (3) [E] the finite  $H^1$  block is already  $\mu_4$ -invariant (v177). So the entire residual collapses to the single *state-invariance* “the raw quasi-free seam state is  $\mu_4$ -invariant” ( $\omega \circ \rho = \omega \Leftarrow$  the Gaussian bulk measure is deck-invariant) — the maximally-operational, non-circular form of QGEO.SYM.01, [O]. *Not* a full closure (a foundational symmetry cannot be derived from nothing, like  $c = \text{const}$ ), but the sharpest falsifiable form, with the circularity gone. Cited in `tfpt_research_contracts`; OpenGates updated. Suite now 197 modules, highest ID v198; Wolfram extension 240/240.

## 2026-06-14 (three review levers: Marks-via-Lefschetz, E\_fail functional, RR→D5 — v195–v197)

- **v195\_marks\_lefschetz\_character.py** [E]/[O] (claim QGEO.MARKS.02). The four seam marks are *forced* (not posited):  $\rho: z \mapsto iz$  fixes only  $\{0, \infty\}$ ,  $H^1 = \chi_1 \oplus \chi_2 \oplus \chi_3$  (the  $A_3$  exponents,  $\text{Tr}(\rho|H^1) = i + i^2 + i^3 = -1$ ) has no trivial component  $\Rightarrow$  no puncture at a fixed point, and  $\text{rank } H^1 = n-1 = 3 \Rightarrow n = 4 \Rightarrow$  a single free  $\mu_4$  orbit (cross-ratio 2). So MARKS reduces from the geometric posit “ $D_4$  marks” to the milder premise “ $H^1$  carries the  $A_3$ -exponent rep” — less circular, still [O]. Cited in `tfpt_research_contracts`.
- **v196\_seam\_energy\_functional.py** [E]/[O] (claim QGEO.VARI.01). The failure functional  $E_{\text{fail}} = \|[\rho, \Lambda_\Sigma]\|_{HS}^2 + \|\rho^4 - 1\|^2 + \|\Theta\rho\Theta - \rho^{-1}\|^2$ , zero locus = the QGEO.SYM.01 conditions; on the finite  $H^1$  block it vanishes at the  $\mu_4$  config [E] and is  $> 0$  for any  $\mu_4$ -breaking perturbation, so  $\mu_4$  is a true minimiser. The full-operator minimisation is the v194 Bisognano–Wichmann residual — a concrete discretisable *target* on the open bedrock, [O]. Cited in `tfpt_research_contracts`.
- **v197\_rr\_carrier\_clifford\_d5.py** [E]/[C] (claim ARCH.RRCAR.02). Hardens the Riemann–Roch carrier from  $g_{\text{car}}$  to  $D_5$  itself:  $\Lambda^{\text{even}}(C_{\text{car}}) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 16 = 2^{g_{\text{car}}-1} = \dim S^+$ , the  $\mathfrak{so}(10) = D_5$  half-spinor — so  $D_5$  is the Clifford spinor of the 5-dim mode space, closing  $\mu_4 \rightarrow g_{\text{car}} \rightarrow D_5$ . [E] arithmetic, [C] identification (as v189). Cited in `origin_theory`.
- All three are validated external-review levers (the axion-hilltop “ $\pi - 3\phi_0$ ” proposal was *rejected* as a  $\pi \approx 3$  coincidence, not built). Suite now 196 modules, highest ID v197; Wolfram extension 239/239.

## 2026-06-14 (subclaim 2 of ENERGY.02 tackled: raw-seam-DtN RP-definability — v194)

- **v194\_raw\_seam\_dtn\_rp.py** [O] (claim QGEO.DTN.01). The next hard lever — “is the raw RP-seam DtN definable from RP data alone?” (sub-claim 2 of QGEO.ENERGY.02) — was tackled, and the honest result is a *reduction*, not a closure. (1) [E] the commutator  $[\rho, \Lambda_\Sigma] = 0$  closes on the finite cohomology  $H^1$  ( $\rho^* w_k = i^k w_k$ , distinct  $\mu_4$  characters; v177). (2) [E] the *hinge* is met: the DtN/transfer operator is RP-canonical (Osterwalder–Schrader, v54) on a quasi-free seam state (v155), so  $\Lambda_\Sigma$  is RP-definable without naming  $\mathbb{P}^1 \setminus \mu_4$ . (3) [O] the residual relocates: the full- $L^2$  commutation  $\Leftrightarrow$  Bisognano–Wichmann geometric modular flow of the quasi-free seam state (which must be intrinsic, else it re-circulates). So QGEO.SYM.01 reduces from a definitional postulate to “intrinsic BW geometric modular covariance” — one notch sharper, still [O]; the last structural fog point sharpens, it does *not* close to [E]. Cited in `tfpt_research_contracts`; OpenGates updated. Suite now 193 modules, highest ID v194.

## 2026-06-14 (QGEO.ENERGY.02 energy-commutator + QFT-wording hardening — v193)

- **v193\_qgeo\_energy\_commutator.py** [O] (claim QGEO.ENERGY.02). The non-circular sharpening of QGEO.ENERGY.01: the energy commutator  $[\rho, \Lambda_\Sigma] = 0$  on the rank-8 Calderón polarisation, with three sub-claims — (1) [E]  $\rho$  is carrier-defined (Coxeter of  $W(A_3)=S_4$ , v117), not imported from the seam; (2) [O] the non-circularity hinge:  $\Lambda_\Sigma$  must be the *raw* RP-seam DtN, not the  $\mu_4$  normal form; (3) [E] the commutator forces the  $(1, 2, 3)$  grading on  $H^1$  (QGEO.COHO.01). Exact [E] rider (Wolfram-mirrored): the induced EH coefficient  $k = (\ln m)/(12\pi)$  (v150) reproduces  $c_3/2$  iff  $\ln m = q(A_3) = \frac{3}{4}$  (*family*), not  $q(D_5) = \frac{5}{4}$  (carrier, off by  $5/3$ ) — gravity is family-geometry-induced. A *proof target*; if sub-claim (2) holds, QGEO.SYM.01 becomes an energy-rigidity theorem. Cited in `tfpt_research_contracts`. Suite now 192 modules, highest ID v193; Wolfram extension 236/236.

- **QFT-wording hardening (review point 7).** `tfpt_2_standard_model` §(4) was retitled to “Constructive QFT closure of the admissible physical sector”, and its standalone opening over-claim replaced by an explicitly scoped sentence (the unprojected ambient metric measure is not constructed;  $G_{\text{metric}}$  frontier). The two superseded standalone phrasings were added to the v188 prose guard so a frozen release cannot re-introduce them.

## 2026-06-14 (geometry round: Riemann-Roch carrier, Nariai bound, branch line, energy clock — v189–v192)

- **v189\_riemann\_roch\_carrier.py** [E]/[C] (claim ARCH.RRCAR.01). The pair  $(g_{\text{car}}, N_{\text{fam}}) = (5, 3)$  are the two canonical invariants of *one* object, the four-point seam divisor  $D = \mu_4$  on  $\mathbb{P}^1$ : the cycle side rank  $H_1(\mathbb{P}^1 \setminus \mu_4) = \deg -1 = 3 = N_{\text{fam}}$  (already QGEO.COHO.01) and the function side  $h^0(\mathbb{P}^1, \mathcal{O}(\mu_4)) = \deg +1 = 5 = g_{\text{car}}$ , matched by  $\text{rank}(D_5) = 5$ . The carrier-as-mode-space reading is [C] (geometric, does not displace the over-determined  $g_{\text{car}} = 5$ ). Cited in `origin_theory`; Wolfram-mirrored.
- **v190\_nariai\_entropy\_bound.py** [E] (claim HOR.NARIAI.02). A variation-free one-line proof of the SdS entropy floor:  $3(x^2+1) - 2\Phi_3(x) = (x-1)^2 \geq 0$ , so  $S_{\text{tot}} \geq \frac{2}{3}S_{dS}$  with equality iff  $x = 1$  (the Nariai merge). Cited in `tfpt_horizon_readouts`; Wolfram-mirrored.
- **v191\_universal\_branch\_line.py** [C] (claim HOR.BRANCHLINE.01). The affine map  $q = \frac{7}{2} + \frac{N_{\text{fam}}^2}{2}m$  carries the SdS branch points  $\pm 1/N_{\text{fam}}$  onto the flavor labels  $\{2, 5\}$ . *Honestly typed* [C] — an exact *relabeling*, not a theorem (a decoy pair  $\{3, 4\}$  admits an equally exact map; the shared content is the known  $2/3$  ramification). Cited in `tfpt_horizon_readouts`; Wolfram-mirrored.
- **v192\_qgeo\_energy\_clock.py** [O] (claim QGEO.ENERGY.01). An operator-checkable restatement of the bedrock:  $\rho^* \Lambda_{\Sigma} \rho = \Lambda_{\Sigma}$  (the clock preserves the DtN energy form). *Honestly typed*: it *relocates* QGEO.SYM.01 to a more falsifiable surface but does *not* eliminate it (plausibly equivalent unless  $\rho$  is independently defined and the invariance proven). Cited in `tfpt_research_contracts`. Suite now 191 modules, highest ID v192 (v186 skipped).
- *All four are validated external-review proposals (problem\_b); each is built with the honest typing determined on review — the branch line is recorded as an alignment, not a theorem, and the energy clock as a relocation, not a closure.*

## 2026-06-14 (Zenodo release 5.1 (rev 120) — F\_transfer firewall, v184–v188)

- **Published to Zenodo.** Version 5.1 (rev 120) of the ten-PDF document set was published as a new version of the TFPT deposit — record 20687564, DOI 10.5281/zenodo.20687564. This version carries the F\_transfer-firewall round (v184–v188): the honest  $\eta_B$  anchored-Boltzmann test (sharper scenario, not a closure), the axion hilltop-relic typing, the ledger v187 and prose v188 guards, and the Route B/Koide/dark-matter wording fixes plus the P1/P2-vs-QGEO.SYM.01 two-layer clarification. The uploader’s DEFAULT\_RECORD\_ID was advanced to the new record for the next release. Suite 187 modules (highest ID v188; v186 skipped). `run_all.py` green, audit clean, website built.

## 2026-06-14 (F\_transfer round: eta\_B honest test v184, axion relic v185, discipline guard v187)

- **v184\_etaB\_anchored\_boltzmann.py** [C] (claim FR.ETAB.03). An honest test of the external-review  $\eta_B$  proposal ( $M_1 = M_R \varphi_0^4$ ,  $\tilde{m}_1 = m_3/A_{\Lambda}$ ). Firewall verdict: *sharper scenario, not a closure*. The washout anchors plausibly ( $\tilde{m}_1 = m_3/10 \approx 5$  meV,  $A_{\Lambda} = 10$  atom, in the strong-washout window) [C]; but  $M_1$  does *not* —  $M_R$  is the seesaw scale  $\sim v^2/m_3$  and the nearest clean TFPT powers of  $\bar{M}_{\text{P1}}$  both miss it ( $c_3/\varphi_0$ -powers off  $\sim 75\%/ \sim 40\%$ ), so

$M_1 = M_R \varphi_0^4$  merely relocates the free input.  $\eta_B$  stays [C], the route viable but with one (not zero) free input. Cited in `tfpt_4_frontier`.

- **v185\_axion\_relic\_solver.py** [C] (claim FR.DM.03). Honest misalignment estimate: the determinant-line axion  $f_a = M_{\text{scal}}/128 \approx 2.39 \times 10^{11}$  GeV ( $m_a \approx 23.8 \mu\text{eV}$ ) CAN be all dark matter, but the harmonic misalignment under-produces ( $\sim 21\%$  at  $\theta \sim O(1)$ ), so the  $\theta_i \approx 170^\circ$  hilltop anharmonic enhancement is *required* — where the relic is exponentially sensitive. Hence a [C] scenario, not a sharp prediction; the relic contract is a kill test for the BRANCH, not the theory. Cited in `tfpt_4_frontier`.
- **v187\_ftransfer\_laws.py** [E] (claim FR.TRANSFER.01). A CI-style firewall guard: the four continuous-transfer frontier observables (Koide,  $\eta_B$ , axion,  $m_p/m_e$ ) are read from the ledger and required to carry a [C]/[O] marker — never a primitive [E] compiler prediction (exact sub-parts like the Koide 53/54 factor or  $b_3 = -7$  may be [E], the physical prediction never is). Records the four interfaces  $F_{\text{pole}}/F_{\text{Boltzmann}}/F_{\text{relic}}/F_{\text{QCD}}$ ; prevents any future edit from quietly promoting a frontier number to a closed output. Cited in `tfpt_4_frontier`.
- **v188\_frontier\_wording\_guard.py** [E] (claim AUDIT.WORDING.01). A prose sentinel (no new physics): v187 protects the ledger *typing*, this guard protects the *prose*. It scans the 10 active documents and forbids five stale phrasings whose semantics earlier rounds superseded (the old [E]-typed Route B leptogenesis claim and its heavy-scale wording  $\rightarrow$  v184 [C]; the old “open relic scale” dark-matter line  $\rightarrow$  v185; the old “near, not exact” Koide line  $\rightarrow$  v183; and the module-count/ID conflation), so a frozen Zenodo release can never re-introduce wording that was true yesterday and is half-true today. The same pass updated the `tfpt_4_frontier` Route B (to [C]), the Koide and dark-matter summary lines, and added the P1/P2-vs-QGEO.SYM.01 two-layer clarification to the introduction. Suite now has **187 modules**, highest verification ID v188 (v186 was intentionally skipped as redundant —  $m_p/m_e$  is already typed [O] as a QCD matching contract, QCD.LAMBDA.01).
- *Review framing points (P1/P2 as projections of the seam-deck postulate,  $v_{\text{geo}}$  as unit gauge,  $G_{\text{net}}$  a corollary of QGEO.SYM.01, grammar tuning = data) were validated as already implemented in the prior rounds;  $m_p/m_e$  is already typed [O] as a QCD matching contract (QCD.LAMBDA.01), so no redundant module was added.*

## 2026-06-14 (operator origin of the Koide 53/54 factor: the missing Sheet-Diamond corner, v183)

- **v183\_koide\_f\_corner\_transfer.py** [E]/[C] (claim FR.KOIDE.07). An external-review proposal, validated exactly: the Koide source $\rightarrow$ pole factor  $\frac{53}{54}$  is not a bare arithmetic guess but an *operator* identity on the carrier,  $u_{\text{pole}}/u_{\text{source}} = a^T(R+Q)\mathbf{1}/(2\mathbf{1}^T R a) = 53/(2\cdot 27) = \frac{53}{54}$ . Here  $\mathbf{1}^T R a = 27$  is the  $E_6 \times A_2$  cubic family block ( $248 = \dim E_6 + \det R + 2(\mathbf{1}^T R a)N_{\text{fam}} = 78 + 8 + 2\cdot 27\cdot 3$ ) and  $a^T(R+Q)\mathbf{1} = 53 = 52 + 1$ , where  $F = R+Q$  is the *missing* Sheet-Diamond corner with  $\det B_F = 52 = \dim F_4$  (v80). Hard negative control:  $a^T M \mathbf{1}$  for  $M \in \{R, Q, K, L\}$  is  $\{32, 21, 35, 56\}$  — only the absent corner  $F$  gives 53. This upgrades FR.KOIDE.03 from “near miss + conjecture” to “operator-sourced source $\rightarrow$ pole transfer”: the *factor* is [E] (exact, mirrored in Wolfram, 232/232); the *mechanism* (the leptonic source $\rightarrow$ pole transfer reads the  $F$  corner) stays [C], an  $F_{\text{transfer}}$  special case. The prediction itself stays [C]. Cited in `tfpt_4_frontier` (Koide section). Suite now 183 modules.
- **Review point 2 noted as already-present:** the claim that flavor and Nariai share one double-cover ClockFunctor (flavor rate  $(2/3)^{2N_{\text{fam}}}$  vs gravity curvature  $(2/3)^{N_{\text{fam}}}$ , exponent ratio  $|\mathbb{Z}_2|$ , “one clock, two geometries”) is exactly HOR.TRISECT.01/v103 — a rediscovery of an existing result, no new module.

## 2026-06-14 (review recommendations 1/3/4 + the four concerns A–D mapped to the named residuals)

- **v182\_reviewer\_residual\_map.py** [E]/[C] (claim REVIEW.MAP.01). A careful external review raised four concerns (A mapping  $2D \rightarrow 4D$ , B UV-gravity abstractness, C grammar-tuning/meta-look-elsewhere, D Koide Möbius flow). This module shows three of them *reduce*, by the same discipline as v175–v181, onto the *two already-named* residuals  $\{\text{QGEO.SYM.01}, F_{\text{transfer}}\}$ : A (Plücker-11 = QBL boundary count v174 + holographic reduction), B ( $\equiv \text{QGEO.SYM.01}$ ), D (Möbius forced by  $\text{Aut}(\mathbb{P}) = \text{PSL}(2, \mathbb{C})$  on the deck; the missing generator =  $F_{\text{transfer}}$ ). Only C is genuinely *experiment-only* (frozen + minimal + over-determined grammar; residual falls to JUNO/CMB-S4). A unification of the concern *structure*, not a closure. Cited in `tfpt_5_redteam`. Suite now 182 modules.
- **Review recommendation #3 — a “Rosetta” translation lexicon** (website RosettaLexicon): TFPT’s compiler-speak (seam, deck, readout, microcode, audit hull, gap, anchor,  $F_{\text{transfer}}$ ,  $v_{\text{geo}}$ , QGEO.SYM.01) mapped onto standard QFT / mathematical-physics language (Wilsonian boundary CFT, orbifold deck, index/intersection number, RG irrelevance rate, renormalization condition, fundamental postulate) — a reading aid for mainstream physicists, not a new claim.
- **Review recommendation #1 — kill-tests up front** (website TrustContract): the frozen pre-data predictions ( $\sin^2\theta_{12} = 0.3067$ , JUNO;  $r \approx 0.004$ , CMB-S4) and the null model ( $P \leq 10^{-30.7}$ ) are now a prominent strip directly under the hero.
- **Review recommendation #4 — the bedrock as a postulate**. QGEO.SYM.01 is now framed confidently as TFPT’s *one fundamental postulate* (the role “ $c = \text{const}$ ” plays in relativity), not a gap — in the `tfpt_research_contracts` box and the website “what is open” card; the achievement is that the whole theory is compressed onto exactly one sharply-located, falsifiable premise.

## 2026-06-14 (external-review response: CSV fix, G\_net 3-level, QGEO.SYM.01 page, claim detox)

- **Ledger CSV bug fixed + guarded**. Three rows (AX.P1.01, AX.P2.01, QGEO.ISO.01) had *unquoted commas* in a status/external field (e.g. “a=(1,1,2)”, “PSL(2,C)”); so a strict CSV parser read 11–12 columns instead of 10 — a real machine-readability defect in the single source of truth. All offending fields are now quoted (every row parses to exactly 10 columns), and `audit_sync.py` gained a `check_csv_wellformed` guard (ledger, registry, clusters, freeze must all parse to constant width) so it cannot regress.
- **$G_{\text{net}}$  typed in three explicit levels** (website “What is open” card): (1) *algebra* [E]  $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1,1) \rangle \cong (E_8)_1$ ; (2) *AQFT machinery* [E] (net existence + full-cone RP, CAR functor, v175); (3) *seam instantiation* [O] (QGEO.SYM.01). The target object’s mathematics is closed; only the physical coupling of the raw seam is open.
- **QGEO.SYM.01 now has its own one-page box** in `tfpt_research_contracts`: what is identified, why it is weaker than CONF/ISO, what follows automatically (the conditional theorem), and *what would falsify it* (genus  $> 0$  double; non-deck transport;  $N_{\text{fam}} \neq 3$ ); with the explicit conditional framing “*given* QGEO.SYM.01, the closure follows” rather than “the theory derives the seam geometry”.
- **Claim-language detox**. “no free fundamental number” softened to “no free *dimensionless* compiler dial beyond the anchor and  $\pi$  (dimensionful scale + transfer physics stay typed)” (`origin_theory`, `introduction`, website); the site title softened from “the Standard Model Derived” to “the Standard-Model *Skeleton* Derived”.

- **External-reviewer map** added to the website verification page: four boxes (exact kernel [E], conditional physics [C], open premises [O], kill tests [X]) so a reviewer sees the attack surface without reading 181 scripts. Suite unchanged at 181.

## 2026-06-14 (Zenodo release 5.1 (rev 114) — v175–v181 bedrock + figures + P1/P2)

- **Published to Zenodo.** Version 5.1 (rev 114) of the ten-PDF document set was published as a new version of the TFPT deposit — record 20686284, DOI 10.5281/zenodo.20686284. This version carries the AQFT-closure-to-bedrock chain (v175–v181, the structural residual reduced to the single definitional premise QGEO.SYM.01), the two overview figures (residual\_chain, script\_timeline), and the P1/P2 anchor-reduction provenance on the axiom surfaces. The README and the Zenodo description were refreshed (181 modules), and the uploader’s DEFAULT\_RECORD\_ID was advanced to the new record for the next release.

## 2026-06-14 (overview figures + P1/P2 reduction reflected on the axiom surfaces)

- **Two overview figures** (make\_figures.py; PDF + website PNG). (i) `residual_chain` — the structural-residual reduction chain v175–v181 as a descending staircase from “build a quantum-gravity measure” to the single definitional bedrock QGEO.SYM.01; embedded in `tfpt_research_contracts`. (ii) `script_timeline` — the eight-phase journey of the ~181 scripts (foundations → SM readouts → seam= horizon → horizon/flavor geometry → R1–R5/premise-(A) → PyR@TE cross-checks → AQFT bridges → AQFT closure), what each did mathematically and physically; embedded in `introduction` ahead of the master script index, and both shown on the website verification page.
- **P1/P2 reduction now reflected on the axiom surfaces.** The reduction of the two axioms to the single parabolic anchor  $a = (1, 1, 2)$  plus  $\pi$  (with  $g_{\text{car}} = 5$  an over-determined bootstrap fixed point — forced three ways, v6, Pascal-unique and Lean-formalised, FORM.LADDER.01) was already established (ANCHOR.GEN.01/v23, ARCH.CORE.01/v53) and stated in the paper-s/glossary, but the *axiom surfaces* still read as bare “declared inputs”. Fixed: the ledger rows AX.P1.01/AX.P2.01 now carry the anchor/bootstrap reduction and point to it, and the website dependency-graph nodes P1/P2 now show the anchor input ( $c_3 = 1/(2e_1(a)\pi)$ ;  $g_{\text{car}} = e_2(a)$ , bootstrap-forced) rather than “declared axiom”. No marker change (still declared inputs), only honest provenance.
- Manifests refreshed (two new figures added to the FIG list). Suite unchanged at 181.

## 2026-06-14 (the final reduction to bedrock v181 + full non-changelog status sweep)

- **v181\_clock\_is\_conformal\_symmetry.py** [E]/[O] (claim QGEO.SYM.01). The isometry premise QGEO.ISO.01 (v180) is weakened one more genuine step — to a conformal-*symmetry* premise — via *equivariant uniformisation* (Nielsen realisation; for cyclic groups Kerékjártó + uniformisation): a *conformal* automorphism (strictly weaker than an isometry — every isometry is conformal, conformal allows any positive scale) already suffices to put the order-4 clock in the form  $z \mapsto iz$ . Since the clock is the Coxeter element of  $W(A_3) = S_4$  (v117), the  $\mu_4$  deck, and a deck transformation preserves the pulled-back conformal structure by definition, the residual collapses to the *definitional* identification QGEO.SYM.01: “the seam’s conformal structure *is* the  $\mu_4$ -deck structure of the carrier clock”. This is bedrock — a physical/definitional statement tying P2 to the seam’s conformal geometry, *not* a finite computation and not reducible further without relabeling. The chain `REALIZE(v175)→⋯→SYM.01(v181)` ends here; we stop honestly. Python-only. Suite now 181 modules.

- **Full non-changelog status sweep.** The current bedrock framing was propagated to every live status surface (not just the changelog): the introduction “latest reductions” prose, `tfpt_5_redteam` Target A, the `tfpt_research_contracts` central-theorem keybox, `origin_theory`; and on the website `OpenGates`, `VerificationDag`, `papers.ts` (Target A  $\times 2$ ,  $G_{\text{net}}$ ), the `glossary`  $G_{\text{net}}$  entry and the `faq` live-residual answer — all now end at `QGEO.SYM.01`.

## 2026-06-14 (the conformal premise reduces to a milder isometry premise v180)

- `v180_clock_is_mobius.py` [E]/[O] (claim `QGEO.ISO.01`). Cracks `QGEO.CONF.01` one level further: the conformal-realisation premise is *replaced*, via three classical theorems, by a strictly *milder, non-circular* premise. (1) *Uniformisation* (Poincaré–Koebe): a genus-0 Riemann surface is conformally  $\mathbb{P}$ , so the genus-0 seam double (P1, v179) with its RP-metric conformal structure *is*  $\mathbb{P}$  — the target is *produced*, not assumed; (2) *Kerékjártó* (1919; Constantin–Kolev 2003): a finite-order orientation-preserving homeomorphism of  $S^2$  is conjugate to a rotation, so the order-4 clock ( $h(A_3)=4$ ) is topologically  $z \mapsto iz$ ; (3) *isometry*  $\Rightarrow$  *conformal*  $\Rightarrow$  *Möbius*: an orientation-preserving isometry of a Riemannian surface is holomorphic,  $\text{Aut}(\mathbb{P}) = \text{PSL}(2, \mathbb{C})$ ; (4) an order-4 Möbius map is  $z \mapsto iz$  (verified: projective order 4, fixed  $\{0, \infty\}$ , multiplier  $i$ , free orbit  $\mu_4$ ). Hence `QGEO.CONF.01` is *implied* by the milder premise `QGEO.ISO.01`: “the carrier clock is an order-4 orientation-preserving *isometry* of the RP seam metric” — which does not name  $\mathbb{P}/\mu_4$  (uniformisation produces them) and is the natural statement that the seam transport is metric-compatible (a holonomy preserving the RP inner product). That milder premise is the single, now milder, structural residual of the whole theory; its surface realisation from the raw seam stays honestly [O]. Cited in `tfpt_research_contracts`; Python-only. Suite now 180 modules.

## 2026-06-14 (the two obligations collapse to ONE geometric premise v179)

- `v179_conformal_realisation.py` [E]/[O] (claim `QGEO.CONF.01`). Investigating the two residual premises left by v178 shows they are *not* independent — they collapse onto a single geometric premise (honest unification; the premise itself stays [O]). MARKS’s residual is not three independent assumptions: (1) genus-0 *is* P1 — the one-sided Gauss–Bonnet  $c_3 = 1/(|\mathbb{Z}_2| 2\pi \chi(S^2)) = 1/(8\pi)$  uses  $\chi(S^2) = 2$  and  $\chi = 2 - 2g \Rightarrow g = 0$ , so “genus-0” is the same  $\chi = 2$  that gives the “8” in  $c_3$  (v58/v73); (2) the order-4 clock *is* the carrier — the Coxeter element of  $W(A_3) = S_4$  (v117), order  $h(A_3) = 4 = |\mu_4|$ ; (3) marks = one orbit is *forced* (the parabolic marks are not the two elliptic fixed points, a free order-4 orbit has exactly  $4 = e_1(a)$  points). KERNEL then *follows* from the same geometry (DtN symbol  $|k|$ , Lee–Uhlmann v156;  $c = 8$  rigidity v158; the cusped-surface residual  $= H^1 = \mathbb{C}^3 = N_{\text{fam}}$ ). So the *entire* structural residual of the theory is the *single* geometric premise `QGEO.CONF.01`: “the raw RP seam normal double is conformally  $(\mathbb{P}, \mu_4)$  with the carrier clock as the order-4 Möbius automorphism”. The two doors of v177/v178 are one geometric realisation, left honestly [O]. Cited in `tfpt_research_contracts`; Python-only. Suite now 179 modules.

## 2026-06-14 (an honest attempt at the two obligations v178 — deeper reduction, not closure)

- `v178_marks_kernel_attempt.py` [E]/[O] (claim `QGEO.REDUCE.01`). An honest attempt at `QGEO.MARKS.01` and `QGEO.KERNEL.01`. These are genuine constructive-geometry/AQFT statements, *not* finite computations — neither is closed. What the module does is push each as far as computation allows: it closes the finite [E] core of each and reduces each to *one sharper* premise. *Marks core* [E]: given a genus-0 double with an order-4 clock  $\rho$  (fixed points  $\{0, \infty\}$ ) and reflection  $\tau$ , the marks are forced to be a generic  $\rho$ -orbit  $\{1, i, -1, -i\} = \mu_4$  (*not* the fixed points), distinct iff  $b_1 = 4 - 1 = 3 = N_{\text{fam}}$  (any collision drops  $b_1$ ), with  $\tau\rho\tau = \rho^{-1}$

giving a faithful  $D_4$  of order 8 — so MARKS reduces to the single topological/clock premise. *Kernel core* [E]: on the 3-dim multiplicity-free  $H^1$  the  $\mu_4$  characters  $(1, 2, 3)$  are distinct, so by Schur a  $\mu_4$ -equivariant operator is forced *diagonal* and completely determined by its eigenvalue per character; two such with the forced transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$  (v162) and the unique residue-normalised basis (v177) are *equal as operators*, not merely isospectral — the spectrum-vs-operator worry *vanishes* on the finite block. So KERNEL reduces to the single full-operator-reduction premise (principal symbol  $|k|$ , Lee-Uhlmann/v156; no drift v158). Net: neither obligation closed; both sharpened to milder premises with their finite cores [E]. Cited in `tfpt_research_contracts`; Python-only (Möbius/Schur symbolic + numeric). Suite now 178 modules.

## 2026-06-14 (QGEO proof tree v177 — the residual sharpened into two named obligations)

- `v177_seam_marking_kernel.py` [E]/[O] (claims QGEO.TREE.01, QGEO.MARKS.01, QGEO.KERNEL.01). A second external residual-class audit pointed out that the v176 statement takes the marks/ $D_4$  as a *hypothesis* — a near-circular trap. This module factors the central theorem honestly as `REALIZE = MARKS · UNIFORM · COHOM · MODULE · KERNEL · NET` and closes *four* nodes *constructively* (all validated symbolically before adoption): `UNIFORM` [E] — an order-4 Möbius map conjugates to  $z \mapsto iz$ , a non-fixed orbit scales to  $\mu_4$ , and  $\sigma\rho\sigma = \rho^{-1}$  for  $\sigma : z \mapsto 1/z$  gives a faithful  $D_4$  (order 8), so a genus-0 four-marked faithful- $D_4$  boundary *is*  $(\mathbb{P}, \mu_4)$  (strengthening v168's cross-ratio to the full uniformisation); `COHOM` [E] —  $\rho^*\omega_k = i^k\omega_k$ , grading  $(1, 2, 3) = A_3$  exponents; `MODULE` [E] — a *unique*  $\mu_4$ -equivariant iso (multiplicity-free + residue normalisation; reflection  $\omega_1 \leftrightarrow \omega_3$ ,  $\omega_2$  fixed); `NET` [E] (v154/v175). The structural residual is thereby sharpened from one diffuse premise into *exactly two* named open obligations, neither a finite computation: QGEO.MARKS.01 (the raw RP seam collar canonically produces the genus-0 four-marked  $D_4$  boundary) and QGEO.KERNEL.01 (the raw seam Calderón operator *is* the  $\mu_4$ -equivariant free gapped contraction, as an operator). Cited in `tfpt_research_contracts` (the central-theorem keybox); Python-only (Möbius/cohomology symbolic; numbers mirrored via v168/v137/v162/v154/v175). Suite now 177 modules.

## 2026-06-13 (Seam Collar Realisation Theorem v176 — the one central proof obligation)

- `v176_seam_collar_realisation.py` [E]/[O] (claim QGEO.REALIZE.02). Following an external residual-class audit, the single remaining structural item is formulated as one central theorem and certified as a *reduction* (not a fabricated proof). *Theorem*: given an oriented one-sided RP seam collar with a sheet-odd Calderón involution, faithful  $D_4$  marks, four parabolic marks with  $\mu_4$  decomposition and admissible  $c=8$  data, its conformal boundary is Möbius-equivalent to  $\mathbb{P} \setminus \mu_4$  with  $H^1$  grading  $(1, 2, 3)$ , the Calderón contraction is the rank-8  $K_\Sigma$  polarisation and the index-4 extension is  $(E_8)_1$ . The proof decomposes into six steps, *five already* [E]: geometry/uniformisation (v168), Calderón DtN symbol  $|k|$  (v156),  $H^1$  character = generation grading  $(1, 2, 3) = A_3$  exponents (v137/v168),  $\mu_4$ -equivariant contraction with gap  $(2/3)^6$  (v162), AQFT closure  $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 = (E_8)_1 + \text{full-cone RP all } m$  (v154/v175). The whole structural residual thus reduces to the *one* open premise QGEO.REALIZE.01: that the physical seam collar's boundary *is* this  $\mathbb{P} \setminus \mu_4$  — a constructive-geometry statement, left honestly [O]. Cited in `tfpt_research_contracts` (the central theorem keybox); assembly/reduction certificate, Python-only (its exact identities are mirrored via v168/v156/v162/v154/v175). Suite now 176 modules. *Audit note*: the metrology residual the same review flags is already discharged — v153 (No-Unit Theorem) is the metrology-line typing ( $v_{\text{geo}} \mapsto \lambda^{-1}v_{\text{geo}}$ ,  $1/G \mapsto v_{\text{geo}}^2$ , dimensionless data invariant), and the Koide  $u \rightarrow \frac{53}{54}u$  source→pole transfer is already typed [C] (FR.KOIDE.03); neither is re-derived.

## 2026-06-13 (Zenodo release 5.1 (rev 105) — v174 + v175 + doc refresh)

- **Published to Zenodo.** Version 5.1 (rev 105) of the ten-PDF document set (introduction, origin\_theory, tfpt\_1–5, horizon, research\_contracts, changelog) was published as a new version of the TFPT deposit — record 20681451, DOI 10.5281/zenodo.20681451. This version carries the AQFT closure round (v170–v175): the seam Fock-space readings and the heavy-artillery discharge of net existence + full-cone reflection positivity to [E]. The README and the Zenodo description were refreshed to the current state (175 modules, Wolfram 116/116 + 231/231), and the uploader’s DEFAULT\_RECORD\_ID was advanced to the new record for the next release.

## 2026-06-13 (the heavy artillery v175: net existence + full-cone RP discharged to [E])

- **v175\_net\_existence\_full\_cone.py [E]/[O] (claim QFT.NETRP.01).** The two structural residuals that the premise-(A) chain had relocated to *already-open* items —  $A_2$  net existence and full-cone reflection positivity — are now *discharged to a theorem*, leaving the seam realisation as the single irreducible open premise (nothing fabricated). (1) *Full-cone RP for all  $m$* : the CAR second-quantisation functor  $\Gamma(t) = \bigoplus_m \Lambda^m(t)$  lifts the one-particle seam contraction  $t$  to the complete Fock space  $\Lambda^\bullet(\mathbb{R}^{16})$  ( $\dim 2^{16} = 65536$ ); by the exterior-power spectrum theorem every eigenvalue is a subset product of  $\text{spec}(t)$ , so all  $2^{16}$  products lie in  $[0, 1]$  (a contraction = full-cone RP), with eigenvalue-1 multiplicity  $2^8 = 256$  (wedges of the rank-8  $K_\Sigma$  polarisation) and sub-leading = the one-particle gap  $(2/3)^6$  — so full-cone RP reduces *for every  $m$*  (not only  $m \leq 6$ , v161) to the one-particle contraction (Shale–Stinespring), discharging the v160 scope premise. (2)  $A_2$  *assembled + verified*: the  $(E_8)_1$  character  $E_4/\eta^8 = 1 + 248q + 4124q^2 + 34752q^3 + 213126q^4 + 1057504q^5$  (level-1 = 248, extends the order-4 v156 check by one order), the embedding  $248 = 120 + 128$ , and the  $E_8$  Cartan matrix even with  $\det 1$  (the unique rank-8 even unimodular lattice, v83); the net *exists* by cited theorems (free-fermion net, lattice VOA, index-4 Q-system, v125). (3) Both then need the *same* input — the seam realisation QGEO.REALIZE.01 (finite half proven, v168) — which is a constructive-geometry/AQFT statement, *not* closeable by a finite computation, and is left honestly [O]. **Net:** net existence and full-cone RP become [E]; the one irreducible structural premise is the seam realisation. Cited in `tfpt_research_contracts` (premise (A) closure) and `tfpt_5_redteam` (Target A); the exact parts (Cartan unimodular, character order 5, functoriality integers) mirrored in the Wolfram extension (231/231). Suite now 175 modules.

## 2026-06-13 (seam Fock-space readings v174: CAR cone, Witten index, g-theorem)

- **v174\_seam\_fock\_readings.py [E] (claim QFT.FOCK.01).** Three more exact audit bridges from the same source note. (1) *CAR cone compression*: the even CAR algebra of the 16 seam Majoranas has  $\dim \Lambda^{\text{even}}(\mathbb{R}^{16}) = 2^{15} = 32768$  directions, but a quasi-free Bogoliubov transport reduces the whole cone to a 16-dim one-particle problem ( $2^{15}/16 = 2^{11}$ ) — the explicit count behind v161. (2) *Witten index = the Plücker norm 11*: the four  $\mu_4$ -corner fermions span a  $2^4 = 16 = \dim S^+$  Fock space; QBL ( $\deg \leq 2$ ) keeps  $\sum_{k \leq 2} \binom{4}{k} = 11 = \dim S^+ - g_{\text{car}}$ , so  $c_u/c_d = 55/117$  reads a topological state-space dimension. (3) *Affleck–Ludwig g-theorem*:  $c_{UV} = 5 + 3 = 8 = c_{IR}((E_8)_1)$ , so  $\Delta c = 0$  forces an exact isometry — the boundary-entropy form of the v157/v158 rigid fixed point. Cited in `tfpt_2` (Exterior Leg Lemma); mirrored in the Wolfram extension (230/230). Suite now 174 modules. (The categorical reframings in the source note remain framing, not asserted claims.)

## 2026-06-13 (CFT/QFT bridges: slice compression v170, OS moment v171, trace-anomaly seed v172, Pfaffian CP v173)

- **v170\_diamond\_slice\_compression.py** [E] (claim E8.SLICE.01). E8 Slice Compression Lemma: the seven  $E_8$  audit slices are *one* projection of two alphabets — anchor power sums  $P = (3, 4, 6, 10)$  ( $p_n = 2 + 2^n$ ) and Sheet-Diamond determinants  $(3, 4, 8, 14, 20, 32)$  summing to  $81 = N_{\text{fam}}^4$ . All seven 248-slices, the  $K_4$  edge graph, and the row-budget cross  $(7, 11, 13) + s(3, 3, 3) + t(1, 2, 3)$  follow;  $78 = \dim E_6 = p_2(p_0 + p_3)$ . Cited in `tfpt_3` (slice atlas). Exact; reading is audit (anti-numerology).
- **v171\_os\_moment\_cluster.py** [E]/[C] (claim QFT.OSMOMENT.01). Atomic OS Moment Lemma:  $\text{spec}(T) = \{1, 64/729, 1/729\}$  makes every correlator a positive moment sequence, so the OS Hankel matrices factorise as Vandermonde Grams (a clean RP certificate); cluster  $729/665$ ,  $\xi = 0.4111$ . Sugawara Gap Safety:  $31 = 1 + h^\vee(E_8)$ ,  $c((E_8)_1) = 248/31 = 8$ ,  $\Delta_{\text{eff}} = 6 \log \frac{3}{2} - 31/(4\pi^2) > 0$ . Cited in `tfpt_research_contracts` (G5).
- **v172\_trace\_anomaly\_seed.py** [E] (claim SEED.ANOMALY.01). The seed prefactor  $\frac{4}{3}$  is the halved  $(E_8)_1$  Weyl anomaly: over the seam  $S^2$  ( $\chi = 2$ ) the  $c = 8$  trace anomaly is  $c\chi/6 = \frac{8}{3}$ , halved by the one-sided  $|\mathbb{Z}_2|$  to  $\frac{4}{3} = 16/12$ . Plus QCD  $b_3 = 7 = \text{scalaron exponent } 48 - 41$  (confinement  $\leftrightarrow$  inflation), and the Volkov–Wipf  $-98/45 = -2 \cdot 7^2 / \dim(D_5)$  factorisation. Cited in `tfpt_2` (seed).
- **v173\_pfaffian\_cp\_car.py** [E] (claim FLAV.STRONGCP.02). Strong CP  $\theta_{\text{eff}} = 0$  as Pfaffian reality in the self-dual 16-Majorana CAR model:  $\{D, \Sigma\} = 0$  pairs the spectrum,  $\gamma_5$ -Hermiticity makes the Pfaffian real, and RP picks the positive branch. Cited in `tfpt_2` (Strong CP section).
- All four validated computationally before adoption; the exact identities (v170/v171/v172) are mirrored in the Wolfram extension (229/229). Suite now 173 modules. (The categorical reframings in the source note — compiler-as-single-functor,  $E_8 = K_0$ , GIT/monadic/Langlands — are recorded as framing, not asserted as machine-checked claims.)

## 2026-06-13 (Zenodo release 5.1 (rev 97))

- **Published to Zenodo.** Version 5.1 (rev 97) of the ten-PDF document set (introduction, `origin_theory`, `tfpt_1–5`, `horizon`, `research_contracts`, `changelog`) was published as a new version of the TFPT deposit — record 20678568, DOI 10.5281/zenodo.20678568. The uploader’s `DEFAULT_RECORD_ID` was advanced to the new record for the next release. Also fixed `build.sh zenodo` on macOS bash 3.2 (empty-array expansion under `set -u`).

## 2026-06-13 (QGEO rigidity theorem v168 + $\eta_B$ leptogenesis interface v169)

- **v168\_qgeo\_rigidity.py** [E]/[C] (claims QGEO.RIGID.01, QGEO.REALIZE.01). Hardens the *finite* half of the one remaining Tier-1 hinge GATE.QGEO (the seam = flavour = horizon identification). Exact, machine-checked:  $\mu_4 = \{1, i, -1, -i\}$  has cross-ratio 2 and a faithful Möbius stabiliser of order  $8 = |D_4|$ ;  $H^1(\mathbb{P}^1 \setminus \mu_4)$  has rank  $4 - 1 = N_{\text{fam}} = 3$ ; and the forms  $\omega_k = z^{k-1} dz / (z^4 - 1)$  ( $k = 1, 2, 3$ ) are  $\mu_4$ -eigenforms with characters  $\chi_1, \chi_2, \chi_3$  of weights  $(1, 2, 3) = A_3$  exponents =  $\text{Spec}(Q_+)$ . So a genus-0 boundary with faithful  $D_4$ , four parabolic marks and this  $\mu_4$ -decomposition is Möbius-equivalent to  $\mathbb{P}^1 \setminus \mu_4$  — the QGEO Rigidity Theorem (QGEO.RIGID.01, [E]). The seam-collar realisation stays the one premise (QGEO.REALIZE.01, [C]). Mirrored in the Wolfram extension (226/226). Cited in `tfpt_research_contracts` (U0). *Not* a new structural class — it hardens the existing hinge (consistent with v167’s completeness claim).
- **v169\_etaB\_boltzmann\_interface.py** [C] (claim FR.ETAB.02). Operationalises the Tier-3  $F_{\text{transfer}}$  leptogenesis path. Type-I thermal leptogenesis  $\eta_B \sim 0.96 \times 10^{-2} \varepsilon_1 \kappa_f$  (sphaleron

28/79), fed by TFPT’s normal-ordered neutrino spectrum and the predicted  $\delta_{CP} = 240^\circ$ . Over natural  $M_1 \in [3 \times 10^9, 3 \times 10^{10}]$  GeV and washout,  $\eta_B$  brackets the observed  $6.1 \times 10^{-10}$  (a canonical point gives  $6.0 \times 10^{-10}$ , untuned) — the route is *viable*. But  $M_1$ /washout are scenario inputs, so  $\eta_B$  stays [C]; if a precise solve excluded the window the leptogenesis *route* falls, not the theory (the downstream  $\Omega_b$  readout is independent). Cited in `tfpt_4_frontier` (Route B). Python-only.

## 2026-06-13 (Higgs quartic from the free seam: SM near-criticality explained, v166)

- `v166_higgs_free_seam.py` [E]/[C] (claim `HIGGS.FREESEAM.01`). A new prediction tying the Higgs mass to premise (A). The seam UV is the free chiral  $c=8$  fixed point (v158, Gaussian, no relevant/marginal interactions), so the one marginal SM scalar coupling vanishes there:  $\lambda(M_{\text{seam}}) = 0$  and  $\beta_\lambda(M_{\text{seam}}) = 0$  — the Shaposhnikov–Wetterich double-criticality, here *derived* from the free seam, not assumed. Integrating the PyR@TE-confirmed *two-loop* SM RGEs from  $M_Z$  up with the measured  $(m_H, m_t)$  gives  $\lambda(\bar{M}_{\text{Pl}}) \approx 0.002$  and  $\beta_\lambda(\bar{M}_{\text{Pl}}) \approx 0$ : the famous SM near-criticality is a *consequence* of the free seam (2-loop sharpens  $\lambda(\bar{M}_{\text{Pl}})$  from  $\sim 0.022$  at one loop to  $\sim 0.002$ ). The double condition predicts  $m_H \sim 129\text{--}134$  GeV; the same condition at the scalaron scale gives  $\sim 107$  GeV (too low), so the boundary condition lives at the reduced Planck scale (seam = horizon = Planck). Fills the Higgs-sector gap (was only  $N_\Phi = 1$ ); cited in `tfpt_4_frontier`.
- PyR@TE-sourced: the reproducer’s SM run supplies the two-loop  $\beta_\lambda$ , reduced to top-only (`verification/pyrate/sm_higgs_betas.txt`). Suite now 166 modules. Python-only (numerical 2-loop ODE).

## 2026-06-13 (PyR@TE $F_{\text{transfer}}$ follow-ups: flavor scale-tagging v163, QCD scale v164)

- `v163_rg_stability_flavor.py` [E]/[C] (claim `FLAV.RGSTAB.01`). Scale-tagging of the flavor sector. The lepton-mixing seeds ( $\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0}{2} = 0.3067$ ,  $\sin^2 \theta_{13} = \varphi_0 e^{-5/6} = 0.0231$ ) are *RG-stable*: the PMNS running is  $y_\tau^2$ -suppressed (the only mixing-rotating Yukawa term is  $\frac{3}{2} Y_e Y_e^\dagger Y_e$ , PyR@TE SM RGEs), so  $\kappa_\tau = y_\tau^2 / (16\pi^2) \ln(M_{\text{scal}}/M_Z) \sim 1.8 \times 10^{-5}$  and  $|\Delta \sin^2 \theta_{ij}| \lesssim 10^{-4}$  — two to three orders below precision. Seam value = measured value; the  $y_t^2$ -driven quark mass ratios ( $m_t/m_b \sim 41$ ) by contrast carry a scale tag. Cited in `tfpt_2` (solar/reactor angle box).
- `v164_qcd_scale.py` [E]/[O] (claim `QCD.LAMBDA.01`). The QCD confinement scale is parameter-free from the carrier  $b_3 = -7$  (v159): running  $\alpha_s(M_Z) = 0.1179$  down (two-loop,  $n_f$  thresholds) reproduces  $\alpha_s(m_b) \simeq 0.22$ ,  $\alpha_s(m_c) \simeq 0.38$  and confinement at  $\sim 0.5$  GeV by dimensional transmutation — so the QCD-scale *numerator* of  $m_p/m_e$  is independently sourced. The  $O(1)$  proton/ $\Lambda$  factor is lattice, so  $m_p/m_e = 1836$  stays the honest “[O] — not forced” frontier ratio. Cited in `tfpt_4_frontier`.
- Both are PyR@TE-backed (the reproducer now also extracts the SM Yukawa  $\beta$ ’s, `verification/pyrate/sm_yukawa_betas.txt`). Suite now 164 modules. Python-only (no Wolfram mirror: magnitude bound / numerical running).

## 2026-06-13 (PyR@TE external cross-check of the $F_{\text{transfer}}$ gauge inputs — v159)

- **New module** `v159_pyrate_gauge_crosscheck.py` [E]/[C] (claim `EM.BUDGET.02`). The one-loop gauge coefficients used by the transfer functor are confirmed by an *independent* third-party RGE generator. Feeding the carrier / Standard-Model field content into the textbook one-loop formula (GUT normalisation) gives  $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$ , and **PyR@TE 3** (`arXiv:2007.12700`, `github.com/LSartore/pyrate`) writes  $\beta_{g_1} = (\frac{41}{10})g_1^3$ ,  $\beta_{g_2} = -\frac{19}{6}g_2^3$ ,

$\beta_{g_3} = -7g_3^3$  verbatim (plus the full standard 2-loop matrix). So  $b_1 = 41/10$  is pinned *three* independent ways — carrier algebra  $10b_1 = g_{\text{car}}2^{g_{\text{car}}-2}+1 = 41$ , the  $U(1)$  hypercharge index  $\sum \frac{3}{5}Y^2$ , and PyR@TE — and the split  $10b_1 = 40_{(\text{ferm.})} + 1_{(\text{Higgs})} = 41$  reproduces the EM-budget Gate 1 of `tfpt_2`. Cited there with `\veri`; the gauge sector of  $F_{\text{transfer}}$  is thus not a free knob.

- **PyR@TE reproducer (not vendored).** `verification/pyrate/reproduce.sh` clones PyR@TE on demand into `experiments/pyrate` (git-ignored), neutralises its interactive `pdflatex` end-command, runs the SM at one and two loops, and re-extracts the gauge  $\beta$  functions; the committed evidence is `verification/pyrate/sm_gauge_betas.txt`. PyR@TE is the right external tool for the transfer/RGE gates (gauge running, and — with a seesaw model — the neutrino-Yukawa running behind  $\eta_B$ ); it is *not* a tool for premise (A) (a 2d boundary-CFT question). Mirrored in the Wolfram extension; suite now 159 modules.

## 2026-06-13 (terminal residual inventory: (A) no longer a structural class; one hinge + $F_{\text{transfer}}$ + irreducible core, **v167**)

- **New module `v167_inventory_post_closure.py` [E]/[C].** The terminal residual inventory (line `v105`→`v123`→`v167`), re-pinned after the (A)-chain. With (A) discharged (`GATE.METRIC.16-19`) the `v123` *five* structural classes collapse: the seam clock (R1), the index-4 net (R2) and the  $Q$ -realisation (R5) all merge into *one* Tier-1 hinge — the seam=flavour=horizon identification =  $A_2$  net existence (assembly of established net constructions) + `GATE.QGEO` (seam boundary =  $\mathbb{P} \setminus \mu_4$ , cohomology  $b_1=N_{\text{fam}}=3$ ); the EH step (R3) folds into  $v_{\text{geo}}$  (**v152**) and the ambient measure into  $G_{\text{net}}$  (**v76**). **Terminal structure:** Tier 0 = the irreducible core  $\{\pi, v_{\text{geo}}\}$  (a theorem, No-Unit **v153**); Tier 1 = one hinge; Tier 2 = folded; Tier 3 =  $F_{\text{transfer}}$ , one functor with four typed conditional interfaces (Koide,  $\eta_B$ , axion,  $m_p/m_e$ ),  $\eta_B$  carrying the most open computational room. Completeness: a sixth independent structural class — or a new irreducible — fails the script (ledger `SYNTH.INVENTORY.03`). A unification + re-typing, NOT an unconditional proof. Python-only (reads the ledger).
- **Bookkeeping.** Suite at 167 modules, all green; Wolfram extension unchanged at 224/224; `origin_theory` (residual-inventory keybox) and `next.txt` moved together.

## 2026-06-13 (premise A maximal honest closure: $A_2$ by assembly, $R_1 = \text{GATE.QGEO}$ , zero independent open gates, **v165**)

- **New module `v165_premise_a_closure.py` [C]/[O].** Pins the two residual gates of the `v160-v162` (A)-chain to their floor.  $A_2$  (**net existence**) **closes by assembly** of established mathematics [C]: the 16 free Majoranas give a free-fermion conformal net (Buchholz–Mack–Todorov), the holomorphic  $c=8$  lattice net is  $(E_8)_1$  (Frenkel–Kac–Segal / Dong–Xu / Kawahigashi–Longo), the  $\mu_4$  extension is a Longo  $Q$ -system of index 4 (**v125**),  $\mu(B)=16/16=1$ ,  $248=120+128$ , character  $E_4/\eta^8=j^{1/3}$  (**v156**); the one TFPT-specific input (seam Calderón kernel = free-fermion two-point function) is done via  $\text{DtN} = |k|$  (**v156/v157**).  $R_1$  (**seam clock**) **reduces to `GATE.QGEO` [C]**: a  $\mu_4$ -equivariant flat gapped generator is *unique* ( $= A_0^*$ , **v115/v116**), so the gap  $(2/3)^6$  is forced and the residual is the one geometric identification “seam boundary =  $\mathbb{P} \setminus \mu_4$ ” = `GATE.QGEO` (cohomology established, **v137**,  $b_1=3=N_{\text{fam}}$ ). **Final accounting:** by the No-Unit Theorem (**v153**) the irreducibles stay  $\{\pi, v_{\text{geo}}\}$ ; premise (A) = ( $A_2$  assembly) + ( $R_1=\text{GATE.QGEO}$ ) + the  $\{\pi, v_{\text{geo}}\}$  core, so it carries *zero independent open gates* and ceases to be its own gate — a unification and re-typing, *not* an unconditional proof (ledger `GATE.METRIC.19`). Python-only (numpy/sympy + stdlib).
- **Bookkeeping.** Suite at 165 modules, all green; Wolfram extension unchanged at 224/224; `tfpt_research_contracts` (premise-A keybox) and `next.txt` moved together.

## 2026-06-13 (the final identification: premise A factors into the already-open A2 + R1, no new open content, **v162**)

- **New module `v162_seam_transport_identification.py`** [E]/[O]. Pushes the **v161** residual as far as it rigorously goes and shows it carries *no new open content*. The one-particle symbol splits into a gap value ( $\beta$ ) and a fixed space ( $\alpha$ ). ( $\beta$ ) **is forced**: a  $\mu_4$ -equivariant generator is diagonal in the cusp basis (the deck average is the diagonal projection,  $\sum_k U^k X U^{-k} = |\mu_4| \text{diag } X$  for  $U = \text{diag}(1, i, -i)$ ; the commutant of  $U$  is exactly the diagonal algebra), so its spectrum is the cusp weights  $\text{spec}(A_0^*) = \{0, \frac{1}{3}, \frac{2}{3}\}$  (**v115**) and the transfer eigenvalue  $(1-\alpha)^{p_2}$  with  $p_2=6=|R_+(A_3)|$  gives  $\{1, (2/3)^6, (1/3)^6\}$ , gap  $6 \log \frac{3}{2}$  — all carrier data; the lift to the 16 Majoranas is the  $A_3$  subnet grading (**v137**). ( $\alpha$ ) is the rank-8 Calderón polarisation of  $\omega_k = |k|$  (**v113/v156**). **Closure**: with **v160+v161**, (A) closes given only  $\mu_4$ -equivariance + admissibility, whose two non-derived inputs are the already-open A2 (net existence, Kawahigashi–Longo) and R1 (seam clock, **HOR.CLOCK**). So premise (A) factors exactly into  $A_2 + R_1$  — it adds *no* independent open item and is closed *conditionally* on them (a unification, not an unconditional proof; ledger **GATE.METRIC.18**). Python-only (numpy/sympy + stdlib).
- **Bookkeeping**. Suite at 162 modules, all green; Wolfram extension unchanged at 224/224; **tftpt\_research\_contracts** (premise-A keybox) and **next.txt** moved together.

## 2026-06-13 (the cone reduces to one particle: premise A’s scope premise discharged to a 16-dim statement, **v161**)

- **New module `v161_one_particle_reduction.py`** [E]/[O]. *Discharges* the scope premise of **v160** (“the transport is gapped on the *full* Schwinger cone”) via Bogoliubov second quantisation. A quasi-free transport is  $T_\Sigma = \Gamma(t)$ , the second quantisation of a one-particle contraction  $t$  on the 16-dim Majorana space;  $\Gamma(t)$  acts on the degree- $m$  correlator sector as the exterior power  $\Lambda^m(t)$ . Machine-checked: (1)  $\text{spec}(\Gamma(t)|_{\deg m}) =$  products of  $m$  one-particle eigenvalues (compound-matrix theorem, all  $m=0..6$ ); (2) **the cone gap equals the one-particle gap** — the sub-leading eigenvalue over the whole cone is  $(2/3)^6$  exactly, so premise 5 on the cumulant space *is* premise 5 on the 16-dim space; (3) the eigenvalue-1 sector is  $\Lambda^*$  of the fixed subspace (disconnected Wick powers), with uniqueness closed by **v113** (one polarisation  $\Leftrightarrow$  one vacuum); (4) quasi-freeness is *forced* — the  $120 = \dim \mathfrak{so}(16)$  chiral currents are the Bogoliubov generators and **v158** makes every non-Gaussian deformation irrelevant. **Consequence**: **v160**’s three full-cone premises collapse to ONE finite (16-dim) one-particle statement (seam symbol = gapped Calderón Bogoliubov map, fixed = rank-8  $K_\Sigma$ , gap  $(2/3)^6$ ); all numbers supplied, only the physical identification (Kawahigashi–Longo class = the A2 net residual) stays open — the infinite-dimensional cone is eliminated (ledger **GATE.METRIC.17**). Python-only (numpy + stdlib).
- **Bookkeeping**. Suite at 161 modules, all green; Wolfram extension unchanged at 224/224; **tftpt\_research\_contracts** (premise-A keybox) and **next.txt** moved together.

## 2026-06-13 (premise A as a conditional fixed-point theorem; the load relocates to one scope premise, **v160**)

- **New module `v160_seam_gaussianity_from_pf.py`** [E]/[O]. Formalises the proposed closure of premise (A) (“the seam bulk is a reflection-positive free field”) as a fixed-point theorem and types it honestly. The *exact* parts: (1) a quasi-free state has *all* higher connected cumulants  $\kappa_{2n}=0$  — verified by the *signed* set-partition/Pfaffian moment  $\leftrightarrow$  cumulant inversion on a concrete pure Majorana covariance (the **v113** “one kernel = net” property at the cumulant level); (2) a kernel-preserving transport fixes the whole quasi-free state by Wick functoriality; (3) the Perron–Frobenius dichotomy (a *fixed*  $\kappa_{2n}$  is a second fixed ray  $\Rightarrow$  breaks uniqueness; a *non-fixed* one decays by the gap) forces Gaussianity; (4) the qualitative claim

needs only  $\text{gap} > 0$ , the value  $(2/3)^6$  only sets the rate. **The honest residual: v56's** gapped PF uniqueness lives on the *three-dimensional* scalar transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$ , while the 16-Majorana seam already carries  $\binom{16}{4}=1820$  and  $\binom{16}{6}=8008$  higher-cumulant directions — so (A) reduces to the single internal, exact-testable statement “the admissible TFPT seam transport is primitive + spectrally gapped on the *full* Schwinger cone, not only the readout spectrum of v56”, which the suite does *not* yet establish (ledger GATE.METRIC.16). Python-only (numpy + stdlib); no new exact-identity content for the Wolfram mirror.

- **Bookkeeping.** Suite at 160 modules, all green; Wolfram extension unchanged at 224/224 (v160 is numerical, Python-only); `tfpt_research_contracts` (the premise-A keybox) and `next.txt` moved together.

### 2026-06-13 (changelog date-order audit + reorder pass)

- **Changelog sort enforced.** `audit_sync.py` now checks that every dated `\subsection × {YYYY-MM-DD...}` block in this file is non-increasing (newest first; range titles such as 2026-06-07 / 2026-06-08 sort by the later date). A manual prepend had left five 2026-06-12 entries above seven 2026-06-13 blocks — reordered; future mis-sorts fail `build.sh audit` as `[B.changelog]`.

### 2026-06-13 (premise A: the destination fixed point is provably stable, v158)

- **The free chiral  $c=8$  fixed point is provably stable (v158, exact dimension count).** The finite core of (A)'s “reaches-the-fixed-point” residual: the chiral Majorana has  $h=\frac{1}{2}$ , the bilinear current  $h=1$ , the quartic  $h=2$ . The relevant window  $0 < h < 1$  for bosonic deformations is *empty*, the  $h=1$  currents are chiral (v157, not  $(1,1)$ -marginal), and the lowest interaction (quartic,  $h=2$ ) is irrelevant — so the fixed point is isolated and stable against all interactions. Premise (A) reduces to a basin-of-attraction statement with a *proven* stable target (ledger GATE.METRIC.15). *Tooling note:* PyR@TE (4d gauge-Yukawa RGEs) is the wrong tool for this 2d boundary-CFT residual, but the right tool for the  $F_{\text{transfer}}$  interfaces ( $\eta_B$  leptogenesis RGEs, gauge-coupling running, the  $b_1=41/10$  cross-check).
- **Bookkeeping.** Suite at 158 modules, all green; Wolfram extension 224/224; contracts / `origin_theory` / `next.txt` / website moved together.

### 2026-06-13 (premise A reframed: freeness is a rigid boundary fixed point, v157)

- **The free-bulk premise becomes a fixed-point property (v157, simplicity-first).** Instead of postulating Gaussianity, three facts click: (i) the DtN/Calderón symbol is *universally*  $|k|$  (Lee-Uhlmann — order-1  $\Psi$ DO, principal symbol  $|\xi|_g$ ; interactions live in the irrelevant lower-order symbol), so the free dispersion is bulk-detail-independent; (ii) a holomorphic  $c=8$  theory is *rigid* — every field has  $\bar{h}=0$ , so no marginal  $(1,1)$  field exists and the fixed point is isolated, with no knob for an interaction; (iii) the same  $|\mathbb{Z}_2|$  that halves  $8\pi$  in  $c_3 = \frac{1}{8\pi}$  is the Ramond/GSO projection giving  $\mu=1$  ( $248=120_{\text{NS}}+128_{\text{R}}$ ). With  $(E_8)_1$  unique (v83), premise (A) shrinks from “prove the seam is Gaussian” to “the gapped ( $\Delta=6\log\frac{3}{2}$ ) boundary flow reaches the holomorphic fixed point” — freeness then forced by rigidity, not fine-tuned (ledger GATE.METRIC.14).
- **Bookkeeping.** Suite at 157 modules, all green; Wolfram extension 223/223; contracts / `origin_theory` / `next.txt` / website moved together.

### 2026-06-13 (Route 3: the seam net constructed and $(E_8)_1$ verified, v156)

- **The constructive route carried to its exact endpoint (v156).** “ $B \cong (E_8)_1$ ” is no longer an assertion: the chiral seam net is built explicitly and the identification *checked*. (a) The

Dirichlet-to-Neumann (Calderón) operator of the 2d Laplacian is the free chiral dispersion  $\omega_k = |k|$  (harmonic extension  $e^{ikx-|k|y}$ ); (b) 16 free Majoranas give  $c = 8 = 5+3$ ; (c) the weight-1 currents are  $248 = 120_{\text{NS}} + 128_{\text{R}}$  ( $SO(16)$  bilinears + Ramond spinor, Frenkel–Kac); (d) the holomorphic character  $E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + 34752q^3 + 213126q^4) = j^{1/3}$  (verified by  $q$ -series) — the 248 at level 1 is  $\dim E_8$ , so the net *is*  $(E_8)_1$ . *Residual*: exact given (i) the free-bulk premise (DtN =  $|k|$ , **v155**) and (ii) the operator-algebraic existence of the net from the seam kernel (a Kawahigashi–Longo-type theorem, not a finite computation) — where TFPT meets constructive QFT, now sharply located (ledger **GATE.METRIC.13**).

- **Bookkeeping.** Suite at 156 modules, all green; Wolfram extension 222/222 (the DtN =  $|k|$  identity, the  $(E_8)_1$  character  $E_4/\eta^8 = j^{1/3}$  and the  $248=120+128$  split mirrored); contracts / redteam (Target A) / **origin\_theory** / **next.txt** / website moved together.

## 2026-06-13 (the seam-side identification reduced to one shared premise, **v155**)

- $G_{\text{net}}$ ’s last premise reduces to a single shared physical premise (**v155**, Python-only). The seam-side identification of the Simple-Current Extension Theorem (**v154**) splits (**v110**) into (a) “the Calderón involution is sheet-odd” (the double-cover deck, structural) and (b) “the seam 2-point kernel is quasi-free”. Part (b) is *not* independent: the boundary marginal of a reflection-positive *Gaussian* bulk is the compression  $PTP$  of the covariance, and the compression of a contraction is a contraction — so the seam state is automatically quasi-free (Wick inherited). Hence (b) *follows* from “the seam bulk is a reflection-positive free field”, the *same* premise already load-bearing in the area law (**v59**) and the EH mechanism (**v150/v151**). **The entire seam-side residual is therefore one physical premise**, shared by  $G_{\text{net}}$ , the area law and R3 — not reducible by a finite computation, but no longer three separate items (ledger **GATE.METRIC.12**).
- **Bookkeeping.** Suite at 155 modules, all green; Wolfram unchanged at 221/221 (**v155** is numerical/numpy, Python-only); contracts / **origin\_theory** / **next.txt** / website moved together.

## 2026-06-13 (external review formalised: No-Unit Theorem + Simple-Current Extension Theorem, **v153–v154**)

- $v_{\text{geo}}$  **re-typed: the No-Unit Theorem (**v153**)**. Every load-bearing compiler datum is mass-dimension 0 and invariant under a unit rescaling  $L \mapsto \lambda L$ , whereas a mass /  $1/G$  is not — so a dimensionless boundary compiler *provably cannot* select an absolute scale.  $v_{\text{geo}}$  is therefore an *irreducible metrology primitive*, not an open gap, and the three apparent residuals collapse onto it ( $U_{\text{point}} \sim v_{\text{geo}}$ ,  $1/G \sim v_{\text{geo}}^2$ ,  $m/\mu = e^{3/4}$  a pure ratio); the irreducibles stay {one unit,  $\pi$ } (ledger **ANCHOR.VGEO.02**).
- $G_{\text{net}}$  **stated as the central closing theorem (**v154**)**. The Simple-Current Extension Theorem:  $A=(D_5)_1 \otimes (A_3)_1$  extended by the isotropic glue  $L=\langle(1,1)\rangle$  has  $[B:A]=|L|=4$ ,  $c=5+3=8$ ,  $\mu(B)=\mu(A)/|L|^2=16/16=1 \Rightarrow$  holomorphic  $\Rightarrow B \cong (E_8)_1$  ( $SO(16)_1$  excluded). Every algebraic ingredient is machine-checked (**v89/v92/v125/v143/v148**); the only residue is the seam-side identification. Pulled to the front of **tfpt\_research\_contracts**, **tfpt\_5\_redteam** (Target A final form) and **origin\_theory** (ledger **GATE.METRIC.11**).
- **Status surfaces re-typed** (review): the canonical status card now reads  $v_{\text{geo}}$  as a metrology *primitive*,  $G_{\text{net}}$  as the simple-current extension  $\cong (E_8)_1$  (**[E]/[C]**), and  $F_{\text{transfer}}$  as one functor with four typed interfaces — “not structural gaps”. The final one-line framing (dimensionless boundary compiler; irreducibles  $\{\pi, v_{\text{geo}}\}$ ; one seam-net theorem; frontier =  $F_{\text{transfer}}$ ) is in **origin\_theory**, the introduction, the README and **next.txt**.
- **Bookkeeping.** Suite at 154 modules, all green; Wolfram extension 221/221; intro/README/**origin\_theory**/contracts/redteam/status-card/website moved together.

## 2026-06-13 (currency sweep: remaining old-series “Paper N” cross-references removed)

- **No old-version framing left in the active docs.** A consistency sweep found a residual layer of old-series “Paper N” attributions and removed them: four literal “Paper 6” references (`tfpt_4_frontier` leptogenesis/axion, `tfpt_2` seam determinant), the old “Paper 4” (OS/admissibility) and “Paper 5” (APS/Hodge) references in `tfpt_2`, the old “Paper 3” transport-kernel attributions in `tfpt_2`, and the internal “Paper 1/2/4” cross-references in `tfpt_3`, `tfpt_4` and `tfpt_horizon_readouts` — all rewritten as content-based phrasing (the flavor sector, the architecture document, the  $E_8$  decoder, the anchor characteristic polynomial, etc.). The active documents now attribute results to the current documents / axioms / scripts only, as the workflow rule requires; the machine-checked sync (scripts  $\leftrightarrow$  papers  $\leftrightarrow$  ledger  $\leftrightarrow$  website) was already current (152 scripts, AUDIT OK).

## 2026-06-13 (fifth round: the R3 normalisation is the dimensionful anchor, **v152**)

- **R3: the  $q(A_3)$  normalisation merges into the one anchor (**v152**).** The replica EH coefficient is a *log-ratio*  $k = \ln(m/\mu)/(12\pi)$  ( $\ln m$  alone is scale-ambiguous, only  $m/\mu$  is physical); pinning  $k = c_3/2$  fixes the dimensionless ratio  $m/\mu = e^{3/4}$ , and in  $d=2$  that ratio *is* the induced Newton constant  $1/G$  (**v68**). So  $v_{\text{geo}}$  (**v78**),  $1/G$  (**v68**) and  $m/\mu$  are *one* dimensionful anchor in three readings — the irreducibles stay {one scale,  $\pi$ }, and SEAM.THEOREM.01’s residual reduces to the kernel-identification premise alone. *Audit (not promoted)*: the log-ratio is numerically  $\frac{3}{4} = q(A_3)$ , the target value the anchor takes in seam units — not a derivation (ledger SEAM.EHMODEL.03).
- **Bookkeeping.** Suite at 152 modules, all green; Wolfram extension 219/219; intro/README/origin\_theory/next.txt/website moved together.

## 2026-06-12 (Zenodo version label: semantic + git rev)

- **Release version string.** `build.sh` now writes `tex-artefacts/release-version.txt` as `{TFPTversion}` (`rev {git count}`) (matches `\TFPTdatestamp` without the date). `scripts/zenodo_upload.py` and `build.sh zenodo/zenodo-publish` use this label on Zenodo; cursor rules extended for when `zenodo_description.html` must move with deposit-facing changes.

## 2026-06-12 (Zenodo API upload workflow)

- **Automated Zenodo versioning.** Added `scripts/zenodo_upload.py` and `.github/workflows/zenodo-release.yml`: creates a new draft version of record 20579275 (DOI 10.5281/zenodo.20579275), uploads the ten active paper PDFs, refreshes `zenodo_description.html`, bumps the version from `tex-artefacts/version.tex`, and optionally publishes via GitHub Actions (ZENODO\_TOKEN secret).

## 2026-06-12 (Overleaf shadow mirror: tfpt5-shadow)

- **One-way GitHub mirror for Overleaf sync.** Added `scripts/export-shadow.sh` and `.github/workflows/shadow-sync.yml`: on every push to main, a filtered copy is pushed to `sthamann/tfpt5-shadow` (papers, `tex-artefacts/`, `figures/`, `verification/`, `experiments/`; excludes `_archive/`, `website/`, all `*.pdf`, `.cursor/`).  $\sim 270$  files /  $\sim 3$  MB — within Overleaf GitHub-sync limits.

## 2026-06-12 (origin\_theory style alignment + box reduction)

- **Fewer boxes, embedded in text.** `origin_theory` from 22 to 16 boxes: the narrative/interpretation callouts (“Why the interpretation is internally consistent”, “Dependency chain of the birefringence test”, “The precise reformulation”, “The three theses honestly graded”, “The whole story”, “One-line summary”) become run-in bold prose; the boxed reformulation becomes a **keyeq**. The exact result keyboxes (**v53–v56**, **v101**, **v105**, **v113**, ...) and the master-principle / Seam–Horizon target boxes are kept.
- **Colour-marked module references + marker hygiene.** `\vref` applied to the inline `vN` references; stale literal `[A]`/`[I]`/`[L]` markers in running text replaced by the 4-class **[O]**/**[E]**.

## 2026-06-12 (tfpt\_5 / horizon / research-contracts pass + build hardening)

- **Forked document-set box removed (horizon).** `tfpt_horizon_readouts` carried a hand-written, *stale* copy of the document-set card (“five short documents”, missing `tfpt_5`) and the canonical `\input{tex-artefacts/tfpt_docset}`; the fork is deleted so the single-source card (all five papers + three companions) appears once.
- **Fewer boxes, embedded in text.** Narrative/result callouts converted to run-in bold prose (with the `\veri{}` citation moved out of the bold title into the body): `tfpt_horizon_readouts` from 22 to 7 boxes (the ten-box “clock” wall becomes prose); `tfpt_research_contracts` from 19 to 10 (the nine progress/result winboxes become prose, the named-theorem keyboxes kept); `tfpt_5_redteam` from 10 to 9 (the outcome summary; the per-target verdict boxes are kept as the structured red-team output).
- **Colour-marked module references + marker hygiene.** `\vref` applied to the inline `vN` references in all three documents; the horizon “Scope/typing” marker line and a stale literal `[A]` in the research contracts updated to the 4-class markers.
- **Build hardening.** `build.sh` now removes a document’s `.aux/.toc/.out` on failure, so a half-written cross-reference file from a crashed compile cannot poison the next build; artefacts are still kept on success.

## 2026-06-12 (fourth round: the Calderón transfer answered — the BFK split, **v151**)

- **R3: the Calderón/DtN kernel is conically clean (**v151**).** Reflection doubling derives that the Kac corner constant is boundary-condition independent ( $C_N = C_D$ ), so by the Burghilea–Friedlander–Kappeler gluing ledger the conical deficit of the two-sided Calderón jump determinant *vanishes identically* ( $C_{\text{cone}}(\gamma) - 2C_D(\gamma/2) = 0$ ): the seam-reduced effective action inherits the **v150** EH form *verbatim*, with all of it localised in the *local* half-space determinants — the “Calderón version” demanded by **v150** needs no separate derivation. SEAM.THEOREM.01 narrows to the  $q(A_3)$  normalisation plus the standing kernel premise; the gate stays **[O]** (ledger SEAM.EHMODEL.02).
- **Bookkeeping.** Suite at 151 modules, all green; Wolfram extension 218/218; intro/README/origin\_theory/next.txt moved together.

## 2026-06-12 (third round: both normals anchor data; the R3 mechanism exists, **v149–v150**)

- **R4': both dual normals are anchor data (**v149**).** In the cusp frame the torsion normal pairs to  $(6, 3, 5) = (p_2, p_0, e_2)(a)$  — one anchor invariant per  $Q_+$  line (the cusp matrix is unimodular, so this determines  $n$  uniquely); with  $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$  the dual normal pair is anchor data through and through. Equivalences to the  $(2, 8, 121)$  frame data and the  $w_0$ -lift exact. R4' residue = *one* discrete assignment (ledger GATE.UWALL.14).

- **R3: the mechanism exists at the gapped-model level (v150) — the first movement on SEAM.THEOREM.01.** The conical heat-trace deficit is exact and  $t$ -independent (Cheeger image sum  $(N^2-1)/(12N)$ ); for a *gapped* operator the  $\zeta$ -regularised replica variation is *finite and cutoff-independent*,  $\Delta \log \det' = 2C(\gamma) \ln m$ , and its linearisation is the Einstein–Hilbert form  $(\ln m/12\pi) \int \sqrt{g} R$ . The v73 normalisation becomes one exact target equation:  $k = c_3/2 \Leftrightarrow \ln m = \frac{3}{4} = q(A_3)$  (audit, not asserted). Remaining: the Calderón (boundary) version and that normalisation — the gate narrows but stays **[O]** (ledger SEAM.EHMODEL.01).
- **Bookkeeping.** Suite at 150 modules, all green; Wolfram extension 217/217; status surfaces (intro reductions + card, `origin_theory`, contracts, `next.txt`, website) moved together.

## 2026-06-12 (R1–R5 second round: every class at its floor, v145–v148)

- **R4': the pairing values are atom identities (v145).** Reading the v139 lift structurally ( $n = w_0(\sigma) + |\mu_4|e_3$ ,  $w_0 =$  the  $A_3$  exponent duality), all three pairings become atom arithmetic — including the *new* closure  $|\mu_4| + g_{\text{car}} = N_{\text{fam}}^2$  ( $4+5=9$ ,  $\sigma \perp w_0(a)$ ) and  $\|\sigma\|^2 = 110 = |\mathbb{Z}_2|g_{\text{car}}\|\text{Pl}(K)\|_1$ . R4' = derive *one* map (ledger GATE.UWALL.13).
- **R5: the Möbius  $D_4$  realisation (v146).** The *full* dihedral symmetry  $\langle z \mapsto iz, z \mapsto 1/z \rangle$  of the seam curve acts on  $H^1$  exactly as the integer model:  $\iota^*$  is the exponent duality with parity +1 on the self-conjugate line ( $= T_A$  in the cusp basis, glue-swap parity), and  $\Sigma$  is the  $\delta\iota$  class — the premise reduces to the module identification itself (ledger FLAV.QGEO.07).
- **R1: the clock is a Gaussian zero-mode integral, and the bend is  $\det'$ -clean (v147).** Variance ratio  $(1 - \alpha)$  (forced by norm = area) gives the Born-squared ratio  $(1 - \alpha)^{p^2}$ , the  $\ln$  series *is* the v127 ring sum; the two quantum weights share *one* Nariai geometry, so the bend  $\log_{3/2} 3$  carries no determinant correction — the v144 obstruction is localised to the  $\alpha = 0$  reference (ledger HOR.CLOCK.13).
- **R2: the NS/R sector census (v148).** The untwisted (Neveu–Schwarz) module carries only the *even* discriminant classes (integer weights), so the odd  $\mu_4$  simple currents have zero NS support — they are Ramond modules; the R zero-mode module (256) splits 128+128 into the odd sectors of the *two* Lagrangian glues: choosing the glue *is* choosing the R-projection ( $248 = 120_{\text{NS}} + 128_{\text{R}}$ ); the one remaining net statement is irreducibly twisted-sector (ledger GATE.METRIC.10). *Correction note (same day)*: the first version of v148 graded the Ramond labels by the sum  $q_D + q_A$  (not the glue label) and called that module untwisted; the census was rewritten as above — conclusion unchanged, support corrected, the R-sector sheet split is new.
- **Bookkeeping.** Suite at 148 modules, all green; Wolfram extension 215/215 (README + GATE.WOLFRAM.02 + website counter in lock-step); status surfaces (intro card, `tfpt_1`, `origin_theory` third update, `next.txt`) moved together.

## 2026-06-12 (external review integrated: $\Lambda$ typing, marker migration, website parity)

- **$\Lambda$  normalisation made unambiguous.** The external review flagged the adjacent reduced/unreduced displays in `origin_theory` as a type trap (the formulas were individually correct —  $\bar{M}_{\text{P1}}$  vs  $M_{\text{P1}}$  — but easy to misread). Both normalisations are now shown explicitly side by side with their order counts ( $3/(4\pi^2) \Rightarrow 120.147$  reduced;  $3/(256\pi^4) \Rightarrow 122.948 = 119.028 + 3.920$  unreduced) plus a typing note.
- **Stale references cleaned.** “How each of the seven papers simplifies”  $\rightarrow$  “How the legacy layers simplify (bridge map)” with the lead-in “(Paper 6)” reference replaced by the scalaron-reheating chain (v86); “(Paper~6/7)” in the `tfpt_3` bridge table reworded; the `tfpt_1` “single remaining geometric input (H2)” passage and the `tfpt_2` “single remaining input (U)” passage

carry dated status updates pointing at the current decomposition (v118–v122, v139, v141, v142, v75).

- **Marker migration completed in the shared figures.** The claim stack, anchor cockpit,  $E_8$  glue,  $2/3$  motif and  $K_5$  figures (`tex-artefacts/tfpt_figures.tex`) now show the four public classes `[E]/[C]/[O]` instead of the legacy `[A]/[I]/[L]/[B]/[P]/[R]` badges.
- **Red-team front table pulled to the final state.** The Target-A residual column now states the one remaining statement (boundary-net holomorphy +  $c=8 \Leftrightarrow$  index-4 inclusion; v83/v87/v89, Lie-level realisation v143); the historical three-residual form is kept below, dated.
- **Website parity.** The Red Team document is now a first-class entry (`papers.ts` number 5, slug `redteam`) and the changelog PDF a download row; Appendix H / Origin Theory / Research Contracts are labelled *companions* (never “Paper 5–7”); the marker system on every website surface migrated to `[E]/[C]/[O]/[X]` (fine types named in prose; the script index keeps quoting the scripts’ internal fine-grained typing); the verification page counters are live (144 checks generated from the registry via `lib/suite.ts`, Wolfram 116/116 + 211/211 — now guarded by the sync audit) and the reproduce block matches the current pipeline. The `papers.ts` section bodies of the affected papers carry the v141–v144 outcomes (sheet question: deck derived;  $G_{net}$ : Lie-level realisation; resummed clock: in-family det-ratio cancellation; selector triangle: two line pairings).
- **Not adopted** (recorded for transparency): the review’s narrative reframings (anchor-algebra trilogy, “one seam clock” framing, one-page structure) and the front-matter/status-card layout changes are editorial decisions left to the authors; the claimed  $\Lambda$  error was in fact a correct-but-treacherous notation switch (see above).

## 2026-06-12 (R1–R5 attack round: four residual classes sharpened, v141–v144)

- **R5 closed at the discrete level (v141, deck selection theorem).** The v140  $\mathbb{Z}_3$  choice is *derived*: the established integer deck  $G = T_A \Sigma$  (v97/v98) pairs the  $Q_+=1$  cusp line with the self-conjugate character 2, so only the sheet-twisted assignment  $(2, 3, 1) = \text{chars}(-U)$  survives; the cusp exponential  $i^{Q_+}$  is excluded (same spectrum, wrong grading pairing). Explicit conjugacy constructed; under  $k \rightarrow -k$  only the established sheet  $\mathbb{Z}_2$  flips. Consequence: Euler grading =  $Q_+ \circ \rho$ . GATE.QGEO keeps only its realisation premise — no discrete freedom left (ledger FLAV.QGEO.06).
- **R4’ narrowed: three pairings  $\rightarrow$  two (v142, frame integrality).** Integer covectors in the  $(1, a, \sigma)$  dual frame form an index-11 sublattice ( $x - 3y + z \equiv 0 \pmod{11}$ ; the index is  $\|\text{Pl}(K)\|_1$ ); with  $(x, y) = (|\mathbb{Z}_2|, \text{rank } E_8)$  the  $\sigma$ -pairing is forced  $\equiv 0 \pmod{11}$ , and primitivity forces the line parameter even. Cramer reductions for both line pairings recorded as restructuring (ledger GATE.UWALL.12).
- **R2 realised at the finite Lie level (v143, graded Frobenius).** The  $\mathbb{Z}_4$ -graded hull carries exact coset duality ( $-C_1 = C_3$ ; Killing pairing nondegenerate per  $g_k \times g_{-k}$ ), the glue average is exactly the carrier projector (index 4), and each glue sector is one Weyl orbit (simple currents, fusion =  $\mathbb{C}[\mathbb{Z}_4]$ ) — the v125 Q-system realised by the hull; residue = one conformal-net statement (ledger GATE.METRIC.09).
- **R1’s det-ratio step derived in-family (v144).**  $e_2$ -rigidity of the traceless SdS cubic gives  $r_b r_c = 1 - \Delta^2/3$  exactly; with the v132 anomaly the non-zero-mode determinant ratio is  $(1 - \Delta^2/3)^{4/3}$  — no first-order term in the horizon split (the v131 cancellation derived, not assumed); deficit coefficient  $\frac{32}{27} = |\mu_4|(\frac{2}{3})^3$  (audit). Finite-weight absorption stays `[C]` with the  $6 \rightarrow \frac{14}{3}$  obstruction stated sharply (ledger HOR.CLOCK.12).
- **R3 attempted, honestly unchanged.** No defensible computation landed for the seam-determinant  $\rightarrow$  EH step (SEAM.THEOREM.01); the gate keeps its `[O]` typing untouched.

- **Wolfram mirror extended to v141–v144:** extension file now 211/211 (GATE.WOLFRAM.02 and the wolfram README updated in lock-step); Python suite at 144 modules, all green.

## 2026-06-12 (sync infrastructure: single-source script index, content maps, one audit)

- **Single-source script index.** The master script→check table (`tex-artefacts/verification.tex`) and the website `ScriptIndex.tsx` are now *generated* from one registry (`script_registry.csv` + `script_clusters.csv` in `verification/`) by `make_script_index.py`; both targets carry a generated-file header and are never edited by hand.
- **Content maps without splitting the papers.** `verification/make_docs_map.py` generates `docs_map.csv` (every paper section with line range, the vN scripts cited there, a content hash and a last-changed date) and `website_map.csv` (which website file mentions which scripts/documents) — the machine-readable enumeration of all sync surfaces.
- **One sync audit.** `verification/audit_sync.py` bundles and extends the previous loose shell checks, now in *both* directions: `suite` ↔ `run_all.py` ↔ `registry` ↔ `ledger`; every script cited in a paper *body* (the master index alone no longer counts); no stale `\veri/\vref` targets; every new module mentioned in this changelog; website mirror byte-identical (PDFs, scripts, `release.ts` hashes, version stamps); wolfram counts; overfull boxes. Frozen exceptions live in `audit_baseline.json` (remove-only).
- **Build pipeline.** `build.sh` gains `gen` / `website` / `audit` / `release` steps; `notes` now regenerates the single-source surfaces first; `website` copies all active PDFs (now incl. `tfpt_5_redteam.pdf` and `changelog.pdf`, with new `release.ts` entries) and the vN scripts, and stamps version, date and git revision into `website/lib/version.ts` (shown in the site header) and `release.ts`.
- **Citation gaps closed.** The audit immediately found three scripts cited only in the master index: `v17` and `v85` are now cited in their `tfpt_2` sections (hexagonal resolvent; master cover), and two short `\veri{v19}` citations were expanded to the full script name. `v91` (spine tetrahedron) had no paper section at all — it is now integrated: a keybox in `tfpt_3` (“Three views of one spine”: anchor closure, edge/face products, volume 120,  $K_6$  negative control, the tautology/sub-grammar caveats) and a closing note in the `tfpt_1` Spine Quotient Lemma; the ARCH.SPINE.01 ledger location is corrected to `tfpt_1;tfpt_3` and the baseline exception list is empty again.
- **Rules.** `.cursor/rules` updated (`tfpt-workflow`, `website-sync`) and a new `sync-maps` rule added describing the agentic procedure: *regenerate* the maps first, enumerate the affected surfaces from them, audit as the exit gate. The mirror map (`website_map.csv`) also covers the root `README.md` and `next.txt` (bare vN ids resolved to full script names), and both files are named explicitly as status surfaces that move with every finding.

## 2026-06-12 (tfpt\_3 / tfpt\_4 style alignment + box reduction)

- **Shared header/footer on tfpt\_3.** `tfpt_3_e8_audit_bootstrap` dropped its custom `\footmarkers` footer (the post-collapse four-[E] artefact) and inherits the shared `tfpt_style` header/footer with the version stamp; the box-orphan `needspace` guards are kept.
- **Fewer boxes, embedded in text.** Narrative/commentary callouts converted to run-in bold paragraphs (boxed one-line formulas to `keyeq`): `tfpt_3` from 39 to 27 boxes (e.g. “Purpose and discipline”, the five “(i)–(v)” audit-lemma boxes, “Net”, “The connection in one sentence”, “The unification”, “The question and the honest answer”, “Net: two axioms”); `tfpt_4` from 18 to 7 (the seven-box Koide stack and the two dark-matter conjecture boxes become prose). The per-item “Honest status of ...” keyboxes are kept — `tfpt_4` is the status authority for the frontier.

- **Colour-marked module references.** `\vref` applied to the inline `vN` references in `tfpt_3` and `tfpt_4` (TikZ node names excluded).
- **Marker hygiene.** A stale literal `[P]` in `tfpt_4` (Koide flow-time) replaced by the 4-class `[C]`; status tables checked against the ledger.

## 2026-06-12 (tfpt\_1 / tfpt\_2 style alignment + box reduction)

- **Shared header/footer everywhere.** `tfpt_1_architecture_e8` dropped its custom `\footmarkers` footer (which after the marker collapse showed four redundant `[E]`) and now inherits the shared `tfpt_style` header/footer with the version stamp, identical to every other document.
- **Fewer boxes, embedded in text.** The narrative/commentary callouts of both documents are converted to run-in bold paragraphs (and the boxed one-line formulas to the colour-marked `keyeq`): `tfpt_1` from 21 to 14 loud boxes (+3 `keyeq`); `tfpt_2` from 52 to 43 (e.g. “In one sentence”, “The one-paragraph version”, the four “Sharpening” notes, “What the simplification means net”, “The disciplined main statement”, “What improves...”, “Scope and honesty”). Only genuine results / theorems / gates keep a box.
- **Colour-marked module references.** `\vref` now also colours the inline `vN` references throughout `tfpt_1` and `tfpt_2` (TikZ node names accidentally matching the pattern were excluded).
- **Status currency.** `tfpt_2`’s inflation-pivot table gains the `v86` scalaron-reheating point ( $N_\star=51.4 \Rightarrow n_s=0.9611, r=0.0045, A_s$ -disfavoured) and states the band-of-record explicitly, matching `COSMO.NSTAR.01` and the introduction; the box is re-typed `[C]`. The other status tables were checked against the ledger and are current under the 4-class markers.

## 2026-06-12 (marker simplification to 4 classes + status reconciliation)

- **Four-class display markers.** The reader-facing marker system is collapsed from eleven symbols to four: `[E]` exact / machine-proven (identity, Lie/lattice, formalised, numerical FP), `[C]` conditional (physical / bridge / readout under named hypotheses), `[O]` open / axiom, and `[X]` kill test. `tex-artefacts/tfpt_style.tex` defines `\mE/\mC/\mO/\mX` and keeps the legacy `\tI ... \tX` names as aliases that render their class. All marker call-sites across the nine documents and the shared fragments were rewritten, collapsing same-class combinations (e.g. `[I]/[L] \rightarrow [E]`, `[N]/[P] \rightarrow [E]/[C]`). The fine-grained per-claim type (Axiom / Formal / Lattice / Numerical / Identity / Physical) is unchanged in `status_ledger.csv` — the single source of truth — so no fidelity is lost. The marker key, badge bar, README marker table and the workflow rule were updated to match.
- **Status tables reconciled against the ledger.** The introduction’s inflation/ $N_\star$  rows are made consistent with `v86/COSMO.NSTAR.01`:  $r$  and  $n_s$  are shown as the frozen *bands* over  $N_\star \in [50, 60]$  with the scalaron-reheating conditional point  $N_\star=51.4$  ( $n_s=0.9611, r=0.0045$ ), replacing the stale point values  $N_\star=55$  (predictions table) and  $N_\star \simeq 57$  (residual card); the closure-ledger inflation row is re-typed `[E] \rightarrow [C]` (conditional on  $M_{\text{Star}}=M_{\text{scal}} + \text{reheating}$ ), matching the other two tables and the ledger `[P]` typing.

## 2026-06-12 (version system, predictions/K5, readability pass B3)

- **Authors, auto-date and auto-version on every paper.** New single source `tex-artefacts/version.tex` defines `\TFPTauthors` (Stefan Hamann, Alessandro Rizzo), a curated `\TFPTversion`, and a title/footer date stamp (`\today + version + revision`). `build.sh` writes the auto build revision (git commit count) into the gitignored `tex-artefacts/version-auto.tex`, so every document’s title page and page footer carry

author, date and v5.2 (rev N) automatically. All nine active documents and `changelog.tex` now set `\author{\TFPTauthors}`.

- **Builder keeps build artefacts.** `build.sh` no longer deletes the `.aux/.log/.out/.toc` after a successful build (the `.log` is needed for the overfull-box audit and the `.aux/.toc` for correct cross-references on the next incremental build).
- **Predictions table outsourced.** The running/upcoming-experiment predictions table is moved from `introduction.tex` into the shared fragment `tex-artefacts/predictions.tex` (`\input` in the predictions section).
- **K5 figure fixed.** The `\TFPTactionkfive` mnemonic ( $K_5$  on the five carrier slots) is redrawn so the complete graph and the Pascal-row reading list sit side by side instead of overlapping.
- **Colour-marked script references.** New `\vref` macro colours inline module references (e.g. `v108–v113`, `v49/v71`) in the introduction, predictions and verification fragments, matching the blue of the `\veri` citation, so machine-checked references are visible at a glance.
- **Lighter, non-breaking, embedded boxes.** The `tcolorbox` families are restyled to a light tint + thin frame + light title band with dark coloured title (no heavy white-on-saturated bars); a neutral `navbox` carries the document map and a colour-marked `keyeq` sets off load-bearing display formulas. The introduction’s callouts use the non-breaking variants (`keyboxnb/winboxnb/gapboxnb`) so no box splits across a page, and the shared status card is non-breaking.
- **Readability / sectioning.** The dense run-in prose of the introduction is broken into `\subsection*` headings (In plain words; one anchor and  $E_8$  as a compiler hull; the companion documents; the compact status formula; the hard reduction; the Pascal action grammar; the closure ledger) with added whitespace.
- **Orphan results linked to scripts.** The Glue theorem, the “Reduction in one line” and the fermion-spectrum callouts now carry inline `\veri{}` citations (`v1`, `v6/v14/v23`, `v18/v20/v46`).

## 2026-06-12 (tex-artefacts refactor + introduction visual pass B2)

- **Shared imports moved to tex-artefacts/.** The shared, imported-only TeX fragments (`tfpt_style.tex`, `tfpt_figures.tex`, `tfpt_docset.tex`, `tfpt_status.tex`) now live in a dedicated `tex-artefacts/` folder; all nine active documents `\input` them via `tex-artefacts/<name>` (and `tfpt_style` loads `tex-artefacts/tfpt_figures`). `changelog.tex` stays at the repo root because it is also a standalone `build.sh` notes deliverable. The `make_manifest.py` TEX list and the workflow rule were updated to the new paths.
- **Master script index outsourced (tex-artefacts/verification.tex).** The long `vN→check` table is moved out of `introduction.tex` into the shared fragment `tex-artefacts/verification.tex` (`\input` in the verification appendix); the paper-consistency audit now also reads that file, so every registered script stays cited exactly once.
- **Introduction box reduction (B2).** The introduction’s eleven `keyboxes` are reduced to the single headline “In one sentence” card; the explanatory `keyboxes` (*In plain words*, *Sharper still*, *compact status formula*, *hard reduction*, *Cosmology Pascal*,  $E_6 \times A_2$ , *closure ledger*, *Plausibility balance*) are converted to inline run-in paragraphs, the *Glue theorem* is re-typed as a `winbox` (a result, not a claim), and the explanatory *plausible/follows next* `winboxes` become prose. The loud boxes now carry only genuinely load-bearing callouts (one `keybox`, five `winboxes`, one `gapbox`) plus the two shared orientation cards.

- **Document-count consistency.** The introduction’s document map gained the missing `tfpt_5_redteam` row; “seven/eight companion documents” and the ledger “six documents” wording are corrected to the true count (nine active documents); `README.md` (“10 ok”, 140 scripts) and the workflow rule (“9 docs”) are reconciled.

## 2026-06-12 (visual pass: box reduction B1)

- **Box reduction (B1).** The 26 non-load-bearing aside boxes (audit / recorded / interpretation: `auditbox/audbox/intbox/resbox/roadbox`) across `tfpt_1`, `tfpt_2`, `tfpt_3` and `origin_theory` are converted to inline bold run-in paragraphs — the reviewer’s prescription that an audit fingerprint must not carry the same visual volume as a theorem. The loud box families are now only `keybox` (claim), `winbox` (result) and `gapbox` (open/caveat).

## 2026-06-12 (visual pass: shared figures, badges, marker typology)

- **Shared figure library `tfpt_figures.tex`** (`\input` by `tfpt_style`): reusable TikZ macros for the architecture *claim stack* (`\TFPTclaimstack[zone]`, B3), the *anchor cockpit*  $a = (1, 1, 2)$  (B4), the  $E_8$  *glue* diagram (B5), the  $2/3$  *motif map* (B6) and the  $K_5$  *action lattice* (B7). The claim stack and a marker *badge bar* are now shown near the front of *every* document, each highlighting that document’s layer; the topical figures are placed in their natural homes (cockpit +  $K_5$  in the introduction, cockpit + glue in `tfpt_1`, motif in the horizon note).
- **Marker typology (A2).** `tfpt_style` adds the sharper sub-types `[E]` (exact algebra), `[E]` (exact number), `[C]` (readout / standard physics in TFPT units) to the existing `[C]` (bridge) and `[X]` (kill test); the marker key and the new `\TFPTbadges` bar show the full typology. (Inline body markers retain `[E]` where they are genuine exact identities.)

## 2026-06-12 (review pass on list.txt)

- **Claim hygiene (review section A).** The introduction’s one-sentence claim is softened from “derives all constants” to “constructs a discrete compiler; physical readouts run through named, status-typed bridges”; the ledger zombie in `FLAV.QRATIO.01` is removed (the statement no longer says “stays [P]” — ratios are `[E]` closed on the derived stratum, only the absolute scale stays `[O]`, with an explicit forbidden-wording note); the seam misalignment is written explicitly as  $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}} = c_3 + 36c_3^4$  (leading  $c_3$ , fourth-order puncture correction exact);  $\theta_{12}$  has a single prediction of record  $0.306747$  with the seam/non-linear values typed as scheme diagnostics; JUNO’s first 59.1-day result ( $0.3092 \pm 0.0087$ , compatible, precision pending) is recorded and NuFIT 6.0 is marked the pre-JUNO baseline; the ACT DR6 birefringence hint carries a systematics caveat (not a validation target); CODATA-2022 is dated; the null-model figure is moved out of the main text and typed as a grammar-internal bound. “Complete solutions of the five open questions” becomes “Closure status of the five former open questions”.
- **Interface language unified (review sections A5/C2/C3/C6/C7/C8).** The live residual is stated everywhere as  $v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$  (historical  $U_{\text{wall}}/G_{\text{metric}}/F_{\text{frontier}}$  kept only for ledger continuity); `tfpt_research_contracts` is retitled and its recommended order rewritten (selector-triangle pairings  $\rightarrow v_{\text{geo}} \rightarrow G_{\text{net}}$ );  $F_{\text{transfer}}$  is named as one missing functor with  $\text{Koide}/\eta_B/\text{axion}/m_p/m_e$  as its four instances; full QG closure is framed as a certification layer, not a prerequisite. Document numbering aligned with file names (the Red Team paper is now in the canonical set; document  $N \equiv \text{tfpt}_N$ ). Website mirrored throughout (`papers.ts`, `predictions.ts`, `faq.ts`, `glossary.ts`, `OpenGates`, `Hero`, orientation components).

## 2026-06-12 (later)

- **Shared style and central status artifact.** Two new single-source fragments: `tftpt_style.tex` (the canonical palette, status markers, marker key, `\veri` reference, the four `tcolorbox` families with back-compat aliases, and one running header/footer) is now `\input` by all eight documents — the per-paper duplicated preambles are gone; and `tftpt_status.tex` (“TFPT status at a glance”) gives one canonical overall-status card imported near the front of every document, replacing the divergent per-paper status statements. Both registered in `build.sh` and the manifest TEX list.

## 2026-06-12

- **Margin theorem (v122).** The established frozen selectors ( $D_4$  annihilator  $n = (5, -9, 6)$ ,  $\det R = 8$ ,  $\det K = 4$ ) pin  $R$  uniquely among hexagon matrices ( $64 \rightarrow 12 \rightarrow 1$ ); the atom margins become theorems — H2 introduces no new residual class.
- **Inventory update (v123).** Residual table re-pinned post v110–v122: exactly five structural classes (R1 clock, R2 index-4 net, R3 EH step, R4' selector establishment — replacing the discharged H2 dictionary — and R5  $Q$  realisation); R2 carries three loads (metric + carrier + QBL: one theorem, three doors).
- **Resummed clock + glue  $Q$ -system (v124/v125).** R1's bend in closed form:  $\text{rate}(n) = -p_2 \ln(1 - n/N_{\text{fam}})$  (pole at  $N_{\text{fam}}$  forces the three-level spectrum); the index-4  $\mu_4$  extension exists explicitly as a Longo  $Q$ -system on  $\mathbb{C}[\mathbb{Z}_4]$  (all Frobenius axioms exact) — R2 sharpens from “find” to “identify”.
- **Clock–wall bridge (v126) and ring resummation (v127).** The resummed-clock weights are the Mehta–Seshadri parabolic weights ( $\text{spec } A_0^*$ );  $\det(\mathbf{1} - A_0^*) = \frac{2}{9}$  = the entropy curvature; the clock is a log-determinant (RPA rings, one tower per hexagon site, coupling = parabolic weight).
- **Graded hull (v128).**  $E_8$  is a  $\mathbb{Z}_4$ -graded Lie algebra over the carrier (grading = glue  $Q$ -system; exact on all 6720 root-pair sums); ring multiplicity 6 = sheet-doubled Ginsparg–Perry zero modes. *Notation fix in the same round:* the clock factor 6 is  $p_2(a)$ , not  $2p_2$  — labels corrected across modules, ledger, papers, website (numbers unchanged).
- **Entropy power law  $\rightarrow$  zeta budget (v129–v133).** The clock is  $\Gamma_n \propto (S_n/S_{dS})^{p_2}$  (Gibbons–Hawking form);  $p_2 = 2h$  with  $h = N_{\text{fam}}$  from mode counting + the Born rule (v130); the zero-mode norm is the area,  $\|Y_{1m}\|^2 = A/(4\pi)$  exactly (v131); the det-ratio anomaly is the Koide constant,  $\zeta(0)|_{\det'} = -\frac{2}{3}$  (v132); both  $\zeta(0)$  budget routes computed exactly — the reduced seam route carries pure anchor ratios ( $-\frac{4}{3}$  = minus the seed gain), the naive 4d route ( $-\frac{109}{45}$ ) carries no atom (v133).
- **Dual anchor + determinant surface (v134/v135).**  $d := a^\top R^{-1} = a^\top L^{-1} = (-\frac{1}{2}, -\frac{1}{2}, 1)$  with  $(1, 1, -2) = -2d$  — the inverse flavor response is the Nariai root, invariance exact via Sherman–Morrison (third algebraic flavor–horizon bridge leg);  $\det M(s, t) = 3s(t+1)(t+2) + t^2 + 5t + 8$  with iso-volume walls at the  $\mathbb{Z}_2/\mu_4$  atoms, winding line  $(8, 14, 20)$ ,  $\det F = 32$ , row-budget axes with  $\text{rowsum}(V) = \text{Spec}(Q_+)$ ; micro-identities  $41 = 16 + 25$ ,  $52 = 16 + 36$ .
- **Dual-normal selector,  $Q_+$  cohomology, Volkov–Wipf firewall (v136–v138).**  $(d, n)$  pins  $R$  *columnwise* (each column the unique lattice point of the address box; kernel  $(6, 8, 7)$ ); the  $A_0^*$  zero mode carries the spectral normal  $\sigma = (2, -9, 5)$  with a  $W(A_3)$  orbit control excluding the naive lift;  $H^1(\mathbb{P}^1 \setminus \mu_4) = \chi_1 \oplus \chi_2 \oplus \chi_3$  with degrees  $(1, 2, 3) = \text{Spec}(Q_+) =$  the  $A_3$  exponents and cusp weights =  $\text{spec } A_0^*$ ; the external full-4d graviton/ghost value  $-\frac{98}{45}$  is pinned (arXiv:2506.02142 = Volkov–Wipf hep-th/0003081) and its *edge* part is exactly two copies of the reduced seam budget — R1 typed closed as an edge-sector object.

- **Selector triangle + canonical map (v139/v140).**  $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$  is pure anchor data (first selector *derived*); the frame  $(\mathbf{1}, a, \sigma)$  has volume 11 and  $n$  is the unique covector with atom pairings  $(2, 8, 121)$  — the  $R4'$  residue is three pairings; the  $H^1 \rightarrow$  generation equivariant map exists and is Schur-rigid (only sign-twisted  $\mu_4$  actions match), so the `GATE.QGEO` residue collapses to one  $\mathbb{Z}_3$  choice.
- **Editorial and consistency passes.** Content-currency sweep (superseded “single remaining input” claims retired across contracts, paper 3, intro, website); introduction: self-explanatory title (spells out Topological Fixed-Point Theory), the old-version comparison removed throughout, redundant diagrams dropped, the unreadable status heatmap removed, the script-index master table converted to a page-breaking `xltabular` (it had silently truncated  $\sim 110$  rows), status card extended with the v134–v140 reductions, predictions table completed (CP phase, cosmic birefringence); the canonical document-set box extracted into the shared `tfpt_docset.tex`; paper 1 cleaned of all 35 old-series references, every section now opens with its machine-check script references; smaller box fonts across all eight documents; this `changelog.tex` introduced as a dual-mode (standalone + imported) document.

## 2026-06-11

- **External-review integration (v83) and release hygiene.** Manifest `-check` mode and the release rule; holomorphy pins  $(E_8)_1$  (the unique even unimodular rank-8 lattice theory).
- **Sheet diamond and centered form (v94/v95).** All four flavor operators are points of one two-parameter surface  $M(s, t)$ ; centered cross  $R, L = C \mp U$ ,  $K, F = C \mp V$  with the cofactor seam normal  $n = (5, -9, 6)$ ; documentation pass (“The Sheet Diamond and the Winding Line” in paper 3).
- **Horizon/anchor round (v101–v103).** The maximal (Nariai) black hole is the anchor: roots  $(1, 1, -2)$ , entropy bound  $\frac{2}{3} = |\mathbb{Z}_2|/N_{\text{fam}}$ ; one repeller/attractor orientation in both sectors; trisection normal form (the canonical cosine coordinate).
- **Clock/ladder round (v104–v108) + Lean hardening.** The classical Nariai clock speaks anchor ( $\chi_{\text{clock}} = (\lambda - 1)(\lambda + 2)$ ); machine-pinned residual inventory; first review validation; quantum-clock target made quantitative; Pascal Ladder Theorem ( $g = 2K + 1 \Leftrightarrow$  the closure).
- **QBL theorem chain (v109–v113).** Sheet-Pairing Lemma; Calderón-sheet selection (a scalar seam datum exists iff the involution is sheet-odd); Quadratic-Transport Theorem (degree exactly 2); Self-Counting Channel (the Pascal closure as an identity); quasi-free kernel (one 2-point kernel determines the whole net, rank =  $c$ ) — the QBL input merges with the  $R2$ /holomorphy premise; communicated across all status surfaces.
- **$U_{\text{wall}}$  made exact (v114–v118).** Torsion normal form and  $\delta = \frac{1}{2}$  theorem; exact anchor residue  $A_0^*$  (diag =  $a/|\mu_4|$ , off-weights  $(8, 0, 5)/144$ ); resonance theorem ( $M_\infty = \mathbf{1} \iff a_{13} = 0$ , one gauge orbit); the holonomy is the exact 24-element  $W(A_3) = S_4$ ; the hyper-charge hexagon is the sign-twisted family spectrum with the lepton coefficients as resolvent determinants.
- **Second review validation + address pinning (v119–v121).** Anchor ratio triad  $(\frac{2}{3}, \frac{4}{3}, \frac{5}{3})$ ; the 121 audit lemma; the lepton words are the compiler atoms  $(8, 5, 3)$  with divmod-hexagon addresses;  $R$  is the unique hexagon matrix with atom margins and  $\det 8$  — the address table carries zero residual information.

## 2026-06-10

- **B/C/D round closed.** Koide attractor toy model (v93), glue uniqueness (v92), the Wolfram extension as the independent second verification path, Lean hardening; papers and website synced with the round.

- **Content rounds v96–v100.** Sheet conjugation bridge, discriminant dictionary, Koide flow time, and the look-elsewhere-corrected numerology null test (v100: joint null probability  $\leq 10^{-30.7}$ , conditional on the declared grammar).

## 2026-06-09

- **Blind registry frozen (v84).** Machine-enforced prediction registry (`predictions_frozen.json`): exactly one  $\theta_{12}$  number of record before JUNO; conditional entries carry their hypotheses by name. Research-contracts document added to the set.

## 2026-06-07 / 2026-06-08

- **Compiler-closure consolidation.** The TFPT development is consolidated into the current document set with the verification suite (modules v1–v82:  $E_8$  glue, carrier Pascal, EM fixed point, flavor matrix, cascade, bootstrap, gravity/cosmology, transport, masses, registry and gate audits) as the single proof layer; manifests (`manifest.sha256`, `lean_manifest.sha256`) as content identity; website mirrors the suite (interactive DAG, in-browser script reproducer, script index).

## 2026-04-27

- **Initial commit.** TFPT papers and website baseline: electromagnetic closure and flavor transport (UFE bridge, birefringence seed, dyonic intercept), PDF-download tracking, accessibility and metadata passes.